Non-Formalized Research Frontiers for the Fabius Function

Consolidated natural-language dossiers awaiting Lean verification

Research quarantine for the `ProveIt` Fabius development
Eleven source notebooks merged without promotion to the formalization-backed exposition

Consolidated repository snapshot: 25 August 2026

Abstract

This volume consolidates every mathematical document formerly stored in a subdirectory of `Analysis/FabiusFunction/docs/non-formalized-research-frontiers/`. The dossiers contain natural-language proofs, symbolic derivations, numerical checks, conjectural extensions, and explanations built on top of the current Lean library. They are gathered here so that no unformalized claim can be mistaken for part of the verified primary exposition or lost while awaiting a machine-checked proof.

The merge is intentionally conservative. Overlapping dossiers and alternative proof routes remain visible, with editorial precedence notes where one source supersedes a weaker one. Standalone preambles, title pages, local tables of contents, and conflicting labels have been normalized, but mathematical bodies, local status warnings, citations, tables, algorithms, plots, and open-problem lists are retained. Editorial corrections made during consolidation are stated locally rather than silently changing the mathematical record.

Formalization boundary. Nothing is promoted merely by appearing in this volume. A claim may enter `docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex` only after an exact Lean theorem has been checked for the full hypotheses and conclusion, the declaration and module have been recorded, and the corresponding frontier obligation has been dispositioned. A related definition, a weaker theorem, numerical agreement, or a paper citation is not sufficient. Some dossiers quote formalized inputs; that does not formalize their later deductions.

Contents

Consolidation map and reading conventions

Each part below is a source dossier. The source-path banner and SHA-256 comment preserve provenance. Cross-reference and bibliography keys have a dossier prefix solely to make the formerly standalone documents coexist. Repeated material is retained when it supplies an independent derivation, a stronger later form, a sharper constant, or a useful warning.

Notation is local to each dossier unless an explicit crosswalk says otherwise. In particular, the Appell polynomials and the saddle logarithmic coefficients both use an A-symbol in their source notebooks; c_n denotes a centered moment in the dyadic web but a finite Thue–Morse row in the repeated-integration dossier; and the signed global extension, Toeplitz weights, and Taylor blocks also have source-local spellings. The Fourier convention printed in each dossier controls its own factors of 2π. These collisions are preserved as provenance, not intended as global identifications.

Dossier	Principal subject	Consolidation status
Integration frontiers	Inverse, probability bridges, discrete engines, saddle machinery	Broad mixed-status notebook; use its local audit boundary.
Repeated integration	Box splines and finite repeated-integration iterates	Distinct construction and sharp-rate research program.
Repeated integration, sharp thresholds	Refined derivative thresholds and exact extremizers	Later formulation; preferred threshold where the two repeated-integration dossiers overlap.
Dyadic calculus	Appell, q-binomial, determinant, transform, and discrete-limit formulas	Earlier proof route; retained for unique derivations.
Dyadic web	Unified exact dyadic and q-Richardson picture	Preferred later formulation where dyadic dossiers overlap.
Dyadic asymptotic bridge	Vertical coefficient extraction, phase locking, and fixed-numerator rays	Independent bridge retained for its coefficient/Bromwich derivations.
Dyadic formulae to asymptotics	Decompression, moment–Laplace transfer, and lower-Lambert asymptotic	Newest and preferred dyadic-to-asymptotic synthesis.
Thue–Morse rate	Elementary finite bounds and binary-phase error	Independent quantitative route.
Thue–Morse rates	Fourier deconvolution and sharp/all-order expansions	Preferred sharper formulation where rate dossiers overlap.
Inverse and saddle	Total inverse, endpoint inversion, and saddle remainder architecture	Mixed Lean inputs and derived frontier consequences.
Small argument	Explicit periodic coefficients, amplitude laws, numerics, and traps	Specialist coefficient notebook and open-problem ledger.

Lean object crosswalk—inputs, not certification

Object	Representative Lean name	Principal module family
Bounded Fabius function	fabiusReal, IsFabius	Basic.lean, PaperStatements.lean
Rvachev up-function	rvachevUp	Basic.lean, Differential.lean
Signed global extension	extendedFabius	GlobalExtension.lean
Exact dyadic evaluator	fabiusDyadicValue	Arithmetic.lean, PaperStatements.lean
Binary and q-binomial series	fabiusReductionSum and global q-series declarations	TaylorReduction.lean, FabiusGlobalQBinomialSeries.lean
Thue–Morse grid samples	correctedPrefixGridSample	ThueMorseApproximation.lean
Totalized inverse	fabiusInv, fabiusInv_differentiableAt_iff, fabiusInv_contDiffAt_infty_iff, fabiusInv_analyticAt_iff	FabiusInverse.lean, InverseNotElementary.lean
Lower-Lambert coordinate	fabiusLambertPhase	FabiusLambertSaddle.lean
All-orders saddle expansion	saddle mass/log coefficient declarations	FabiusFullAsymptoticExpansion.lean and related modules

Explicit nonclaims. The crosswalk identifies formal inputs and nearby APIs; it is not a claim that every formula in the corresponding dossier is formalized. In particular, the sharp quantitative Thue–Morse rates, repeated-integration norm identities, inverse endpoint asymptotic, general ordered-composition coefficient formulas, Fourier-amplitude estimates, and any claim marked derived, numerical, conjectural, or open remain in this frontier until an exact Lean declaration is audited.

Part I
Fabius Integration Research Frontiers

Source provenance and status. Former path: `Fabius_Integration_Research_Frontiers/Fabius_Integration_Research_Frontiers.tex`. Broad mixed-status dossier. It supplies inverse, probability, discrete-engine, and all-orders saddle machinery. Claims already represented in Lean are inputs; the dossier-level generalizations and consequences remain quarantined here.

Original source SHA-256: `21222ae5a8c64cf556dac562fd66943ae0b6ed881408e23e37edfa9113bdecbd`.

Dossier abstract

The long article *The Fabius Function and Rvachev's Up-Function* gives a broad human-readable account of the formal development in `ProveIt`. Its breadth is also its unavoidable weakness: several new branches of the Lean library postdate the article, and several generic proof engines are named only in the module guide because they have no counterpart in the exposition. This companion supplies those missing parts.

The first part develops the total inverse of the bounded Fabius function, transports infinite flatness into infinite endpoint steepness, and proves the strong non-elementarity results: no elementary function can agree with the Fabius function, Rvachev's up-function, or the signed global extension on any set with nonempty interior in the relevant support. The second part fills in the measure-theoretic bridges between the product probability law, its survival function, tilted Laplace moments, finite atomic laws, and half-weighted step approximants. The third part presents two reusable discrete engines omitted from the main article: weighted path sums for triangular recurrences and a finite-row Toeplitz convergence theorem. The opening audit also records a newly formalized redundancy in the original compact-support characterization: positivity of its dilation parameter follows from the remaining hypotheses. The final and largest part exposes the formal and analytic machinery behind the all-orders saddle expansion: parameter-dependent Poincaré expansions, recursive exponential and logarithmic coefficients, exact finite polynomial remainders, Gaussian contraction, uniform polynomial tails, the closed Stirling description of the saddle jets, and the explicit second periodic correction.

The purpose is complementarity rather than repetition. Definitions and headline theorems already proved in detail in the main article are recalled only when they are needed as input. Every substantial assertion below is either a direct human-readable rendering of a current Lean declaration, a proof-level explanation of a generic module that the main article explicitly omits, or a clearly identified elementary corollary of those declarations.

Formalization boundary. This is a research-frontier document. It preserves the source draft used during integration, including exact Lean-backed results, proof sketches, ordinary-analysis corollaries, stronger generalizations, and formulas that still lack exact declarations. Its presence under `non-formalized-research-frontiers` is intentional: no mathematical assertion in this notebook should be attributed to the formal library merely because a related definition or weaker theorem exists. The primary exposition contains the audited, declaration-backed subset. Among the obligations retained here are Hölder consequences beyond the now-formalized inverse-curvature identities, generic moment-matrix consequences, arbitrary-family Toeplitz wording, generic Gaussian coefficient bounds, and the passage from the mass top jet to the logarithmic highest-derivative law.

1 Scope, conventions, and the audit boundary

Let $F : \mathbb{R} \to [0,1]$ denote the bounded Fabius function, totalized in the usual way by $F(x) = 0$ for $x \le 0$ and $F(x) = 1$ for $x \ge 1$. Let

$$\mathrm{up}(x) = F(1 - |x|) \tag{1}$$

be Rvachev's compactly supported up-function, and let $\widetilde{F}$ denote the signed global extension used in the Lean development. On $[0,1]$ the basic identities are

$$F(1-x) = 1 - F(x), \qquad F'(x) = 2\,\mathrm{up}(2x-1), \qquad F'(x) = 2F(2x) \quad (0 < x < \tfrac{1}{2}). \tag{2}$$

These are inputs here, not topics to be reproved.

The repository snapshot underlying this supplement is the following commit on the `codex/fabius-both-papers` branch, dated 25 August 2026.

`531338d79f2f640041562138c7b80772ee2f0490`.

The main article itself records an earlier pinned snapshot and, near the beginning of its module guide, explicitly says that four clusters have "no counterpart anywhere in the article": Gaussian contraction and Gaussian tails, the formal saddle exponential and logarithm algebra, the abstract Toeplitz row lemma, and the generic weighted-path solver. Those four clusters receive complete mathematical chapters below. Two additional clusters, the inverse and non-elementarity developments, are absent because they were added after the article's pinned snapshot. The current snapshot also adds a proof that positivity of the dilation parameter in the original compact-support characterization is redundant. Finally, several bridges are present only as one- or two-sentence transitions; they are expanded here into proof-level arguments.

1.1 Three levels of coverage

To keep the audit precise, each topic belongs to one of the following levels.

Status	Meaning in this companion
New branch	A current Lean module whose mathematical content does not occur in the main article: principally `FabiusInverse.lean`, `ElementaryFunction.lean`, `NotElementary.lean`, and `ProbabilityLaplaceMoments.lean`.
Missing engine	A generic theorem family that the main article deliberately leaves in the module guide: path sums, Toeplitz rows, formal coefficient recurrences, finite remainders, and Gaussian polynomial estimates.
Compressed bridge	A statement whose endpoints occur in the main article but whose intermediate measure theory, topology, or asymptotic algebra is suppressed.

1.2 Notation that must not collide

The exact half moments are denoted by d_n, following the main article. The bounded saddle exponent jets are instead denoted by

$$Z_n(t), \tag{3}$$

although their Lean declaration is `negativeLaplaceBoundedExponentJet`. This avoids using the same letter for two wholly different sequences. We write $L = \log 2$, Ψ for the smooth one-periodic correction in the negative-Laplace exponent, and $\lambda(x)$ for the exact lower-Lambert phase at small positive x.

Formal boundary convention. When a statement is a theorem in the current Lean library, the corresponding declaration is named in the margin of the discussion or in the concordance in section 5.13. When a statement is an immediate classical calculus corollary but is not presently packaged as a declaration, it is called a *corollary in ordinary analysis*. The inverse second-derivative formula and its strict closed-half shape are now packaged formally; the distinction remains relevant for general higher inverse derivatives and for the transfer of the highest-jet coefficient from mass coefficients to logarithmic coefficients.

1.3 The positivity hypothesis in the original characterization is redundant

The compact-support characterization used in the first Arias de Reyna paper [3] assumes a smooth function $\phi : \mathbb{R} \to \mathbb{R}$, a real dilation parameter k, and

$$\operatorname{tsupp} \phi = [-1, 1], \qquad \phi(x) > 0 \quad (-1 < x < 1), \qquad \phi(0) = 1, \tag{4}$$

together with the derivative law

$$\phi'(x) = k\big(\phi(2x+1) - \phi(2x-1)\big). \tag{5}$$

The paper states $k > 0$ as a hypothesis. In the current formal development that field is retained for source fidelity, but a smart constructor proves that its positivity follows from the other assumptions.

Theorem 1.1 (Redundancy of scale positivity)**.** *Suppose $\phi \in C^\infty(\mathbb{R})$ satisfies (4) and (5) for some $k \in \mathbb{R}$. Then $k > 0$. Consequently these data determine an instance of the full original-characterization predicate without a separate positivity proof.*

Proof. The support identity and continuity imply $\phi(-1) = 0$: the function vanishes on $(-\infty, -1)$, and -1 is a limit point of that interval. The mean-value theorem on $[-1, 0]$ therefore gives a point $c \in (-1, 0)$ such that

$$\phi'(c) = \frac{\phi(0) - \phi(-1)}{0 - (-1)} = 1. \tag{6}$$

Now $2c - 1 < -1$, so the support condition gives $\phi(2c - 1) = 0$, whereas $-1 < 2c + 1 < 1$, so interior positivity gives $\phi(2c + 1) > 0$. Evaluating (5) at c yields

$$1 = k\phi(2c+1). \tag{7}$$

The second factor is positive, hence $k > 0$. ☐

This is `IsOriginalFabius.mk_of_derivative_law`. The proof is small, but it clarifies the logical content of the original theorem: the sign of the scale is forced before the later Fourier argument identifies its exact value $k = 2$.

2 The inverse and the elementary-function obstruction

2.1 The total inverse of the bounded Fabius function

The main article proves that F is a continuous strictly increasing bijection of $[0, 1]$ onto itself. The inverse development turns that qualitative fact into a reusable global object whose endpoint behavior can be estimated exactly.

2.1.1 Order-isomorphism construction and totalization

Let $F_{[0,1]}$ be the restriction of F to the subtype $[0,1]$. Strict monotonicity and surjectivity package it as an order isomorphism

$$\mathfrak{F} : [0,1] \simeq_o [0,1]. \tag{8}$$

Define the clamping map

$$\pi(y) = \min(1, \max(0, y)) \tag{9}$$

and the total inverse

$$I(y) = \mathrm{val}(\mathfrak{F}^{-1}(\pi(y))). \tag{10}$$

The declarations `fabiusOrderIso` and `fabiusInv` formalize this construction. Lean uses `Set.IccExtend` rather than spelling out (9).

Theorem 2.1 (Global inverse package)**.** *The function $I : \mathbb{R} \to \mathbb{R}$ has the following properties.*

1. $0 \le I(y) \le 1$ *for every* $y \in \mathbb{R}$.
2. I *is continuous and nondecreasing on* $\mathbb{R}$.
3. I *is strictly increasing on* $[0,1]$ *and maps that interval bijectively onto itself.*
4. *For* $x, y \in [0,1]$,
$$F(I(y)) = y, \qquad I(F(x)) = x. \tag{11}$$
5. *The clamped tails are*
$$I(y) = 0 \quad (y \le 0), \qquad I(y) = 1 \quad (y \ge 1). \tag{12}$$

Proof. The inverse of an order isomorphism is again an order isomorphism, hence continuous for the order topologies and strictly increasing. The interval projection π is continuous and nondecreasing, so the composite (10) is continuous and nondecreasing on all of $\mathbb{R}$. Its values lie in $[0,1]$ because the codomain of $\mathfrak{F}^{-1}$ is that subtype. On $[0,1]$ the projection is the identity, and the two inverse identities are the usual *apply–inverse–apply* laws for $\mathfrak{F}$. If $y \le 0$, then $\pi(y) = 0$ and $\mathfrak{F}^{-1}(0) = 0$ because $F(0) = 0$; the right tail is identical. □

The comparison laws are often more useful than direct substitution.

Proposition 2.2 (Comparison calculus)**.** *For* $x, y \in [0,1]$,

$$I(y) \le x \iff y \le F(x), \qquad I(y) < x \iff y < F(x), \tag{13}$$

$$x \le I(y) \iff F(x) \le y, \qquad x < I(y) \iff F(x) < y. \tag{14}$$

Proof. Apply the strictly increasing map F to the first comparison, or the strictly increasing map I to the second, and use (11). The strict versions are the same argument with $<$ in place of $\le$. □

2.1.2 Reflection and exact inverse values

The reflection symmetry of F survives totalization without a domain restriction.

Theorem 2.3 (Global reflection)**.** *For every* $y \in \mathbb{R}$,

$$I(1-y) = 1 - I(y). \tag{15}$$

In particular

$$I(0) = 0, \qquad I(1) = 1, \qquad I\left(\frac{1}{2}\right) = \frac{1}{2}. \tag{16}$$

Proof. If $y \le 0$ or $y \ge 1$, the identity follows directly from the two constant tails. Suppose $0 < y < 1$. Both $I(1-y)$ and $1 - I(y)$ lie in $[0,1]$. Applying F and using the reflection law for F gives

$$F(1 - I(y)) = 1 - F(I(y)) = 1 - y = F(I(1-y)).$$

Injectivity of F on $[0,1]$ proves (15). Substituting $y = 1/2$ then forces the midpoint identity. □

Exact dyadic evaluation of F immediately becomes exact evaluation of I at a dense family of rational ordinates. Let $D_{n,a}$ denote the exact bounded dyadic evaluator, so that for $0 \le a \le 2^n$,

$$D_{n,a} = F\left(\frac{a}{2^n}\right) \in \mathbb{Q}. \tag{17}$$

The evaluator is clamped for $a > 2^n$.

Theorem 2.4 (Inverting the exact dyadic evaluator)**.** *For all $n, a \in \mathbb{N}$,*

$$I(D_{n,a}) = \frac{\min(a, 2^n)}{2^n}. \tag{18}$$

In particular,

$$F\left(\frac{1}{4}\right) = \frac{5}{72} \quad \Longrightarrow \quad I\left(\frac{5}{72}\right) = \frac{1}{4}. \tag{19}$$

Proof. For $a \le 2^n$, combine (17) with the left inverse identity in (11). For $a \ge 2^n$, the clamped evaluator equals 1 and the result is $I(1) = 1$. □

2.1.3 Formal interior calculus and curvature

The inverse is constructed using order topology alone. Once the positivity theorem for F' in the open unit interval is imported, the local inverse theorem supplies a sharper interior picture; the formal development now packages both the reciprocal derivative and full interior smoothness directly.

Corollary 2.5 (Interior derivative of the inverse)**.** *For $0 < y < 1$,*

$$I'(y) = \frac{1}{F'(I(y))} = \frac{1}{2\,\mathrm{up}(2I(y) - 1)}. \tag{20}$$

Thus I is C^∞ on $(0,1)$.

Proof. The point $x = I(y)$ lies in $(0,1)$, and strict positivity of $F'(x)$ there makes F a local C^∞ diffeomorphism. Differentiate $F(I(y)) = y$ and substitute the second identity in (2). □

Because F is strictly convex on $[0,1/2]$ and strictly concave on $[1/2,1]$, inversion reverses those two shapes.

Corollary 2.6 (Curvature of the inverse; formalized)**.** *The inverse I is strictly concave on $[0,1/2]$ and strictly convex on $[1/2,1]$. For every $0 < y < 1$,*

$$I''(y) = -\frac{F''(I(y))}{F'(I(y))^3}. \tag{21}$$

The midpoint is the reflection-symmetric change of curvature.

Remark 2.7. The first identity in corollary 2.5, its explicit up-form, strict positivity, and the resulting interior smoothness are the named Lean declarations `fabiusInv_hasDerivAt`, `deriv_fabiusInv`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, and `deriv_fabiusInv_pos`, together with `fabiusInv_contDiffOn_Ioo`. The reciprocal-cubic rule in corollary 2.6 is `deriv_fabiusInv_hasDerivAt` together with `deriv_deriv_fabiusInv`; its two strict closed-half conclusions are `strictConcaveOn_fabiusInv_firstHalf` and `strictConvexOn_fabiusInv_secondHalf`. These shape theorems use derivatives only on the interval interiors, so they do not assert finite endpoint derivatives.

2.2 Flatness of F becomes steepness of I

The inverse is most interesting at the two endpoints, where the classical formula (20) ceases to have a finite limit. The formal development proves this without differentiating the inverse: exact upper bounds for F are transported through monotonicity and then solved as inequalities for I.

2.2.1 Root-free inverse bounds

For $n \in \mathbb{N}$, the effective flatness theorem for F says that on the dyadic window $0 \le x \le 2^{-n}$,

$$F(x) \le 2^{\binom{n+1}{2}} x^n. \tag{22}$$

The sharp factorial refinement is

$$F(x) \le \frac{2^{\binom{n+1}{2}}}{n!} x^n. \tag{23}$$

Substitute $x = I(y)$ and use $F(I(y)) = y$.

Theorem 2.8 (Flatness transported through the inverse)**.** *Let $n \in \mathbb{N}$ and $y \in [0,1]$. If*

$$2^n I(y) \le 1, \tag{24}$$

then

$$y \le 2^{\binom{n+1}{2}} I(y)^n, \tag{25}$$

$$y \le \frac{2^{\binom{n+1}{2}}}{n!} I(y)^n. \tag{26}$$

It is enough to assume the argument-side condition

$$0 \le y \le F(2^{-n}). \tag{27}$$

Proof. Under (24), the point $x = I(y)$ belongs to the window of (22) and (23). Their left sides become $F(I(y)) = y$, giving the two conclusions. Under (27), the comparison calculus (13) gives $I(y) \le 2^{-n}$, which is exactly (24). □

If roots are allowed, (26) reads

$$I(y) \ge \left(\frac{n!}{2^{\binom{n+1}{2}}} y \right)^{1/n} \qquad (0 \le y \le F(2^{-n})). \tag{28}$$

The Lean statements retain the root-free form because all constants remain exact and no case split for real powers is required.

2.2.2 An effective infinite-slope estimate

The case $n = 2$ already forces the difference quotient to diverge. Since $2^{\binom{3}{2}} = 8$ and $F(1/4) = 5/72$, (25) gives

$$y \le 8 I(y)^2 \qquad \left(0 \le y \le \frac{5}{72}\right). \tag{29}$$

Theorem 2.9 (Effective domination of every slope)**.** *Let $M \in \mathbb{R}$ and $y > 0$. If*

$$y \le F(1/4) = \frac{5}{72}, \qquad 8M^2 y < 1, \tag{30}$$

then

$$My < I(y). \tag{31}$$

Proof. Assume contrariwise that $I(y) \le My$. The left side is nonnegative, so squaring preserves the inequality and gives $I(y)^2 \le M^2y^2$. Combining this with (29) yields

$$y \le 8M^2y^2.$$

Since $y > 0$, division by y gives $1 \le 8M^2y$, contradicting (30). □

No positivity assumption on M is necessary: for $M \le 0$ the conclusion is automatic from $I(y) > 0$, and the same algebra proves it uniformly.

Theorem 2.10 (Infinite one-sided endpoint slopes)**.** *The endpoint difference quotients diverge:*

$$\frac{I(y)}{y} \longrightarrow +\infty \qquad (y \downarrow 0), \tag{32}$$

$$\frac{1 - I(y)}{1 - y} \longrightarrow +\infty \qquad (y \uparrow 1). \tag{33}$$

Proof. Fix a target slope M. Replace it by $|M| + 1$ and choose $y > 0$ below both $F(1/4)$ and $[8(|M| + 1)^2]^{-1}$. Then theorem 2.9 gives $(|M| + 1)y < I(y)$, hence $M < I(y)/y$. This is exactly convergence to $+\infty$ in the order topology. For the right endpoint, put $z = 1 - y$ and use $I(1 - z) = 1 - I(z)$. □

Corollary 2.11 (Failure of endpoint Lipschitz and Hölder bounds)**.** *The inverse is not locally Lipschitz at* 0 *or* 1*. More strongly, for every $\alpha > 0$ and every $C > 0$, no estimate*

$$I(y) \le Cy^{\alpha} \tag{34}$$

can hold throughout a punctured right neighborhood of 0*; the reflected statement holds at* 1*.*

Proof. The Lipschitz claim follows immediately from (32). For the Hölder claim, choose n with $1/n < \alpha$. On the shrinking window $0 < y \le F(2^{-n})$, (28) gives $I(y) \ge c_n y^{1/n}$ with $c_n > 0$. If (34) held, then $c_n \le Cy^{\alpha - 1/n}$, whose right side tends to zero as $y \downarrow 0$, a contradiction. □

The windows $y \le F(2^{-n})$ shrink super-exponentially as n grows. One must therefore not read (28) as a lower bound uniform in n. Its strength is quantifier order: for each prescribed root exponent one obtains a sufficiently small window. That is exactly what is needed to defeat every fixed positive Hölder exponent.

2.3 Elementary functions through their analytic locus

The non-elementarity proof does not attempt differential algebra or Liouville theory. It uses a topological-analytic invariant that is both stronger and easier to formalize: every one-variable elementary function is analytic on a dense open set, whereas the Fabius family is nonanalytic at every point of the relevant interval.

2.3.1 The exact formal class

Definition 2.12 (The class Elem)**.** Let Elem be the smallest class of total functions $\mathbb{R} \to \mathbb{R}$ containing all constant functions and the identity and closed under

$$+, \quad \cdot, \quad -, \quad (\cdot)^{-1}, \quad f \mapsto f^r \ (r \in \mathbb{R}), \quad \exp, \ \log, \ \sin, \ \cos, \ \arcsin, \ \arctan, \tag{35}$$

where every unary operation is applied pointwise.

This is the inductive predicate `IsElementary`. Composition is deliberately not a constructor.

Proposition 2.13 (Closure derived from the generators)**.** *The class* Elem *is closed under composition. It is consequently closed under subtraction, division, natural powers, polynomial and rational functions, square roots, absolute values, tangent, hyperbolic functions,* arccos, arsinh*, and signed odd roots.*

Proof. Induct on the construction of the outer function g. Precomposing a constant or the identity with an elementary f is elementary. Every remaining constructor commutes with precomposition: for example $(g_1 + g_2) \circ f = (g_1 \circ f) + (g_2 \circ f)$, and similarly for the unary generators. This proves closure under composition. The remaining operations are identities in the generated algebra. For example

$$|f| = \sqrt{f^2}, \qquad \tan f = \frac{\sin f}{\cos f}, \qquad \sinh f = \frac{\mathrm{e}^f - \mathrm{e}^{-f}}{2}.$$

For an odd positive integer n, the signed root may be written

$$\frac{f}{|f|}|f|^{1/n},$$

with value zero at $f = 0$ under the total reciprocal convention. □

Remark 2.14 (Totalization makes the theorem stronger). Mathlib treats reciprocal, logarithm, real power, and inverse trigonometric functions as total maps on $\mathbb{R}$; outside their classical domains they receive specified extension values. Thus definition 2.12 is at least as large as the class represented by classical expressions on open domains. Proving that F does not belong to this larger class is a stronger statement, not a domain-convention loophole.

2.3.2 Analytic loci

For a function $f : \mathbb{R} \to \mathbb{R}$, define

$$\mathcal{A}(f) = \{x \in \mathbb{R} : f \text{ is real analytic at } x\}. \tag{36}$$

Analyticity is local, hence $\mathcal{A}(f)$ is open for every f. The substantive theorem is density for elementary functions.

The only difficult induction step is composition with a unary generator that has finitely many exceptional values. The following lemma isolates it.

Lemma 2.15 (Finite-bad-value composition principle)**.** *Let $f, g : \mathbb{R} \to \mathbb{R}$. Suppose that $\mathcal{A}(f)$ is dense and that there is a finite set $B \subset \mathbb{R}$ such that g is analytic at every point of $\mathbb{R} \setminus B$. Then $\mathcal{A}(g \circ f)$ is dense.*

Proof. Let U be a nonempty open interval. Because $\mathcal{A}(f)$ is dense and open, choose $x_0 \in U \cap \mathcal{A}(f)$. We prove that every punctured neighborhood of x_0 contains a point where $g \circ f$ is analytic.

Fix $c \in B$. The analytic function $f - c$ near x_0 has two possibilities.

1. It vanishes identically on a neighborhood of x_0. Then f is locally constant equal to c, so $g \circ f$ is locally constant and therefore analytic at x_0, regardless of whether g is analytic at c.
2. It is not locally zero. The isolated-zero theorem gives a punctured neighborhood of x_0 on which $f(x) \neq c$.

If the first alternative occurs for any $c \in B$, we are done. Otherwise intersect the finitely many punctured neighborhoods. The intersection still contains points arbitrarily close to x_0; intersect once more with the open set $U \cap \mathcal{A}(f)$. At any resulting point x, the value $f(x)$ avoids B, so g is analytic at $f(x)$ and f is analytic at x. The analytic chain rule makes $g \circ f$ analytic at x. Thus every nonempty open U meets $\mathcal{A}(g \circ f)$. □

Theorem 2.16 (Dense open analytic locus)**.** *If $f \in \mathsf{Elem}$, then $\mathcal{A}(f)$ is dense and open. Equivalently, the nonanalytic locus*

$$\mathbb{R} \setminus \mathcal{A}(f) \tag{37}$$

is closed and nowhere dense.

Proof. Openness is true for every analytic locus. Density follows by induction on the elementary construction.

Constants and the identity are analytic everywhere. For sums and products, the intersection of the two dense open analytic loci is dense and each point in the intersection is analytic for the result. Negation is immediate. For a unary generator, apply lemma 2.15 with the following exceptional sets:

Generator	Exceptional values	Analytic elsewhere
$t \mapsto t^{-1}$	$\{0\}$	yes
$t \mapsto t^r$	$\{0\}$	yes for every fixed $r \in \mathbb{R}$
$\log t$	$\{0\}$	yes under the total real-log convention
$\arcsin t$	$\{-1, 1\}$	yes
$\exp t, \sin t, \cos t, \arctan t$	$\varnothing$	everywhere

The complement of a dense open set is closed with empty interior, hence nowhere dense. □

2.3.3 The reusable obstruction

Theorem 2.17 (No agreement on an interior)**.** *Let $h : \mathbb{R} \to \mathbb{R}$ be nonanalytic at every point of $\operatorname{int}(S)$, where $S \subset \mathbb{R}$ has nonempty interior. No elementary function g can agree with h on all of S.*

Proof. If g were elementary, theorem 2.16 would give a point $x \in \operatorname{int}(S)$ at which g is analytic. Equality on S implies equality on a whole neighborhood of x contained in S, so analyticity transfers from g to h at x, a contradiction. □

The theorem is stronger than saying that h itself is not elementary: it rules out any piecewise repair outside S and any elementary surrogate on a substantial subset.

2.4 Strong non-elementarity of the Fabius family

The main article proves nowhere analyticity of F on $[0, 1]$, of up on $[-1, 1]$, and of the signed extension on its active interval. Combining those results with theorem 2.17 gives the new non-elementarity branch almost for free.

Theorem 2.18 (Strong local non-elementarity of F)**.** *Let $U \subset [0, 1]$ have nonempty interior. There is no $g \in \mathsf{Elem}$ such that*

$$g(x) = F(x) \qquad (x \in U). \tag{38}$$

In particular, no elementary function agrees with F on $(0, 1)$, and F is not an elementary function on $\mathbb{R}$.

Proof. The function F is nonanalytic at every point of $[0,1]$, hence at every point of $\operatorname{int}(U)$. Apply theorem 2.17. □

Theorem 2.19 (Strong local non-elementarity of up). *If $U \subset [-1,1]$ has nonempty interior, no elementary g agrees with* up *on U. Consequently* up *is not elementary.*

Theorem 2.20 (Strong local non-elementarity of the signed extension). *If $U \subset [0,2)$ has nonempty interior, no elementary g agrees with the signed global extension $\widetilde{F}$ on U. Consequently $\widetilde{F}$ is not elementary.*

Proof of the last two theorems. The corresponding nowhere-analyticity theorems supply the hypothesis of theorem 2.17. The global conclusions follow by taking a nonempty open subinterval of the active support and observing that an elementary representation on all of $\mathbb{R}$ would in particular agree there. □

Remark 2.21 (Why compact support alone is not the argument). A compactly supported nonzero function cannot be globally real analytic, but elementary functions need not be analytic everywhere: $|x|$, x^{-1}, and totalized logarithms already have exceptional points. What defeats elementarity is not compact support by itself; it is nonanalyticity on an entire interval, while elementary exceptional sets are nowhere dense.

Remark 2.22 (Why this is sharper than a differential-algebra slogan). The conclusion does not depend on choosing a particular syntax for "closed form" beyond the explicit inductive class. More importantly, it is local: changing the function outside a small open interval cannot make it elementary on that interval. Thus the theorem rules out not only a global elementary formula for F, but also an elementary formula valid on any nontrivial open patch of its natural domain.

3 Probability laws and the bridges the headline formulas conceal

3.1 The product probability law in measure-theoretic form

The random-series representation in the main article is concise because it speaks in the usual probabilistic shorthand. The formal construction must establish continuity, measurability, product independence, pushforward identities, and support before it may use the same shorthand. This section records that chain explicitly.

3.1.1 Sample space and uniform convergence

Let

$$\Omega = [0,1]^{\mathbb{N}} \tag{39}$$

with its product topology and product Borel probability measure

$$\mathbf{P} = \bigotimes_{n\geq 0} \mathrm{Unif}[0,1]. \tag{40}$$

Write ω_n for the nth coordinate and define

$$X(\omega) = \sum_{n=0}^{\infty} \frac{\omega_n}{2^{n+1}}. \tag{41}$$

For every ω,

$$0 \leq \frac{\omega_n}{2^{n+1}} \leq 2^{-n-1}, \qquad \sum_{n\geq 0} 2^{-n-1} = 1. \tag{42}$$

The Weierstrass test therefore gives uniform convergence on Ω. Every partial sum is continuous, so X is continuous and hence measurable. Termwise comparison in (42) gives the pointwise support statement

$$0 \le X(\omega) \le 1. \tag{43}$$

Let

$$\mu = \mathbf{P} \circ X^{-1} \tag{44}$$

be the pushforward law. It is a probability measure and

$$\mu([0,1]) = 1, \qquad \mu(\mathbb{R} \setminus [0,1]) = 0. \tag{45}$$

The Lean declarations are `weightedCoordinateSum`, `continuous_weightedCoordinateSum`, `weightedSumDistribution`, and the support lemmas surrounding them.

3.1.2 Deleting the head coordinate

Define the left shift

$$(T\omega)_n = \omega_{n+1}. \tag{46}$$

The product measure is invariant under this deletion in the sense that $T_{\#}\mathbf{P} = \mathbf{P}$, and the head coordinate ω_0 is independent of the entire tail $T\omega$. Reindexing the series gives the exact pointwise split

$$X(\omega) = \frac{\omega_0 + X(T\omega)}{2}. \tag{47}$$

Consequently the pair $(\omega_0, X(T\omega))$ has law

$$\mathrm{Unif}[0,1] \otimes \mu. \tag{48}$$

Pushing (47) through this joint law gives the self-similarity identity

$$\mu = (\mathrm{Unif}[0,1] \otimes \mu) \circ \left[(u,x) \mapsto \frac{u+x}{2}\right]^{-1}. \tag{49}$$

Proposition 3.1 (The smoothing equation for the CDF)**.** *Let $H(y) = \mu((-\infty, y])$. Then for every $y \in \mathbb{R}$,*

$$H(y) = \int_0^1 H(2y-u)\,\mathrm{d}u. \tag{50}$$

For $0 \le y \le 1/2$ this reduces to

$$H(y) = \int_0^{2y} H(t)\,\mathrm{d}t. \tag{51}$$

Proof. Condition on the head coordinate in (47). For a fixed u,

$$\mathbb{P}\left(\frac{u+X'}{2} \le y\right) = \mathbb{P}(X' \le 2y-u) = H(2y-u),$$

where X' has law μ and is independent of u. Integrate over $u \in [0,1]$. If $0 \le y \le 1/2$, support implies $H(2y-u) = 0$ when $u > 2y$; substitute $t = 2y - u$ and use the reflection identity established below, or reverse the interval directly. □

3.1.3 Coordinate reflection

Let $R : \Omega \to \Omega$ be

$$(R\omega)_n = 1 - \omega_n. \tag{52}$$

Lebesgue measure on $[0, 1]$ is invariant under $u \mapsto 1 - u$, so the infinite product measure is invariant under R. Moreover

$$X(R\omega) = \sum_{n\geq 0} \frac{1-\omega_n}{2^{n+1}} = 1 - X(\omega). \tag{53}$$

Thus the law is reflection invariant:

$$(x \mapsto 1 - x)_{\#}\mu = \mu. \tag{54}$$

Conditioning on the tail X', the head coordinate U_0 is uniform, so $(U_0 + X')/2$ has no atoms; equivalently, the law is absolutely continuous after convolution with the head uniform law. Thus H is continuous. Reflection invariance then gives

$$H(1 - y) = 1 - H(y). \tag{55}$$

The fixed-point uniqueness theorem for the bounded Fabius equation now identifies

$$H = F \quad \text{on all of } \mathbb{R}. \tag{56}$$

The phrase "on all of $\mathbb{R}$" includes the two constant tails; it is not a reference to the signed extension $\widetilde{F}$.

Theorem 3.2 (Probability representation, global form)**.** *For every $y \in \mathbb{R}$,*

$$F(y) = \mathbb{P}\left(\sum_{n\geq 0} \frac{U_n}{2^{n+1}} \leq y\right), \tag{57}$$

where the U_n are independent uniform variables on $[0, 1]$. Equivalently,

$$\mathrm{up}(x) = \mu((-\infty, 1 - |x|]) \tag{58}$$

for every $x \in \mathbb{R}$.

3.2 The survival-function Fubini identity

A particularly useful bridge, absent from the main article's probability chapter, identifies integration against the Fabius law with an ordinary integral against up. It is the compact support version of the layer-cake or tail-integration formula.

For $t \geq 0$, support and (56) give

$$\mu((t, \infty)) = 1 - F(t) = \mathrm{up}(t). \tag{59}$$

The equality remains correct for $t > 1$, when both sides are zero.

Theorem 3.3 (Expectation from the survival function)**.** *Let $g, g' : \mathbb{R} \to \mathbb{R}$ be continuous and suppose*

$$g(x) - g(0) = \int_0^x g'(t)\,\mathrm{d}t \qquad (0 \leq x \leq 1). \tag{60}$$

Then

$$\int_{[0,1]} g(x)\,\mathrm{d}\mu(x) = g(0) + \int_0^1 g'(t)\,\mathrm{up}(t)\,\mathrm{d}t. \tag{61}$$

Proof. Put

$$A = \{(t,x) \in [0,1] \times \mathbb{R} : t < x\}, \qquad K(t,x) = \mathbf{1}_A(t,x) g'(t). \tag{62}$$

Continuity of g' on the compact interval makes K integrable with respect to $\mathrm{Leb}_{[0,1]} \otimes \mu$. Integrating first in t and using (60) gives, for $x \in [0,1]$,

$$\int_0^1 K(t,x)\,\mathrm{d}t = \int_0^x g'(t)\,\mathrm{d}t = g(x) - g(0). \tag{63}$$

Integrating first in x gives

$$\int K(t,x)\,\mathrm{d}\mu(x) = g'(t)\mu((t,\infty)) = g'(t)\,\mathrm{up}(t) \tag{64}$$

by (59). Fubini equates the two iterated integrals. Since μ is a probability measure supported on $[0,1]$, the integral of the constant $g(0)$ is $g(0)$, and rearrangement yields (61). □

Remark 3.4 (Endpoint conventions disappear under integration). The survival function uses the open ray (t,∞). The CDF uses $(-\infty,t]$. Their complements are exact, and no endpoint atom enters because μ has a continuous CDF. Likewise, replacing $(0,1)$ by any of the standard half-open versions changes no Lebesgue integral. The formal proof nevertheless records these null-set conversions explicitly.

3.3 Raw, endpoint, and tilted Laplace moments

The main article develops half moments and the negative generating function from the analytic side. The new probability–Laplace module proves that those quantities are exactly the moments of the product law.

3.3.1 Three moment families

For a probability measure ν supported on $[0,1]$, define

$$U_n(\nu) = \int_{[0,1]} (1-x)^n\,\mathrm{d}\nu(x), \tag{65}$$

$$L_k(\nu;s) = \int_{[0,1]} \mathrm{e}^{-sx} x^k\,\mathrm{d}\nu(x), \tag{66}$$

$$J_k(s) = \int_0^1 t^k\,\mathrm{up}(t)\mathrm{e}^{-st}\,\mathrm{d}t. \tag{67}$$

For real s, the integrand in (66) is nonnegative, so

$$L_k(\nu;s) \geq 0. \tag{68}$$

Apply theorem 3.3 to $g(x) = \mathrm{e}^{-sx}x^k$. For $k = 0$ one obtains

$$L_0(\mu;s) = 1 - sJ_0(s). \tag{69}$$

For $k \geq 0$, applying it to $g(x) = \mathrm{e}^{-sx}x^{k+1}$ gives

$$L_{k+1}(\mu;s) = (k+1)J_k(s) - sJ_{k+1}(s). \tag{70}$$

These equations are exactly the defining clauses of `fabiusLaplaceMoment`.

Theorem 3.5 (Law moments equal analytic Fabius moments)**.** *For every $k \in \mathbb{N}$ and $s \in \mathbb{R}$,*

$$L_k(\mu; s) = M_k(s), \tag{71}$$

where

$$M_0(s) = G(-s), \tag{72}$$

$$M_{k+1}(s) = (k+1)J_k(s) - sJ_{k+1}(s) \tag{73}$$

are the tilted Fabius moments and G is the generating function of the main article. In particular $M_k(s) \geq 0$ for real s.

Proof. The case $k = 0$ is the integration-by-parts identity (69), which is also the negative generating-function formula. The successor case is (70). □

3.3.2 Differential recurrence

Differentiation under the compact integral in (67) is justified by a uniform exponential majorant in a neighborhood of each real s. It gives

$$J_k'(s) = -J_{k+1}(s). \tag{74}$$

Substituting into (72)–(73) yields the cleaner recurrence

$$M_k'(s) = -M_{k+1}(s). \tag{75}$$

Iteration gives

$$M_k^{(n)}(s) = (-1)^n M_{k+n}(s). \tag{76}$$

Thus all M_k are C^∞ on $\mathbb{R}$, and the negative generating function is completely monotone:

$$(-1)^n \frac{\mathrm{d}^n}{\mathrm{d}s^n} G(-s) = M_n(s) \geq 0. \tag{77}$$

3.3.3 Reflection and the half moments

Reflection invariance (54) gives

$$\int (1-x)^n \,\mathrm{d}\mu(x) = \int x^n \,\mathrm{d}\mu(x). \tag{78}$$

At zero tilt the analytic moment normalization is the exact rational half-moment sequence d_n:

$$M_n(0) = d_n. \tag{79}$$

Combining the identifications yields the full probability interpretation.

Theorem 3.6 (Probability meaning of the half moments)**.** *For every $n \in \mathbb{N}$,*

$$d_n = \mathbb{E}[X^n] = \mathbb{E}[(1-X)^n] = \int_{[0,1]} x^n \,\mathrm{d}\mu(x) = \int_{[0,1]} (1-x)^n \,\mathrm{d}\mu(x). \tag{80}$$

Moreover

$$\left.\frac{\mathrm{d}^n}{\mathrm{d}s^n} G(-s)\right|_{s=0} = (-1)^n d_n. \tag{81}$$

Proof. Use (71) at $s = 0$, then (79) and (78). The derivative identity is (76) with $k = 0$ and $s = 0$. □

The equality $d_n = \mathbb{E}[X^n]$ is not merely an interpretation of a previously known rational sequence. It is the bridge that licenses positivity, log-convexity, moment-matrix positivity, and Laplace comparison arguments without rebuilding them from the recurrence for d_n. The formal library keeps the bridge in a separate module precisely so the probabilistic and saddle branches can import it independently.

3.4 From atomic laws to half-endpoint histograms

The main article introduces the polynomial step approximants $\mathcal{W}_n$ and then moves quickly to their convergence. The formal proof has a nontrivial middle layer: the coefficients first define an atomic probability measure; each atom is smeared across a centered cell; interval integrals of the resulting histogram are sandwiched between masses of slightly shrunken and enlarged intervals.

3.4.1 Half-endpoint indicators are ordinary indicators almost everywhere

For $a < b$, define

$$\mathbf{1}^*_{[a,b]}(x) = \begin{cases} 1, & a < x < b, \\ \frac{1}{2}, & x = a \text{ or } x = b, \\ 0, & \text{otherwise.} \end{cases} \tag{82}$$

The two endpoint values are essential for pointwise gluing of adjacent cells, but irrelevant to Lebesgue integration.

Lemma 3.7 (Almost-everywhere replacement)**.** *For every $a, b \in \mathbb{R}$,*

$$\mathbf{1}^*_{[a,b]} = \mathbf{1}_{(a,b]} \quad \textit{Lebesgue-almost everywhere.} \tag{83}$$

Consequently

$$\int_{\mathbb{R}} \mathbf{1}^*_{[a,b]}(x)\,\mathrm{d}x = \max(b-a, 0), \tag{84}$$

and, for $c \le d$,

$$\int_c^d \mathbf{1}^*_{[a,b]}(x)\,\mathrm{d}x = \max\bigl(\min(d,b) - \max(c,a), 0\bigr). \tag{85}$$

Proof. The functions differ only at a and b, a null set. The first integral is the length of $(a, b]$. The second is the length of $(c, d] \cap (a, b] = (\max(c, a), \min(d, b)]$, with truncation at zero when the interval is empty. □

3.4.2 Atomic measure, cell geometry, and exact normalization

At level n, let $x_{n,m}$ be the location of the mth polynomial atom and let $p_{n,m} \ge 0$ be its normalized mass, with

$$\sum_m p_{n,m} = 1. \tag{86}$$

The corresponding atomic law is

$$\nu_n = \sum_m p_{n,m}\delta_{x_{n,m}}. \tag{87}$$

Let

$$h_n = 2^{-n-1} \tag{88}$$

be the half-width of a cell and

$$C_{n,m} = [x_{n,m} - h_n, x_{n,m} + h_n]. \tag{89}$$

Its full width is $2h_n = 2^{-n}$. Smearing the atom uniformly over its cell produces the histogram

$$\mathcal{W}_n(x) = 2^n \sum_m p_{n,m}\mathbf{1}^*_{C_{n,m}}(x). \tag{90}$$

By (84), each term has integral $p_{n,m}$, hence

$$\int_{\mathbb{R}} \mathcal{W}_n(x)\,\mathrm{d}x = 1. \tag{91}$$

The pointwise half weights ensure that when two cells share an endpoint, the two half contributions add to the correct plateau value.

3.4.3 The overlap sandwich

For a fixed interval $[a,b]$ with $a \le b$, let

$$\ell_{n,m}(a,b) = |[a,b] \cap C_{n,m}| = \max(\min(b, x_{n,m} + h_n) - \max(a, x_{n,m} - h_n), 0). \tag{92}$$

Then

$$\int_a^b \mathcal{W}_n(x)\,\mathrm{d}x = \sum_m p_{n,m}\, 2^n \ell_{n,m}(a,b). \tag{93}$$

The normalized overlap $2^n \ell_{n,m}$ lies in $[0,1]$.

If $x_{n,m} \in (a+h_n, b-h_n]$, then the entire cell lies inside $[a,b]$, so the normalized overlap is 1. If the cell meets $[a,b]$, then its center lies in $(a-h_n, b+h_n]$. Therefore, term by term,

$$\mathbf{1}_{(a+h_n, b-h_n]}(x_{n,m}) \le 2^n \ell_{n,m}(a,b) \le \mathbf{1}_{(a-h_n, b+h_n]}(x_{n,m}). \tag{94}$$

Multiplication by $p_{n,m}$ and summation gives the central bridge.

Theorem 3.8 (Atomic–histogram interval sandwich)**.** *For $a \le b$,*

$$\nu_n((a+h_n, b-h_n]) \le \int_a^b \mathcal{W}_n(x)\,\mathrm{d}x \le \nu_n((a-h_n, b+h_n]). \tag{95}$$

This is the exact content of the inner- and outer-overlap lemmas in `StepMeasureBridge.lean`. Its importance is conceptual: it converts weak convergence of measures into convergence of integrals of discontinuous histograms without claiming pointwise convergence of densities from weak convergence alone.

3.4.4 Portmanteau plus a moving-boundary squeeze

The atomic laws ν_n coincide with the finite convolution laws and converge weakly to the absolutely continuous law with density up. Because that limit gives zero mass to every singleton, the frontier of $(a,b]$ is null. Portmanteau therefore gives

$$\nu_n((a,b]) \longrightarrow \int_a^b \mathrm{up}(x)\,\mathrm{d}x. \tag{96}$$

The intervals in (95) move with n, so one more squeeze is needed. Fix $\varepsilon > 0$ and choose $\delta > 0$ with $4\delta < \varepsilon$. Eventually $h_n \le \delta$, and hence

$$(a+\delta, b-\delta] \subset (a+h_n, b-h_n], \tag{97}$$

$$(a-h_n, b+h_n] \subset (a-\delta, b+\delta]. \tag{98}$$

The bound $0 \le \mathrm{up} \le 1$ gives

$$0 \le \int_a^b \mathrm{up} - \int_{a+\delta}^{b-\delta} \mathrm{up} \le 2\delta, \tag{99}$$

$$0 \le \int_{a-\delta}^{b+\delta} \mathrm{up} - \int_a^b \mathrm{up} \le 2\delta. \tag{100}$$

Apply (96) to the two fixed comparison intervals and combine with (95).

Theorem 3.9 (Interval-integral convergence of the step approximants)**.** *For every $a, b \in \mathbb{R}$,*

$$\int_a^b \mathcal{W}_n(x)\,\mathrm{d}x \longrightarrow \int_a^b \mathrm{up}(x)\,\mathrm{d}x. \tag{101}$$

Proof. The preceding fixed-δ sandwich proves the result for $a \le b$. If $b < a$, reverse the orientation of both interval integrals. □

Warning 3.10 (The logical order cannot be shortened). Weak convergence of ν_n does not by itself imply pointwise convergence of the histograms $\mathcal{W}_n$, nor even convergence of their values at a single point. The formal route is

$$\begin{aligned}\text{weak convergence} &\Longrightarrow \text{null-boundary interval masses}\\ &\Longrightarrow \text{moving-cell sandwich} \Longrightarrow \text{interval-integral convergence.}\end{aligned}$$

Other parts of the development then use shape, refinement, Fourier decay, or difference quotients to obtain stronger function-level convergence.

4 The reusable discrete engines

4.1 Weighted paths solve triangular recurrences

The exact inverse-power recurrence for $F(2^{-n})$ is triangular. The main article states its composition formula, but the Lean proof first establishes a generic solver for every semiring-valued triangular recurrence. That abstraction explains both the form of the answer and the uniqueness argument.

4.1.1 Compositions as increasing paths

A composition of n is an ordered list of positive integers

$$c = (r_1, \dots, r_m), \qquad r_1 + \cdots + r_m = n. \tag{102}$$

Its partial sums

$$s_0 = 0, \qquad s_j = r_1 + \cdots + r_j \quad (1 \le j \le m) \tag{103}$$

form a strictly increasing path

$$0 = s_0 < s_1 < \cdots < s_m = n. \tag{104}$$

Conversely, every such path determines the positive increments $r_j = s_j - s_{j-1}$.

Let R be a semiring and let

$$w : \mathbb{N} \times \mathbb{N} \to R \tag{105}$$

be an edge-weight function. Define

$$W_w(c) = \prod_{j=1}^{m} w(s_{j-1}, s_j), \tag{106}$$

$$P_w(n) = \sum_{c \models n} W_w(c). \tag{107}$$

For $n = 0$, there is one empty composition and its empty product is 1, so

$$P_w(0) = 1. \tag{108}$$

4.1.2 Last-edge decomposition

Every nonempty path to n has a unique last predecessor $k < n$. Deleting its final edge gives a path to k, and appending the edge $k \to n$ reverses the operation. Therefore

Theorem 4.1 (Last-edge decomposition)**.** *For $n > 0$,*

$$P_w(n) = \sum_{k=0}^{n-1} P_w(k) w(k,n). \tag{109}$$

Proof. Partition the finite set of compositions of n by the sum k of all blocks except the last. For fixed k, deleting the final block is a bijection with compositions of k. Path-weight multiplicativity gives

$$W_w(c \parallel (n-k)) = W_w(c) w(k,n).$$

Summing first over compositions of k and then over $k < n$ gives (109). □

One can instead partition by the number of edges. Let $P_{w,r}(n)$ be the contribution of compositions with exactly r blocks. Uniformly in n,

$$P_w(n) = \sum_{r=0}^{n} P_{w,r}(n), \tag{110}$$

and for $n > 0$ the zero-edge term vanishes, so the range may be written $1 \le r \le n$.

4.1.3 The generic triangular solver

Theorem 4.2 (Path-sum solution of a triangular recurrence)**.** *Let $x : \mathbb{N} \to R$ satisfy*

$$x_0 = b, \tag{111}$$

$$x_n = \sum_{k=0}^{n-1} x_k w(k,n) \qquad (n > 0). \tag{112}$$

Then

$$x_n = bP_w(n) \qquad (n \ge 0). \tag{113}$$

Consequently the recurrence and initial value determine x uniquely.

Proof. Use strong induction on n. At $n = 0$, (108) gives $bP_w(0) = b = x_0$. For $n > 0$, insert the induction hypotheses into (112):

$$x_n = \sum_{k<n} bP_w(k)w(k,n) = b \sum_{k<n} P_w(k)w(k,n) = bP_w(n),$$

where the last equality is theorem 4.1. Any two solutions therefore equal the same path sum at each index. □

Remark 4.3. No subtraction, division, topology, or order is used. The theorem works in an arbitrary semiring. This is why the formal development separates it from the rational Fabius specialization.

4.2 The nonrecursive composition formula for $F(2^{-n})$

Let a_n be the normalized inverse-dyadic recurrence sequence used in the main article, so

$$F(2^{-n}) = 2^{-\binom{n}{2}} a_n. \tag{114}$$

Its triangular recurrence has edge weights

$$w(k,n) = \frac{1}{(2^n - 1)(n-k+1)!} \qquad (0 \le k < n). \tag{115}$$

Because $a_0 = 1$, theorem 4.2 gives $a_n = P_w(n)$.

For a composition $c = (r_1, \ldots, r_m)$ of n, the edge from s_{j-1} to s_j has $n-k$ replaced by r_j, hence

$$w(s_{j-1}, s_j) = \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \tag{116}$$

Therefore:

Theorem 4.4 (Nonrecursive inverse-dyadic formula)**.** *For every $n \in \mathbb{N}$,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{(r_1,\ldots,r_m)\models n} \prod_{j=1}^{m} \frac{1}{(2^{r_1+\cdots+r_j} - 1)(r_j+1)!}. \tag{117}$$

At $n = 0$, the unique empty composition contributes the empty product 1.

Proof. Apply theorem 4.2 with (115), translate paths into compositions using (103), and multiply by the outer normalization in (114). □

The formula may be grouped by the number m of blocks:

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{m=1}^{n} \sum_{\substack{r_1+\cdots+r_m=n \\ r_j \ge 1}} \prod_{j=1}^{m} \frac{1}{(2^{r_1+\cdots+r_j} - 1)(r_j+1)!}, \qquad n > 0. \tag{118}$$

4.2.1 Small cases as structural checks

Example 4.5 (The value at $1/2$)**.** For $n = 1$ the sole composition is (1), with weight

$$\frac{1}{(2^1 - 1)2!} = \frac{1}{2}.$$

The outer power is $2^0 = 1$, so $F(1/2) = 1/2$.

Example 4.6 (The value at $1/4$)**.** The two compositions of 2 are (2) and $(1,1)$:

$$W(2) = \frac{1}{(4-1)3!} = \frac{1}{18},$$
$$W(1,1) = \frac{1}{(2-1)2!}\frac{1}{(4-1)2!} = \frac{1}{12}.$$

Their sum is $5/36$ and the outer factor is 2^{-1}, giving

$$F(1/4) = \frac{5}{72}. \tag{119}$$

Example 4.7 (The value at 1/8)**.** The four compositions of 3 have weights

$$\frac{1}{168}, \qquad \frac{1}{84}, \qquad \frac{1}{252}, \qquad \frac{1}{168},$$

for (3), (1, 2), (2, 1), and (1, 1, 1) respectively. Their sum is 1/36; multiplying by $2^{-\binom{3}{2}} = 1/8$ gives

$$F(1/8) = \frac{1}{288}. \tag{120}$$

The composition formula is not a mysterious closed form guessed from the first values. It is the universal unrolling of a triangular recurrence. Every block records a jump, every partial sum records the endpoint of that jump, and the product records the chronological edge weights. The formal path layer makes this explanation exact and reusable.

4.3 A finite-row Toeplitz convergence theorem

The complex-shift and discrete-limit formulas involve a row of coefficients depending on n, not a fixed summability kernel. The main article verifies the Fabius specialization but omits the abstract theorem. The needed result is a finite-row version of the Toeplitz summability principle in a normed ring.

4.3.1 Abstract statement

Let K be a normed ring. For each n, let J_n be a finite index set, let $w_{n,j} \in K$, and let $q(n,j) \in \mathbb{N}$ be the sampled sequence index. Assume:

$$\sum_{j \in J_n} w_{n,j} = 1, \tag{121}$$

$$\sum_{j \in J_n} \|w_{n,j}\| \le C \tag{122}$$

for one constant $C \ge 0$, and assume that the sampled indices move uniformly into every tail:

$$\forall N \; \exists n_0 \; \forall n \ge n_0 \; \forall j \in J_n, \qquad q(n,j) \ge N. \tag{123}$$

Theorem 4.8 (Finite-row Toeplitz convergence)**.** *If $H_m \to L$ in K, then*

$$\sum_{j \in J_n} w_{n,j} H_{q(n,j)} \longrightarrow L. \tag{124}$$

Proof. Fix $\varepsilon > 0$. Since $H_m \to L$, choose N such that

$$\|H_m - L\| < \frac{\varepsilon}{C+1} \qquad (m \ge N). \tag{125}$$

Choose n_0 from (123). For $n \ge n_0$, use the row-mass identity to subtract L:

$$\sum_{j \in J_n} w_{n,j} H_{q(n,j)} - L = \sum_{j \in J_n} w_{n,j} \big(H_{q(n,j)} - L\big).$$

Therefore

$$\begin{aligned}\left\|\sum_{j\in J_n} w_{n,j}H_{q(n,j)}-L\right\| &\le \sum_{j\in J_n}\|w_{n,j}\|\,\|H_{q(n,j)}-L\| \\ &< \frac{\varepsilon}{C+1}\sum_{j\in J_n}\|w_{n,j}\| \\ &\le \varepsilon\frac{C}{C+1}<\varepsilon.\end{aligned}$$

This proves convergence. □

Remark 4.9. The weights need not be positive, the rows need not be nested, and no entrywise limit of $w_{n,j}$ is required. The theorem uses exactly three facts: row mass one, uniformly bounded total variation, and uniform migration of all sampled indices into the tail.

4.4 The Fabius Toeplitz rows and corrected discrete prefixes

4.4.1 The half-base q-binomial weights

For $0 \le j \le n$, define

$$w_{n,j}=\frac{(-1)^j\left[{n\atop j}\right]_{1/2}(1/2)^{\binom{j+1}{2}}}{(1/2;1/2)_n}. \tag{126}$$

The half-base finite q-binomial theorem evaluated at $z=1/2$ gives

$$\sum_{j=0}^{n} w_{n,j}=1. \tag{127}$$

Taking absolute values removes the signs and changes the finite product from $(1/2;1/2)_n$ to $(-1/2;1/2)_n$:

$$\sum_{j=0}^{n}|w_{n,j}|=\frac{(-1/2;1/2)_n}{(1/2;1/2)_n}. \tag{128}$$

The formal proof uses the elementary uniform lower bound

$$(1/2;1/2)_n\ge\frac{1}{4} \tag{129}$$

and the product comparison

$$(-1/2;1/2)_n(1/2;1/2)_n\le 1. \tag{130}$$

Hence

$$\sum_{j=0}^{n}|w_{n,j}|\le 16. \tag{131}$$

The constant 16 is deliberately coarse but uniform and algebraically simple.

The reindexing used by the discrete formula is

$$q(n,j)=2n-j. \tag{132}$$

Since $0\le j\le n$,

$$q(n,j)=2n-j\ge n. \tag{133}$$

Thus every index in row n lies beyond the common threshold n.

Corollary 4.10 (Specialized Fabius row convergence)**.** *For every convergent sequence $H_m \to L$ in $\mathbb{R}$ or $\mathbb{C}$,*

$$\sum_{j=0}^{n} w_{n,j} H_{2n-j} \longrightarrow L. \tag{134}$$

Proof. Apply theorem 4.8 with $C = 16$ and use (127), (131), and (133). □

4.4.2 Why the inner range uses a half-cell correction

The finite discrete approximant contains an inner prefix whose source-facing notation has inclusive upper endpoint

$$\left\lfloor 2^p x - \frac{1}{2} \right\rfloor. \tag{135}$$

A direct natural-number range cannot represent the value -1, which should mean an empty sum. The safe range length is

$$\ell_p(x) = \left\lfloor 2^p x + \frac{1}{2} \right\rfloor_{\mathbb{N}}, \tag{136}$$

where the natural floor clamps negative results to zero. For $x \geq 0$,

$$\ell_p(x) = \left\lfloor 2^p x - \frac{1}{2} \right\rfloor + 1, \tag{137}$$

so the natural range $0 \leq r < \ell_p(x)$ exactly represents the intended inclusive sum. Moreover

$$\ell_p(x) = 0 \iff 2^p x < \frac{1}{2}. \tag{138}$$

In particular every prefix is empty for $x \leq 0$, so every finite approximant vanishes there before taking a limit.

Warning 4.11 (The half-cell is not cosmetic). Replacing (136) by $\lfloor 2^p x \rfloor$ shifts every jump and breaks the centered nearest-integer convention. Replacing it by a signed inclusive endpoint inside a natural range mishandles the empty case. The formula $\lfloor 2^p x + 1/2 \rfloor_{\mathbb{N}}$ simultaneously fixes both issues and is the exact convention used by the formal development.

4.4.3 Where the abstract theorem enters the discrete limit

After the algebraic q-binomial reindexing, the nth discrete approximant takes the form

$$A_n(x) = \sum_{j=0}^{n} w_{n,j} H_{2n-j}(x), \tag{139}$$

where, for each fixed x, the inner objects $H_p(x)$ converge to the desired translated Fabius value. The entire outer-limit argument is then

$$H_p(x) \to L(x) \quad + \quad (127), (131), (133) \quad \Longrightarrow \quad A_n(x) \to L(x).$$

All Fabius-specific work lies in identifying H_p and its limit; all row summation is contained in theorem 4.8.

5 The formal and analytic machinery of the all-orders saddle expansion

5.1 Parameter-dependent Poincaré expansions

The endpoint asymptotic is not an ordinary expansion with constant coefficients. Its coefficients retain a bounded one-periodic dependence on the exact Lambert phase. The formal library therefore uses an asymptotic API in which coefficient families may depend on the asymptotic parameter, provided that dependence remains bounded.

Let ℓ be a filter on a parameter space A, let $\sigma : A \to \mathbb{K}$ be a scale tending to zero, and let $f : A \to E$ take values in a normed vector space over the normed field $\mathbb{K}$. For coefficient families $c_k : A \to E$, put

$$S_N[f](a) = \sum_{k=0}^{N-1} \sigma(a)^k c_k(a). \tag{140}$$

Definition 5.1 (Full parameter-dependent expansion)**.** We write

$$f(a) \sim_\ell \sum_{k\geq 0} \sigma(a)^k c_k(a) \tag{141}$$

when

1. every c_k is bounded along ℓ, equivalently $c_k = O_\ell(1)$;
2. for every $N \in \mathbb{N}$,
$$f - S_N[f] = O_\ell(\sigma^N). \tag{142}$$

This is `HasAsymptoticExpansion`. Boundedness of the coefficients is not redundant: if arbitrary unbounded parameter dependence were allowed, powers of σ could be moved between the "coefficient" and the scale, destroying the meaning of order.

5.1.1 Elementary algebra of full expansions

Suppose f and g have expansions with the same scale. Direct manipulation of finite partial sums proves:

$$f + g \sim \sum_{k\geq 0} \sigma^k (c_k + d_k), \tag{143}$$

$$f - g \sim \sum_{k\geq 0} \sigma^k (c_k - d_k), \tag{144}$$

$$\alpha f \sim \sum_{k\geq 0} \sigma^k (\alpha c_k). \tag{145}$$

The API also supports transport across eventual equality and pullback along a map tending to the source filter. These apparently routine lemmas are what let the final proof switch between complex and real saddle ratios, replace exact identities after a threshold, and move from the logarithmic coordinate to $x \downarrow 0$ without reopening every remainder estimate.

5.1.2 Flat functions

Definition 5.2 (Flat relative to a scale)**.** A function r is flat relative to σ along ℓ if

$$r = O_\ell(\sigma^N) \qquad \text{for every } N \in \mathbb{N}. \tag{146}$$

Theorem 5.3 (Flat remainders are invisible to all coefficients)**.** *A flat r has the full expansion with every coefficient zero. Consequently, if*

$$f \sim \sum_{k\geq 0} \sigma^k c_k, \tag{147}$$

then

$$f + r \sim \sum_{k\geq 0} \sigma^k c_k, \qquad f - r \sim \sum_{k\geq 0} \sigma^k c_k. \tag{148}$$

The implications hold in both directions.

Proof. For the zero-coefficient expansion, the Nth partial sum is identically zero, so the remainder condition is exactly (146). Add or subtract this expansion coefficientwise from the expansion of f. Reversibility follows by moving r to the other side. □

This theorem is the formal reason that the forward negative-Laplace tail can be restored after the central saddle expansion without changing any algebraic-order coefficient.

5.1.3 Exact logarithmic coordinates

Define

$$x(t) = 2^{-t}, \qquad t(x) = -\log_2 x \quad (x > 0). \tag{149}$$

These maps are exact inverses on $x > 0$, and

$$t \to +\infty \iff x(t) \downarrow 0, \qquad x \downarrow 0 \iff t(x) \to +\infty. \tag{150}$$

Theorem 5.4 (Full-expansion coordinate equivalence)**.** *For functions f, σ, c_k on the positive half-line,*

$$f(2^{-t}) \sim_{t\to\infty} \sum_{k\geq 0} \sigma(2^{-t})^k c_k(2^{-t}) \tag{151}$$

if and only if

$$f(x) \sim_{x\downarrow 0} \sum_{k\geq 0} \sigma(x)^k c_k(x). \tag{152}$$

Proof. Pull back the full expansion along either coordinate map. The two composites are eventually identical on their respective filters by (149); boundedness and every remainder estimate therefore transfer in both directions. □

5.2 Recursive coefficients of a formal exponential

The saddle integrand contains the exponential of a formal perturbation. Rather than invoke Bell polynomials or repeatedly expand a universal power series, the Lean development uses a finite recurrence whose nth output depends only on the first n input coefficients.

Let R be a commutative $\mathbb{Q}$-algebra, and let

$$E(u) = \sum_{j\geq 1} E_j u^j \tag{153}$$

be a formal exponent with zero constant term. Define $a_n \in R$ recursively by

$$a_0 = 1, \tag{154}$$

$$a_{n+1} = \frac{1}{n+1} \sum_{j=0}^{n} (j+1) E_{j+1} a_{n-j}. \tag{155}$$

Theorem 5.5 (Correctness and uniqueness of the exponential recurrence)**.** *The power series*

$$A(u) = \sum_{n \geq 0} a_n u^n \tag{156}$$

satisfies

$$A'(u) = E'(u)A(u), \qquad A(0) = 1. \tag{157}$$

It is the unique formal power series with these two properties, and therefore

$$A(u) = \exp(E(u)). \tag{158}$$

Proof. The coefficient of u^n in A' is $(n+1)a_{n+1}$. The coefficient of u^n in $E'A$ is

$$\sum_{j=0}^{n} (j+1)E_{j+1}a_{n-j},$$

so (157) is coefficientwise equivalent to (155). Conversely the differential equation and initial value determine a_0, then a_1, then a_2, and so on, proving uniqueness. The universal formal exponential satisfies the same equation and initial value. □

The first coefficients are

$$a_1 = E_1, \tag{159}$$

$$a_2 = E_2 + \frac{1}{2}E_1^2, \tag{160}$$

$$a_3 = E_3 + E_1E_2 + \frac{1}{6}E_1^3, \tag{161}$$

$$a_4 = E_4 + E_1E_3 + \frac{1}{2}E_2^2 + \frac{1}{2}E_1^2E_2 + \frac{1}{24}E_1^4. \tag{162}$$

Equation (162) is the exact finite identity used for the second saddle correction.

5.2.1 Structural laws

The recurrence is preferable to a black-box exponential because it exposes several laws needed downstream.

Proposition 5.6 (Finite dependence)**.** *The coefficient a_n depends only on $E_1, \ldots, E_n$. If two exponent families agree through order n, their nth exponential coefficients agree.*

Proposition 5.7 (Naturality)**.** *For every morphism of commutative $\mathbb{Q}$-algebras $\varphi : R \to S$,*

$$\varphi(a_n(E)) = a_n(\varphi \circ E). \tag{163}$$

In particular, evaluation of polynomial- or function-valued coefficients commutes with the recurrence.

Proposition 5.8 (Rescaling, sums, and parity)**.** *Let $c \in R$ be central.*

1. *If $\widetilde{E}_j = c^j E_j$, then*

$$a_n(\widetilde{E}) = c^n a_n(E). \tag{164}$$

2. *If E and F are two exponent families, then*

$$a_n(E + F) = \sum_{r+s=n} a_r(E)a_s(F). \tag{165}$$

3. If $E_j(-v) = (-1)^j E_j(v)$, then

$$a_n(-v) = (-1)^n a_n(v). \tag{166}$$

Proof architecture. Naturality is induction through finite sums and rational scalar multiplication. Rescaling follows from the recurrence because every summand has total degree $(j+1)+(n-j) = n+1$. The addition law is the coefficient form of $\exp(E+F) = \exp(E)\exp(F)$. Parity is the specialization $c = -1$ of rescaling after pointwise reflection in v. □

5.3 Recursive coefficients of a formal logarithm

Let

$$A(u) = 1 + \sum_{n\geq 1} a_n u^n. \tag{167}$$

We seek

$$B(u) = \log A(u) = \sum_{n\geq 1} b_n u^n. \tag{168}$$

Instead of expanding the universal logarithm, use

$$A(u)B'(u) = A'(u). \tag{169}$$

Solving the coefficient of u^n for b_{n+1} gives

$$b_0 = 0, \tag{170}$$

$$b_{n+1} = a_{n+1} - \frac{1}{n+1}\sum_{j=0}^{n-1}(n-j)b_{n-j}a_{j+1}. \tag{171}$$

The empty sum at $n = 0$ gives $b_1 = a_1$.

Theorem 5.9 (Correctness and uniqueness of the logarithm recurrence)**.** *If $a_0 = 1$, the series generated by* (171) *is the unique series with zero constant term satisfying* (169)*; hence it is the formal logarithm of A.*

Proof. The coefficient of u^n in AB' contains the term $(n+1)b_{n+1}$ from $a_0 b'$, plus terms involving only $b_1, \ldots, b_n$. Equating it with $(n+1)a_{n+1}$ and dividing by $n+1$ gives (171). This triangular relation proves uniqueness. The formal logarithm satisfies $B(0) = 0$ and $B'A = A'$. □

The first coefficients are

$$b_1 = a_1, \tag{172}$$

$$b_2 = a_2 - \frac{1}{2}a_1^2, \tag{173}$$

$$b_3 = a_3 - a_1a_2 + \frac{1}{3}a_1^3, \tag{174}$$

$$b_4 = a_4 - a_1a_3 - \frac{1}{2}a_2^2 + a_1^2a_2 - \frac{1}{4}a_1^4. \tag{175}$$

As with the exponential recurrence, b_n depends only on $a_1, \ldots, a_n$, commutes with $\mathbb{Q}$-algebra morphisms and evaluation, obeys the rescaling law

$$a_j \mapsto c^j a_j \quad \Longrightarrow \quad b_n \mapsto c^n b_n, \tag{176}$$

and preserves parity.

Remark 5.10 (Why a_0 is a hypothesis, not an input). The recursive definition (171) never reads a_0; it is meaningful for every sequence. Correctness as a logarithm, however, requires $a_0 = 1$. Separating the recurrence from its normalization makes finite coefficient calculations easier while keeping the theorem honest.

5.3.1 From mass expansions to logarithmic expansions

Suppose a positive real function has a full expansion

$$R(a) \sim 1 + \sum_{n\geq 1} \sigma(a)^n a_n(a), \qquad \sigma(a) \to 0, \tag{177}$$

with bounded coefficient families. Finite polynomial remainders, discussed in the next section, show that for each N,

$$\log R(a) = \sum_{n=1}^{N-1} \sigma(a)^n b_n(a) + O(\sigma(a)^N), \tag{178}$$

where b_n is generated by (171). The positivity hypothesis fixes the real branch and is eventually automatic when $R \to 1$.

5.4 Exact finite remainders from formal coefficient identities

Formal power-series equality says that two expressions have the same coefficients below a chosen order. An asymptotic proof needs an evaluated identity with an explicit factor of the first omitted power. The module `SaddleExpansionFiniteRemainder.lean` provides exactly that bridge without invoking an infinite series.

Fix $L \geq 1$ and truncate the exponent:

$$E_{<L}(X) = \sum_{k=1}^{L-1} E_k X^k. \tag{179}$$

Truncate the scalar exponential polynomial at the same order and substitute:

$$T_L(X) = \sum_{q=0}^{L-1} \frac{E_{<L}(X)^q}{q!}. \tag{180}$$

Let

$$A_{<L}(X) = \sum_{k=0}^{L-1} a_k X^k \tag{181}$$

be the polynomial formed from the recursive exponential coefficients.

Theorem 5.11 (Exact divisibility of the finite defect)**.** *If $E_0 = 0$, then*

$$X^L \mid (T_L(X) - A_{<L}(X)). \tag{182}$$

Equivalently, there is a canonical polynomial $Q_L(X)$ such that

$$T_L(X) = A_{<L}(X) + X^L Q_L(X). \tag{183}$$

Proof. The coefficient of X^k in T_L agrees with that of the full formal substitution $\exp(E(X))$ for every $k < L$: terms of scalar exponential degree $q \geq L$ cannot contribute below degree L because E has zero constant term, and truncating E itself does not alter coefficients below L. By theorem 5.5, that common coefficient is a_k. Hence the first L coefficients of the defect vanish, which is equivalent to divisibility by X^L. The formal construction takes Q_L to be the result of applying polynomial division by X exactly L times. □

Evaluating at $x \in R$ gives the identity actually used in estimates:

$$\sum_{q=0}^{L-1} \frac{1}{q!} \left(\sum_{k=1}^{L-1} E_k x^k \right)^q = \sum_{k=0}^{L-1} a_k x^k + x^L Q_L(x). \tag{184}$$

The quotient Q_L is a polynomial in x whose coefficients are algebraic expressions in the finitely many parameters $E_1, \dots, E_{L-1}$. If those parameter families are uniformly bounded, then Q_L is uniformly bounded on a fixed compact x-window. Thus the exact factorization (184) turns formal coefficient agreement into a uniform $O(x^L)$ estimate with no appeal to convergence of an infinite formal series.

5.5 Gaussian contraction: integration becomes polynomial algebra

After the saddle variable is scaled by $v = \sqrt{\lambda}\,\theta$, the leading kernel is the standard Gaussian

$$\gamma(v) = \mathrm{e}^{-v^2/2}. \tag{185}$$

Every finite reference term is $\gamma(v)$ times a complex polynomial. The generic contraction module turns its whole-line integral into an algebraic linear functional.

5.5.1 Gaussian moments

Define the unnormalized moments

$$M_n^{\mathrm{G}} = \int_{\mathbb{R}} \mathrm{e}^{-v^2/2} v^n \,\mathrm{d}v. \tag{186}$$

Every moment is absolutely integrable. Symmetry gives

$$M_{2j+1}^{\mathrm{G}} = 0. \tag{187}$$

Integration by parts, with $u = v^{n+1}$ and $\mathrm{d}w = -v\mathrm{e}^{-v^2/2}\,\mathrm{d}v$, gives

$$M_{n+2}^{\mathrm{G}} = (n+1)M_n^{\mathrm{G}}. \tag{188}$$

Since

$$M_0^{\mathrm{G}} = \sqrt{2\pi}, \tag{189}$$

the normalized moments

$$\mu_n = \frac{M_n^{\mathrm{G}}}{\sqrt{2\pi}} \tag{190}$$

satisfy

$$\mu_0 = 1, \quad \mu_1 = 0, \quad \mu_{n+2} = (n+1)\mu_n. \tag{191}$$

Thus

$$\mu_{2j} = (2j-1)!!, \qquad \mu_{2j+1} = 0. \tag{192}$$

The first even values needed below are

$$\mu_4 = 3, \quad \mu_6 = 15, \quad \mu_8 = 105, \quad \mu_{10} = 945, \quad \mu_{12} = 10395. \tag{193}$$

5.5.2 The contraction functional

For $p(X) = \sum_k p_k X^k \in \mathbb{C}[X]$, define

$$\mathcal{G}[p] = \sum_k p_k \mu_k. \tag{194}$$

This is a $\mathbb{C}$-linear map $\mathbb{C}[X] \to \mathbb{C}$.

Theorem 5.12 (Gaussian polynomial contraction). *For every complex polynomial p,*

$$\int_{\mathbb{R}} \mathrm{e}^{-v^2/2} p(v)\,\mathrm{d}v = \sqrt{2\pi}\,\mathcal{G}[p]. \tag{195}$$

In particular

$$\mathcal{G}[X^{2j}] = (2j-1)!!, \qquad \mathcal{G}[X^{2j+1}] = 0. \tag{196}$$

Proof. Expand p as a finite sum of monomials, interchange that finite sum with the integral, and use (190). Integrability follows coefficientwise from the absolute Gaussian moments. □

The same coefficientwise expansion yields the norm estimate

$$|\mathcal{G}[p]| \leq \sum_k |p_k| \mu_{2\lceil k/2 \rceil}, \tag{197}$$

with odd coefficients contributing zero in the exact functional. In the formal proof a slightly more uniform support sum is used so it composes smoothly with polynomial families.

Remark 5.13 (Parity is an algebraic annihilator). The vanishing of odd Gaussian moments is not merely a simplification after integration. It is the mechanism that removes every odd power of the small parameter $\varepsilon = \lambda^{-1/2}$ from the mass expansion. The exponent coefficient of order m has parity m in v, the formal exponential recurrence preserves total parity, and (196) kills every odd result. The final expansion is therefore in integer powers of λ^{-1}.

5.6 Uniform Gaussian tails for polynomial references

Whole-line contraction is useful only after the central-window integral has been replaced by a whole-line integral with a negligible error. The omitted Gaussian-tail modules provide a single coefficient weight that controls every polynomial reference.

5.6.1 A factorial coefficient weight

For $p(X) = \sum_k p_k X^k$, define

$$\mathfrak{W}(p) = \sum_{k \in \operatorname{supp} p} |p_k|\, 8k!. \tag{198}$$

It is finite and nonnegative.

Lemma 5.14 (Monomial Gaussian tail). *For $A \geq 4$ and $k \in \mathbb{N}$,*

$$\int_{|v|>A} \mathrm{e}^{-v^2/2} |v|^k\,\mathrm{d}v \leq 8k!\,\frac{\mathrm{e}^{-A^2/4}}{A}. \tag{199}$$

Proof architecture. Split the kernel as $\mathrm{e}^{-v^2/2} = \mathrm{e}^{-v^2/4}\mathrm{e}^{-v^2/4}$. The factor $|v|^k \mathrm{e}^{-v^2/4}$ is controlled uniformly by a factorial bound, while the remaining Gaussian tail is estimated by integrating the derivative of $\mathrm{e}^{-v^2/4}$ and using $|v| \geq A$. The constants are chosen so the result holds simultaneously for every $k \geq 0$ once $A \geq 4$; no optimization is needed in the saddle application. □

Theorem 5.15 (Polynomial Gaussian tail). *For every $p \in \mathbb{C}[X]$ and $A \geq 4$,*

$$\int_{|v|>A} \left|\mathrm{e}^{-v^2/2} p(v)\right| \mathrm{d}v \leq \mathfrak{W}(p)\frac{\mathrm{e}^{-A^2/4}}{A}. \tag{200}$$

Proof. Use $|p(v)| \leq \sum_{k\in\operatorname{supp} p} |p_k||v|^k$, integrate the finite sum, and apply lemma 5.14 coefficientwise. □

5.6.2 The all-orders central radius

For expansion order N and scale parameter $b \geq 1$, set

$$A_N(b) = \sqrt{32(N+1)\log b}. \tag{201}$$

Then

$$\mathrm{e}^{-A_N(b)^2/4} = b^{-8(N+1)}. \tag{202}$$

The exponent $8(N+1)$ is much stronger than the requested algebraic order $N+1$, leaving ample room for the factor $1/A_N(b)$ and for bounded coefficient weights.

Theorem 5.16 (All-orders polynomial tail)**.** *Let $b(a) \to +\infty$ along a filter and let $p_a \in \mathbb{C}[X]$ satisfy*

$$\mathfrak{W}(p_a) = O(1). \tag{203}$$

Then for every fixed N,

$$\int_{|v|>A_N(b(a))} \left|\mathrm{e}^{-v^2/2} p_a(v)\right| \,\mathrm{d}v = O\big(b(a)^{-(N+1)}\big). \tag{204}$$

Proof. Eventually $A_N(b(a)) \geq 4$ and $b(a) \geq 1$. Apply theorem 5.15, substitute (202), and use

$$b^{-8(N+1)} A_N(b)^{-1} \leq b^{-(N+1)}.$$

Multiplication by the bounded family (203) preserves the big-O rate. □

Remark 5.17. The theorem is formulated for a moving polynomial family. This is essential: the saddle reference polynomial depends periodically on the exact phase. Periodicity and smoothness make its finitely many coefficient functions bounded, hence its factorial weight is bounded.

5.7 Closed saddle jets and shifted Stirling numbers

The main article states several low-order jets. The new closed-form modules identify the entire jet sequence and explain the combinatorial coefficients.

Let $L = \log 2$ and let Ψ be the smooth one-periodic fluctuation in the exact negative Laplace exponent. The bounded exponent jets $Z_n(t)$ satisfy

$$Z_0(t) = -\frac{1}{2} + \frac{\Psi'(t)}{L}, \tag{205}$$

$$Z_{n+1}(t) = \frac{Z_n'(t)}{L} - (n+1)Z_n(t) + \frac{(-1)^{n+1} n!}{L}. \tag{206}$$

Define integers $c(n,m)$ by

$$\prod_{k=1}^{n}(z-k) = \sum_{m=0}^{n} c(n,m) z^m. \tag{207}$$

With $H_n = 1 + 1/2 + \cdots + 1/n$ and $H_0 = 0$, the recurrence solves explicitly.

Theorem 5.18 (Closed form for every bounded exponent jet)**.** *For every $n \geq 0$,*

$$Z_n(t) = (-1)^n n! \left(\frac{H_n}{L} - \frac{1}{2}\right) + \sum_{m=0}^{n} c(n,m) \frac{\Psi^{(m+1)}(t)}{L^{m+1}}. \tag{208}$$

Moreover

$$c(n,m) = s(n+1, m+1), \tag{209}$$

where $s(\cdot,\cdot)$ denotes the signed Stirling number of the first kind.

Proof. Let $D = L^{-1}\,\mathrm{d}/\,\mathrm{d}t$. The derivative-dependent part of (206) is generated by the operator product

$$(D-n)(D-(n-1))\cdots(D-1)\frac{\Psi'}{L}. \tag{210}$$

Expanding the product gives the sum in (208). The constant part C_n satisfies

$$C_{n+1} = -(n+1)C_n + \frac{(-1)^{n+1}n!}{L}, \quad C_0 = -\frac{1}{2}, \tag{211}$$

whose solution is $C_n = (-1)^n n!(H_n/L - 1/2)$, using $H_{n+1} = H_n + 1/(n+1)$. Finally, the coefficients of $\prod_{k=1}^{n}(z-k)$ satisfy

$$c(n+1,m) = c(n,m-1) - (n+1)c(n,m), \tag{212}$$

which is exactly the shifted signed-Stirling recurrence; the initial rows agree. □

The first four jets are

$$Z_0 = -\frac{1}{2} + \frac{\Psi'}{L}, \tag{213}$$
$$Z_1 = \frac{1}{2} - \frac{1}{L} - \frac{\Psi'}{L} + \frac{\Psi''}{L^2}, \tag{214}$$
$$Z_2 = -1 + \frac{3}{L} + \frac{2\Psi'}{L} - \frac{3\Psi''}{L^2} + \frac{\Psi^{(3)}}{L^3}, \tag{215}$$
$$Z_3 = 3 - \frac{11}{L} - \frac{6\Psi'}{L} + \frac{11\Psi''}{L^2} - \frac{6\Psi^{(3)}}{L^3} + \frac{\Psi^{(4)}}{L^4}. \tag{216}$$

All derivatives are evaluated at the common argument t. These formulas make boundedness, one-periodicity, and C^∞ regularity immediate.

5.7.1 The exponent coefficient at arbitrary order

Let $\varepsilon = \lambda^{-1/2}$ and v be the Gaussian variable. The coefficient of ε^m in the centered saddle exponent is, for $m \geq 1$,

$$E_m(t,v) = \mathrm{i}^m\left(\frac{Z_{m-1}(t)}{m!}v^m + \frac{(-1)^{m+1}}{m+2}v^{m+2}\right). \tag{217}$$

The second monomial is universal: it is independent of Ψ and of every jet. It arises when the jet slope $(-1)^m(m+1)!$, the factorial $(m+2)!$, and the extra factor $\mathrm{i}^{m+2} = -\mathrm{i}^m$ cancel.

By (208), each E_m is therefore completely explicit in terms of L^{-1} and $\Psi', \ldots, \Psi^{(m)}$. It has the parity law

$$E_m(t,-v) = (-1)^m E_m(t,v). \tag{218}$$

5.8 From exponent coefficients to periodic mass and log coefficients

Let $a_n(t,v)$ be the recursive exponential coefficient

$$\exp\left(\sum_{m\geq 1}\varepsilon^m E_m(t,v)\right) \sim \sum_{n\geq 0}\varepsilon^n a_n(t,v). \tag{219}$$

Parity preservation gives

$$a_n(t,-v) = (-1)^n a_n(t,v). \tag{220}$$

Define the mass coefficients by Gaussian contraction:

$$B_j(t) = \mathcal{G}[a_{2j}(t,\cdot)]. \tag{221}$$

Odd orders disappear. The normalized saddle mass therefore has the formal pattern

$$1 + \frac{B_1(t)}{\lambda} + \frac{B_2(t)}{\lambda^2} + \cdots . \tag{222}$$

Apply the logarithm recurrence to $1 + B_1 u + B_2 u^2 + \cdots$ and define

$$A_j(t) = b_j(B_0, B_1, \ldots), \qquad B_0 = 1. \tag{223}$$

Then

$$A_0 = 0, \qquad A_1 = B_1, \qquad A_2 = B_2 - \frac{1}{2}B_1^2. \tag{224}$$

Every A_j and B_j is bounded, one-periodic, and C^∞.

5.8.1 The highest derivative has a universal coefficient

The jet Z_{2j-1} first appears at mass order j. The formalized leading-jet theorem says that its coefficient in B_j is

$$\frac{(-1)^j}{2^j j!}. \tag{225}$$

The reason is rigid: among all monomials produced by the exponential recurrence, exactly one can contain the newest jet, namely the linear occurrence of the exponent coefficient E_{2j}. Its jet monomial is

$$\mathrm{i}^{2j}\frac{Z_{2j-1}}{(2j)!}v^{2j} = (-1)^j \frac{Z_{2j-1}}{(2j)!}v^{2j}. \tag{226}$$

Gaussian contraction multiplies by $(2j-1)!! = (2j)!/(2^j j!)$, giving (225).

Because lower B_k involve only $Z_0, \ldots, Z_{2k-1}$, every nonlinear term in the logarithm recurrence for A_j uses jets strictly below Z_{2j-1}. Thus the newest jet survives with the same coefficient. Combining with the leading term $\Psi^{(2j)}/L^{2j}$ in (208) yields the ordinary-analysis corollary

$$[\Psi^{(2j)}]A_j = \frac{(-1)^j}{2^j j! L^{2j}}. \tag{227}$$

The mass version is a named Lean theorem; the final triangular transfer to the logarithmic coefficient is presently a short prose corollary rather than a separate declaration.

5.9 The second periodic correction, line by line

The main article records the first correction and announces the general coefficient engine. The current development now computes the second correction completely. This is the first order at which every layer of the machinery is visible at once: four exponent polynomials, the order-four exponential coefficient, five Gaussian moments, and the mass-to-logarithm conversion.

Fix the phase t and abbreviate

$$z_j = Z_j(t). \tag{228}$$

From (217), the first four exponent polynomials are

$$E_1 = \mathrm{i}\left(z_0 X + \frac{X^3}{3}\right), \tag{229}$$

$$E_2 = -\frac{z_1}{2}X^2 + \frac{1}{4}X^4, \tag{230}$$

$$E_3 = \mathrm{i}\left(-\frac{z_2}{6}X^3 - \frac{1}{5}X^5\right), \tag{231}$$

$$E_4 = \frac{z_3}{24}X^4 - \frac{1}{6}X^6. \tag{232}$$

At order four in ε, (162) gives

$$a_4 = E_4 + E_3E_1 + \frac{1}{2}E_2^2 + \frac{1}{2}E_2E_1^2 + \frac{1}{24}E_1^4. \tag{233}$$

Every occurrence of i cancels because the odd exponent polynomials carry one common square root of -1. Expanding and collecting even monomials gives

$$a_4 = c_4X^4 + c_6X^6 + c_8X^8 + c_{10}X^{10} + c_{12}X^{12}, \tag{234}$$

where

$$c_4 = \frac{z_0^4}{24} + \frac{z_0^2 z_1}{4} + \frac{z_0 z_2}{6} + \frac{z_1^2}{8} + \frac{z_3}{24}, \tag{235}$$

$$c_6 = \frac{z_0^3}{18} - \frac{z_0^2}{8} + \frac{z_0 z_1}{6} + \frac{z_0}{5} - \frac{z_1}{8} + \frac{z_2}{18} - \frac{1}{6}, \tag{236}$$

$$c_8 = \frac{z_0^2}{36} - \frac{z_0}{12} + \frac{z_1}{36} + \frac{47}{480}, \tag{237}$$

$$c_{10} = \frac{z_0}{162} - \frac{1}{72}, \tag{238}$$

$$c_{12} = \frac{1}{1944}. \tag{239}$$

The five rational constants in these formulas are the visible trace of the universal jet-free monomials in (217).

5.9.1 Gaussian contraction

Using (193), the second mass coefficient is

$$B_2 = 3c_4 + 15c_6 + 105c_8 + 945c_{10} + 10395c_{12}. \tag{240}$$

A direct simplification gives

$$\begin{aligned} B_2 = {} & \frac{z_0^4}{8} + \frac{5z_0^3}{6} + \frac{3z_0^2 z_1}{4} + \frac{25z_0^2}{24} + \frac{5z_0 z_1}{2} + \frac{z_0 z_2}{2} + \frac{z_0}{12} \\ & + \frac{3z_1^2}{8} + \frac{25z_1}{24} + \frac{5z_2}{6} + \frac{z_3}{8} + \frac{1}{288}. \end{aligned} \tag{241}$$

The first mass coefficient, already logarithmic at first order, is

$$B_1 = A_1 = -\frac{z_0^2}{2} - z_0 - \frac{z_1}{2} - \frac{1}{12}. \tag{242}$$

Substitution of (213)–(214) reduces this to the compact formula from the main article,

$$A_1(t) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(t) + \Psi'(t)^2}{2L^2}. \tag{243}$$

5.9.2 Taking the logarithm

By (224),

$$A_2 = B_2 - \frac{1}{2}B_1^2. \tag{244}$$

The quartic and many lower-degree terms cancel. The result is

Theorem 5.19 (Closed second periodic correction)**.** *With $z_j = Z_j(t)$,*

$$\begin{aligned} A_2(t) = \frac{1}{24}\big(&8z_0^3 + 12z_0^2 z_1 + 12z_0^2 + 48z_0 z_1 + 12z_0 z_2 \\ &+ 6z_1^2 + 24z_1 + 20z_2 + 3z_3\big). \end{aligned} \tag{245}$$

It is a bounded smooth one-periodic function.

Proof. Insert (241) and (242) into (244), expand the square, and collect. Periodicity and smoothness follow from the same properties of $Z_0, \dots, Z_3$; boundedness follows from periodic continuity. □

Substitution of (213)–(216) makes A_2 an explicit differential polynomial in

$$\Psi', \ \Psi'', \ \Psi^{(3)}, \ \Psi^{(4)} \tag{246}$$

with coefficients in $\mathbb{Q}[L^{-1}]$. The jet form (245) is shorter, more stable under formal manipulation, and displays the triangular structure that generalizes to all orders.

At order two the mass coefficient (241) contains a quartic term $z_0^4/8$. It disappears from A_2 because the logarithm removes the disconnected square $B_1^2/2$. This is the same connected-versus-disconnected cancellation familiar from cumulants. The formal logarithm recurrence performs that cancellation over an arbitrary commutative $\mathbb{Q}$-algebra, before any analytic interpretation is imposed.

5.10 Assembly of the full endpoint expansion

We now put the separate engines into the order in which the formal proof uses them. This section does not rederive the negative-Laplace product or the Bromwich formula from the main article; it explains every transition from that exact representation to the all-orders statement.

5.10.1 Exact Lambert phase and exact normalized mass

Let $\lambda = \lambda(x)$ be the large positive solution of

$$\lambda 2^{-\lambda} = x, \qquad x \downarrow 0, \tag{247}$$

so

$$r = 2^{\lambda} = \frac{\lambda}{x}. \tag{248}$$

Write $\Lambda(r) = \log G(-r)$ for the negative log-transform. Define the dimensionless normalized saddle mass

$$\mathcal{S}(x) = \frac{F(x)\sqrt{2\pi\lambda}}{\exp(rx + \Lambda(r))}. \tag{249}$$

It is positive and satisfies the exact identity

$$\log F(x) = rx + \Lambda(r) - \frac{1}{2}\log(2\pi\lambda) + \log \mathcal{S}(x). \tag{250}$$

No asymptotic estimate has entered yet.

Let

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda). \tag{251}$$

The exact negative-Laplace decomposition gives

$$rx + \Lambda(r) - \frac{1}{2}\log(2\pi\lambda) = \mathcal{M}(x) + E(r), \tag{252}$$

where the forward tail satisfies

$$0 \le E(2^\lambda) \le 4\exp(-2^\lambda). \tag{253}$$

Therefore

$$\log F(x) - \mathcal{M}(x) = \log \mathcal{S}(x) + E(2^\lambda). \tag{254}$$

The final term is flat relative to λ^{-1} by elementary exponential domination and will be reattached using theorem 5.3.

5.10.2 The central integral and the moving window

Bromwich inversion and the scaling $\theta = v/\sqrt{\lambda}$ give

$$\mathcal{S}(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp \Phi_\lambda(v/\sqrt{\lambda})\, \mathrm{d}v. \tag{255}$$

The exact phase separates as

$$\Phi_\lambda(v/\sqrt{\lambda}) = -\frac{v^2}{2} + \sum_{m=1}^{M} \varepsilon^m E_m(\lambda, v) + R_{M+1}(\lambda, v), \qquad \varepsilon = \lambda^{-1/2}, \tag{256}$$

inside an order-dependent central window

$$|v| \le A_N(\lambda) = \sqrt{32(N+1)\log\lambda}. \tag{257}$$

The all-orders Taylor modules prove, uniformly on this moving window, that the truncated perturbation tends to zero and that the analytic remainder is dominated by the first omitted power times a polynomial in v. The slow growth of A_N is crucial:

$$\lambda^{-1/2} A_N(\lambda)^d \longrightarrow 0 \qquad \text{for every fixed } d. \tag{258}$$

5.10.3 Finite exponential substitution

Fix the requested logarithmic order N. Choose the exponent and scalar-exponential truncation orders high enough that every omitted term is $O(\varepsilon^{2N}) = O(\lambda^{-N})$. Apply the exact finite-remainder identity (184) pointwise with

$$x = \varepsilon, \qquad E_m = E_m(\lambda, v). \tag{259}$$

This yields, on the central window,

$$\exp\left(\sum_{m<M} \varepsilon^m E_m\right) = \sum_{k<2N} \varepsilon^k a_k(\lambda, v) + O\Big(\varepsilon^{2N} P_N(v)\Big), \tag{260}$$

where P_N is a fixed-degree polynomial with bounded periodic coefficient families. The analytic remainder R_{M+1} is absorbed by a finite scalar exponential estimate because the whole perturbation is eventually bounded by one on the central window.

5.10.4 Gaussian contraction and the disappearance of half-orders

Multiply (260) by $\mathrm{e}^{-v^2/2}$ and integrate. Parity gives

$$\mathcal{G}[a_{2j+1}(\lambda, \cdot)] = 0. \tag{261}$$

For the even terms,

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \mathrm{e}^{-v^2/2} a_{2j}(\lambda, v)\, \mathrm{d}v = B_j(\lambda). \tag{262}$$

The central-window integral may be replaced by the whole-line integral because theorem 5.16 makes the complementary Gaussian-polynomial tail $O(\lambda^{-N})$.

The genuine saddle integrand outside the central window is controlled separately. The minor-arc estimates combine strict off-axis decay of the negative-Laplace product with far-tail polynomial bounds. They show that the noncentral part of (255) is also $O(\lambda^{-N})$ for every fixed N. Consequently

$$\mathcal{S}(x) = 1 + \sum_{j=1}^{N-1} \frac{B_j(\lambda)}{\lambda^j} + O(\lambda^{-N}). \tag{263}$$

This is the full mass expansion in the sense of definition 5.1.

5.10.5 Logarithmic transfer and restoration of the flat tail

The ratio in (263) tends to 1 and is positive. Apply the formal logarithm recurrence and its finite analytic remainder theorem:

$$\log \mathcal{S}(x) = \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + O(\lambda^{-N}). \tag{264}$$

Now restore $E(2^\lambda)$ in (254). It is flat by (253), so theorem 5.3 leaves every coefficient unchanged.

Theorem 5.20 (Full exact-phase logarithmic expansion)**.** *For every $N \geq 1$, as $x \downarrow 0$,*

$$\boxed{\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} + O\Big(\lambda(x)^{-N}\Big).} \tag{265}$$

The coefficient functions A_j are bounded, smooth, and one-periodic; A_1 and A_2 are given by (242) *and* (245)*.*

Proof. The preceding five subsections establish the expansion on the dyadic logarithmic coordinate. Use theorem 5.4 to transport it to the small-positive filter. The exact phase identity $\lambda(2^{-t})$ is retained throughout, so the coefficient families pull back without approximation. □

Because $\lambda(x)$ is comparable to $-\log x$, the remainder may be weakened to

$$O((-\log x)^{-N}), \tag{266}$$

but the coefficients still remain evaluated at the exact Lambert phase.

5.11 Why the exact phase cannot be casually re-expanded

The Lambert phase itself has an all-orders expansion in $T = -\log x$ and $\log T$. It is tempting to substitute that expansion into every $A_j(\lambda(x))$ and claim an ordinary nested-log series. The current formal theorem does not do so, for a good reason.

Let $\lambda_N(x)$ be a finite asymptotic approximation to $\lambda(x)$ and write

$$\Delta_N(x) = \lambda(x) - \lambda_N(x). \tag{267}$$

For a nonconstant periodic coefficient,

$$A_j(\lambda) - A_j(\lambda_N) = A_j'(\lambda_N)\Delta_N + \frac{1}{2}A_j''(\xi)\Delta_N^2. \tag{268}$$

To preserve an $O(\lambda^{-N})$ total remainder, the phase approximation must be accurate enough after multiplication by every prefactor λ^{-j} and after repeated Taylor composition through all coefficient families. A statement merely of the form $\Delta_N = O(\lambda^{-N})$ at one fixed order is not automatically stable under the entire triangular composition.

Warning 5.21 (The formalized boundary). The theorem proved in `FabiusFullAsymptoticExpansion.lean` is (265): every periodic coefficient is evaluated at the exact lower-Lambert phase. A separate branch expands the phase itself to all orders. What is not currently claimed is a single all-orders composition theorem that substitutes the phase series through every oscillatory coefficient while preserving the requested remainder. This is a missing formal composition layer, not a defect in the exact-phase expansion.

The exact-phase form is in any case mathematically preferable. It isolates the implicit saddle inversion once, avoids secular phase error, and makes phase-locked subsequences transparent. For any fixed $u \in \mathbb{R}$ and all sufficiently large integers n, the sequence

$$x_n = (n+u)2^{-(n+u)} \tag{269}$$

has $n + u > 0$ on the large lower-Lambert branch and therefore has exact phase $\lambda(x_n) = n + u$. Hence every periodic coefficient is literally constant along it:

$$A_j(\lambda(x_n)) = A_j(u). \tag{270}$$

This is the cleanest way to separate algebraic decay from periodic oscillation.

Supplementary material from this dossier

5.12 Executable recurrences in mathematical pseudocode

The following algorithms are not substitutes for the proofs above. They are compact implementations of the finite algebra that appears repeatedly in the formal development. All arrays are zero-indexed.

Algorithm 5.22 (Weighted path sum)**.** The dynamic program below computes $P_w(n)$ without enumerating compositions. It is the triangular recurrence proved equivalent to the composition sum in theorem 4.2.

```
function PathSums(N, weight):
    P[0] := 1
    for n := 1 to N:
        P[n] := 0
        for k := 0 to n - 1:
            P[n] := P[n] + P[k] * weight(k, n)
    return P

function FabiusInverseDyadic(N):
    weight(k, n) := 1 / ((2^n - 1) * factorial(n - k + 1))
    a := PathSums(N, weight)
    for n := 0 to N:
        value[n] := 2^(-binomial(n, 2)) * a[n]
    return value
```

Algorithm 5.23 (Formal exponential coefficients)**.** Given $E[1], \dots, E[N]$ in a commutative $\mathbb{Q}$-algebra, this returns the first N coefficients of $\exp(\sum_{j\geq 1} E[j]u^j)$.

```
function ExpCoeff(E, N):
    a[0] := 1
    for n := 0 to N - 1:
        s := 0
        for j := 0 to n:
            s := s + (j + 1) * E[j + 1] * a[n - j]
        a[n + 1] := s / (n + 1)
    return a
```

Algorithm 5.24 (Formal logarithm coefficients)**.** The input is a unit series $1 + \sum_{j\geq 1} a[j]u^j$. The output satisfies $\log(1 + \sum a[j]u^j) = \sum b[j]u^j$.

```
function LogCoeff(a, N):
    b[0] := 0
    for n := 0 to N - 1:
        s := 0
        for j := 0 to n - 1:
            s := s + (n - j) * b[n - j] * a[j + 1]
        b[n + 1] := a[n + 1] - s / (n + 1)
    return b
```

Algorithm 5.25 (Normalized Gaussian contraction)**.** The moment table can be built once to the degree of the input polynomial.

```
function GaussianMoments(D):
    mu[0] := 1
    if D >= 1: mu[1] := 0
    for n := 0 to D - 2:
        mu[n + 2] := (n + 1) * mu[n]
    return mu

function GaussianContract(coeff, D):
    mu := GaussianMoments(D)
    result := 0
    for k := 0 to D:
        result := result + coeff[k] * mu[k]
    return result
```

Algorithm 5.26 (Saddle coefficient pipeline)**.** This is the finite algebra at fixed logarithmic order N. The analytic proof is what justifies applying it to the central integral with a uniform remainder.

```
function SaddleCoefficients(N, phase):
    # 1. Build bounded exponent jets Z[0 .. 2N - 1].
    Z := ClosedJets(2*N - 1, phase)

    # 2. Build exponent polynomials E[1 .. 2N].
    for m := 1 to 2*N:
        E[m] := i^m * (Z[m - 1] * X^m / factorial(m)
                       + (-1)^(m + 1) * X^(m + 2) / (m + 2))

    # 3. Exponentiate formally in epsilon.
    a := ExpCoeff(E, 2*N)

    # 4. Contract even epsilon orders against the normalized Gaussian.
    B[0] := 1
    for j := 1 to N:
        B[j] := GaussianContract(coefficients(a[2*j]), degree(a[2*j]))

    # 5. Convert mass coefficients to logarithmic coefficients.
    A := LogCoeff(B, N)
    return (B, A)
```

5.13 Lean declaration and module concordance

File names below are relative to

`Analysis/FabiusFunction/Lean/FabiusFunction/`.

The table lists the principal declarations, not every supporting lemma.

Module	Content explained in this companion
`OriginalCharacterization.lean`	`IsOriginalFabius.mk_of_derivative_law`: the support/mean-value argument showing that scale positivity follows from the other original-characterization assumptions; section 1.3.

Module	Content explained in this companion
`FabiusInverse.lean`	`fabiusOrderIso`, `fabiusInv`, `fabiusReal_fabiusInv`, `fabiusInv_fabiusReal`, `continuous_fabiusInv`, `strictMonoOn_fabiusInv`, `fabiusInv_one_sub`, `fabiusInv_fabiusDyadicUnit`, `fabiusInv_hasDerivAt`, `deriv_fabiusInv`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, `deriv_fabiusInv_pos`, `fabiusInv_contDiffOn_Ioo`, `deriv_deriv_fabiusInv`, `strictConcaveOn_fabiusInv_firstHalf`, `strictConvexOn_fabiusInv_secondHalf`, the transported root-free bounds, `mul_lt_fabiusInv`, `tendsto_fabiusInv_div_atTop`, and reflected companions; sections 2.1 and 2.2.
`ElementaryFunction.lean`	The inductive predicate `IsElementary`, composition closure, derived elementary operations, analytic loci, the finite-bad-value composition theorem, and the dense-open analytic-locus theorem; section 2.3.
`NotElementary.lean`	No elementary function agrees with a bounded Fabius function, up, or the signed extension on a set with nonempty interior; section 2.4.
`ProbabilityRepresentation.lean`	The product sample space, uniformly convergent coordinate sum, pushforward law, head–tail split, coordinate reflection, and identification of the CDF with F; section 3.1.
`ProbabilityLaplaceMoments.lean`	Almost-sure support, restriction to $[0, 1]$, survival function, the compact-support Fubini identity, raw and endpoint moment identifications, zero-tilt normalization, and positivity; sections 3.2 and 3.3.
`LaplaceMoments.lean`	`tiltedSurvivalMoment`, `fabiusLaplaceMoment`, the differential recurrence, iterated derivatives, smoothness, and evaluation at zero tilt; section 3.3.
`StepMeasureBridge.lean`	Almost-everywhere replacement of half-endpoint indicators, exact overlap integrals, atomic measure comparisons, the shrunken/enlarged interval sandwich, portmanteau, and convergence of every interval integral; section 3.4.
`FabiusInverseDyadicClosedForm.lean`	Compositions, path weights, last-edge decomposition, generic triangular solver, uniqueness, and the exact composition formula for $F(2^{-n})$; sections 4.1 and 4.2.
`FabiusDiscreteLimitToeplitz.lean`	Corrected prefix length, exact q-binomial row mass, total-variation bound 16, the generic finite-row Toeplitz theorem, and the specialization $q(n, j) = 2n - j$; sections 4.3 and 4.4.
`SaddleExpansionAlgebra.lean`	`expCoeff`, its differential equation, uniqueness, naturality, rescaling, addition, parity, and `HasAsymptoticExpansion`; sections 5.1 and 5.2.
`SaddleLogExpansionAlgebra.lean`	`logCoeff`, the identity $AB' = A'$, low-order formulas, uniqueness, naturality, rescaling, and parity; section 5.3.
`SaddleExpansionFiniteRemainder.lean`	Finite exponent substitution, equality of coefficients below the truncation order, divisibility of the defect by X^L, and the evaluated exact remainder; section 5.4.
`SaddleExpansionSmallArgument.lean`	Exact transfer of full expansions between $t \to \infty$ and $x = 2^{-t} \downarrow 0$; section 5.1.
`SaddleExpansionFlat.lean`	Zero-coefficient expansion of all-orders-flat errors and invariance of every coefficient under addition or subtraction of such an error; sections 5.1 and 5.10.

Module	Content explained in this companion
`GaussianPolynomialContraction.lean`	Gaussian integrability, odd moments, integration-by-parts recurrence, normalized moments, linear polynomial contraction, and the whole-line integral identity; section 5.5.
`GaussianPolynomialTail.lean`	The factorial coefficient weight and explicit complementary-interval tail bound; section 5.6.
`GaussianPolynomialTailAllOrders.lean`	The radius $\sqrt{32(N+1)\log b}$ and arbitrary algebraic tail order for bounded polynomial families; section 5.6.
`FabiusSaddleJetClosedForm.lean`	Closed formula for the periodic and bounded exponent jets, including harmonic constants and the shifted falling-factorial coefficients; section 5.7.
`FabiusSaddleJetStirling.lean`	Identification $c(n,m) = s(n+1, m+1)$ with signed Stirling numbers of the first kind; section 5.7.
`FabiusSaddleExponentClosedForm.lean`	Cancellation of factorials and the extra power of i, yielding the two-monomial formula (217); section 5.7.
`FabiusSaddleLeadingCoefficient.lean`	The unique newest-jet monomial and coefficient $(-1)^j/(2^j j!)$ in the jth mass coefficient; section 5.8.
`FabiusSecondSaddleCorrection.lean`	Low-order formal exponential identities, explicit order-four polynomial, Gaussian contraction, closed B_2, and closed A_2; section 5.9.
`FabiusSecondSaddleExpansion.lean`	The explicit expansion through A_2/λ^2 with $O(\lambda^{-3})$ and the weakened $O((-\log x)^{-3})$ form; sections 5.9 and 5.10.
`FabiusSaddleMassAllOrders.lean`	Finite exponential substitution, bounded jet families, central reference polynomial, Gaussian whole-line contraction, and all-orders normalized saddle mass; section 5.10.
`FabiusFullAsymptoticExpansion.lean`	Transfer from complex kernel mass to the real saddle ratio, logarithmic coefficient transfer, addition of the flat forward tail, exact-phase full expansion, and small-argument transport; sections 5.10 and 5.11.

5.14 Audit matrix against the main article

Topic	Status before this companion	Added here
Original scale positivity	Assumed in the source statement	Mean-value proof that the dilation parameter is automatically positive before its exact value $k = 2$ is derived.
Total inverse I	Absent	Global clamped construction, order calculus, reflection, exact dyadic inverse values, interior calculus consequences, endpoint steepness, and non-Hölder corollaries.
Elementary function class	Absent	Exact inductive closure, totalization convention, finite-bad-value composition, and dense-open analytic locus.
Non-elementarity	Absent	No local agreement on any set with nonempty interior for F, up, and the signed extension.
Product probability law	Headline proof only	Uniform convergence and measurability, head–tail pushforward identity, coordinate reflection, and global support bookkeeping.
Probability–Laplace bridge	Absent	Survival-function Fubini identity, identification of raw, endpoint, and tilted moments, complete monotonicity, and half-moment interpretation.

Topic	Status before this companion	Added here
Atomic-to-step bridge	Compressed	Half-endpoint null-set conversion, exact cell overlap, shrunken/enlarged interval sandwich, portmanteau, and moving-boundary squeeze.
Weighted path solver	Explicitly omitted	Generic semiring theorem, uniqueness, edge-count decomposition, Fabius specialization, and worked values.
Finite-row Toeplitz theorem	Explicitly omitted	Abstract normed-ring proof, exact Fabius row mass, uniform variation bound, tail-index condition, and corrected prefix rounding.
Parameter-dependent full expansions	Only summarized	Exact API, coefficient boundedness, closure, flat errors, and coordinate transfer.
Formal exponential/logarithm	Explicitly omitted	Recurrences, formal differential equations, uniqueness, low orders, naturality, rescaling, addition, and parity.
Finite formal-to-analytic remainder	Absent	Exact divisibility by the first omitted power and evaluated polynomial remainder.
Gaussian contraction/tails	Explicitly omitted	Moment recurrence, contraction functional, factorial coefficient weight, explicit tail, and all-orders radius.
Closed saddle jets	Stated, lightly motivated	Operator-product solution, harmonic constants, shifted signed Stirling identification, and first four bounded jets.
Second saddle correction	Not yet in pinned snapshot	Complete E_1–E_4 expansion, order-four polynomial, Gaussian contraction, B_2, and closed A_2.
Full saddle assembly	Proof architecture only	Exact sequence of finite remainder, central window, contraction, Gaussian tail, minor arc, log transfer, flat-tail restoration, and filter transport.
Exact phase boundary	Warning only	Quantified explanation of why all-orders substitution through oscillatory coefficients needs a separate composition theorem.

The following substantial topics are intentionally not repeated because the main article already treats them at proof level: construction and uniqueness of F and up; the original Rvachev characterization apart from the redundancy isolated in section 1.3; the infinite sinc product and Fourier decay; support, positivity, shape, and nowhere analyticity; exact moment recurrences and denominator arithmetic; dyadic rational evaluation, q-Pochhammer and q-binomial formulas; Thue–Morse interpolation and iterated difference identities; Legendre expansions and least-squares approximation; and the computational API. Their use as input above is always signposted.

5.15 Notation crosswalk

This companion	Main-article notation	Principal Lean declaration
$I(y)$	inverse of bounded F	`fabiusInv`
μ	law of the weighted uniform sum	`weightedSumDistribution`
$J_k(s)$	tilted survival moment	`tiltedSurvivalMoment`
$M_k(s)$	tilted probability/Fabius moment	`fabiusLaplaceMoment`

This companion	Main-article notation	Principal Lean declaration
d_n	exact half moment	`halfMoment`
$P_w(n)$	weighted composition/path sum	generic path-sum definition in `FabiusInverseDyadicClosedForm.lean`
σ	abstract small scale	first argument of `HasAsymptoticExpansion`
a_n	formal exponential coefficient	`SaddleExpansion.expCoeff`
b_n	formal logarithm coefficient	`SaddleExpansion.logCoeff`
$\mathcal{G}[p]$	normalized Gaussian contraction	`SaddleExpansion.gaussianPolynomialContraction`
$Z_n(t)$	bounded exponent jet $\widetilde{\mathcal{Z}}_n$	`negativeLaplaceBoundedExponentJet`
$B_j(t)$	normalized saddle mass coefficient	`fabiusSaddleMassCoefficient`
$A_j(t)$	logarithmic saddle coefficient	`fabiusSaddleLogCoefficient`
$\lambda(x)$	exact lower-Lambert phase	`fabiusLambertPhase`
$\mathcal{M}(x)$	corrected sharp Lambert main	`fabiusSharpLambertMain`
$\mathcal{S}(x)$	normalized real saddle mass/ratio	`fabiusSaddleRatio` and the kernel-mass bridge

5.16 A compact dependency diagram

The proof dependencies of the material unique to this companion can be read from the following diagram. An arrow means that the target uses the source as a mathematical input, not merely as a Lean import.

support + interior positivity + mean-value theorem + derivative law $\longrightarrow$ $k > 0$,

strict monotonicity + surjectivity + flatness $\longrightarrow$ total inverse $\longrightarrow$ endpoint steepness,

nowhere analyticity + dense elementary analytic locus $\longrightarrow$ strong non-elementarity,

product law $\longrightarrow$ survival identity $\longrightarrow$ Laplace/raw/endpoint moment bridge,

weak convergence + null boundaries + cell geometry $\longrightarrow$ step-integral convergence,

triangular recurrence $\longrightarrow$ weighted paths $\longrightarrow$ composition formula,

q-binomial row mass + variation + tail indices $\longrightarrow$ discrete Toeplitz limit,

closed jets $\longrightarrow$ exponent coefficients $\longrightarrow$ formal exponential
$\longrightarrow$ Gaussian contraction $\longrightarrow$ mass coefficients,

mass coefficients $\longrightarrow$ formal logarithm $\longrightarrow$ log coefficients,

finite remainders + central Taylor + Gaussian tails + minor arcs
$\longrightarrow$ all-orders saddle mass,

all-orders mass + exact reduction + flat tail + coordinate transfer
$\longrightarrow$ full exact-phase expansion.

Dossier references

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[2] J. Arias de Reyna, On the Fabius function, *The American Mathematical Monthly* **89** (1982), 291–298.

[3] J. Arias de Reyna, An infinitely differentiable function with compact support: Definition and properties, arXiv:1702.05442.

[4] J. Arias de Reyna, Arithmetic of the Fabius function, arXiv:1702.06487, version 3 (2017).

[5] V. Reshetnikov, *The Fabius Function and Rvachev's Up-Function: Exact Arithmetic, Probability, Fourier Analysis, and Corrected Sharp Asymptotics*, `ProveIt`, `Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/`.

[6] V. Reshetnikov, `ProveIt`: formal development of the Fabius function and Rvachev's up-function, GitHub repository, snapshot `531338d79f2f640041562138c7b80772ee2f0490`, 25 August 2026, https://github.com/VladimirReshetnikov/ProveIt/tree/531338d79f2f640041562138c7b80772ee2f0490/Analysis/FabiusFunction.

Part II
Repeated Integration and Rvachev's Up-Function

Source provenance and status. Former path: `Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex`. Distinct box-spline and repeated-integration dossier preserving the 2012 Mathematics Stack Exchange provenance, exact finite-stage formulas, sharp-error claims, and formalization roadmap. Its new mathematical statements are natural-language proofs awaiting formalization; the all-orders theorem is supported here by a proof sketch, and the variance-matched 16^{-n} scheme is only a conjectural direction.

Original source SHA-256: `eadcb4b414ac93723a91acbd5062d44340f78134a2cecbc18f7d7ba67eb2c9be`.

Dossier abstract

A 2012 Mathematics Stack Exchange question starts from the discontinuous density $f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x)$ and repeatedly applies

$$f_n(x) = 2\int_{-1}^{x} \big(f_{n-1}(2t+1) - f_{n-1}(2t-1)\big)\,\mathrm{d}t.$$

The graphs rapidly approach Rvachev's compactly supported up-function, but the precise finite-stage structure and the rate of convergence were left unexplained. This report gives both in exact form.

The integral first collapses to $f_n(x) = \int_{2x-1}^{2x+1} f_{n-1}(t)\,\mathrm{d}t$. Consequently f_n is the density of a sum of $n+1$ independent uniform variables with dyadic half-widths $2^{-1}, 2^{-2}, \ldots, 2^{-n}, 2^{-n}$: the smallest box occurs twice. Thus f_n is a compact box spline of degree n, supported on $[-1,1]$, and its Fourier transform is a finite sinc product. Expanding the box spline gives the exact Thue–Morse formula

$$f_n(x) = \frac{2^{\binom{n+1}{2}-1}}{n!}\sum_{j=0}^{2^n}\big(c_n(j) - c_n(j-1)\big)\left(x+1-\frac{j}{2^{n-1}}\right)_+^n, \qquad c_n(j) = (-1)^{\mathrm{wt}_2(j)}$$

inside the natural range, with $c_n(j) = 0$ outside it.

The main new quantitative conclusion is exact, not merely asymptotic. If up denotes the normalization used in the companion document, then

$$\|f_n - \mathrm{up}\|_\infty = \begin{cases} \frac{1}{2}, & n=0,\\ \frac{13}{72}, & n=1,\\ \frac{8}{9}\,4^{-n}, & n\ge 2.\end{cases}$$

For every fixed derivative order $m \ge 0$, an exact analogue holds once the finite spline has enough smoothness:

$$\left\|f_n^{(m)} - \mathrm{up}^{(m)}\right\|_\infty = \frac{2^{\binom{m+3}{2}}}{9}\,4^{-n}, \qquad n \ge m+3.$$

The proof factors the error through a nonnegative Peano kernel of mass 1/9. Probabilistically, the iteration replaces the true dyadic tail $2^{-n}Y$ by a uniform random variable on the same

support. The two tails have equal mass and mean, while their variance differs by $\frac{2}{9}4^{-n}$; convex order and an exact Thue–Morse plateau in a derivative of the finite prefix spline turn this second-moment discrepancy into the sharp norm identity. A Fourier analysis further gives an all-orders expansion beginning $f_n = \mathrm{up} + 4^{-n}\,\mathrm{up}''/9 + 2\,16^{-n}\,\mathrm{up}^{(4)}/2025 + \cdots$.

6 The overlooked construction

The detailed Fabius–Rvachev exposition in the `ProveIt` repository develops the bounded Fabius distribution function, its signed global continuation, Rvachev's even bump, the infinite box convolution, the sinc product, exact dyadic arithmetic, and sharp derivative bounds [9, 10]. It also records the standard finite convolution obtained by *truncating* the random series. A closely related construction, however, begins from a single piecewise constant function and repeatedly integrates a rescaled difference. It was posed independently on Mathematics Stack Exchange in 2012 [7]; Nathan Grigg found an exact Heaviside/Thue–Morse expression for every finite stage [8], and other answers identified the limit with the Fabius–Rvachev function.

What remained missing is the structural explanation of the finite stages and, especially, a sharp convergence theorem. The repeated-integration family is not exactly the usual partial box convolution. It is better: it appends one additional uniform box whose radius is exactly the radius of the omitted infinite tail. This makes every stage have the final support $[-1, 1]$, preserves mass and symmetry, and cancels the first-order tail error. The remaining error is controlled by a variance defect and therefore has scale 4^{-n}.

The following display is the final norm law proved below.

Exact convergence, including the constant. Let $f_0 = \frac{1}{2}\mathbf{1}_{(-1,1)}$ and let f_n be generated by the repeated integration. Then, for every $n \geq 2$,

$$\boxed{\|f_n - \mathrm{up}\|_\infty = \frac{8}{9\,4^n}.}$$

The maximum absolute error is attained at all four points $x = \pm\frac{1}{4}, \pm\frac{3}{4}$; the signs are negative at $\pm\frac{1}{4}$ and positive at $\pm\frac{3}{4}$. Hence every further integration divides the uniform error by exactly four.

The proof is elementary once the correct factorization is visible, but it combines four viewpoints that are usually presented separately:

1. an affine Markov recursion;
2. a finite nonuniform box spline;
3. a Thue–Morse truncated-power formula;
4. a positive Peano kernel encoding convex order.

7 Normalization and the integration operator

Throughout, up is the even probability density supported on $[-1, 1]$ and normalized by $\mathrm{up}(0) = 1$. Its Fourier transform under

$$\widehat{g}(\xi) = \int_{\mathbb{R}} g(x) e^{-\mathrm{i}\xi x}\,\mathrm{d}x$$

is

$$\widehat{\mathrm{up}}(\xi) = \prod_{k=1}^{\infty} \mathrm{sinc}(2^{-k}\xi), \qquad \mathrm{sinc}\, z = \frac{\sin z}{z}, \tag{271}$$

and it is the density of

$$Y = \sum_{k=1}^{\infty} 2^{-k} U_k, \qquad U_k \overset{\text{iid}}{\sim} \mathrm{Unif}[-1,1]. \tag{272}$$

This is the centered normalization used in the companion article [9]; on $0 \le x \le 1$ it is related to the bounded Fabius function by

$$\mathrm{up}(x) = F(1-x), \qquad \mathrm{up}(-x) = \mathrm{up}(x). \tag{273}$$

Define the initial density

$$\rho(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x) \tag{274}$$

and, for a function g supported on $[-1,1]$, define

$$(Tg)(x) = 2\int_{-1}^{x} \big(g(2t+1) - g(2t-1)\big)\,\mathrm{d}t. \tag{275}$$

Endpoint values of ρ are immaterial for integration and convolution; the convention in (274) agrees with the original question.

Lemma 7.1 (The recursive integral is a moving integral)**.** *For every integrable g supported on $[-1,1]$,*

$$(Tg)(x) = \int_{2x-1}^{2x+1} g(t)\,\mathrm{d}t. \tag{276}$$

Proof. In the first term of (275), put $u = 2t+1$; in the second, put $u = 2t-1$. This gives

$$(Tg)(x) = \int_{-1}^{2x+1} g(u)\,\mathrm{d}u - \int_{-3}^{2x-1} g(u)\,\mathrm{d}u.$$

The lower limit -3 in the second integral may be replaced by -1 because g vanishes to the left of -1. Subtracting the two primitives gives (276), with the usual oriented-integral interpretation when the endpoints are reversed. □

Proposition 7.2 (Elementary invariants)**.** *Suppose g is a nonnegative even probability density supported on $[-1,1]$. Then Tg has the same four properties. Moreover Tg is continuous whenever $g \in L^1$, and*

$$(Tg)'(x) = 2\big(g(2x+1) - g(2x-1)\big) \tag{277}$$

at every point at which the right side is continuous.

Proof. Nonnegativity is immediate from (276). If $|x| \ge 1$, the integration interval meets the support only in a null endpoint, so $Tg(x) = 0$. Evenness follows by the substitution $t \mapsto -t$. For the mass, either use Fubini or use the probabilistic interpretation in Proposition 8.1 below. Continuity is absolute continuity of the indefinite integral, and differentiating its two affine endpoints gives (277). □

The repeated-integration sequence is therefore

$$f_0 = \rho, \qquad f_n = T^n \rho. \tag{278}$$

Every f_n is an even probability density supported on $[-1,1]$. Also $f_n(0) = 1$ for every $n \ge 1$, since

$$f_n(0) = \int_{-1}^{1} f_{n-1}(t)\,\mathrm{d}t = 1. \tag{279}$$

8 The affine random recursion

The most economical interpretation of T is probabilistic.

Proposition 8.1 (The operator as an affine smoothing map)**.** *Let X have density g, let $U \sim \mathrm{Unif}[-1,1]$ be independent of X, and put*

$$Z = \frac{X+U}{2}. \tag{280}$$

Then Z has density Tg.

Proof. Conditioning on $X = t$, scaling the density of $X + U$, and then integrating gives

$$f_Z(x) = 2\int_{\mathbb{R}} g(t)\frac{1}{2}\mathbf{1}_{[-1,1]}(2x-t)\,\mathrm{d}t = \int_{2x-1}^{2x+1} g(t)\,\mathrm{d}t.$$

Use Lemma 7.1. □

Thus the original iteration is the law recursion

$$X_0 \sim \mathrm{Unif}[-1,1], \qquad X_n \overset{d}{=} \frac{X_{n-1}+U_n}{2}. \tag{281}$$

Unrolling and relabeling the independent variables yields the exact finite sum below.

Theorem 8.2 (Exact random-sum representation)**.** *For every $n \geq 1$,*

$$X_n \overset{d}{=} \sum_{k=1}^{n} 2^{-k}U_k + 2^{-n}V, \tag{282}$$

where $U_1, \ldots, U_n, V$ are independent and uniform on $[-1,1]$. Consequently

$$X_n \longrightarrow Y = \sum_{k=1}^{\infty} 2^{-k}U_k \tag{283}$$

almost surely under the coupling in which the same U_k are used and V is fixed.

Proof. Repeated substitution in (281) gives weights $2^{-1}, 2^{-2}, \ldots, 2^{-n}$ for the newly introduced variables and another weight 2^{-n} for the original X_0. Since all variables have the same law, they may be relabeled to obtain (282). The difference between its right side and (272) is bounded by

$$2^{-n}|V| + \sum_{k=n+1}^{\infty} 2^{-k}|U_k| \leq 2^{1-n},$$

which tends to zero. □

Remark 8.3 (The natural Markov chain need not converge pathwise)*.* The almost-sure convergence just used is a coupling of the *laws*. The forward recursion $X_n = (X_{n-1} + U_n)/2$ continually inserts fresh noise of size $1/2$, so it should be viewed as a Markov chain converging to its stationary law. Reordering the i.i.d. summands produces the nested series coupling (283).

The stationary law of (281) satisfies $Y \overset{d}{=} (Y + U)/2$. At the density level this is $T\,\mathrm{up} = \mathrm{up}$. Differentiating recovers Rvachev's equation

$$\mathrm{up}'(x) = 2\big(\mathrm{up}(2x+1) - \mathrm{up}(2x-1)\big). \tag{284}$$

Thus the repeated integration does not merely resemble the up-function: it is iteration of the natural affine smoothing operator for which up is the stationary density.

9 The finite stages are box splines

For $a > 0$, write

$$\beta_a(x) = \frac{1}{2a}\mathbf{1}_{[-a,a]}(x), \tag{285}$$

the uniform probability density on the interval of half-width a.

Theorem 9.1 (The smallest dyadic box occurs twice)**.** *For every $n \geq 1$,*

$$\boxed{f_n = \beta_{2^{-1}} * \beta_{2^{-2}} * \cdots * \beta_{2^{-n}} * \beta_{2^{-n}}.} \tag{286}$$

In particular, f_n is a compact box spline of degree n, belongs to $C^{n-1}(\mathbb{R})$, is not C^n, and has support exactly $[-1, 1]$.

Proof. The convolution identity is the density form of (282). A convolution of $n+1$ interval indicators is piecewise polynomial of degree n and gains one order of continuity at each convolution after the first; the nth piecewise derivative retains jumps. The support radius is

$$\sum_{k=1}^{n} 2^{-k} + 2^{-n} = 1.$$

Nonnegative compactly supported factors have convolution support equal to the Minkowski sum of their interval supports, so no smaller support is possible. □

The standard partial convolution in the companion article is

$$p_n = \beta_{2^{-1}} * \beta_{2^{-2}} * \cdots * \beta_{2^{-n}}, \tag{287}$$

which is the density of the truncated sum $S_n = \sum_{k=1}^{n} 2^{-k} U_k$ and has support radius $1 - 2^{-n}$. The present iteration is therefore

$$f_n = p_n * \beta_{2^{-n}}. \tag{288}$$

The additional box exactly fills the missing support radius. This single observation is the key difference between the repeated-integration family and ordinary truncation.

9.1 The first stages

The first three functions already display the regularity gain:

$$f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x), \tag{289}$$

$$f_1(x) = (1 - |x|)_+, \tag{290}$$

$$f_2(x) = \begin{cases} 1 - 2x^2, & |x| \leq \frac{1}{2}, \\ 2(1 - |x|)^2, & \frac{1}{2} \leq |x| \leq 1, \\ 0, & |x| \geq 1. \end{cases} \tag{291}$$

The next stage is a piecewise cubic, and in general the degree equals the iteration index.

10 Fourier product and exact identification of the limit

Under the Fourier convention fixed above,

$$\widehat{\beta_a}(\xi) = \operatorname{sinc}(a\xi). \tag{292}$$

Therefore Theorem 9.1 gives the transform conjectured in the original question, without any induction through piecewise polynomials.

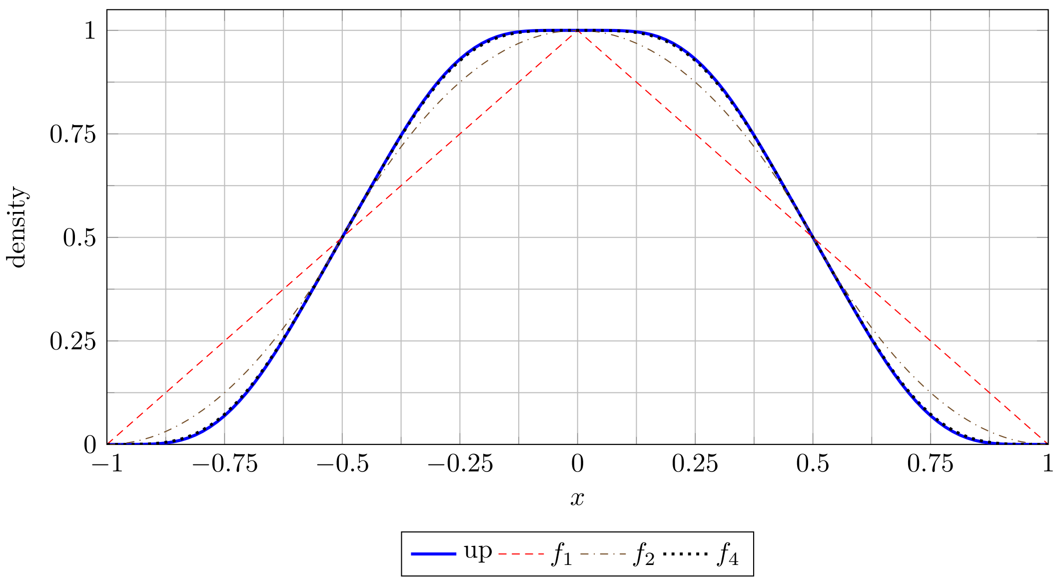

Figure 1: The first smooth finite box splines and their limit. The plotted up-function is a high-stage numerical evaluation used only for illustration.

Theorem 10.1 (Finite Fourier product)**.** *For every $n \geq 1$,*

$$\widehat{f}_n(\xi) = \operatorname{sinc}(2^{-n}\xi) \prod_{k=1}^{n} \operatorname{sinc}(2^{-k}\xi) \tag{293}$$

$$= \left(\prod_{k=1}^{n-1} \operatorname{sinc}(2^{-k}\xi) \right) \operatorname{sinc}^2(2^{-n}\xi). \tag{294}$$

As $n \to \infty$, this converges pointwise to (271)*.*

The extra front factor in the Stack Exchange formula is now transparent: it is the Fourier transform of the duplicated smallest box. It is not an arbitrary normalization correction; it records the support-filling tail replacement.

A useful formula for the operator itself is

$$\widehat{Tg}(\xi) = \operatorname{sinc}(\xi/2)\widehat{g}(\xi/2). \tag{295}$$

Iterating (295) from $\widehat{\rho}(\xi) = \operatorname{sinc}\xi$ gives (293) directly. Passing to the limit gives the unique continuous transform with value 1 at the origin that solves the fixed-point equation.

Corollary 10.2 (Strong convergence in every fixed smoothness order)**.** *For every fixed $m \geq 0$,*

$$\max_{0 \leq r \leq m} \left\| f_n^{(r)} - \mathrm{up}^{(r)} \right\|_\infty \longrightarrow 0, \tag{296}$$

where n is taken large enough that the indicated finite-stage derivatives exist.

Proof. Fix $M > m + 2$. For $n \geq M$, the first M sinc factors in (293) supply an integrable majorant after multiplication by $|\xi|^m$, while every remaining sinc factor has modulus at most one. Dominated convergence and Fourier inversion give uniform convergence of each derivative through order m. A much sharper and exact estimate is proved in Section 15. □

11 The exact nth stage in truncated powers

The positive-part notation is

$$(y)_+^r = \begin{cases} y^r, & y > 0, \\ 0, & y \le 0. \end{cases}$$

For $r > 0$ the value at $y = 0$ is immaterial. The following standard box-spline identity is the closed form behind the repeated integrations.

Lemma 11.1 (A convolution of arbitrary boxes)**.** *Let $a_1, \dots, a_N > 0$, put $A = \sum_{i=1}^N a_i$, and let β_{a_i} be as in (285). Then*

$$(\beta_{a_1} * \cdots * \beta_{a_N})(x) = \frac{1}{(N-1)! \prod_{i=1}^N (2a_i)} \sum_{S \subseteq \{1,\dots,N\}} (-1)^{|S|} \left(x + A - 2 \sum_{i \in S} a_i \right)_+^{N-1}. \tag{297}$$

Proof. Write $\beta_a = (2a)^{-1}(H(\cdot + a) - H(\cdot - a))$, where H is the Heaviside distribution. Expanding the N binary differences gives one term for every subset S. The N-fold convolution of Heaviside functions is $(x)_+^{N-1}/(N-1)!$. Translating by the sum of the selected endpoints gives (297). The same proof can be written as an induction by integrating one more moving interval at each step. □

For the dyadic widths in (286), the total half-width is $A = 1$ and

$$\left(\prod_{k=1}^n 2 \cdot 2^{-k} \right) (2 \cdot 2^{-n}) = 2^{1 - \binom{n+1}{2}}. \tag{298}$$

This gives a direct subset formula.

Theorem 11.2 (Exact subset-sum formula)**.** *For $n \ge 1$, let*

$$a_1 = 2^{-1}, \dots, a_n = 2^{-n}, \qquad a_{n+1} = 2^{-n}.$$

Then

$$f_n(x) = \frac{2^{\binom{n+1}{2} - 1}}{n!} \sum_{S \subseteq \{1,\dots,n+1\}} (-1)^{|S|} \left(x + 1 - 2 \sum_{i \in S} a_i \right)_+^n. \tag{299}$$

The binary subset sums of the first n distinct boxes are unique. This is where the Thue–Morse signs enter.

Definition 11.3 (Finite Thue–Morse row)**.** For a fixed $n \ge 1$, define

$$c_n(j) = \begin{cases} (-1)^{\mathrm{wt}_2(j)}, & 0 \le j < 2^n, \\ 0, & \text{otherwise}. \end{cases} \tag{300}$$

Write $\Delta c_n(j) = c_n(j) - c_n(j-1)$.

Theorem 11.4 (Exact Thue–Morse closed form)**.** *For every $n \ge 1$ and $x \in \mathbb{R}$,*

$$\boxed{f_n(x) = \frac{2^{\binom{n+1}{2} - 1}}{n!} \sum_{j=0}^{2^n} \Delta c_n(j) \left(x + 1 - \frac{j}{2^{n-1}} \right)_+^n.} \tag{301}$$

Equivalently, away from the dyadic breakpoints, the sum may be stopped at $j = \lfloor 2^{n-1}(x+1) \rfloor$.

Proof. For a subset of the first n boxes, the quantity $2\sum a_i$ equals $j/2^{n-1}$, where the binary digits of j record the chosen indices and the sign is $(-1)^{\mathrm{wt}_2(j)} = c_n(j)$. If the duplicated last box is not chosen, the coefficient at j is $c_n(j)$. If it is chosen, its extra binary unit shifts $j-1$ to j and reverses the sign, contributing $-c_n(j-1)$. Their sum is $\Delta c_n(j)$. Substitute into (299). □

Formula (301) is algebraically equivalent to the 2012 answer [8]. The present derivation shows why the coefficient is a *difference* of two consecutive Thue–Morse signs: it is exactly the collision caused by the duplicated smallest box.

11.1 Knots, cells, and the highest derivative

All possible breakpoints lie on the grid

$$x_j = -1 + \frac{j}{2^{n-1}}, \qquad 0 \le j \le 2^n. \tag{302}$$

Some grid points are inactive because $\Delta c_n(j)$ can vanish. Differentiating (301) r times for $0 \le r \le n-1$ gives

$$f_n^{(r)}(x) = \frac{2^{\binom{n+1}{2}-1}}{(n-r)!} \sum_{j=0}^{2^n} \Delta c_n(j) \left(x + 1 - \frac{j}{2^{n-1}}\right)_+^{n-r}. \tag{303}$$

The nth derivative is a step function away from the grid.

Corollary 11.5 (The nth derivative is a Thue–Morse step function)**.** *If $-1 < x < 1$ and x is not on the grid* (302)*, put*

$$J = \left\lfloor 2^{n-1}(x+1) \right\rfloor. \tag{304}$$

Then

$$f_n^{(n)}(x) = 2^{\binom{n+1}{2}-1}(-1)^{\mathrm{wt}_2(J)}. \tag{305}$$

Proof. After n derivatives, each positive-part power becomes a Heaviside step. On the indicated cell, the active coefficients telescope:

$$\sum_{j=0}^{J} \Delta c_n(j) = c_n(J) = (-1)^{\mathrm{wt}_2(J)}.$$

□

This identity explains the derivative plots in the original question exactly: the finest derivative is a rescaled Thue–Morse row, and every lower derivative is its successive piecewise-polynomial integral.

12 Tail replacement rather than truncation

The random series makes the approximation mechanism exact. Recall the prefix

$$S_n = \sum_{k=1}^{n} 2^{-k} U_k, \qquad \text{density } p_n. \tag{306}$$

Split the infinite series at level n:

$$Y = S_n + 2^{-n} Y', \tag{307}$$

where Y' is an independent copy of Y. On the other hand,

$$X_n = S_n + 2^{-n}V, \tag{308}$$

where $V \sim \mathrm{Unif}[-1,1]$ is independent. Thus the finite iteration keeps the first n dyadic boxes exactly and replaces the true self-similar tail by a uniform law on the same support.

The two tail laws agree in their zeroth and first moments. Their second moments are

$$\mathbb{E}V^2 = \frac{1}{3}, \qquad \mathbb{E}Y^2 = \frac{1}{3}\sum_{k=1}^{\infty} 4^{-k} = \frac{1}{9}. \tag{309}$$

Consequently

$$\mathrm{Var}(X_n) - \mathrm{Var}(Y) = \frac{2}{9}\,4^{-n}. \tag{310}$$

This already predicts a second-order error with coefficient $1/9$: a centered smoothing error acts to first order like one half of the variance defect times a second derivative. The next section turns this heuristic into an exact norm identity.

13 A positive Peano kernel

Let

$$h = \rho - \mathrm{up}\,. \tag{311}$$

It is an even compactly supported signed density with zero total mass and zero first moment. Define its twice-integrated kernel by

$$K(x) = \frac{1}{2}\int_{\mathbb{R}} |x-t|h(t)\,\mathrm{d}t. \tag{312}$$

Then $K'' = h$ in the sense of distributions. The essential fact is that K is nonnegative.

Proposition 13.1 (The tail Peano kernel)**.** *The function K has the following properties:*

1. *K is even, belongs to $C^1(\mathbb{R})$, is supported on $[-1,1]$, and satisfies $K'' = \rho - \mathrm{up}$;*
2. *$K(x) \geq 0$ for all x, and K is nonincreasing on $[0,1]$;*
3. $$\int_{\mathbb{R}} K(x)\,\mathrm{d}x = \frac{1}{9}; \tag{313}$$
4. $$K(0) = \frac{1}{9}. \tag{314}$$

Proof. Evenness and C^1 regularity follow directly from (312); two distributional derivatives of $|x-t|/2$ give δ_t. For $x > 1$, one has $|x-t| = x-t$ on the support of h, so $K(x) = \frac{1}{2}(x\int h - \int th) = 0$. The same argument applies for $x < -1$.

For $0 \leq x \leq 1$, differentiation gives the difference of the two cumulative distribution functions:

$$K'(x) = \frac{x}{2} - \int_0^x \mathrm{up}(t)\,\mathrm{d}t. \tag{315}$$

For $0 < x < 1$, the differential equation (284) and the support of up give $\mathrm{up}'(x) = -2\,\mathrm{up}(2x-1) \leq 0$. Thus the up-function is nonincreasing on $[0,1]$; it has integral $1/2$ there by evenness and normalization. Hence the average of up over $[0,x]$ is at least its average over $[0,1]$, so $\int_0^x \mathrm{up}(t)\,\mathrm{d}t \geq x/2$. Thus $K'(x) \leq 0$. Since $K(1) = 0$, this proves $K \geq 0$ on $[0,1]$, and evenness handles the other half.

Integrating $K'' = h$ twice by parts against x^2 and using the compact support of K gives

$$2\int K(x)\,\mathrm{d}x = \int x^2 h(x)\,\mathrm{d}x = \mathbb{E}V^2 - \mathbb{E}Y^2 = \frac{1}{3} - \frac{1}{9} = \frac{2}{9},$$

which proves (313).

Finally,

$$K(0) = \frac{1}{2}(\mathbb{E}|V| - \mathbb{E}|Y|).$$

Clearly $\mathbb{E}|V| = 1/2$. From the stationary identity $Y \overset{d}{=} (U + Y')/2$, with U uniform and Y' an independent copy, and the elementary formula

$$\mathbb{E}_U|U+y| = \frac{1}{2}\int_{-1}^{1} |u+y|\,\mathrm{d}u = \frac{1+y^2}{2} \qquad (|y| \le 1),$$

we obtain

$$\mathbb{E}|Y| = \frac{1}{4}(1 + \mathbb{E}Y^2) = \frac{1}{4}\left(1 + \frac{1}{9}\right) = \frac{5}{18}.$$

Therefore $K(0) = \frac{1}{2}(\frac{1}{2} - \frac{5}{18}) = 1/9$. $\square$

Remark 13.2 (Convex order). The nonnegativity of K is equivalent to

$$Y \preceq_{\mathrm{cx}} V, \tag{316}$$

meaning $\mathbb{E}\phi(Y) \le \mathbb{E}\phi(V)$ for every convex ϕ for which the expectations exist. After adding the independent prefix and scaling, this gives $Y \preceq_{\mathrm{cx}} X_n$. The iterates are therefore systematically more dispersed than the limiting law, even though they share the same support and mean.

Scale the signed tail and its Peano kernel by

$$h_n(t) = 2^n h(2^n t), \qquad K_n(t) = 2^{-n} K(2^n t). \tag{317}$$

Then

$$K_n'' = h_n, \qquad \operatorname{supp} K_n \subseteq [-2^{-n}, 2^{-n}], \qquad K_n \ge 0, \qquad \int K_n = \frac{4^{-n}}{9}. \tag{318}$$

The two exact tail factorizations (307)–(308) now give

$$f_n - \mathrm{up} = p_n * h_n = p_n * K_n'' = p_n'' * K_n \tag{319}$$

in the sense of distributions, and as an ordinary convolution whenever the displayed derivatives are functions. This is the central error identity.

Corollary 13.3 (A sharp weak error bound). *For every $\phi \in C^2(\mathbb{R})$ with bounded second derivative,*

$$|\mathbb{E}\phi(X_n) - \mathbb{E}\phi(Y)| \le \frac{4^{-n}}{9}\,\|\phi''\|_\infty\,. \tag{320}$$

If ϕ is convex, the difference on the left before taking absolute values is nonnegative. The constant is sharp, as seen from $\phi(x) = x^2$.

Proof. Pair (319) with ϕ, move the two derivatives onto ϕ, and use that p_n is a probability density and K_n is nonnegative of mass $4^{-n}/9$. For $\phi(x) = x^2$, equality is exactly the variance defect (310). $\square$

14 A finite-prefix derivative plateau

To turn (319) into a sharp uniform estimate, one needs an exact bound for derivatives of the prefix spline p_n. The relevant constant is the same triangular power of two that occurs in the sharp derivative bounds for up.

Lemma 14.1 (Summatory Thue–Morse cancellation). *Let $\varepsilon_j = (-1)^{\mathrm{wt}_2(j)}$. Then for every $J \geq 0$,*

$$\sum_{j=0}^{J} \varepsilon_j = \begin{cases} \varepsilon_{J/2}, & J \text{ even}, \\ 0, & J \text{ odd}. \end{cases} \tag{321}$$

In particular, every prefix sum has absolute value at most one.

Proof. The identities $\varepsilon_{2q} = \varepsilon_q$ and $\varepsilon_{2q+1} = -\varepsilon_q$ make every complete adjacent pair cancel. An even endpoint leaves the last even term unpaired. $\square$

Theorem 14.2 (Exact derivative norm and a saturated interval). *Let $r \geq 0$ and $n \geq r+1$. Put*

$$C_r = 2^{\binom{r+1}{2}}, \qquad x_r = -1 + 2^{-r}. \tag{322}$$

Then the rth derivative of p_n exists as a bounded piecewise function and satisfies

$$\left\| p_n^{(r)} \right\|_\infty = C_r. \tag{323}$$

More precisely,

$$p_n^{(r)}(x) = C_r \quad \text{for almost every } x \in [x_r - 2^{-n}, x_r + 2^{-n}]. \tag{324}$$

Proof. First take $n = r+1$. Applying Lemma 11.1 to the $r+1$ distinct boxes in p_{r+1} and differentiating r times gives a step function. Its possible knots are

$$-\left(1 - 2^{-(r+1)}\right) + \frac{j}{2^r}, \qquad 0 \leq j < 2^{r+1},$$

and its value on the cell after the Jth knot is

$$2^{\binom{r+1}{2}} \sum_{j=0}^{J} (-1)^{\mathrm{wt}_2(j)}.$$

By Lemma 14.1, its absolute value is at most C_r. On the first cell the prefix sum is 1, so the value is C_r. That first cell is centered at $x_r = -1 + 2^{-r}$ and has half-width $2^{-(r+1)}$.

For larger n, factor

$$p_n = p_{r+1} * \beta_{2^{-(r+2)}} * \cdots * \beta_{2^{-n}} = p_{r+1} * q_{r,n},$$

where $q_{r,n}$ is a probability density supported in the interval of radius

$$\sum_{k=r+2}^{n} 2^{-k} = 2^{-(r+1)} - 2^{-n}.$$

Therefore $p_n^{(r)} = p_{r+1}^{(r)} * q_{r,n}$ has norm at most C_r. If $|x - x_r| \leq 2^{-n}$, then every point $x - t$ sampled by the convolution remains in the first cell of $p_{r+1}^{(r)}$, except possibly at its null endpoints. Hence the convolution equals C_r there. This proves both the upper bound and its attainment. $\square$

Remark 14.3. The limiting sharp bound in the companion document is $\left\| \mathrm{up}^{(r)} \right\|_\infty = C_r$, attained at the same dyadic point x_r [9]. Theorem 14.2 explains this without taking a limit: every sufficiently long finite prefix already has the exact extremal height, and the plateau shrinks to the extremal point.

15 The exact convergence rate

We now combine the positive kernel (319) with the saturated derivative interval (324).

Theorem 15.1 (Exact C^m error)**.** *Let $m \geq 0$ and $n \geq m+3$. Then*

$$\boxed{\left\| f_n^{(m)} - \mathrm{up}^{(m)} \right\|_\infty = \frac{2^{\binom{m+3}{2}}}{9} 4^{-n}.} \tag{325}$$

The positive maximum is attained at

$$x = -1 + 2^{-(m+2)}. \tag{326}$$

Proof. Differentiate (319) m times and put $r = m+2$:

$$f_n^{(m)} - \mathrm{up}^{(m)} = p_n^{(r)} * K_n. \tag{327}$$

The kernel is nonnegative and has mass $4^{-n}/9$, while $\left\| p_n^{(r)} \right\|_\infty = C_r = 2^{\binom{m+3}{2}}$ by Theorem 14.2. Therefore

$$\left\| f_n^{(m)} - \mathrm{up}^{(m)} \right\|_\infty \leq C_r \int K_n = \frac{C_r}{9} 4^{-n}.$$

At $x = x_r = -1 + 2^{-r}$, the interval $x_r - \operatorname{supp} K_n$ is exactly contained in the saturated interval (324). Hence the integrand in (327) equals C_r almost everywhere on the support of K_n, and

$$(f_n^{(m)} - \mathrm{up}^{(m)})(x_r) = C_r \int K_n = \frac{C_r}{9} 4^{-n}.$$

The upper bound is therefore an equality. □

For $m = 0$, the theorem directly covers $n \geq 3$. The second stage has one derivative too few for the preceding L^∞ argument, but the distributional formula can be evaluated explicitly.

Proposition 15.2 (The exceptional second stage)**.** *One has*

$$\|f_2 - \mathrm{up}\|_\infty = \frac{1}{18} = \frac{8}{9} 4^{-2}. \tag{328}$$

Proof. The prefix $p_2 = \beta_{1/2} * \beta_{1/4}$ is a trapezoid, and its distributional second derivative is

$$p_2'' = 2(\delta_{-3/4} - \delta_{-1/4} - \delta_{1/4} + \delta_{3/4}). \tag{329}$$

Thus (319) gives

$$\begin{aligned} f_2(x) - \mathrm{up}(x) = 2\big(&K_2(x+3/4) - K_2(x+1/4) \\ &- K_2(x-1/4) + K_2(x-3/4)\big). \end{aligned} \tag{330}$$

The four translates have disjoint interiors because K_2 is supported on $[-1/4, 1/4]$. Since K is even and decreases on $[0,1]$, its maximum is $K(0) = 1/9$, so $\max K_2 = 2^{-2}K(0) = 1/36$. Therefore the maximum absolute value in (330) is $2/36 = 1/18$, attained at the four translation centers. □

The first two stages can also be evaluated exactly. The initial error is plainly $1/2$. For f_1, use (273): on $0 \leq x \leq 1$,

$$f_1(x) - \mathrm{up}(x) = 1 - x - F(1-x) = F(x) - x. \tag{331}$$

On $[0, 1/2]$ its derivative is $2F(2x) - 1$, which changes sign only at $x = 1/4$. The already established dyadic value $F(1/4) = 5/72$ then gives the extremal value $-13/72$; symmetry about $1/2$ gives $+13/72$ at $3/4$.

Theorem 15.3 (Complete exact uniform-error sequence)**.** *For every $n \geq 0$,*

$$\boxed{\|f_n - \mathrm{up}\|_\infty = \begin{cases} \frac{1}{2}, & n = 0, \\ \frac{13}{72}, & n = 1, \\ \frac{8}{9}\,4^{-n}, & n \geq 2. \end{cases}} \tag{332}$$

For $n \geq 2$,

$$\begin{aligned} f_n(\pm 3/4) - \mathrm{up}(\pm 3/4) &= +\frac{8}{9}\,4^{-n}, \\ f_n(\pm 1/4) - \mathrm{up}(\pm 1/4) &= -\frac{8}{9}\,4^{-n}. \end{aligned} \tag{333}$$

Proof. The cases $n = 0, 1$ were just discussed, $n = 2$ is Proposition 15.2, and $n \geq 3$ is Theorem 15.1 with $m = 0$. The four signed extrema follow from the four plateaux of height ± 8 in the second derivative of the three-box prefix, preserved by all later probability convolutions; equivalently they follow directly from (330) and the same plateau argument used in Theorem 15.1. □

Since $\mathrm{up}(3/4) = F(1/4) = 5/72$ and $\mathrm{up}(1/4) = F(3/4) = 67/72$, the extremal values themselves are

$$f_n(3/4) = \frac{5}{72} + \frac{8}{9\,4^n}, \qquad f_n(1/4) = \frac{67}{72} - \frac{8}{9\,4^n} \qquad (n \geq 2). \tag{334}$$

For example,

n	$f_n(3/4)$	$4^n(f_n(3/4) - \mathrm{up}(3/4))$
2	1/8	8/9
3	1/12	8/9
4	7/96	8/9
5	9/128	8/9
6	107/1536	8/9

15.1 Certified iteration counts

The exact norm law immediately gives a best possible stopping rule. To guarantee $\|f_n - \mathrm{up}\|_\infty \leq \eta$ with $n \geq 2$, it is necessary and sufficient that

$$n \geq \left\lceil \log_4 \frac{8}{9\eta} \right\rceil. \tag{335}$$

For the mth derivative, once $n \geq m + 3$, the corresponding condition is

$$n \geq \left\lceil \log_4 \frac{2^{\binom{m+3}{2}}}{9\eta} \right\rceil. \tag{336}$$

No unspecified constants or asymptotic safety factors are needed.

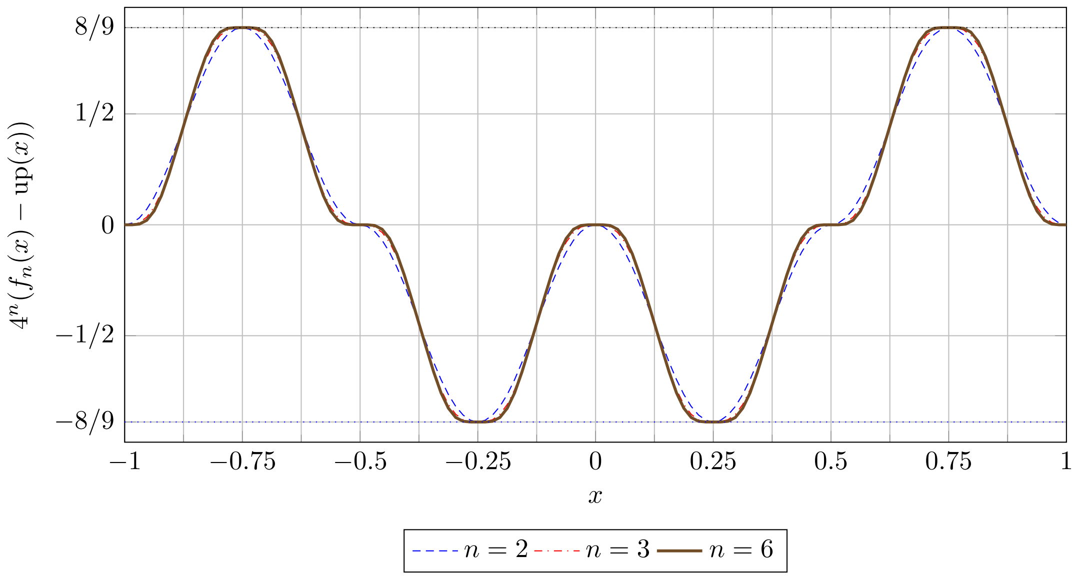

Figure 2: The errors after multiplication by 4^n. Their shape approaches $\mathrm{up}''/9$, while the exact extrema remain fixed at $\pm 8/9$ from the second stage onward. Numerical curves are shown only as an illustration of the proved identities.

16 Fourier asymptotics and the spectral meaning of $1/4$

The exact norm theorem has a complementary Fourier interpretation. Put

$$P_n(\xi) = \prod_{k=1}^{n} \operatorname{sinc}(2^{-k}\xi), \qquad z = 2^{-n}\xi, \qquad \Phi(z) = \prod_{k=1}^{\infty} \operatorname{sinc}(2^{-k}z). \tag{337}$$

Then

$$\widehat{f_n}(\xi) = P_n(\xi)\operatorname{sinc} z, \qquad \widehat{\mathrm{up}}(\xi) = P_n(\xi)\Phi(z). \tag{338}$$

Near the origin, consider the analytic germ

$$R(z) = \frac{\operatorname{sinc} z}{\Phi(z)}. \tag{339}$$

Its logarithm is

$$\log R(z) = -\sum_{r=1}^{\infty} \frac{2^{2r-1}|B_{2r}|}{r(2r)!}\,\frac{4^r - 2}{4^r - 1}\, z^{2r}, \tag{340}$$

where B_{2r} are Bernoulli numbers. Exponentiation gives

$$R(z) = 1 - \frac{z^2}{9} + \frac{2z^4}{2025} + \frac{z^6}{2679075} + \frac{58z^8}{10247461875} + \cdots. \tag{341}$$

Theorem 16.1 (All-orders expansion; proof sketch and formal obligation)**.** *Let $m, J \geq 0$ be fixed. If $R(z) = \sum_{j\geq 0} \rho_j z^{2j}$ near 0 and $a_j = (-1)^j \rho_j$, then*

$$\left\| f_n^{(m)} - \sum_{j=0}^{J-1} a_j 4^{-jn}\, \mathrm{up}^{(m+2j)} \right\|_\infty = \mathcal{O}_{m,J}(4^{-Jn}). \tag{342}$$

In particular,

$$f_n = \mathrm{up} + \frac{4^{-n}}{9}\,\mathrm{up}'' + \frac{2\,16^{-n}}{2025}\,\mathrm{up}^{(4)} - \frac{64^{-n}}{2679075}\,\mathrm{up}^{(6)} + \mathcal{O}_{C^m}(256^{-n}) \tag{343}$$

for every fixed m.

Proof architecture. Subtract the truncated series from (338) without dividing by Φ globally:

$$P_n(\xi)\left[\operatorname{sinc} z - \Phi(z)\sum_{j=0}^{J-1}\rho_j z^{2j}\right].$$

The bracket is $\mathcal{O}(z^{2J})$ near zero and is bounded globally by a constant times $|z|^{2J}$. After multiplying by $|\xi|^m$, factor out 4^{-Jn}. A fixed number of the first sinc factors in P_n supplies an integrable majorant with arbitrarily high polynomial decay. Fourier inversion then gives (342). □

The leading term gives

$$4^n\left(f_n^{(m)} - \mathrm{up}^{(m)}\right) \longrightarrow \frac{1}{9}\,\mathrm{up}^{(m+2)} \tag{344}$$

uniformly. Since the companion development proves

$$\left\|\mathrm{up}^{(m+2)}\right\|_\infty = 2^{\binom{m+3}{2}}, \tag{345}$$

the asymptotic constant predicted by (344) agrees with the exact finite-n constant in (325).

There is also an operator-theoretic explanation. From (295) and $T\,\mathrm{up} = \mathrm{up}$,

$$T\left(\mathrm{up}^{(r)}\right) = 2^{-r}\,\mathrm{up}^{(r)}. \tag{346}$$

Thus up'' is an eigenfunction with eigenvalue $1/4$. The equal-mass and equal-mean tail replacement annihilates the modes of orders zero and one; the first surviving mode is the second derivative, and each iteration multiplies it by $1/4$.

Corollary 16.2 (L^p asymptotics)**.** *For every fixed $m \ge 0$ and $1 \le p < \infty$,*

$$\lim_{n\to\infty} 4^n \left\|f_n^{(m)} - \mathrm{up}^{(m)}\right\|_{L^p} = \frac{1}{9}\left\|\mathrm{up}^{(m+2)}\right\|_{L^p}. \tag{347}$$

For $p = \infty$, the corresponding equality is already exact for every $n \ge m+3$.

Proof. Uniform convergence in (344), together with the common compact support, gives the finite-p limits. The $p = \infty$ statement is Theorem 15.1. □

17 Why the exact constant is visible at dyadic points

The extremal point in (326) is dyadic of level $m+2$. The companion document proves that every derivative of up of order strictly above the level of a reduced dyadic point vanishes there. Therefore, at $x = -1 + 2^{-(m+2)}$,

$$\mathrm{up}^{(m+2j)}(x) = 0 \qquad (j \ge 2). \tag{348}$$

The all-orders series (342) consequently has only its first correction term at that point. The Peano-kernel proof shows something stronger than a formal or asymptotic termination: the finite-stage identity is exact because the relevant derivative of p_n is constant over the whole support of the tail kernel.

This connects three appearances of Thue–Morse structure:

1. the coefficient differences $\Delta c_n(j)$ in the exact spline formula;
2. the prefix cancellation that bounds $p_n^{(r)}$ by one triangular power of two;
3. the terminating derivative tower of up at dyadic points.

They are not separate coincidences. All arise from unique binary subset sums of dyadic box widths.

18 Comparison with ordinary truncation

The standard prefix density p_n and the repeated-integration density f_n approximate the same limit in different ways:

	Ordinary truncation p_n	Repeated integration f_n
Random variable	S_n	$S_n + 2^{-n}V$
Number of boxes	n	$n+1$
Last widths	$\ldots, 2^{-n}$	$\ldots, 2^{-n}, 2^{-n}$
Support radius	$1-2^{-n}$	1 exactly
Tail model	omitted	uniform on the exact tail support
Mean error	zero by symmetry	zero by symmetry
Leading smooth error	support/tail-scale without correction	variance defect, scale 4^{-n}
Exact L^∞ law	not the law studied here	$8/(9\,4^n)$ for $n \geq 2$

The extra uniform box should therefore not be described as a negligible finite-product artifact. It is a deliberate support correction. It buys a full extra power of the tail radius because the substituted and true tails have the same mass and center. Had one also matched the variance, the leading up'' term would disappear and the next Fourier mode would produce a 16^{-n} scheme. This conjecturally suggests a general family of higher-order tail corrections; no such positive compactly supported scheme is constructed here, and the additional moment constraints remain open.

19 Computational forms

Three exact forms serve different computational purposes.

19.1 Recurrence by antiderivatives

Given a piecewise-polynomial representation of f_{n-1}, compute a continuous primitive A_{n-1} and use

$$f_n(x) = A_{n-1}(2x+1) - A_{n-1}(2x-1). \tag{349}$$

This preserves exact rational coefficients and increases the degree by one. It is convenient for producing all cells of moderate n.

19.2 Point evaluation by the Thue–Morse sum

For a single rational x, (301) is finite and exact. One may stop at the last active positive part. A direct implementation is:

Listing 1: Exact evaluation of the nth repeated-integration stage

```
function iterate_value(n, x):
    if n == 0:
        return 1/2 if abs(x) < 1 else 0
    total := 0
    for j from 0 to 2^n:
        c  := (-1)^(binary_weight(j)) if 0 <= j < 2^n else 0
        cm := (-1)^(binary_weight(j-1)) if 0 <= j-1 < 2^n else 0
        y  := x + 1 - j / 2^(n-1)
        if y > 0:
```

```
        total := total + (c-cm) * y^n
    return 2^(n(n+1)/2 - 1) * total / n!
```

All arithmetic is rational when x is rational. The formula is cancellation-heavy for large n, so exact integer or rational arithmetic is preferable.

19.3 Certified approximation of up

For a uniform approximation, use any representation of f_n and the exact certificate

$$\operatorname{up}(x) \in \left[f_n(x) - \frac{8}{9\,4^n}, f_n(x) + \frac{8}{9\,4^n}\right] \tag{350}$$

for every x and every $n \geq 2$. At the four extremal dyadic points, one endpoint of this interval is exact. Derivative bands follow from (325).

20 A roadmap for Lean formalization

The repository already contains the probability series, finite box convolutions, weak convergence, Thue–Morse machinery, dyadic derivative identities, and sharp bounds for $\operatorname{up}^{(m)}$ [10]. The present argument can be added in a relatively modular way. The following is a proof-dependency outline rather than a claim that these new statements are already formalized.

1. Define the operator T and prove (276) for functions supported on $[-1, 1]$.
2. Identify T with the pushforward of the product law under $(x, u) \mapsto (x + u)/2$.
3. Prove the finite random-sum identity (282) and derive the box convolution (286).
4. Instantiate a general truncated-power formula for convolutions of interval indicators; coalesce the duplicated last weight to obtain (301).
5. Define K by (312). Use the established monotonicity of up to prove $K \geq 0$, and calculate its mass from the second moments.
6. Prove the summatory Thue–Morse lemma (321) and the finite-prefix plateau (324).
7. Combine convolution differentiation, positivity, and the plateau to prove the exact norm identity (325); handle $n = 2$ through the atomic second derivative (329).

The only mildly delicate formal point is that $p_{r+1}^{(r)}$ is a step function defined only almost everywhere, while the final error derivative is continuous. A measure/distribution formulation, or an almost-everywhere convolution lemma followed by continuity, avoids choosing arbitrary values at knots.

21 Conclusions

The repeated integration in the 2012 question is exactly solvable at every finite stage and has an unusually clean limiting theory.

1. The apparently two-term recursive integral is the moving-average operator $Tg(x) = \int_{2x-1}^{2x+1} g$.
2. Its nth iterate is a dyadic box spline with widths $2^{-1}, \ldots, 2^{-n}, 2^{-n}$.
3. The duplicated smallest width explains the consecutive difference of Thue–Morse signs in the closed formula.

4. The limit is Rvachev's up-function because T is the affine smoothing map $X \mapsto (X+U)/2$ and up is its stationary density.
5. The finite stage replaces the true self-similar tail by a uniform tail on the same support. A positive Peano kernel of mass $1/9$ measures this replacement.
6. A saturated Thue–Morse plateau in the appropriate derivative of the finite prefix turns the kernel bound into an equality. Hence the uniform error is exactly $8/(9\,4^n)$ from $n = 2$ onward, and every fixed derivative has the exact law (325) once enough integrations have been performed.

Thus the rapid visual convergence is neither mysterious nor merely qualitative. Its ratio, constant, extremal points, derivative analogues, and full Fourier expansion are all dictated by the same dyadic box structure.

Supplementary material from this dossier

22 Algebraic equivalence with the Stack Exchange formula

The formula displayed in [8] is

$$\sum_{j=0}^{2^n} \frac{c_n(j) - c_n(j-1)}{2} \frac{(2^n x + 2^n - 2j)_+^n}{n!\, 2^{n(n-1)/2}}. \tag{351}$$

Because

$$2^n x + 2^n - 2j = 2^n \left(x + 1 - \frac{j}{2^{n-1}} \right),$$

the total power of two in each term of (351) is

$$2^{n^2 - 1 - n(n-1)/2} = 2^{n(n+1)/2 - 1}.$$

This is exactly the prefactor in (301). The two displays therefore agree term by term, including normalization.

23 A few exact finite-stage formulas

The positive-part formula gives compact global expressions. In addition to (290)–(291),

$$\begin{aligned} f_3(x) = \frac{16}{3} \big(&(x+1)_+^3 - 2\,(x+3/4)_+^3 + 2\,(x+1/4)_+^3 - 2\,(x)_+^3 \\ &+ 2\,(x-1/4)_+^3 - 2\,(x-3/4)_+^3 + (x-1)_+^3 \big). \end{aligned} \tag{352}$$

Its third derivative, away from the grid, is 32 times the length-8 Thue–Morse row. The zero coefficients at some nominal grid points explain why adjacent cells can share the same polynomial branch.

24 Notation crosswalk

Symbol	Meaning here	Relation to the companion document
F	bounded Fabius distribution function	same normalization
up	even density supported on $[-1,1]$	same normalization
ρ	initial uniform density on $[-1,1]$	a single box of half-width 1
T	smoothing operator defined by repeated integration	fixed by up
p_n	convolution of the first n distinct dyadic boxes	the ordinary finite prefix
f_n	p_n convolved with a second box of half-width 2^{-n}	support-corrected finite spline
K	positive Peano kernel with $K'' = \rho - \mathrm{up}$	encodes the tail convex order
$c_n(j)$	finite Thue–Morse sign $(-1)^{\mathrm{wt}_2(j)}$	same binary-weight convention

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Part III
Repeated Integration and Rvachev's Up-Function: Sharp Thresholds

Source provenance and status. Former path: `Rvachev_Up_from_Repeated_Integration-2/Rvachev_Up_from_Repeated_Integration.tex`. Later, sharper treatment of the repeated-integration construction. It improves the derivative-norm threshold from the earlier dossier's $n \geq r+3$ presentation to $n \geq r+2$, while separately resolving the first admissible stage. Prefer this threshold where the dossiers overlap. Its new sharp identities and all-orders expansions remain natural-language proofs and formal obligations.

Original source SHA-256: `fad16072df30d9e6eb5df03da57dd217769180730581f1dfd19fa5be0d16b262`.

Dossier abstract

A Math Stack Exchange question asks for a closed form for the sequence

$$f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x), \qquad f_n(x) = 2\int_{-1}^{x} \left(f_{n-1}(2t+1) - f_{n-1}(2t-1)\right) \mathrm{d}t.$$

The first integration produces the piecewise linear tent $f_1(x) = (1-|x|)_+$, and the subsequent iterates visually converge very rapidly to Rvachev's compactly supported C^∞ up-function. This report gives a complete finite-stage description and a sharp convergence theorem.

The integral operator is exactly a probabilistic averaging operator: if g is the density of Y and U is uniform on $[-1,1]$, then the next density is that of $(Y+U)/2$. Consequently, for $n \geq 1$,

$$f_n = u_{2^{-1}} * u_{2^{-2}} * \cdots * u_{2^{-n}} * u_{2^{-n}}, \qquad u_a(x) = \frac{1}{2a}\mathbf{1}_{[-a,a]}(x).$$

This identifies f_n as a nonuniform box spline. Collapsing its general inclusion–exclusion formula with the Thue–Morse generating polynomial yields, for $n \geq 1$,

$$f_n(x) = \frac{2^{n(n+1)/2-1}}{n!} \sum_{j=0}^{2^n} (\tau_n(j) - \tau_n(j-1)) \left(x + 1 - \frac{j}{2^{n-1}}\right)_+^n,$$

where $\tau_n(j) = (-1)^{\mathrm{wt}_2(j)}$ on $0 \leq j < 2^n$ and is zero outside that range. The nth derivative is therefore a rescaled Thue–Morse step function.

The principal quantitative result derived here is exact, rather than merely asymptotic. For every fixed $r \geq 0$ and every $n \geq r+2$,

$$\boxed{\left\| f_n^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9\, 4^n}.}$$

Thus, once the rth derivative has passed its first admissible spline stage, every further integration divides its uniform error by exactly four. In the indexing that starts with the piecewise linear tent $g_0 = f_1$, this becomes

$$\|g_m - \mathrm{up}\|_\infty = \frac{2}{9\, 4^m} \qquad (m \geq 1),$$

while the exceptional initial error is $\|g_0 - \mathrm{up}\|_\infty = 13/72$. The proof is based on an exact Peano-kernel identity. The comparison kernel between a uniform random variable and the up-distributed random variable has both integral and maximum equal to 1/9; the finite partial density has disjoint extremal plateaux whose height is known exactly. Their combination gives both the upper bound and matching extremizers.

25 The problem, the indexing, and the answer

The recurrence considered in the Math Stack Exchange question [1, 2] is

$$f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x), \qquad f_n(x) = 2\int_{-1}^{x} \left(f_{n-1}(2t+1) - f_{n-1}(2t-1)\right) \mathrm{d}t. \tag{353}$$

Changing the values of f_0 at $x = \pm 1$ changes neither any later iterate nor any of the norm statements below. We therefore use whichever endpoint convention is convenient.

There is a small indexing issue in the informal description of the construction. The literal seed f_0 in (353) is piecewise constant; the first iterate is the piecewise linear function

$$f_1(x) = (1 - |x|)_+ . \tag{354}$$

When one says that the up-function arises by "repeatedly integrating a piecewise linear function," the natural indexing is therefore

$$g_m = f_{m+1}, \qquad g_0(x) = (1 - |x|)_+ . \tag{355}$$

Both indexings will be retained: f_n matches the question, while g_m matches the piecewise-linear wording.

25.1 Main finite-stage formula

Write $\mathrm{wt}_2(j)$ for the number of 1's in the binary expansion of j. For $n \geq 0$ define the zero-extended finite Thue–Morse word

$$\tau_n(j) = \begin{cases} (-1)^{\mathrm{wt}_2(j)}, & 0 \leq j < 2^n, \\ 0, & \text{otherwise,} \end{cases} \qquad \Delta\tau_n(j) = \tau_n(j) - \tau_n(j-1). \tag{356}$$

The accepted Stack Exchange answer gives a formula equivalent to the following one; the probabilistic and spline derivation in section 28 explains why its coefficients are first differences of the Thue–Morse word.

Theorem 25.1 (Exact nth iterate)**.** *For every $n \geq 1$ and $x \in \mathbb{R}$,*

$$\boxed{f_n(x) = \frac{2^{n(n+1)/2-1}}{n!} \sum_{j=0}^{2^n} \Delta\tau_n(j) \left(x + 1 - \frac{j}{2^{n-1}}\right)_+^n .} \tag{357}$$

Equivalently,

$$f_n(x) = \frac{1}{2n!\, 2^{n(n-1)/2}} \sum_{j=0}^{2^n} \Delta\tau_n(j)\, (2^n x + 2^n - 2j)_+^n . \tag{358}$$

The knots are

$$-1, -1 + 2^{1-n}, \ldots, 1 - 2^{1-n}, 1. \tag{359}$$

On every open knot interval f_n is a polynomial of degree n, and globally $f_n \in C^{n-1}(\mathbb{R})$.

In the tent indexing (355), theorem 25.1 reads

$$\boxed{g_m(x) = \frac{2^{m(m+3)/2}}{(m+1)!} \sum_{j=0}^{2^{m+1}} \Delta\tau_{m+1}(j) \left(x + 1 - \frac{j}{2^m}\right)_+^{m+1}.} \tag{360}$$

Thus the mth repeated integration of the tent is a degree-$(m+1)$ spline on the dyadic mesh of width 2^{-m}.

25.2 Main sharp convergence theorem

For a bounded function h let $\|h\|_\infty = \sup_{x\in\mathbb{R}} |h(x)|$. When $h \in C^r$, we use the max version of the C^r norm,

$$\|h\|_{C^r} = \max_{0\le j\le r} \left\|h^{(j)}\right\|_\infty. \tag{361}$$

Theorem 25.2 (Exact derivative error)**.** *Let $r \ge 0$. The first iterate belonging to C^r is f_{r+1}, and its rth-derivative error is*

$$\left\|f_{r+1}^{(r)} - \mathrm{up}^{(r)}\right\|_\infty = \frac{13}{72} 2^{\binom{r+1}{2}}. \tag{362}$$

For every $n \ge r+2$,

$$\boxed{\left\|f_n^{(r)} - \mathrm{up}^{(r)}\right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9\,4^n}.} \tag{363}$$

All of these bounds are attained at the same 2^{r+2} points

$$c_{r+2,j} = -1 + \frac{2j+1}{2^{r+2}}, \qquad 0 \le j < 2^{r+2}. \tag{364}$$

At the first admissible stage,

$$f_{r+1}^{(r)}(c_{r+2,j}) - \mathrm{up}^{(r)}(c_{r+2,j}) = (-1)^{\mathrm{wt}_2(j)} \frac{13}{72} 2^{\binom{r+1}{2}}, \tag{365}$$

and for $n \ge r+2$,

$$f_n^{(r)}(c_{r+2,j}) - \mathrm{up}^{(r)}(c_{r+2,j}) = (-1)^{\mathrm{wt}_2(j)} \frac{2^{\binom{r+3}{2}}}{9\,4^n}. \tag{366}$$

The exact quartering begins immediately after the exceptional first C^r spline:

$$\frac{\left\|f_{r+2}^{(r)} - \mathrm{up}^{(r)}\right\|_\infty}{\left\|f_{r+1}^{(r)} - \mathrm{up}^{(r)}\right\|_\infty} = \frac{4}{13}, \qquad \frac{\left\|f_{n+1}^{(r)} - \mathrm{up}^{(r)}\right\|_\infty}{\left\|f_n^{(r)} - \mathrm{up}^{(r)}\right\|_\infty} = \frac{1}{4} \quad (n \ge r+2). \tag{367}$$

For the tent indexing this yields the particularly simple uniform result

$$\|g_0 - \mathrm{up}\|_\infty = \frac{13}{72}, \qquad \boxed{\|g_m - \mathrm{up}\|_\infty = \frac{2}{9\,4^m} \quad (m \ge 1).} \tag{368}$$

Thus every integration after the first one gains exactly two binary digits in the sharp uniform error.

The approximation in this report is distinct from the centered finite-CDF spline already discussed in the companion article. Here f_n is itself a probability *density*, and the last unresolved tail is replaced by a scaled uniform variable. The true tail and the uniform replacement have equal mass and equal mean. The first surviving discrepancy is therefore quadratic in the tail scale 2^{-n}, which explains the factor 4^{-n}; the comparison-kernel argument below determines the coefficient exactly.

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26 The integration operator is a probability operator

For $a > 0$, let

$$u_a(x) = \frac{1}{2a}\mathbf{1}_{[-a,a]}(x) \tag{369}$$

be the density of a random variable uniform on $[-a, a]$. Let T denote the operator in (353):

$$(Th)(x) = 2\int_{-1}^{x} \big(h(2t+1) - h(2t-1)\big)\,\mathrm{d}t. \tag{370}$$

Lemma 26.1 (Moving-window and convolution forms)**.** *Suppose h is integrable and supported in $[-1, 1]$. Then*

$$(Th)(x) = \int_{2x-1}^{2x+1} h(s)\,\mathrm{d}s = 2(h * u_1)(2x). \tag{371}$$

Proof. In the first term of (370), put $s = 2t + 1$; in the second, put $s = 2t - 1$. This gives

$$(Th)(x) = \int_{-1}^{2x+1} h(s)\,\mathrm{d}s - \int_{-3}^{2x-1} h(s)\,\mathrm{d}s.$$

The support assumption allows the lower endpoint -3 in the second integral to be replaced by -1, after which the difference is the integral over $[2x - 1, 2x + 1]$. The convolution identity follows directly from the definition of u_1. □

Proposition 26.2 (Probabilistic meaning)**.** *If h is the density of a random variable Y, and U is independent and uniform on $[-1, 1]$, then Th is the density of*

$$\frac{Y + U}{2}. \tag{372}$$

In particular, T preserves nonnegativity, total mass one, evenness, and support in $[-1, 1]$.

Proof. The density of $Y + U$ is $h * u_1$. Scaling a random variable by $1/2$ multiplies its density by 2 and evaluates it at $2x$, giving exactly (371). The listed properties are immediate. □

Let $U_1, U_2, \dots$ be independent random variables, each uniform on $[-1, 1]$. Start with $X_0 = U_1$ and define recursively

$$X_n = \frac{X_{n-1} + U_{n+1}}{2}. \tag{373}$$

Then f_n is the density of X_n.

Theorem 26.3 (Finite random-sum representation)**.** *For every $n \geq 1$,*

$$X_n \overset{d}{=} \sum_{k=1}^{n} 2^{-k} U_k + 2^{-n} U_{n+1}. \tag{374}$$

Consequently,

$$f_n = u_{2^{-1}} * u_{2^{-2}} * \cdots * u_{2^{-n}} * u_{2^{-n}}. \tag{375}$$

Proof. Expanding (373) gives weights $2^{-n}, 2^{-n}, 2^{-(n-1)}, \dots, 2^{-1}$ on $n+1$ independent copies of U. Because the variables are identically distributed, the two copies of 2^{-n} may be placed last, giving (374). Convolution of the corresponding scaled uniform densities gives (375). □

Several facts that are less transparent in the integral recursion are now immediate:

$$f_n \geq 0, \qquad \int_{\mathbb{R}} f_n(x)\,\mathrm{d}x = 1, \qquad f_n(-x) = f_n(x), \qquad \operatorname{supp} f_n = [-1, 1]. \tag{376}$$

Moreover, for $n \geq 1$,

$$f_n(0) = \int_{-1}^{1} f_{n-1}(s)\,\mathrm{d}s = 1. \tag{377}$$

27 The limit is Rvachev's up-function

Consider the almost surely convergent random series

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k. \tag{378}$$

Its support is $[-1, 1]$. Denote its density by up. Existence and smoothness also follow from the Fourier product below; they are classical properties of the Rvachev–Arias de Reyna construction [3].

27.1 Self-similarity and the differential equation

Splitting off the first summand in (378) gives the distributional fixed-point identity

$$X \overset{d}{=} \frac{U + X'}{2}, \tag{379}$$

where X' is an independent copy of X. By proposition 26.2, its density is a fixed point of T:

$$\mathrm{up}(x) = \int_{2x-1}^{2x+1} \mathrm{up}(s)\, \mathrm{d}s. \tag{380}$$

Differentiating gives

$$\mathrm{up}'(x) = 2\big(\mathrm{up}(2x+1) - \mathrm{up}(2x-1)\big). \tag{381}$$

Also $\mathrm{up}(0) = \int_{-1}^{1} \mathrm{up} = 1$. The even compactly supported smooth solution of (381) normalized by $\mathrm{up}(0) = 1$ is Rvachev's up-function. Thus the repeated-integration limit is not merely analogous to up; it is exactly up with the normalization used in the companion article.

27.2 Fourier product

Use the non-unitary characteristic Fourier transform

$$\mathscr{F}h(t) = \int_{\mathbb{R}} h(x) \mathrm{e}^{-\mathrm{i}tx}\, \mathrm{d}x, \qquad \operatorname{sinc} z = \begin{cases} \sin z / z, & z \neq 0, \\ 1, & z = 0. \end{cases} \tag{382}$$

Then $\mathscr{F}u_a(t) = \operatorname{sinc}(at)$. From (375),

$$\mathscr{F} f_n(t) = \operatorname{sinc}(2^{-n}t) \prod_{k=1}^{n} \operatorname{sinc}(2^{-k}t) = \left(\prod_{k=1}^{n-1} \operatorname{sinc}(2^{-k}t) \right) \operatorname{sinc}^2(2^{-n}t). \tag{383}$$

Under the symmetric convention used in the Stack Exchange question, the right side is multiplied by $(2\pi)^{-1/2}$, exactly reproducing the conjectured formula there. Taking the limit gives

$$\mathscr{F}\, \mathrm{up}(t) = \prod_{k=1}^{\infty} \operatorname{sinc}(2^{-k}t). \tag{384}$$

The product is rapidly decreasing together with all its derivatives; Fourier inversion therefore gives a C^{∞} density. The product also makes the finite-depth correction transparent: f_n has the first n factors of the limiting product, with the last factor repeated once.

27.3 Relation with the bounded Fabius function

Let $\xi_k = (U_k + 1)/2$, so that the ξ_k are independent uniform variables on $[0, 1]$. Then

$$Y = \sum_{k=1}^{\infty} 2^{-k} \xi_k = \frac{X + 1}{2}. \tag{385}$$

The cumulative distribution function of Y is the bounded Fabius function F. Hence

$$\mathbb{P}(X \le x) = F\left(\frac{x+1}{2}\right), \tag{386}$$

and, using $F(1 - y) = 1 - F(y)$,

$$\mathbb{P}(X > t) = F\left(\frac{1-t}{2}\right), \qquad 0 \le t \le 1. \tag{387}$$

On the two halves of the support,

$$\mathrm{up}(x) = \begin{cases} F(x+1), & -1 \le x \le 0, \\ F(1-x), & 0 \le x \le 1. \end{cases} \tag{388}$$

This is exactly the normalization used in the existing exposition.

28 Exact finite splines and the Thue–Morse collapse

The convolution formula (375) already gives a closed form in the language of box splines. To obtain the explicit truncated-power expression, we first record the general one-dimensional inclusion–exclusion formula.

28.1 A general nonuniform box-spline formula

Lemma 28.1 (Density of a sum of centered uniforms)**.** *Let V_i be independent and uniform on $[-a_i, a_i]$, where $a_i > 0$ and $1 \le i \le m$. Put $A = a_1 + \cdots + a_m$. The density $b_{\boldsymbol{a}}$ of $V_1 + \cdots + V_m$ is*

$$b_{\boldsymbol{a}}(x) = \frac{1}{2^m (m-1)! \prod_{i=1}^m a_i} \sum_{\varepsilon \in \{0,1\}^m} (-1)^{|\varepsilon|} \left(x + A - 2\sum_{i=1}^m \varepsilon_i a_i\right)_+^{m-1}. \tag{389}$$

Proof. Translate V_i to $W_i = V_i + a_i$, which is uniform on $[0, 2a_i]$. The distributional derivative of the density of W_i is $(\delta_0 - \delta_{2a_i})/(2a_i)$. After convolving the m factors and integrating $m - 1$ times from the left endpoint, one obtains the usual inclusion–exclusion formula for the density of $W_1 + \cdots + W_m$. Translating the sum back by A gives (389). Equivalently, one may verify the formula by taking Laplace transforms. □

For f_n, the $m = n + 1$ half-widths are

$$2^{-1}, 2^{-2}, \ldots, 2^{-(n-1)}, 2^{-n}, 2^{-n}. \tag{390}$$

Their sum is 1, and their product is $2^{-n(n+3)/2}$. Thus the scalar prefactor in (389) becomes

$$\frac{2^{n(n+1)/2-1}}{n!}. \tag{391}$$

It remains to combine subsets that produce the same knot.

28.2 The subset-sum polynomial

After multiplication by 2^{n-1}, the possible increments $2a_i$ in (389) are

$$2^{n-1}, 2^{n-2}, \ldots, 2, 1, 1. \tag{392}$$

The signed subset-sum generating polynomial is therefore

$$\begin{aligned} B_n(z) &= (1-z)^2 \prod_{r=1}^{n-1} (1 - z^{2^r}) \\ &= (1-z) \prod_{r=0}^{n-1} (1 - z^{2^r}). \end{aligned} \tag{393}$$

The classical finite Thue–Morse identity is

$$\prod_{r=0}^{n-1} (1 - z^{2^r}) = \sum_{j=0}^{2^n-1} (-1)^{\mathrm{wt}_2(j)} z^j = \sum_{j \in \mathbb{Z}} \tau_n(j) z^j. \tag{394}$$

Multiplication by $1 - z$ takes first differences of the coefficient sequence:

$$B_n(z) = \sum_{j=0}^{2^n} \Delta\tau_n(j) z^j. \tag{395}$$

This explains the otherwise rather mysterious coefficient $c_n(j) - c_n(j-1)$ in the Stack Exchange answer: the duplicated smallest uniform interval contributes the extra factor $1 - z$, turning the Thue–Morse word into its first difference.

Proof of theorem 25.1. Insert the widths (390) into lemma 28.1. The subset sum indexed by ε equals $j/2^{n-1}$ after grouping terms with the same integer j; by (395), the total sign of that group is $\Delta\tau_n(j)$. The scalar is (391). This gives (357). Factoring 2^{-n} from each truncated power gives (358). □

28.3 Derivatives, knots, and exact smoothness

For $0 \le r \le n-1$, differentiating (357) gives

$$f_n^{(r)}(x) = \frac{2^{n(n+1)/2-1}}{(n-r)!} \sum_{j=0}^{2^n} \Delta\tau_n(j) \left(x + 1 - \frac{j}{2^{n-1}} \right)_+^{n-r}. \tag{396}$$

The formula is also valid away from the knots for $r = n$. If

$$-1 + \frac{j}{2^{n-1}} < x < -1 + \frac{j+1}{2^{n-1}}, \qquad 0 \le j < 2^n, \tag{397}$$

then the first-difference sum telescopes and gives

$$\boxed{f_n^{(n)}(x) = 2^{n(n+1)/2-1} \tau_n(j).} \tag{398}$$

Thus the highest ordinary derivative is literally a Thue–Morse step word. Because this step function has jumps at the active knots, f_n is C^{n-1} but not C^n globally. Each application of T adds exactly one degree and exactly one order of classical smoothness.

28.4 The first iterates

The first three functions are

$$f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x), \tag{399}$$

$$f_1(x) = (1 - |x|)_+ \,, \tag{400}$$

$$f_2(x) = \begin{cases} 1 - 2x^2, & |x| \le \frac{1}{2}, \\ 2(1-|x|)^2, & \frac{1}{2} \le |x| \le 1, \\ 0, & |x| \ge 1. \end{cases} \tag{401}$$

The next one already displays the full Thue–Morse organization compactly:

$$f_3(x) = \frac{16}{3}\sum_{j=0}^{8} \Delta\tau_3(j) \left(x + 1 - \frac{j}{4}\right)_+^3 . \tag{402}$$

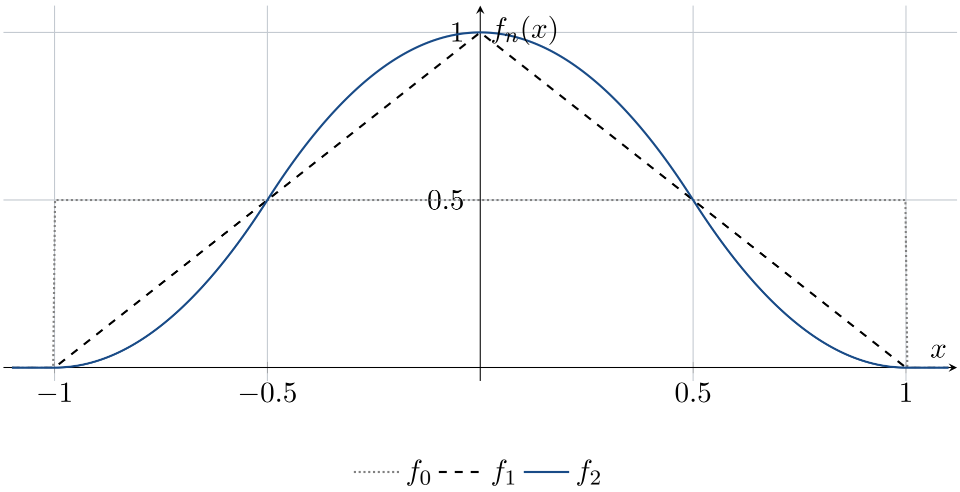

Figure 3: The rectangle, the piecewise linear tent, and the quadratic spline. Later iterates add one polynomial degree and one smoothness order at each step.

29 Replacing the last tail: the exact error decomposition

Put

$$S_n = \sum_{k=1}^{n} 2^{-k} U_k, \qquad a_n = 2^{-n}, \tag{403}$$

and let p_n be the density of S_n. Let X' be an independent copy of the limiting random variable X. The finite and infinite laws split as

$$X_n \overset{d}{=} S_n + a_n U, \qquad X \overset{d}{=} S_n + a_n X'. \tag{404}$$

If

$$\rho_a(x) = a^{-1}\,\mathrm{up}(x/a) \tag{405}$$

then (404) is exactly

$$f_n = p_n * u_{a_n}, \qquad \mathrm{up} = p_n * \rho_{a_n}. \tag{406}$$

The repeated-integration approximation therefore does one very specific thing: it keeps the first n dyadic uniform coordinates exact and replaces the remaining self-similar tail $a_n X'$ by a single uniform coordinate $a_n U$.

29.1 The comparison kernel

Definition 29.1 (Uniform-versus-up Peano kernel)**.** For $t \in \mathbb{R}$, define

$$K(t) = \mathbb{E}\,(U - t)_+ - \mathbb{E}\,(X - t)_+\,. \tag{407}$$

The kernel contains every constant needed for the sharp error theorem. We first calculate its basic invariants.

Lemma 29.2 (Second and first absolute moments)**.** *The limiting random variable X satisfies*

$$\mathbb{E}X^2 = \frac{1}{9}, \qquad \mathbb{E}|X| = \frac{5}{18}. \tag{408}$$

Consequently,

$$F\left(\frac{1}{4}\right) = \frac{5}{72}. \tag{409}$$

Proof. Independence and centering in (378) give

$$\mathbb{E}X^2 = \frac{1}{3}\sum_{k=1}^{\infty} 4^{-k} = \frac{1}{9}.$$

For fixed $x \in [-1,1]$ and U uniform on $[-1,1]$,

$$\mathbb{E}(|x+U| \mid x) = \frac{1}{2}\int_{-1}^{1} |x+u|\,\mathrm{d}u = \frac{1+x^2}{2}.$$

Using $X \overset{d}{=} (X'+U)/2$ gives

$$\mathbb{E}|X| = \frac{1}{2}\mathbb{E}|X'+U| = \frac{1}{4}(1+\mathbb{E}X^2) = \frac{5}{18}.$$

By symmetry, $\mathbb{E}X_+ = \frac{1}{2}\mathbb{E}|X| = 5/36$. On the other hand, (387) and $F'(y) = 2F(2y)$ imply

$$\mathbb{E}X_+ = \int_0^1 F\left(\frac{1-t}{2}\right)\mathrm{d}t = 2\int_0^{1/2} F(y)\,\mathrm{d}y = 2F\left(\frac{1}{4}\right),$$

which proves (409). ☐

Theorem 29.3 (Exact kernel data)**.** *The kernel K is even, continuous, supported in $[-1,1]$, and nonnegative. For $0 \le t \le 1$,*

$$K(t) = \frac{(1-t)^2}{4} - 2F\left(\frac{1-t}{4}\right). \tag{410}$$

Moreover,

$$\boxed{\int_{-1}^{1} K(t)\,\mathrm{d}t = \frac{1}{9}, \qquad \|K\|_\infty = K(0) = \frac{1}{9},} \tag{411}$$

while

$$\|K'\|_\infty = \frac{13}{72}. \tag{412}$$

Proof. Evenness follows from symmetry and the fact that U and X have the same mean. Both variables are supported in $[-1,1]$, so K vanishes outside that interval.

For nonnegativity, let ϕ be convex. From the fixed-point identity $X \overset{d}{=} (X'+U)/2$ and convexity,

$$\mathbb{E}\phi(X) \le \frac{1}{2}\mathbb{E}\phi(X') + \frac{1}{2}\mathbb{E}\phi(U).$$

Since X' has the same law as X, this rearranges to $\mathbb{E}\phi(X) \le \mathbb{E}\phi(U)$. Thus X is below U in convex order. Applying this to $\phi_t(y) = (y-t)_+$ gives $K(t) \ge 0$.

For $0 \le t \le 1$,

$$\mathbb{E}\,(U-t)_+ = \frac{(1-t)^2}{4}.$$

Also, by (387),

$$\begin{aligned}\mathbb{E}\,(X-t)_+ &= \int_t^1 \mathbb{P}(X>s)\,\mathrm{d}s = \int_t^1 F\left(\frac{1-s}{2}\right)\mathrm{d}s\\ &= 2\int_0^{(1-t)/2} F(y)\,\mathrm{d}y = 2F\left(\frac{1-t}{4}\right),\end{aligned}$$

which proves (410).

The Peano identity proved in lemma 29.4 below, applied to $h(y) = y^2/2$, gives

$$\int K(t)\,\mathrm{d}t = \frac{1}{2}(\mathbb{E}U^2 - \mathbb{E}X^2) = \frac{1}{2}\left(\frac{1}{3}-\frac{1}{9}\right) = \frac{1}{9}.$$

At the origin, lemma 29.2 gives

$$K(0) = \mathbb{E}U_+ - \mathbb{E}X_+ = \frac{1}{4} - \frac{5}{36} = \frac{1}{9}.$$

It remains to see that this is the maximum. Differentiating (410) gives

$$K'(t) = F\left(\frac{1-t}{2}\right) - \frac{1-t}{2}. \tag{413}$$

On $[0,1/2]$, the Fabius function is convex: $F'(y) = 2F(2y)$ is increasing. Since $F(0)=0$ and $F(1/2) = 1/2$, convexity places its graph below the chord, so $F(y) \le y$. Therefore $K'(t) \le 0$ for $t \ge 0$, and $K(0)$ is the maximum.

Finally, put $y = (1-t)/2$ in (413). The magnitude of K' is the maximum of $y - F(y)$ on $[0,1/2]$. Its derivative is $1 - 2F(2y)$, which vanishes exactly at $y = 1/4$ because $F(1/2) = 1/2$. Thus

$$\|K'\|_\infty = \frac{1}{4} - F\left(\frac{1}{4}\right) = \frac{1}{4} - \frac{5}{72} = \frac{13}{72}.$$

□

The equality between the mass and the maximum in (411) is the key numerical coincidence behind the clean final rate. The mass controls all sufficiently smooth stages; the maximum controls the borderline stage at which a distributional box-spline derivative becomes an ordinary continuous error function.

29.2 Peano identity and the pointwise error formula

Lemma 29.4 (Peano identity)**.** *Let h' be absolutely continuous on $[-1,1]$. Then*

$$\mathbb{E}h(U) - \mathbb{E}h(X) = \int_{-1}^{1} K(t)h''(t)\,\mathrm{d}t. \tag{414}$$

Proof. For $y \in [-1, 1]$,

$$h(y) = h(-1) + h'(-1)(y+1) + \int_{-1}^{1} (y-t)_+ \, h''(t) \, \mathrm{d}t.$$

Take expectations first with $y = U$ and then with $y = X$. The constant terms cancel, and the linear terms cancel because both random variables have mean zero. Fubini's theorem then gives (414). □

Apply the lemma with $h(y) = p_n^{(r)}(x - a_n y)$. Whenever $n \geq r+3$, the needed weak second derivative is bounded and the result is an ordinary integral.

Proposition 29.5 (Exact pointwise error identity)**.** *Let $r \geq 0$ and $n \geq r+3$. Then for every $x \in \mathbb{R}$,*

$$f_n^{(r)}(x) - \mathrm{up}^{(r)}(x) = a_n^2 \int_{-1}^{1} K(t) p_n^{(r+2)}(x - a_n t) \, \mathrm{d}t. \tag{415}$$

Proof. From (406),

$$f_n^{(r)}(x) = \mathbb{E} p_n^{(r)}(x - a_n U), \qquad \mathrm{up}^{(r)}(x) = \mathbb{E} p_n^{(r)}(x - a_n X).$$

Use lemma 29.4; differentiating twice with respect to its argument contributes the factor a_n^2. □

There is also a distributional form that remains valid at the borderline stage. Define

$$G_a(x) = aK(x/a). \tag{416}$$

Since K'' is the difference between the densities of U and X in the distributional sense,

$$u_a - \rho_a = \mathrm{D}^2 G_a. \tag{417}$$

Consequently,

$$f_n - \mathrm{up} = p_n * \mathrm{D}^2 G_{a_n}. \tag{418}$$

This identity will handle the first two smoothness thresholds without any appeal to a classical second derivative of p_n.

30 Sharp derivative geometry of the partial densities

The pointwise identity (415) becomes sharp because the derivatives of p_n are disjoint signed translates of a single tail density, each having a flat top of known width and height.

For $m \geq 0$, define the dyadic centers

$$c_{m,j} = -1 + \frac{2j+1}{2^m}, \qquad 0 \leq j < 2^m. \tag{419}$$

For $m = 0$, this is the singleton $\{0\}$. At the top piecewise-polynomial derivative order, derivatives below are understood through their bounded almost-everywhere representatives; values at the finitely many knots may be chosen arbitrarily between the adjacent limits and do not affect the L^∞ norm used in this section.

Theorem 30.1 (Derivative decomposition and exact plateaux)**.** *Let $0 \le m \le n-1$. Let $q_{m,n}$ be the density of the tail sum*

$$R_{m,n} = \sum_{k=m+1}^{n} 2^{-k} U_k. \tag{420}$$

Then, almost everywhere,

$$p_n^{(m)}(x) = 2^{\binom{m}{2}} \sum_{j=0}^{2^m-1} (-1)^{\mathrm{wt}_2(j)} q_{m,n}(x - c_{m,j}). \tag{421}$$

The translated supports in this sum have disjoint interiors. Moreover,

$$\left\| p_n^{(m)} \right\|_\infty = 2^{\binom{m+1}{2}}, \tag{422}$$

and on the central interval of each translated tail,

$$p_n^{(m)}(x) = (-1)^{\mathrm{wt}_2(j)} 2^{\binom{m+1}{2}} \quad \text{whenever} \quad |x - c_{m,j}| < 2^{-n}. \tag{423}$$

Proof. In the distributional sense,

$$\mathrm{D}u_{2^{-k}} = 2^{k-1}\big(\delta_{-2^{-k}} - \delta_{2^{-k}}\big). \tag{424}$$

Differentiate the first m factors in $p_n = u_{2^{-1}} * \cdots * u_{2^{-n}}$. The product of the scalar factors is

$$\prod_{k=1}^{m} 2^{k-1} = 2^{\binom{m}{2}}.$$

The 2^m signed sums of the locations $\pm 2^{-k}$ are precisely the centers $c_{m,j}$, in increasing order, and the sign is $(-1)^{\mathrm{wt}_2(j)}$. Convolution with the undifferentiated tail gives (421).

Adjacent centers are separated by 2^{1-m}. The support radius of $q_{m,n}$ is

$$\sum_{k=m+1}^{n} 2^{-k} = 2^{-m} - 2^{-n},$$

so its support diameter is $2^{1-m} - 2^{1-n}$, strictly smaller than the center spacing. Hence the translated interiors are disjoint.

Scaling shows

$$q_{m,n}(x) = 2^m p_{n-m}(2^m x). \tag{425}$$

The density p_s has supremum one: its first factor $u_{1/2}$ has height one, and convolution with a probability density cannot increase the supremum. It actually equals one on $|x| < 2^{-s}$, because the remaining factors have total support radius $1/2 - 2^{-s}$. Thus $q_{m,n}$ has height 2^m and is constant at that height on $|x| < 2^{-n}$. Multiplying by $2^{\binom{m}{2}}$ proves (422) and (423). □

At the missing endpoint $m = n$, the same calculation gives a signed Dirac comb:

$$\mathrm{D}^n p_n = 2^{\binom{n}{2}} \sum_{j=0}^{2^n-1} (-1)^{\mathrm{wt}_2(j)} \delta_{c_{n,j}}. \tag{426}$$

The centers are spaced by $2^{1-n} = 2a_n$.

30.1 A sharp derivative norm for the up-function

The finite plateau geometry also recovers a useful sharp fact about the limit itself.

Corollary 30.2 (Exact derivative peaks of up)**.** *For every $m \geq 0$,*

$$\boxed{\left\|\mathrm{up}^{(m)}\right\|_\infty = 2^{\binom{m+1}{2}}.} \tag{427}$$

More precisely,

$$\mathrm{up}^{(m)}(c_{m,j}) = (-1)^{\mathrm{wt}_2(j)} 2^{\binom{m+1}{2}} \qquad (0 \leq j < 2^m). \tag{428}$$

Proof. Take $n \geq m+1$. Since $\mathrm{up} = p_n * \rho_{a_n}$,

$$\mathrm{up}^{(m)}(x) = \int p_n^{(m)}(x-y)\rho_{a_n}(y)\,\mathrm{d}y.$$

Convolution with a probability density cannot exceed $\left\|p_n^{(m)}\right\|_\infty$, giving the upper bound. At $x = c_{m,j}$, the whole support $[-a_n, a_n]$ of ρ_{a_n} lies in the constant plateau (423), so the convolution equals the signed plateau height. This proves both statements. □

Corollary 30.3 (Higher-order flatness at the peak grid)**.** *For every $q \geq 0$, every $0 \leq j < 2^q$, and every $k > q$,*

$$\mathrm{up}^{(k)}(c_{q,j}) = 0. \tag{429}$$

Proof. At $q = 0$, the only point is $c_{0,0} = 0$. On the right of zero, $\mathrm{up}(x) = F(1-x)$, and the smooth bounded Fabius function is constant on $[1, \infty)$; therefore all positive-order derivatives of up vanish at zero.

Let $q \geq 1$, put $a = 2^{-q}$, and write $\mathrm{up} = p_q * \rho_a$. For $\ell = k - q \geq 1$,

$$\mathrm{up}^{(k)} = (\mathrm{D}^q p_q) * \rho_a^{(\ell)}.$$

Insert the signed Dirac comb (426). At $x = c_{q,j}$ every off-diagonal translate is evaluated at distance at least $2a$, outside the support $[-a, a]$ of ρ_a. The diagonal translate is $\rho_a^{(\ell)}(0) = a^{-\ell-1}\,\mathrm{up}^{(\ell)}(0) = 0$ by the first paragraph. Thus every term vanishes. □

31 The exact convergence rate

We now combine the nonnegative kernel of section 29 with the exact plateaux of section 30.

31.1 All stages after the first admissible one

Theorem 31.1 (Sharp L^∞ error for every derivative)**.** *Let $r \geq 0$ and $n \geq r+2$. Then*

$$\left\|f_n^{(r)} - \mathrm{up}^{(r)}\right\|_\infty = \frac{4^{-n}}{9}\left\|\mathrm{up}^{(r+2)}\right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9\,4^n}. \tag{430}$$

At the grid (364), the signed equality (366) holds.

Proof. Put $m = r+2$.

First suppose $n \geq m+1$. From (415), the nonnegativity and mass of K, and (422),

$$\begin{aligned}|f_n^{(r)}(x) - \mathrm{up}^{(r)}(x)| &\leq a_n^2 \int K(t)\,\mathrm{d}t\,\left\|p_n^{(m)}\right\|_\infty \\ &= 4^{-n} \cdot \frac{1}{9} \cdot 2^{\binom{m+1}{2}}.\end{aligned}$$

At $x = c_{m,j}$, every point $c_{m,j} - a_n t$ with $-1 < t < 1$ lies in the constant plateau (423). Hence no inequality remains:

$$f_n^{(r)}(c_{m,j}) - \mathrm{up}^{(r)}(c_{m,j}) = (-1)^{\mathrm{wt}_2(j)} 4^{-n} 2^{\binom{m+1}{2}} \int K(t)\, \mathrm{d}t.$$

This proves the result for $n \geq m+1$.

It remains to treat the borderline case $n = m$. Use (418) and move all $m = r+2$ derivatives to p_m:

$$f_m^{(r)} - \mathrm{up}^{(r)} = (\mathrm{D}^m p_m) * G_{a_m}.$$

By (426),

$$f_m^{(r)}(x) - \mathrm{up}^{(r)}(x) = 2^{\binom{m}{2}} a_m \sum_{j=0}^{2^m-1} (-1)^{\mathrm{wt}_2(j)} K\left(\frac{x - c_{m,j}}{a_m}\right). \tag{431}$$

The copies of K have support radius a_m and their centers are $2a_m$ apart, so their interiors are disjoint. Therefore

$$\left\| f_m^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = 2^{\binom{m}{2}} a_m \|K\|_\infty = 2^{\binom{m}{2} - m} \frac{1}{9}.$$

Since $\binom{m}{2} - m = \binom{m+1}{2} - 2m$, this is exactly the right side of (430) with $n = m$. At $x = c_{m,j}$, use $K(0) = 1/9$ to obtain the signed formula. □

The two cases of the proof use different geometric data:

- for $n \geq r+3$, the error averages a constant extremal plateau and uses $\int K = 1/9$;
- for $n = r+2$, the error is a disjoint sum of rescaled kernel copies and uses $\|K\|_\infty = 1/9$.

The equality of those two kernel constants is exactly what makes the same formula valid at the threshold and beyond it.

31.2 The first C^r spline

The first iterate with r continuous derivatives is f_{r+1}. Its error has a different constant, controlled by K' rather than by K.

Theorem 31.2 (Exceptional first stage)**.** *For every $r \geq 0$,*

$$\left\| f_{r+1}^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = 2^{\binom{r+1}{2}} \|K'\|_\infty = \frac{13}{72} 2^{\binom{r+1}{2}}. \tag{432}$$

The supremum is attained at every point $c_{r+2,k}$, and the signed value is (365).

Proof. Set $n = r+1$. In (418), the total derivative order is $r+2 = n+1$. Move n derivatives to p_n and one to G_{a_n}. Since $G'_{a_n}(x) = K'(x/a_n)$, (426) gives

$$f_n^{(r)}(x) - \mathrm{up}^{(r)}(x) = 2^{\binom{n}{2}} \sum_{j=0}^{2^n-1} (-1)^{\mathrm{wt}_2(j)} K'\left(\frac{x - c_{n,j}}{a_n}\right). \tag{433}$$

Again the supports have disjoint interiors, so the supremum is $2^{\binom{n}{2}} \|K'\|_\infty$. Substitute $n = r+1$ and (412).

For the signed values, (413) gives $K'(-1/2) = 13/72$ and $K'(1/2) = -13/72$. The two points $c_{n,j} \mp a_n/2$ are respectively $c_{n+1,2j}$ and $c_{n+1,2j+1}$; using $\tau_{n+1}(2j) = \tau_n(j)$ and $\tau_{n+1}(2j+1) = -\tau_n(j)$ in (433) proves (365). □

31.3 Exact C^r convergence

Because the constants in (430) increase strictly with the derivative order, the highest derivative controls the max norm (361).

Corollary 31.3 (Exact C^r norm)**.** *For every $r \geq 0$,*

$$\|f_{r+1} - \mathrm{up}\|_{C^r} = \frac{13}{72} 2^{\binom{r+1}{2}}, \tag{434}$$

and for every $n \geq r+2$,

$$\boxed{\|f_n - \mathrm{up}\|_{C^r} = \frac{2^{\binom{r+3}{2}}}{9\,4^n}.} \tag{435}$$

Thus, for every fixed r, the tail of the sequence lies in $C^r(\mathbb{R})$ and converges to up *in the C^r norm; the error is exactly geometric from the second admissible stage onward. No finite f_n is itself C^∞, so this is deliberately stated as eventual convergence in every finite smoothness norm rather than literally as convergence of a sequence inside the Fréchet space $C^\infty(\mathbb{R})$.*

Proof. For $n \geq r+2$, the constants in (430) grow strictly with the derivative order, since the ratio of the order-$(j+1)$ constant to the order-j constant is 2^{j+3}. Hence the rth derivative controls the maximum in (361). At the exceptional stage $n = r+1$, the assertion is immediate for $r = 0$. For $r \geq 1$, the largest lower-order error is the order-$(r-1)$ error, and its ratio to the order-r error in (432) is

$$\frac{2^{\binom{r+2}{2}}/(9\,4^{r+1})}{(13/72)2^{\binom{r+1}{2}}} = \frac{2^{2-r}}{13} < 1.$$

Thus the highest available derivative controls the max norm at the first stage as well. □

For the tent indexing $g_m = f_{m+1}$, this becomes the following direct answer to the piecewise-linear formulation.

Corollary 31.4 (Repeated integration starting from the tent)**.** *Let $g_0(x) = (1-|x|)_+$ and $g_{m+1} = Tg_m$. For every $r \geq 0$,*

$$\|g_r - \mathrm{up}\|_{C^r} = \frac{13}{72} 2^{\binom{r+1}{2}}, \tag{436}$$

and, for every $m \geq r+1$,

$$\boxed{\|g_m - \mathrm{up}\|_{C^r} = \frac{2^{\binom{r+3}{2}-2}}{9\,4^m}.} \tag{437}$$

In particular, (368) *holds.*

31.4 The uniform errors and exact extremal values

For $r = 0$, the complete startup behavior is

$$\|f_0 - \mathrm{up}\|_\infty = \frac{1}{2}, \qquad \|f_1 - \mathrm{up}\|_\infty = \frac{13}{72}, \qquad \|f_n - \mathrm{up}\|_\infty = \frac{8}{9\,4^n} \quad (n \geq 2). \tag{438}$$

The value for f_0 is immediate from $\mathrm{up}(0) = 1$ and $0 \leq \mathrm{up} \leq 1$. For f_1, put $y = 1 - |x|$. By (388),

$$f_1(x) - \mathrm{up}(x) = y - F(y), \tag{439}$$

whose absolute maximum is attained at $y = 1/4$ and equals $1/4 - F(1/4) = 13/72$.

The four extremizers from (364) are $\pm 1/4, \pm 3/4$. Since

$$\mathrm{up}(\pm 3/4) = F(1/4) = \frac{5}{72}, \qquad \mathrm{up}(\pm 1/4) = 1 - F(1/4) = \frac{67}{72}, \tag{440}$$

we obtain exact finite-stage values for every $n \geq 2$:

$$f_n(\pm 3/4) = \frac{5}{72} + \frac{8}{9\,4^n}, \tag{441}$$

$$f_n(\pm 1/4) = \frac{67}{72} - \frac{8}{9\,4^n}. \tag{442}$$

The first few are shown in table 7.

Table 7: Sharp uniform errors and two extremal values.

n	$\|f_n - \mathrm{up}\|_\infty$	$f_n(3/4)$	$f_n(1/4)$
0	1/2	–	–
1	13/72	1/4	3/4
2	1/18	1/8	7/8
3	1/72	1/12	11/12
4	1/288	7/96	89/96
5	1/1152	9/128	119/128
6	1/4608	107/1536	1429/1536

32 The asymptotic error profile

The exact norm formula gives the rate, but the Peano identity also identifies the whole limiting error shape.

Theorem 32.1 (Uniform leading profile)**.** *For every fixed $r \geq 0$,*

$$4^n\big(f_n^{(r)} - \mathrm{up}^{(r)}\big) \longrightarrow \frac{1}{9}\,\mathrm{up}^{(r+2)} \qquad \textit{uniformly on } \mathbb{R}. \tag{443}$$

Proof. Put $m = r + 2$. From (415),

$$4^n\big(f_n^{(r)}(x) - \mathrm{up}^{(r)}(x)\big) = \int K(t) p_n^{(m)}(x - a_n t)\,\mathrm{d}t.$$

For fixed m, $p_n^{(m)} \to \mathrm{up}^{(m)}$ uniformly. One direct estimate follows from $\mathrm{up} = p_n * \rho_{a_n}$ and the mean-value theorem:

$$\left\| p_n^{(m)} - \mathrm{up}^{(m)} \right\|_\infty \leq a_n \mathbb{E}|X| \left\| p_n^{(m+1)} \right\|_\infty,$$

for all sufficiently large n. The same derivative bound shows that replacing $p_n^{(m)}(x - a_n t)$ by $p_n^{(m)}(x)$ costs $O(a_n)$ uniformly. Finally use $\int K = 1/9$. $\square$

Combining (443) with (427) predicts the sharp constant in (430). The exact theorem says more: the norm of the leading profile is already the exact norm of the full error for every $n \geq r + 2$.

32.1 A full even-derivative expansion

The Fourier product provides an all-orders refinement. Put

$$\Phi(z) = \prod_{k=1}^{\infty} \operatorname{sinc}(z/2^k), \qquad A(z) = \frac{\operatorname{sinc} z}{\Phi(z)}. \tag{444}$$

The quotient is analytic near the origin (indeed on $|z| < 2\pi$) and meromorphic globally. Wherever $\Phi(2^{-n}t) \neq 0$ one has

$$\mathscr{F} f_n(t) = \mathscr{F}\,\mathrm{up}(t) A(2^{-n}t), \tag{445}$$

and at the exceptional frequencies the product on the right is interpreted by its continuous finite-product extension. Only the Taylor germ at zero is used below. It is convenient to encode that germ by

$$B(w) = A(\mathrm{i}w) = \frac{\sinh w/w}{\displaystyle\prod_{k=1}^{\infty} \frac{\sinh(w/2^k)}{w/2^k}} = \sum_{m=0}^{\infty} \beta_m w^{2m}. \tag{446}$$

The first coefficients are

$$\beta_0 = 1, \qquad \beta_1 = \frac{1}{9}, \qquad \beta_2 = \frac{2}{2025}, \qquad \beta_3 = -\frac{1}{2679075}, \qquad \beta_4 = \frac{58}{10247461875}. \tag{447}$$

Proposition 32.2 (All-orders C^r expansion)**.** *For fixed integers $r \geq 0$ and $M \geq 1$,*

$$f_n^{(r)} = \sum_{m=0}^{M-1} \beta_m 4^{-mn}\,\mathrm{up}^{(r+2m)} + O_{r,M}(4^{-Mn}) \tag{448}$$

uniformly on $\mathbb{R}$ as $n \to \infty$.

Proof. Choose R with $1 < R < 2\pi$ and put $a = 2^{-n}$. On $|z| \leq R$, Taylor's theorem gives

$$A(z) = \sum_{m=0}^{M-1} \beta_m (-1)^m z^{2m} + O_{M,R}(|z|^{2M}). \tag{449}$$

On the low-frequency region $|t| \leq R/a$, multiply this by $(\mathrm{i}t)^r \mathscr{F}\,\mathrm{up}(t)$. Since $\mathscr{F}\,\mathrm{up}$ is a Schwartz function, the L^1 norm of the resulting remainder is at most

$$C_{r,M,R} a^{2M} \int_{\mathbb{R}} |t|^{r+2M} |\mathscr{F}\,\mathrm{up}(t)|\,\mathrm{d}t = O_{r,M}(a^{2M}).$$

The sign in (449) is exactly the sign in $(\mathrm{i}t)^{2m} = (-1)^m t^{2m}$, so Fourier inversion produces the displayed derivatives of up.

It remains to control $|t| > R/a$. From (383) and $|\operatorname{sinc} y| \leq \min(1, |y|^{-1})$, for $|t| \geq R2^n$,

$$|\mathscr{F} f_n(t)| \leq 2^{n(n+1)/2+n} |t|^{-(n+1)}. \tag{450}$$

Hence, for fixed r and large n,

$$\int_{|t|>R2^n} |t|^r |\mathscr{F} f_n(t)|\,\mathrm{d}t \leq C_r R^{r-n} 2^{-n^2/2+(r+3/2)n},$$

which is smaller than a^{2M} for all sufficiently large n. The high-frequency tails of the finitely many $\mathrm{up}^{(r+2m)}$ terms are also $O_{r,M}(a^{2M})$, again because $\mathscr{F}\,\mathrm{up}$ is Schwartz. Fourier inversion now proves the uniform remainder in (448). □

For explicit coefficient generation, write

$$\log B(w) = \sum_{m=1}^{\infty} \lambda_m w^{2m}. \tag{451}$$

With the standard Bernoulli numbers B_{2m},

$$\lambda_m = \frac{2^{2m-1} B_{2m}}{m(2m)!} \frac{4^m - 2}{4^m - 1}. \tag{452}$$

Indeed,

$$\log \frac{\sinh w}{w} = \sum_{m \geq 1} \frac{2^{2m-1} B_{2m}}{m(2m)!} w^{2m}, \qquad \sum_{k \geq 1} 4^{-mk} = \frac{1}{4^m - 1};$$

subtracting the logarithms of the denominator factors in (446) gives (452). The coefficients β_m are then obtained recursively from

$$\beta_0 = 1, \qquad \beta_m = \frac{1}{m} \sum_{j=1}^{m} j \lambda_j \beta_{m-j}. \tag{453}$$

Thus the first terms of (448) are

$$f_n^{(r)} = \mathrm{up}^{(r)} + \frac{\mathrm{up}^{(r+2)}}{9\, 4^n} + \frac{2\, \mathrm{up}^{(r+4)}}{2025\, 16^n} - \frac{\mathrm{up}^{(r+6)}}{2679075\, 64^n} + O_r(256^{-n}). \tag{454}$$

At the extremal grid (364), all higher derivative terms vanish by corollary 30.3; there the leading term is not merely asymptotic but exactly equal to the full error, as (366) states.

33 Exact computation and numerical use

33.1 A direct exact-arithmetic evaluator

Formula (357) immediately gives an exact rational evaluator at rational arguments. The following compact Python implementation uses arbitrary-precision fractions. It is intended as executable notation rather than as the fastest large-n algorithm.

Listing 2: Exact evaluation of the finite iterates.

```
from fractions import Fraction
from math import factorial


def tau(n: int, j: int) -> int:
    if j < 0 or j >= (1 << n):
        return 0
    return -1 if (j.bit_count() & 1) else 1


def positive_power(x: Fraction, n: int) -> Fraction:
    return x**n if x > 0 else Fraction(0)


def f(n: int, x: Fraction) -> Fraction:
    x = Fraction(x)
    if n == 0:
        return Fraction(1, 2) if abs(x) < 1 else Fraction(0)
```

```
    coefficient = Fraction(
        1 << (n * (n + 1) // 2 - 1), factorial(n)
    )
    mesh = Fraction(1, 1 << (n - 1))

    return coefficient * sum(
        (tau(n, j) - tau(n, j - 1))
        * positive_power(x + 1 - j * mesh, n)
        for j in range((1 << n) + 1)
    )
```

Only indices satisfying $j < 2^{n-1}(x+1)$ contribute, so the loop may be truncated at the active cell. Derivatives are obtained by replacing $n!$ with $(n-r)!$ and the power n with $n-r$, as in (396).

33.2 Conditioning and stable evaluation

The truncated-power formula is exact but can exhibit severe cancellation at large n: individual terms are much larger than their sum. Three alternatives are preferable in floating-point work.

(i) Use the convolution description (375) and propagate a piecewise-polynomial representation from one level to the next.

(ii) Use a B-spline/de Boor representation on the dyadic knot sequence rather than the truncated-power basis.

(iii) In Fourier space, evaluate the finite sinc product (383) and invert it with a quadrature adapted to a compactly supported smooth target.

For certified approximation of up, the sharp error gives an immediate stopping rule. To obtain uniform error at most ε from the tent-indexed sequence, it is necessary and sufficient that

$$m \geq \left\lceil \frac{1}{2} \log_2 \left(\frac{2}{9\varepsilon} \right) \right\rceil \qquad (m \geq 1). \tag{455}$$

For a fixed derivative order r, replace the numerator $2/9$ by $2^{\binom{r+3}{2}-2}/9$ and also require $m \geq r+1$.

34 How this fits the existing ProveIt development

The present approximation is closely related to, but not identical with, two families already developed in the repository.

- The finite convolution and polynomial machinery in `FabiusFunction/EarlyApproximants.lean` provides reusable measure-theoretic and coefficient infrastructure.
- The Thue–Morse spline machinery in `FabiusFunction/FabiusUniformSpline.lean` already contains many of the finite-power sum and block-translation lemmas needed to identify the top derivative pattern.
- The centered finite-CDF spline in the main article approximates the bounded Fabius function. The sequence here approximates the up-density itself and has a different tail replacement and a different sharp rate.

A natural formalization could be organized as follows.

(1) Define the probability operator T and prove $Th(x) = \int_{2x-1}^{2x+1} h$ for supported integrable h.

(2) Define X_n recursively and prove the finite random-sum identity (374); derive (375) at the level of measures.

(3) Introduce the nonuniform box spline and prove the general inclusion–exclusion lemma (389).

(4) Reuse the finite Thue–Morse product to prove (395), then obtain the exact iterate and derivative formulas.

(5) Define K by (407). Prove convex domination with the one-line Jensen argument, then formalize its mass, maximum, and derivative maximum.

(6) Prove the derivative decomposition (421), support separation, and the exact plateaux.

(7) Combine the Peano identity with the plateau lemma, splitting the proof into the ordinary case $n \geq r+3$ and the distributional threshold $n = r+2$.

The analytic core of the sharp theorem is therefore modest. The technically delicate parts for Lean are likely to be bookkeeping for weak derivatives of interval densities and the transfer between measure convolution and classical derivatives. One can avoid most of that friction by proving the borderline formulas directly from finite signed Dirac combs and using ordinary absolutely continuous derivatives only after the threshold.

35 Conclusions

The repeated-integration construction has three mutually reinforcing exact descriptions:

(i) an averaging recursion $X \mapsto (X+U)/2$;

(ii) a convolution of dyadically scaled uniform densities;

(iii) a dyadic polynomial spline whose top derivative is a Thue–Morse word.

These descriptions identify the limit as Rvachev's up-function and reduce the finite-stage closed form to a single first-difference Thue–Morse sum. More unexpectedly, the same probabilistic representation yields a completely sharp convergence theorem. The finite stage and the limit differ only in the last scaled tail: a uniform variable replaces an up-distributed variable. Their mass and mean agree, so the error starts at second order in 2^{-n}. The comparison kernel quantifies this replacement exactly, while the partial box spline supplies disjoint extremal plateaux. The outcome is the exact identity

$$\left\| f_n^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+3}{2}}}{9\,4^n} \qquad (n \geq r+2),$$

not just an upper bound or an asymptotic equivalent.

In the most literal piecewise-linear formulation, starting from the tent and integrating again and again, the answer is especially crisp: after the initial tent, the uniform error is exactly $2/(9\,4^m)$. Every additional integration divides the worst-case error by four, and the points at which that error is attained are fixed dyadic points rather than moving with the iteration.

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Part IV
Dyadic Calculus for the Fabius Function

Source provenance and status. Former path: `Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex`. Earlier dyadic/Appell/q-binomial dossier. Its coarse shifted-spline and $O(2^{-n})$ Toeplitz estimates remain valid historical corollaries; the following Dyadic Web dossier's sharper shifted estimate and explicit $O_Q(4^{-n})$ row estimate are preferred when the constructions overlap. The determinant, Appell compression, superexponential tail, prefix rearrangement, limiting-filter, and integral-transform displays remain formal obligations unless an exact declaration is named locally.

Original source SHA-256: `81a0a911ac7ab28e12c6b87ccaf76ffccaf32e48f873abff18fb7a2d01bcf3e5`.

Dossier abstract

The exact formulae for the values of the Fabius function at dyadic rationals, the binary-reduction series at an arbitrary argument, and the discrete-limit formula involving Gaussian binomial coefficients look at first like three rather different constructions. They are, in fact, organized by one finite polynomial identity and one regular summability argument.

The finite identity is most transparent after introducing the Appell moment polynomials

$$A_n(\theta) = \mathbb{E}(X+\theta)^n,$$

where X has the Fabius distribution. The Taylor-reduction polynomial at scale n is exactly $2^{-\binom{n}{2}} A_n(2^n y)/n!$. Consequently, the global binary expansion becomes a rapidly convergent, digit-sparse Appell series; for a dyadic rational it terminates, so the finite dyadic formula is literally the terminating case of the arbitrary-argument expansion. Replacing each Appell polynomial by its finite Thue–Morse/Gaussian-binomial representation gives the nested q-series term by term, not merely after taking a limit.

The discrete-limit formula uses the same finite Gaussian stencil in a different way. After an exact reindexing it is a signed Toeplitz average of shifted uniform-spline approximants. Its row-generating polynomial is

$$\frac{(z/2;1/2)_n}{(1/2;1/2)_n},$$

which simultaneously explains unit row mass, bounded total variation, and the appearance of q-Pochhammer symbols. This article develops these connections, derives quantitative error bounds, and adds several alternative exact representations: a finite composition sum obtained directly from the moment product, a lower-Hessenberg determinant, a Bernoulli–Bell-polynomial formula, and Fourier and Laplace inversion formulae.

36 Orientation and notation

36.1 What is being connected

The companion article and the Lean development establish, among many other facts, the following three families of formulae.

1. Exact rational expressions for $F(m/2^n)$, including formulae in which a Gaussian binomial coefficient at base $1/2$ multiplies a finite signed Thue–Morse power sum.

2. An exact infinite expansion for $F(x)$, or for the signed global extension $\mathcal{F}(x)$, obtained by repeatedly reducing along the binary expansion of x. Substituting the finite q-binomial expression into every reduction block produces a nested infinite q-series.
3. A discrete limit in which a finite q-binomial sum of increasingly fine Thue–Morse splines converges to $\mathcal{F}(x)$ at an arbitrary real argument.

The purpose here is not to repeat the proofs already presented there. We instead isolate the common algebraic mechanism, compress the formulae into a small collection of reusable objects, and then unfold those objects in several new directions.

Throughout,

$$\rho = \frac{1}{2}, \qquad (a;\rho)_n = \prod_{j=0}^{n-1}(1-a\rho^j), \qquad \begin{bmatrix} n \\ k \end{bmatrix}_\rho = \frac{(\rho;\rho)_n}{(\rho;\rho)_k\,(\rho;\rho)_{n-k}}.$$

The empty product is 1. We write

$$\varepsilon_r = (-1)^{\mathrm{wt}_2(r)},$$

where $\mathrm{wt}_2(r)$ is the number of 1's in the binary expansion of r. Thus $(\varepsilon_r)_{r\geq 0}$ is the $\{\pm 1\}$ Thue–Morse sequence.

36.2 The half-base notation is Mersenne arithmetic

At the special base $\rho = 1/2$, the q-Pochhammer symbol is simply a compact normalization of a product of Mersenne numbers:

$$(1/2;1/2)_n = \prod_{j=1}^{n}(1-2^{-j}) = 2^{-\binom{n+1}{2}}\prod_{j=1}^{n}(2^j-1). \tag{456}$$

In particular,

$$2^{n^2}\,(1/2;1/2)_n = 2^{\binom{n}{2}}\prod_{j=1}^{n}(2^j-1). \tag{457}$$

The factors $2^j - 1$ appearing separately in moment recurrences and composition formulae are therefore the same arithmetic content that the q-Pochhammer symbol packages in the closed Gaussian-binomial formula. Likewise,

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = 2^{-k(n-k)}\begin{bmatrix} n \\ k \end{bmatrix}_2. \tag{458}$$

Thus the half-base Gaussian coefficients are ordinary integral Gaussian coefficients at base 2, accompanied by an explicit power of 2. The q-notation is not introducing mysterious irrational constants; it records the geometry of the dyadic interpolation grid and keeps its Mersenne normalizations factored.

36.3 The probabilistic normalization

Let $U_1, U_2, \ldots$ be independent random variables, each uniformly distributed on $[0,1]$, and put

$$X = \sum_{j=1}^{\infty}\frac{U_j}{2^j}. \tag{459}$$

Then $0 \le X \le 1$, and the distribution function of X is the Fabius function:

$$F(x) = \mathbb{P}(X \le x).$$

This representation fixes all normalizations used below. Define the moments

$$d_n = \mathbb{E}X^n, \qquad d_0 = 1, \tag{460}$$

and their exponential generating function

$$G(z) = \mathbb{E}e^{zX} = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!}. \tag{461}$$

Independence in (459) gives the entire product

$$G(z) = \prod_{j=1}^{\infty} E\left(\frac{z}{2^j}\right), \qquad E(w) = \frac{e^w - 1}{w}. \tag{462}$$

Shifting the product by one factor yields

$$G(2z) = E(z)G(z). \tag{463}$$

Comparing coefficients gives the basic moment recurrence

$$(2^n - 1)\frac{d_n}{n!} = \sum_{k=0}^{n-1} \frac{d_k}{k!(n-k+1)!} \qquad (n \ge 1). \tag{464}$$

One of the fundamental exact dyadic identities is

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{d_n}{n!}. \tag{465}$$

It will reappear in nearly every section, but in different disguises.

The product (462), the recurrence (464), the dyadic identity (465), the finite q-binomial formulae, and the spline limits are all formally proved in the cited Lean development. Some of the regroupings below—notably the Appell compression, determinant form, and explicit error estimate for the Toeplitz limit—are elementary deductions from those formalized identities and are presented here in human-readable form.

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37 The master Taylor block

37.1 The reduction polynomial

For $n \ge 0$, define

$$R_n(y) = \sum_{k=0}^{n} 2^{\binom{k+1}{2} - \binom{n-k}{2}} \frac{d_{n-k}}{(n-k)!} \frac{y^k}{k!}. \tag{466}$$

This is the polynomial that occurs in the exact Taylor reduction. On the appropriate dyadic shell, if $y = x - 2^{-n}$, the global Fabius extension satisfies an identity of the form

$$\mathcal{F}(x) = -\mathcal{F}(y) + R_n(y).$$

Only the polynomial itself will be needed here.

Putting $y = 0$ in (466) gives

$$R_n(0) = 2^{-\binom{n}{2}} \frac{d_n}{n!} = F(2^{-n}),$$

so every inverse-dyadic value is the constant term of one reduction block. The less obvious fact is that the whole block has an equally simple meaning.

37.2 Appell compression

Definition 37.1 (Moment Appell polynomials)**.** For $n \geq 0$ and $\theta \in \mathbb{C}$, let

$$A_n(\theta) = \mathbb{E}(X+\theta)^n = \sum_{k=0}^{n} \binom{n}{k} d_{n-k}\theta^k. \tag{467}$$

Equivalently,

$$\sum_{n=0}^{\infty} A_n(\theta)\frac{z^n}{n!} = e^{\theta z}G(z). \tag{468}$$

The family is Appell in the classical sense:

$$A_n'(\theta) = nA_{n-1}(\theta), \qquad A_n(\theta+c) = \sum_{j=0}^{n} \binom{n}{j} A_{n-j}(\theta)c^j. \tag{469}$$

Since X and $1-X$ have the same distribution,

$$A_n(\theta) = (-1)^n A_n(-1-\theta). \tag{470}$$

Theorem 37.2 (Appell compression of the reduction block)**.** *For every $n \geq 0$ and $\theta \in \mathbb{C}$,*

$$R_n(2^{-n}\theta) = 2^{-\binom{n}{2}}\frac{A_n(\theta)}{n!}. \tag{471}$$

Equivalently, for arbitrary y,

$$R_n(y) = 2^{-\binom{n}{2}}\frac{A_n(2^n y)}{n!}. \tag{472}$$

Proof. Substitute $y = 2^{-n}\theta$ into (466). The power of 2 in the kth term becomes

$$\binom{k+1}{2} - \binom{n-k}{2} - nk = -\binom{n}{2},$$

independently of k. Factoring out $2^{-\binom{n}{2}}$ leaves

$$\sum_{k=0}^{n} \frac{d_{n-k}\theta^k}{(n-k)!k!} = \frac{1}{n!}\sum_{k=0}^{n}\binom{n}{k} d_{n-k}\theta^k = \frac{A_n(\theta)}{n!}.$$

□

This elementary cancellation is the main compression used throughout the article. A triangular-looking polynomial with apparently irregular powers of 2 is simply a rescaled shifted moment.

37.3 Immediate consequences

For $0 \leq \theta < 1$, positivity and bounded support give

$$0 \leq A_n(\theta) \leq 2^n. \tag{473}$$

Consequently,

$$0 \leq R_n(2^{-n}\theta) \leq \frac{2^{n-\binom{n}{2}}}{n!}. \tag{474}$$

This estimate is much sharper than the geometric estimate normally used to prove convergence of the binary-reduction series.

The first few Appell polynomials are

$$\begin{aligned}
A_0(\theta) &= 1,\\
A_1(\theta) &= \theta + \frac{1}{2},\\
A_2(\theta) &= \theta^2 + \theta + \frac{5}{18},\\
A_3(\theta) &= \theta^3 + \frac{3}{2}\theta^2 + \frac{5}{6}\theta + \frac{1}{6},\\
A_4(\theta) &= \theta^4 + 2\theta^3 + \frac{5}{3}\theta^2 + \frac{2}{3}\theta + \frac{143}{1350}.
\end{aligned}$$

They encode all Taylor blocks of the corresponding orders by a single change of scale.

38 The binary expansion as a sparse Appell series

38.1 Prefixes, bits, and tails

Fix $x \in [0, 1)$. Use the canonical binary expansion that is not eventually all 1's,

$$x = 0.b_1b_2b_3\ldots{}_2, \qquad b_m \in \{0, 1\}. \tag{475}$$

Let

$$a_m = \lfloor 2^m x\rfloor, \qquad \theta_m = \{2^m x\}, \qquad y_m = x - \frac{a_m}{2^m} = 2^{-m}\theta_m. \tag{476}$$

Then

$$a_m = 2a_{m-1} + b_m, \qquad \theta_m = 0.b_{m+1}b_{m+2}\ldots{}_2. \tag{477}$$

The binary-reduction series in the formal development has coefficient

$$c_m(x) = \varepsilon_{a_m}(2a_{m-1} - a_m) \qquad (m \geq 1). \tag{478}$$

Using (477), this simplifies completely:

$$c_m(x) = \begin{cases} 0, & b_m = 0,\\ \varepsilon_{a_{m-1}}, & b_m = 1. \end{cases} \tag{479}$$

Indeed, when $b_m = 1$ one has $\varepsilon_{a_m} = -\varepsilon_{a_{m-1}}$ and $2a_{m-1} - a_m = -1$. Thus only the positions of the 1-bits contribute.

38.2 The digit–Appell formula

Theorem 38.1 (Digit-sparse Appell expansion)**.** *For every $x \in [0, 1)$,*

$$\boxed{F(x) = \sum_{m=1}^{\infty} b_m\,(-1)^{b_1+\cdots+b_{m-1}}\,\frac{2^{-\binom{m}{2}}}{m!}\,A_m(\theta_m).} \tag{480}$$

The series converges absolutely, and in fact uniformly with respect to $x \in [0, 1)$.

Proof. The exact binary-reduction series is

$$F(x) = \sum_{m\geq 1} c_m(x) R_m(y_m).$$

Insert (479), use $\varepsilon_{a_{m-1}} = (-1)^{b_1+\cdots+b_{m-1}}$, and then apply theorem 37.2 with $y_m = 2^{-m}\theta_m$. Absolute and uniform convergence follow from (473) and the Weierstrass test. □

Formula (480) is a compact arbitrary-argument representation that contains no q-notation at all. The q-series will emerge by replacing each A_m by an exactly equal finite Gaussian-binomial expression.

38.3 A superexponential remainder bound

Let $S_N(x)$ denote the truncation of (480) after $m = N$. From (473),

$$|F(x) - S_N(x)| \le \sum_{m=N+1}^{\infty} \frac{2^{m-\binom{m}{2}}}{m!}.$$

The ratio of consecutive majorants is

$$\frac{2^{m+1-\binom{m+1}{2}}/(m+1)!}{2^{m-\binom{m}{2}}/m!} = \frac{2^{1-m}}{m+1} \le \frac{1}{2} \qquad (m \ge 1).$$

Therefore, for $N \ge 0$,

$$\boxed{|F(x) - S_N(x)| \le 2\,\frac{2^{N+1-\binom{N+1}{2}}}{(N+1)!}.} \tag{481}$$

The right-hand side decays faster than geometrically in N. In practical computation, the binary digit series is therefore far more rapidly convergent than the elementary residual estimate $O(2^{-N})$ suggests.

38.4 Dyadic rationals are the terminating case

Suppose

$$x = \frac{M}{2^N} = 0.b_1 b_2 \dots b_N 000\dots_2, \qquad 0 \le M < 2^N.$$

Then $b_m = 0$ for $m > N$, and (480) terminates:

$$\boxed{F\left(\frac{M}{2^N}\right) = \sum_{\substack{1 \le m \le N \\ b_m = 1}} (-1)^{b_1 + \cdots + b_{m-1}} \frac{2^{-\binom{m}{2}}}{m!} A_m(0.b_{m+1} \dots b_{N\,2}).} \tag{482}$$

Thus the exact bit-recursive evaluator, the finite exact dyadic formula, and the arbitrary-argument series are not separate phenomena: they are the finite, recursive, and infinite readings of the same identity.

Example 38.2 (The value at 5/16)**.** The terminating binary expansion is

$$\frac{5}{16} = 0.0101_2.$$

Only $m = 2$ and $m = 4$ contribute. At $m = 2$ the previous prefix has even weight and $\theta_2 = 1/4$; at $m = 4$ the previous prefix is 010_2, so the sign is negative and $\theta_4 = 0$. Hence

$$\begin{aligned} F\left(\frac{5}{16}\right) &= \frac{2^{-1}}{2!} A_2\left(\frac{1}{4}\right) - \frac{2^{-6}}{4!} A_4(0) \\ &= \frac{1}{4}\left(\frac{5}{18} + \frac{1}{4} + \frac{1}{16}\right) - \frac{1}{1536}\,\frac{143}{1350} \\ &= \frac{305857}{2073600}. \end{aligned}$$

The first contribution accounts for the prefix 01 together with the remaining tail 01; the second contribution corrects for the final 1-bit.

39 Why Thue–Morse and Gaussian binomial coefficients appear

Two independent annihilation mechanisms are superposed in the nested exact formula. The inner Thue–Morse sum is an additive finite-difference operator; the outer Gaussian-binomial sum is a divided-difference operator on a geometric grid. Each kills lower-degree terms. Their composition isolates precisely the coefficient needed by the Taylor block.

39.1 The Thue–Morse difference operator

For a polynomial or entire function P, define

$$(\mathfrak{T}_k P)(c) = \sum_{r=0}^{2^k-1} \varepsilon_r P(c+r). \tag{483}$$

Writing $D = \frac{\mathrm{d}}{\mathrm{d}c}$ and using the binary digits of r gives the operator factorization

$$\mathfrak{T}_k = \prod_{j=0}^{k-1} \left(1 - e^{2^j D}\right). \tag{484}$$

Indeed,

$$\sum_{r<2^k} \varepsilon_r e^{rz} = \prod_{j=0}^{k-1} (1 - e^{2^j z}).$$

Since every factor in (484) begins with $-2^j D$, the operator annihilates all polynomials of degree less than k and satisfies

$$\sum_{r=0}^{2^k-1} \varepsilon_r r^k = (-1)^k 2^{\binom{k}{2}} k!. \tag{485}$$

More generally, for fixed n,

$$c \longmapsto \frac{1}{(n+k)!} \sum_{r<2^k} \varepsilon_r (r+c)^{n+k} \tag{486}$$

has degree at most n, and its leading coefficient is

$$[c^n] \left(\frac{1}{(n+k)!} \sum_{r<2^k} \varepsilon_r (r+c)^{n+k} \right) = \frac{(-1)^k 2^{\binom{k}{2}}}{n!}. \tag{487}$$

A symmetric form of the operator is also useful. With $\operatorname{sinhc}(z) = \sinh(z)/z$,

$$\mathfrak{T}_k = (-1)^k 2^{\binom{k}{2}} D^k e^{(2^k-1)D/2} \prod_{j=0}^{k-1} \operatorname{sinhc}(2^{j-1} D). \tag{488}$$

This shows that the Thue–Morse block is a kth derivative, followed by an even differential correction and evaluation about the midpoint of the block. It also explains why centered and translated versions of the formula are so natural.

39.2 Geometric divided differences

Let $0 < q < 1$ for a moment, and consider the geometric nodes

$$1, q^{-1}, q^{-2}, \ldots, q^{-n}.$$

The leading coefficient of any polynomial P of degree at most n is its nth divided difference on these nodes. Lagrange interpolation gives

$$[t^n]\, P(t) = \frac{(-1)^n q^{\binom{n+1}{2}}}{(q;q)_n} \sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q P(q^{-k}). \tag{489}$$

For completeness, the barycentric denominator is

$$\frac{1}{\prod_{j \neq k} (q^{-k} - q^{-j})} = \frac{(-1)^{n-k} q^{\binom{n+1}{2} + \binom{k}{2}}}{(q;q)_n} \begin{bmatrix} n \\ k \end{bmatrix}_q. \tag{490}$$

At $q = 1/2$, the nodes are $1, 2, 4, \ldots, 2^n$ and the interpolation weights are exactly Gaussian binomial coefficients at base $1/2$, decorated by triangular powers of 2.

The same identity follows immediately from the finite q-binomial theorem

$$\sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k = (z;q)_n\,. \tag{491}$$

If $z = q^{-m}$, the right-hand side is zero for $m < n$, while

$$(q^{-n};q)_n = (-1)^n q^{-\binom{n+1}{2}}\, (q;q)_n\,. \tag{492}$$

Thus the stencil kills every lower monomial and normalizes the top one.

39.3 The two annihilators together

The inner sum (486) first lowers degree by k. Its remaining dependence is a polynomial of degree at most n. The outer geometric stencil (489) then kills all but one prescribed coefficient of that degree-n polynomial. Schematically,

$$\boxed{\text{power of degree } n+k \xrightarrow{\mathfrak{T}_k} \text{polynomial of degree } n \xrightarrow{\mathfrak{Q}_n} \text{one coefficient.}}$$

This is the structural reason that a formula about a smooth distribution function involves both Thue–Morse signs and Gaussian binomial coefficients. Neither ingredient is ornamental: each is the coefficient vector of a finite annihilator.

Translation invariance is now transparent. Translating the argument of the intermediate degree-n polynomial changes only lower coefficients after the target coefficient is fixed. The geometric extractor annihilates precisely those changes. The Lean proof strengthens this observation by treating the translation as an indeterminate and proving that the complete numerator polynomial is constant.

40 The finite q-formula as an Appell identity

40.1 A centered master formula

The exact inverse-dyadic formula is most revealing when one retains the Taylor variable instead of setting it to zero.

Theorem 40.1 (Centered Gaussian-binomial representation of A_n)**.** *For every $n \geq 0$ and $\theta \in \mathbb{C}$,*

$$\boxed{\frac{A_n(\theta)}{n!} = \frac{2^{-\binom{n+1}{2}}}{(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r \big(r - 2^k(1+\theta)\big)^{n+k}.} \tag{493}$$

Proof. The polynomial version of the formalized q-binomial Taylor identity is

$$R_n(y) = \frac{1}{2^{n^2}\,(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r<2^k} \varepsilon_r \big(r - 2^k - 2^{n+k}y\big)^{n+k}.$$

Set $y = 2^{-n}\theta$ and use theorem 37.2. Multiplying by $2^{\binom{n}{2}}$ changes the outer power 2^{-n^2} into $2^{-\binom{n+1}{2}}$, while the inner shift becomes $-2^k(1+\theta)$. □

At $\theta = 0$, multiplying (493) by $2^{-\binom{n}{2}}$ gives the exact centered inverse-dyadic formula

$$\boxed{F(2^{-n}) = \frac{1}{2^{n^2}\,(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k)^{n+k}.} \tag{494}$$

Thus the formula for one special value is merely the constant-term specialization of a formula for the entire Appell polynomial.

40.2 The arbitrary-translation form

The companion development proves an even stronger polynomial identity. A common translation parameter may be inserted into all the raw Thue–Morse blocks without changing the result.

Theorem 40.2 (Raw translated representation)**.** *For every $n \geq 0$ and every $\theta, \eta \in \mathbb{C}$,*

$$\boxed{\frac{A_n(\theta)}{n!} = \frac{(-1)^n 2^{-\binom{n+1}{2}}}{(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r \big(r + \eta + 2^k\theta\big)^{n+k}.} \tag{495}$$

The right-hand side is independent of η.

Putting $\theta = 0$ gives

$$F(2^{-n}) = \frac{1}{(-2)^{n^2}\,(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r + \eta)^{n+k}. \tag{496}$$

The identity $(-1)^n = (-1)^{n^2}$ accounts for the denominator $(-2)^{n^2}$.

Remark 40.3 (A scale-dependent translation)*.* Because (495) is an identity separately for each n, one may choose a different common translation η_n at every outer scale in an infinite expansion. This does not change a single term. In exact arithmetic this is merely a gauge freedom; in floating-point arithmetic it may be used to reduce the size of intermediate powers. The translation must remain common to the inner k-sum for a fixed n.

41 One-shot and digitwise formulae for a general dyadic rational

The formula (494) treats the numerator 1. There is also a one-shot formula for a general numerator. For $0 \le M < 2^N$,

$$\boxed{F\left(\frac{M}{2^N}\right) = \frac{1}{2^{N^2}\,(1/2;1/2)_N}\sum_{k=0}^{N}\frac{\begin{bmatrix}N\\k\end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(N+k)!}\sum_{r=0}^{M2^k-1}\varepsilon_r(r-M2^k)^{N+k}.} \tag{497}$$

A common translation may be inserted in all the inner powers; the complete outer numerator is unchanged. Formula (497) uses the denominator exponent N once and for all. In contrast, (482) uses one lower-order block for each 1-bit of M. The two organizations are related by the canonical dyadic decomposition of the prefix interval.

Write

$$\frac{M}{2^N} = 0.b_1\dots b_N{}_2, \qquad a_m = \lfloor 2^m M/2^N\rfloor.$$

For every function P on the nonnegative integers and every $k \ge 0$,

$$\sum_{r=0}^{M2^k-1}\varepsilon_r P(r) = \sum_{\substack{1\le m\le N\\ b_m=1}}\varepsilon_{a_{m-1}}\sum_{s=0}^{2^{N-m+k}-1}\varepsilon_s P\Big(a_{m-1}2^{N-m+k+1}+s\Big). \tag{498}$$

Indeed, the interval $[0,M)$ is the disjoint union, over its 1-bits, of

$$\Big[a_{m-1}2^{N-m+1}, a_{m-1}2^{N-m+1}+2^{N-m}\Big), \qquad b_m = 1.$$

After multiplication by 2^k, every block starts at a multiple of twice its length. Hence adding the local coordinate s creates no binary carry and

$$\varepsilon_{a_{m-1}2^{N-m+k+1}+s} = \varepsilon_{a_{m-1}}\varepsilon_s.$$

This proves (498).

Substituting (498) into the one-shot formula splits every long Thue–Morse prefix into complete blocks carrying exactly the signs $\varepsilon_{a_{m-1}}$ that occur in the digit formula. The remaining outer Gaussian stencil is still of order N; applying the polynomial Taylor identity to each rescaled block lowers it to its natural bit scale m and produces (482). Thus:

- (497) groups the computation by polynomial degree $N+k$;
- (482) groups it by the occupied binary scales of the numerator;
- (498) is the combinatorial bridge between the two groupings.

For $M = 5$ and $N = 4$, the one-shot expression has one outer sum of order 4 and inner prefixes of lengths $5 \cdot 2^k$. The digitwise expression has only two active blocks, at scales $m = 2$ and $m = 4$, as shown in example 38.2. Their equality is a nontrivial finite rearrangement, not a limiting assertion.

42 From exact dyadic values to the arbitrary-argument q-series

The connection can now be made without any further analysis: substitute theorem 40.1 or theorem 40.2 into the digit–Appell series term by term.

42.1 The centered nested series

Theorem 42.1 (Centered digit–q expansion)**.** *For $x \in [0,1)$, with notation (475)–(476),*

$$F(x) = \sum_{m=1}^{\infty} b_m \varepsilon_{a_{m-1}} \frac{1}{2^{m^2}\,(1/2;1/2)_m} \sum_{k=0}^{m} \frac{\left[{m \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(m+k)!} \times \sum_{r=0}^{2^k-1} \varepsilon_r \big(r - 2^k(1+\theta_m)\big)^{m+k}. \tag{499}$$

The equality holds term by term with (480).

Proof. In the mth term of (480), multiply the prefactor $2^{-\binom{m}{2}}$ by the prefactor $2^{-\binom{m+1}{2}}$ in (493). Their product is 2^{-m^2}. Everything else is a direct substitution. □

If $x = M/2^N$ has a terminating binary expansion, then (499) terminates after $m = N$. The "infinite" global q-series is therefore exactly a finite dyadic formula whenever its argument is dyadic. Conversely, allowing infinitely many nonzero binary digits turns the same finite block identity into the arbitrary-argument series.

42.2 The raw translated nested series

For any fixed $\eta \in \mathbb{C}$, the raw form is

$$F(x) = \sum_{m=1}^{\infty} b_m \varepsilon_{a_{m-1}} \frac{1}{(-2)^{m^2}\,(1/2;1/2)_m} \sum_{k=0}^{m} \frac{\left[{m \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(m+k)!} \times \sum_{r=0}^{2^k-1} \varepsilon_r \big(r + \eta + 2^k\theta_m\big)^{m+k}. \tag{500}$$

Again, every individual mth summand equals the corresponding Appell summand in (480); no interchange of infinite sums is needed to prove the identity.

42.3 A useful commutative diagram

The relations may be summarized as follows:

$$\begin{tikzcd}
R_m(2^{-m}\theta) \arrow[rr, leftrightarrow, "\text{Appell compression}"] \arrow[d, leftrightarrow, "\text{finite Gaussian/Thue–Morse identity}"] & & 2^{-\binom{m}{2}} A_m(\theta)/m! \arrow[d, "\text{insert binary tail } \theta=\theta_m"] \\
\text{finite nested } q\text{-block} \arrow[rr, "\text{sum over the 1-bits}"] & & F(x).
\end{tikzcd}$$

Setting $\theta = 0$ in the top row gives $F(2^{-m})$. Taking a terminating binary expansion makes the right-hand summation finite. Taking an arbitrary binary expansion leaves an absolutely convergent infinite series. These are specializations, not analogies.

43 Alternative exact representations of inverse-dyadic values

The q-binomial formula is only one exact representation of $F(2^{-n})$. The moment product (462) leads to several others, each exposing a different combinatorial structure.

43.1 Coefficient extraction from the moment product

Let

$$a_n = \frac{d_n}{n!} = 2^{\binom{n}{2}} F(2^{-n}). \tag{501}$$

Then

$$a_n = [z^n] \prod_{j=1}^{\infty} \left(\sum_{r=0}^{\infty} \frac{z^r}{2^{jr}(r+1)!} \right). \tag{502}$$

Expanding by the set of scales at which a nonconstant term is chosen gives

$$a_n = \sum_{\ell=1}^{n} \sum_{\substack{r_1+\cdots+r_\ell=n \\ r_i \geq 1}} \frac{1}{\prod_{i=1}^{\ell}(r_i+1)!} \sum_{1 \leq j_1 < \cdots < j_\ell} 2^{-\sum_{i=1}^{\ell} j_i r_i}, \qquad (n \geq 1). \tag{503}$$

The infinite inner sum is an elementary geometric simplex sum. If $t_h = r_h + \cdots + r_\ell$, then

$$\sum_{1 \leq j_1 < \cdots < j_\ell} 2^{-\sum_i j_i r_i} = \prod_{h=1}^{\ell} \frac{1}{2^{t_h} - 1}. \tag{504}$$

To see this, put $j_i = i + g_1 + \cdots + g_i$ with $g_h \geq 0$; the resulting independent geometric series contribute $(1 - 2^{-t_h})^{-1}$, and their powers of 2 cancel the initial offset.

Reversing the composition $(r_1, \ldots, r_\ell)$ changes tail sums into prefix sums. We obtain the finite composition formula

Theorem 43.1 (Composition representation)**.** *Let $\rho = (r_1, \ldots, r_\ell) \vDash n$ run over all ordered compositions of n, and let $s_j = r_1 + \cdots + r_j$. Then*

$$\boxed{F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \vDash n} \prod_{j=1}^{\ell} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}.} \tag{505}$$

This derivation explains the factors $2^{s_j} - 1$: they are the closed forms of geometric sums over the infinitely many binary scales in (462).

43.2 A lower-Hessenberg determinant

The recurrence (464) may be written

$$a_n = \sum_{k=0}^{n-1} \beta_{n,k} a_k, \qquad \beta_{n,k} = \frac{1}{(2^n - 1)(n - k + 1)!}. \tag{506}$$

Define the $n \times n$ lower-Hessenberg matrix

$$H_n = \begin{pmatrix} \beta_{1,0} & -1 & 0 & \cdots & 0 \\ \beta_{2,0} & \beta_{2,1} & -1 & \ddots & \vdots \\ \beta_{3,0} & \beta_{3,1} & \beta_{3,2} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ \beta_{n,0} & \beta_{n,1} & \beta_{n,2} & \cdots & \beta_{n,n-1} \end{pmatrix}. \tag{507}$$

Proposition 43.2 (Determinant representation)**.** *For $n \geq 1$,*

$$\boxed{F(2^{-n}) = 2^{-\binom{n}{2}} \det H_n.} \tag{508}$$

Proof. Let $D_0 = 1$ and $D_n = \det H_n$. Expanding the determinant along its last row, with the -1 superdiagonal supplying the compensating cofactor signs, gives

$$D_n = \sum_{k=0}^{n-1} \beta_{n,k} D_k.$$

This is exactly (506), with the same initial value; hence $D_n = a_n$. □

Expanding the determinant by paths through the nonzero Hessenberg band recovers the composition sum (505). Thus the composition and determinant formulae are two standard expansions of the same triangular resolvent.

43.3 Cumulants, Bernoulli numbers, and Bell polynomials

Taking the logarithm of (462) gives another exact description. The classical expansion

$$\log \frac{e^z - 1}{z} = \frac{z}{2} + \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!} z^{2m} \tag{509}$$

uses the Bernoulli numbers B_{2m}. Summing over the dyadic scales yields

$$\log G(z) = \frac{z}{2} + \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2^{2m} - 1)} \frac{z^{2m}}{(2m)!}. \tag{510}$$

Therefore the cumulants of X are

$$\kappa_1 = \frac{1}{2}, \qquad \kappa_{2m} = \frac{B_{2m}}{2m(2^{2m} - 1)}, \qquad \kappa_{2m+1} = 0 \quad (m \geq 1). \tag{511}$$

The vanishing odd cumulants beyond the first are another expression of the symmetry $X \stackrel{d}{=} 1 - X$.

Let $\mathcal{B}_n$ denote the complete exponential Bell polynomial, normalized by

$$\exp\left(\sum_{j=1}^{\infty} x_j \frac{z^j}{j!}\right) = \sum_{n=0}^{\infty} \mathcal{B}_n(x_1, \ldots, x_n) \frac{z^n}{n!}. \tag{512}$$

Then $d_n = \mathcal{B}_n(\kappa_1, \ldots, \kappa_n)$, and hence

Theorem 43.3 (Bernoulli–Bell representation)**.**

$$\boxed{F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} \mathcal{B}_n(\kappa_1, \ldots, \kappa_n),} \tag{513}$$

where the κ_j *are given by* (511)*.*

Equivalently, expanding the complete Bell polynomial gives the finite partition sum

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\substack{\nu_1, \ldots, \nu_n \geq 0 \\ \nu_1 + 2\nu_2 + \cdots + n\nu_n = n}} \prod_{j=1}^{n} \frac{1}{\nu_j!} \left(\frac{\kappa_j}{j!}\right)^{\nu_j}. \tag{514}$$

Only κ_1 and the even cumulants are nonzero, so the apparent n-variable sum is substantially sparse.

For example,

$$d_2 = \kappa_1^2 + \kappa_2 = \frac{5}{18}, \qquad d_4 = \kappa_1^4 + 6\kappa_1^2\kappa_2 + 3\kappa_2^2 + \kappa_4 = \frac{143}{1350}.$$

Consequently,

$$F\left(\frac{1}{4}\right) = \frac{5}{72}, \qquad F\left(\frac{1}{16}\right) = \frac{143}{2073600}.$$

The Bell-polynomial formula replaces Thue–Morse cancellation by cumulant addition: the dyadic scales contribute independently to $\log G$.

44 The discrete limit as a q-Toeplitz transform

The discrete limit uses the same finite Gaussian coefficients as the exact formula, but its inner sums are truncated at an x-dependent point rather than taken over a complete Thue–Morse block. Exact coefficient extraction is therefore replaced by convergence of splines and regular summability.

44.1 Shifted finite splines

For $p \geq 0$, $\eta \in \mathbb{C}$, and real x, define

$$S_{p,\eta}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{0 \leq r < \lfloor 2^p x + 1/2 \rfloor} \varepsilon_r (r - 2^p x + \eta)^p. \tag{515}$$

An empty sum is zero. The centered choice is $\eta = 1/2$. The formal development proves the uniform estimate

$$|S_{p,1/2}(x) - \mathcal{F}(x)| \leq 2^{-p} \qquad (x \in \mathbb{R}), \tag{516}$$

and, for fixed complex shift and $p \geq 1$,

$$|S_{p,\eta}(x) - S_{p,1/2}(x)| \leq 2^{1-p} e^{|\eta - 1/2|} \qquad (x \geq 0). \tag{517}$$

On the negative half-line the relevant sums and the global extension vanish, so the convergence statement extends without difficulty.

44.2 The discrete q-sum

Define

$$\begin{aligned} D_n(\eta, x) = {} & \frac{1}{2^{n^2} (1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ & \times \sum_{0 \leq r < \lfloor 2^{n+k} x + 1/2 \rfloor} \varepsilon_r (r - 2^{n+k} x + \eta)^{n+k}. \end{aligned} \tag{518}$$

This looks like a direct arbitrary-x analogue of (494). The decisive simplification is an exact reindexing.

Theorem 44.1 (Exact Toeplitz decomposition)**.** *Put*

$$w_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}}{(1/2;1/2)_n} \qquad (0 \leq j \leq n). \tag{519}$$

Then

$$\boxed{D_n(\eta, x) = \sum_{j=0}^{n} w_{n,j} S_{2n-j,\eta}(x).} \tag{520}$$

Proof. Set $j = n - k$ and $p = n + k = 2n - j$ in (518). Replacing the inner sum by the definition of $S_{p,\eta}$ leaves the power of 2

$$\binom{2n-j}{2} - n^2 - 2\binom{n-j}{2} = -\binom{j+1}{2},$$

and the sign $(-1)^{2n-j} = (-1)^j$. Symmetry of the Gaussian binomial coefficient gives $\begin{bmatrix} n \\ n-j \end{bmatrix}_{1/2} = \begin{bmatrix} n \\ j \end{bmatrix}_{1/2}$. □

The indices $2n - j$ lie between n and $2n$. Thus every spline entering the nth row is already at resolution at least n.

44.3 One polynomial explains the entire row

Define the row-generating polynomial

$$W_n(z) = \sum_{j=0}^{n} w_{n,j} z^j. \tag{521}$$

The finite q-binomial theorem gives the compact product

Theorem 44.2 (Toeplitz row polynomial)**.**

$$\boxed{W_n(z) = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}.} \tag{522}$$

Three important facts are immediate.

First, the row mass is one:

$$\sum_{j=0}^{n} w_{n,j} = W_n(1) = 1. \tag{523}$$

Second, the signs alternate, so the exact total variation is

$$V_n := \sum_{j=0}^{n} |w_{n,j}| = W_n(-1) = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n}. \tag{524}$$

The sequence V_n is bounded; the formal proof uses the convenient uniform bound $V_n \le 16$.

Third, differentiating at $z = 1$ gives the first signed moment of the row:

$$\sum_{j=0}^{n} j\, w_{n,j} = W_n'(1) = -\sum_{\ell=1}^{n} \frac{1}{2^\ell - 1}. \tag{525}$$

Higher factorial moments follow by further logarithmic differentiation of (522). Thus the entire Toeplitz stencil is encoded by one finite q-Pochhammer ratio.

The weights also admit the symmetric form

$$w_{n,j} = \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_j\, (1/2; 1/2)_{n-j}}. \tag{526}$$

For fixed j they converge to

$$w_{\infty,j} = \frac{(-1)^j 2^{-\binom{j+1}{2}}}{(1/2; 1/2)_j\, (1/2; 1/2)_\infty}, \tag{527}$$

whose generating function is

$$W_\infty(z) = \frac{(z/2; 1/2)_\infty}{(1/2; 1/2)_\infty}. \tag{528}$$

This limiting object is a stationary signed q-filter of unit mass and finite variation.

44.4 A quantitative discrete-limit estimate

The exact Toeplitz decomposition makes an explicit error bound immediate.

Theorem 44.3 (Uniform rate for fixed shift)**.** *For $n \geq 1$, $x \geq 0$, and $\eta \in \mathbb{C}$,*

$$\boxed{|D_n(\eta, x) - \mathcal{F}(x)| \leq V_n \left(2^{-n} + 2^{1-n} e^{|\eta - 1/2|}\right),} \tag{529}$$

where V_n is given by (524). In particular,

$$|D_n(\eta, x) - \mathcal{F}(x)| \leq 16 \left(2^{-n} + 2^{1-n} e^{|\eta - 1/2|}\right). \tag{530}$$

Proof. By (523) and theorem 44.1,

$$D_n(\eta, x) - \mathcal{F}(x) = \sum_{j=0}^{n} w_{n,j} (S_{2n-j,\eta}(x) - \mathcal{F}(x)).$$

For $p = 2n - j \geq n$, combine (516) and (517):

$$|S_{p,\eta}(x) - \mathcal{F}(x)| \leq 2^{-p} + 2^{1-p} e^{|\eta - 1/2|} \leq 2^{-n} + 2^{1-n} e^{|\eta - 1/2|}.$$

Sum against $|w_{n,j}|$ and use (524). □

For the centered shift one should not insert $\eta = 1/2$ into the deliberately coarse shift estimate. The shift difference is then exactly zero, and the sharper bound is

$$|D_n(1/2, x) - \mathcal{F}(x)| \leq V_n 2^{-n} \leq 16\, 2^{-n}. \tag{531}$$

This argument shows precisely why the translation parameter disappears in the limit. It is not cancelled algebraically as in the finite exact formula; rather, every fixed shift perturbs a degree-p spline by $O(2^{-p})$, and the Toeplitz rows have uniformly bounded total variation.

44.5 Exact formula versus discrete limit

The two constructions may now be compared without ambiguity.

	Exact dyadic/Appell identity	Discrete limit
Inner Thue–Morse sum	Complete block $0 \leq r < 2^k$; exact polynomial annihilation	Prefix $0 \leq r < \lfloor 2^{n+k} x + 1/2 \rfloor$; spline approximation
Outer Gaussian stencil	Extracts one coefficient exactly	Averages spline degrees $n, \ldots, 2n$ with unit mass
Translation parameter	Cancels identically in the full numerator polynomial	Produces an $O(2^{-p})$ perturbation that vanishes in the limit
Passage to arbitrary x	Sum exact blocks over the binary 1-digits of x	Let spline degree tend to infinity through a regular Toeplitz transform

The formulae share the same algebraic stencil, but the reason they are valid is different. Conflating coefficient extraction with Toeplitz convergence hides the most interesting part of the story.

45 Integral-transform representations

The previous formulae read the entire function G through its Taylor coefficients. Fourier and Laplace inversion read the same product globally. They provide useful alternative representations at arbitrary, not necessarily dyadic, arguments.

45.1 Fourier–sinc inversion

Let $Y = 2X - 1$. Its density is Rvachev's up-function on $[-1, 1]$. With the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i \xi t} \, \mathrm{d}t,$$

independence gives

$$\widehat{\mathrm{up}}(\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi\xi}{2^j}\right), \qquad \operatorname{sinc} u = \begin{cases} \sin u/u, & u \neq 0, \\ 1, & u = 0. \end{cases} \tag{532}$$

Since $F(x) = \mathbb{P}(Y \leq 2x - 1)$ and up is even, Fourier inversion followed by one integration gives, for $0 < x < 1$,

$$\boxed{F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \frac{\sin\bigl(2\pi\xi(2x-1)\bigr)}{\xi} \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi\xi}{2^j}\right) \mathrm{d}\xi.} \tag{533}$$

The rapid decay of the infinite sinc product makes this a genuine convergent integral, not merely a distributional formula.

The dyadic exact formulae are invisible term by term in (533); their rationality results from highly nonlocal cancellation in the integral. This contrast is useful: the Fourier form is well adapted to smoothness and global shape, while the Appell and q-forms are adapted to arithmetic at dyadic points.

45.2 Bromwich inversion of the moment product

The Laplace transform of X is

$$L_X(s) = \mathbb{E}e^{-sX} = G(-s) = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}. \tag{534}$$

For $\sigma > 0$ and $0 < x < 1$, inversion of the transform of the distribution function gives

$$\boxed{F(x) = \frac{1}{2\pi i} \lim_{T\to\infty} \int_{\sigma - iT}^{\sigma + iT} \frac{e^{sx}}{s} \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j} \, \mathrm{d}s.} \tag{535}$$

The same product therefore supports three complementary readings:

- coefficient extraction gives moments and exact dyadic values;
- logarithmic expansion gives cumulants and Bell-polynomial formulae;
- contour inversion gives $F(x)$ directly at an arbitrary argument.

46 Computational consequences

46.1 Exact dyadic evaluation

For exact rational computation at $x = M/2^N$, the shortest route is usually:

1. compute $d_0, \ldots, d_N$ from the triangular recurrence (464);
2. form $A_m(\theta)$ by Horner evaluation or by the binomial formula (467);
3. sum only over the 1-bits in (482).

This uses $O(N^2)$ rational arithmetic operations in a straightforward implementation; it is not a bit-complexity bound. The method avoids the exponentially long inner Thue–Morse sums that appear in the closed q-formula. The latter remain invaluable as structural identities and formal certificates, but are not generally the numerically cheapest way to obtain a rational value.

For inverse powers $x = 2^{-N}$, only one bit is active, so (482) collapses at once to (465). The recurrence, composition sum, determinant, Bell formula, and Gaussian-binomial formula then provide five exactly equivalent ways to compute the same rational number.

46.2 Arbitrary arguments

For a non-dyadic x, the digit–Appell series (480) is attractive because:

- zero bits cost nothing;
- each active term needs one polynomial value $A_m(\theta_m)$;
- the uniform remainder bound (481) is superexponentially small;
- no cancellation inside a 2^k-term Thue–Morse block is required.

The centered nested q-series (499) is mathematically equivalent term by term, but may be badly conditioned in ordinary floating point because large powers cancel. The raw translation freedom (500) can improve conditioning, although exact rational or ball arithmetic is preferable when evaluating the nested sums directly.

The discrete limit (518) supplies an independent numerical route and a global approximation to $\mathcal{F}$. Its error is controlled by (529); the centered shift $\eta = 1/2$ removes the shift-error term entirely, giving the sharper estimate (531), and is normally the best-conditioned canonical choice.

47 Synthesis

The apparent diversity of the formulae is reduced by separating three layers.

The moment layer. The random series (459) gives the entire moment product (462). Its coefficients are d_n, its logarithm gives the cumulants (511), and its dilation gives the recurrence (464).

The finite polynomial layer. The reduction block is the rescaled Appell polynomial

$$R_n(2^{-n}\theta) = 2^{-\binom{n}{2}} A_n(\theta)/n!.$$

The finite Thue–Morse operator removes k degrees, and the geometric Gaussian-binomial stencil removes the remaining lower-degree terms. This produces the centered and translated finite q-formulae for the same Appell polynomial.

The passage to arbitrary arguments. There are two distinct routes.

1. Follow the binary digits of x and sum exact Appell blocks. This yields (480); substitution of the finite q-identity yields (499) and (500). Dyadic rationals are exactly the terminating cases.

2. Truncate Thue–Morse power sums at the spatial point x, interpret them as splines, and apply a bounded-variation Toeplitz row. The row polynomial (522) makes the q-Pochhammer structure explicit, while (529) proves convergence quantitatively.

The central identities can be compressed into the following chain:

$$\boxed{\begin{aligned} F(2^{-n}) &= 2^{-\binom{n}{2}}\frac{A_n(0)}{n!}, \\ R_n(2^{-n}\theta) &= 2^{-\binom{n}{2}}\frac{A_n(\theta)}{n!}, \\ F(x) &= \sum_{m\geq 1} b_m\varepsilon_{a_{m-1}}2^{-\binom{m}{2}}\frac{A_m(\theta_m)}{m!}, \\ \frac{A_m(\theta)}{m!} &= \text{one finite Gaussian-binomial/Thue–Morse block}, \\ D_n(\eta,x) &= \sum_{j=0}^{n} w_{n,j}S_{2n-j,\eta}(x), \qquad \sum_{j=0}^{n} w_{n,j}z^j = \frac{(z/2;1/2)_n}{(1/2;1/2)_n}. \end{aligned}} \tag{536}$$

The first four lines are exact finite algebra assembled along the binary expansion; the last line is regular summability of convergent spline approximants. Together they explain both the arithmetic exactness at dyadic rationals and the analytic convergence at arbitrary arguments.

Supplementary material from this dossier

48 A compact proof of the geometric simplex sum

For positive integers $r_1, \dots, r_\ell$, put $t_h = r_h + \cdots + r_\ell$. Every increasing sequence $1 \le j_1 < \cdots < j_\ell$ has a unique gap representation

$$j_i = i + g_1 + \cdots + g_i, \qquad g_h \in \mathbb{N} \cup \{0\}.$$

Therefore

$$\begin{aligned}\sum_{j_1<\cdots<j_\ell} 2^{-\sum_i j_i r_i} &= 2^{-\sum_i i r_i} \prod_{h=1}^{\ell} \sum_{g_h=0}^{\infty} 2^{-g_h t_h} \\ &= 2^{-\sum_i i r_i} \prod_{h=1}^{\ell} \frac{1}{1-2^{-t_h}}.\end{aligned}$$

But $\sum_h t_h = \sum_i i r_i$, so

$$2^{-\sum_i i r_i} \prod_h \frac{2^{t_h}}{2^{t_h}-1} = \prod_h \frac{1}{2^{t_h}-1},$$

which is (504).

49 First moments and inverse-dyadic values

The recurrence (464) gives:

n	$d_n = \mathbb{E}X^n$	$F(2^{-n}) = 2^{-\binom{n}{2}} d_n/n!$
0	1	1
1	1/2	1/2
2	5/18	5/72
3	1/6	1/288
4	143/1350	143/2073600

The $n = 0$ row uses the standard convention $2^0 = 1$ and $0! = 1$.

50 Map to the formal development

The principal formal ingredients used here are organized in the following source files in the `Analysis/FabiusFunction` directory of the `ProveIt` repository:

File	Mathematical role
`HalfQBinomial.lean`	finite q-Pochhammer identities, Gaussian binomial coefficients at $q = 1/2$, and the finite q-binomial annihilator
`FabiusQBinomialFormula.lean`	centered exact formulae for inverse and general dyadic arguments
`FabiusRawQBinomialFormula.lean`	raw-coordinate formula with arbitrary common translation
`FabiusQBinomialTaylor.lean`	polynomial identity equating the q-binomial block with the Taylor reduction polynomial
`TaylorReduction.lean`	exact reduction polynomial R_n and dyadic-shell Taylor identities
`FabiusBinaryReductionSeries.lean`	finite telescoping identity, residual estimate, and global binary series
`FabiusGlobalQBinomialSeries.lean`	termwise substitution of the finite q-block into the binary series
`FabiusUniformSpline.lean`	centered finite splines and uniform convergence
`FabiusComplexShiftSpline.lean`	complex translation of the spline and quantitative shift stability
`FabiusDiscreteLimitToeplitz.lean`	Toeplitz weights, row mass, and total-variation bound
`FabiusDiscreteLimitIntegration.lean`	exact reindexing of the discrete formula and convergence to the global Fabius extension

Dossier references

[1] V. Reshetnikov, *The Fabius Function and Rvachev's up-Function*, companion article in the `ProveIt` repository, 2026.

[2] V. Reshetnikov, *ProveIt: formal developments for the Fabius function*, https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction.

[3] G. Gasper and M. Rahman, *Basic Hypergeometric Series*, 2nd ed., Cambridge University Press, 2004.

[4] L. Comtet, *Advanced Combinatorics*, D. Reidel, 1974.

[5] G. H. Hardy, *Divergent Series*, Oxford University Press, 1949.

Part V
The Dyadic Web of the Fabius Function

Source provenance and status. Former path: `Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex`. Preferred dyadic backbone: alternative exact representations, q-Richardson filtering, quantitative limits, and exact stabilization. All claims lacking an exact audited Lean declaration remain frontier obligations.

Original source SHA-256: `462276b10fcd32b0446deb7cfedc4ec07c2ae55dbd333d5ff9b1d98f07df89e1`.

Dossier abstract

Exact values of the Fabius function at dyadic rationals appear in several forms that look unrelated: a Thue–Morse sum corrected by the even moments of Rvachev's up-function, a finite half-base q-binomial expression with a seemingly arbitrary complex translation, a bit-recursive Taylor reduction, an absolutely convergent binary series at arbitrary argument, and a finite q-weighted discrete approximant whose limit is the signed global Fabius function. This companion article identifies the common algebra behind these formulae and makes their connections explicit.

The central object is the coefficient

$$[X^n]\big(B_m(X)\mathscr{A}(X)\big), \qquad B_m(X) = \sum_{h<m} (-1)^{\mathrm{wt}_2(h)} \mathrm{e}^{(m-1-h)X},$$

where $\mathscr{A}$ is the exponential moment series of the Fabius probability law. Direct expansion gives the classical moment formula; probabilistic interpretation gives an expectation formula; Cauchy's formula gives a contour representation; and geometric divided differences through $-1, -2, \ldots, -2^n$ give the q-binomial formula. The same coefficient also produces new finite identities: an exact moment-corrected expansion in centered splines, its reciprocal deconvolution, and a residue formula for the Gaussian interpolation row.

At arbitrary arguments the two infinite constructions have different architectures. The binary series is a genuine telescoping sum whose summands are individually independent of the complex shift. The discrete formula is instead a Toeplitz transform of shifted splines; its finite rows generally depend on the shift, although their limit does not. Writing the row generating polynomial as

$$W_n(z) = \frac{(z/2;1/2)_n}{(1/2;1/2)_n}$$

reveals it as a q-Richardson filter: it preserves constants and annihilates the geometric error modes $2^{-p}, 2^{-2p}, \ldots, 2^{-np}$. This yields an explicit uniform $\mathcal{O}(4^{-n})$ error bound. Most importantly, at a dyadic input $m/2^s$ the discrete approximant stabilizes exactly for every row $n \ge s$; thus the "limit formula" literally becomes the exact dyadic q-formula after the denominator has been resolved. A synchronized nearest-dyadic version provides a second, termwise translation-invariant discrete limit.

51 Scope, conventions, and the formula map

This article is a companion rather than a replacement for the comprehensive exposition [12]. It assumes the construction and basic analytic theory developed there and concentrates on one cluster of results: exact dyadic evaluation, binary reduction, finite half-base q-calculus, and discrete limits. The proofs formalized in Lean under `Analysis/FabiusFunction`[11] are the source of the exact identities used below. Several consequences in this article are then derived directly from those identities; a formalization map is given in section 59.

We use three related functions, with the same convention as the companion article:

- F is the bounded Fabius distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported Rvachev density on $[-1, 1]$;
- $\mathcal{F}$ is the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on $\mathbb{R}$.

They agree on $[0, 1]$. Put

$$\varepsilon_r = (-1)^{\mathrm{wt}_2(r)}, \qquad c_k = \int_{-1}^{1} u^{2k}\,\mathrm{up}(u)\,\mathrm{d}u,$$

and let d_n denote the half moments normalized by

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{d_n}{n!}. \tag{537}$$

The normalized moment series is

$$\mathscr{A}(X) = \sum_{n \geq 0} d_n \frac{X^n}{n!}. \tag{538}$$

We reserve q for the fixed base $q = 1/2$ and use $Q \in \mathbb{C}$ for a freely chosen translation. This distinction is essential: in the expressions $(a; q)_n$ and $(r - 2^k + Q)^{n+k}$ the two symbols play completely different roles.

51.1 Five presentations of the same value

The identities studied below fall into five families. Their superficial differences are summarized in table 8; the rest of the article explains the arrows between rows.

Presentation	Range and status	Principal coefficients	Mechanism
Moment/Thue–Morse formula	finite and exact at $m/2^n$	c_k and ε_h	coefficient expansion in the centered probability law
Half-base q-formula	finite and exact at $m/2^n$	$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ and $(1/2; 1/2)_n$	geometric divided difference after block cancellation
Binary Taylor series	infinite and absolutely convergent for $x \geq 0$; finite at dyadics	binary prefixes and d_j	exact residual telescope along the binary expansion
Global q-binary series	same outer series as the preceding row	a finite q-identity inside each scale	termwise replacement of a Taylor polynomial
Discrete q-limit	one finite row for each n, convergent for all $x \in \mathbb{R}$	normalized Gaussian row $\omega_{n,j}$	Toeplitz transform of shifted finite splines

Table 8: The formula families. The two infinite constructions are not partial-sum versions of one another.

The main structural conclusions can be stated before any calculation.

The common core. The exact dyadic formulae are all coefficient extractors applied to the same entire function $B_m\mathscr{A}$. The binary series repeatedly reuses its Taylor coefficients at successive binary scales. The discrete formula applies a normalized q-Pochhammer polynomial to the sequence of finite-spline degrees. At a dyadic input these two arbitrary-argument mechanisms both become finite: the binary series terminates, while the discrete rows stabilize exactly.

51.2 The exact dyadic formula taken as input

For $0 \le m \le 2^n$ the moment formula proved in the formal development is

$$\boxed{F\left(\frac{m}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2m-2h-1)^{n-2k}.} \tag{539}$$

The same finite expression, interpreted with $\mathcal{F}$ on the left, is valid for every $m, n \in \mathbb{N}$. The q-binomial version is

$$\begin{aligned} \mathcal{F}(m/2^n) = {} & \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ & \times \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + Q)^{n+k}, \end{aligned} \tag{540}$$

valid for every $Q \in \mathbb{C}$ and independent of Q as a complete finite expression. A central goal below is to explain why (539) and (540) are two readings of one coefficient.

52 The master coefficient and its probabilistic meaning

52.1 The probability law behind the moment series

Let $U_1, U_2, \ldots$ be independent random variables, each uniformly distributed on $[0,1]$, and set

$$Y = \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \qquad Z = 2Y - 1. \tag{541}$$

Then $Y \in [0,1]$, its distribution function is F, and Z has density up on $[-1,1]$. Consequently

$$d_n = \mathbb{E}(Y^n), \qquad c_k = \mathbb{E}(Z^{2k}), \qquad \mathbb{E}(Z^{2k+1}) = 0. \tag{542}$$

The moment generating function of one uniform variable is

$$E_0(X) = \frac{\mathrm{e}^X - 1}{X} = \int_0^1 \mathrm{e}^{Xu}\,\mathrm{d}u,$$

with the removable value $E_0(0) = 1$. Independence gives the locally uniformly convergent entire product

$$\mathscr{A}(X) = \mathbb{E}(\mathrm{e}^{XY}) = \prod_{j=1}^{\infty} E_0(X/2^j). \tag{543}$$

Separating the first factor, or merely replacing X by $2X$ in the product, gives the refinement equation

$$\mathscr{A}(2X) = E_0(X)\mathscr{A}(X). \tag{544}$$

The centered moment series

$$\mathscr{C}(T) = \mathbb{E}(\mathrm{e}^{TZ}) = \sum_{k\geq 0} c_k \frac{T^{2k}}{(2k)!} \tag{545}$$

satisfies the exact change of coordinates

$$\boxed{\mathscr{A}(X) = \mathrm{e}^{X/2}\mathscr{C}(X/2).} \tag{546}$$

The familiar hyperbolic-sine form follows by writing $E_0(t) = \mathrm{e}^{t/2}\sinh(t/2)/(t/2)$:

$$\mathscr{A}(X) = \mathrm{e}^{X/2}\prod_{j=1}^{\infty}\frac{\sinh(X/2^{j+1})}{X/2^{j+1}}, \tag{547}$$

$$\mathscr{C}(T) = \prod_{j=1}^{\infty}\frac{\sinh(T/2^{j})}{T/2^{j}}. \tag{548}$$

Thus the moment series used in exact arithmetic is the real-exponential counterpart of the sinc product in Fourier analysis.

52.2 The Thue–Morse prefix series

For $m \in \mathbb{N}$ define

$$B_m(X) = \sum_{h=0}^{m-1} \varepsilon_h \mathrm{e}^{(m-1-h)X}, \qquad B_0(X) = 0. \tag{549}$$

This finite exponential polynomial records the numerator m and nothing else. Its two factors have distinct roles:

- B_m carries the finite Thue–Morse prefix and therefore the binary numerator;
- $\mathscr{A}$ carries the universal Fabius moments and therefore the analytic law.

Their product is the promised master object.

Theorem 52.1 (Master coefficient identity)**.** *For every $m, n \in \mathbb{N}$,*

$$\boxed{\mathcal{F}\left(\frac{m}{2^n}\right) = 2^{-\binom{n}{2}}[X^n]\big(B_m(X)\mathscr{A}(X)\big).} \tag{550}$$

If $0 \leq m \leq 2^n$, the left side is the bounded value $F(m/2^n)$.

Proof. Put $t_h = 2m - 2h - 1$. By (545),

$$\sum_{k=0}^{\lfloor n/2\rfloor}\binom{n}{2k} c_k t_h^{n-2k} = n![W^n]\big(\mathrm{e}^{t_h W}\mathscr{C}(W)\big).$$

Using (546) at $X = 2W$ gives $\mathscr{C}(W) = \mathrm{e}^{-W}\mathscr{A}(2W)$. Since $t_h - 1 = 2(m-1-h)$,

$$\begin{aligned} n![W^n]\big(\mathrm{e}^{t_h W}\mathscr{C}(W)\big) &= n![W^n]\big(\mathrm{e}^{2(m-1-h)W}\mathscr{A}(2W)\big) \\ &= n!2^n[X^n]\big(\mathrm{e}^{(m-1-h)X}\mathscr{A}(X)\big). \end{aligned}$$

Multiply by ε_h, sum over $h < m$, and insert the result in (539). The power of two becomes $2^{n-\binom{n+1}{2}} = 2^{-\binom{n}{2}}$. ∎

The theorem compresses the double sum in (539) into one coefficient, but it also expands naturally in several different directions.

Corollary 52.2 (Expectation form). *For every $m, n \in \mathbb{N}$,*

$$\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \, \mathbb{E}\big(m-1-h+Y\big)^n \tag{551}$$
$$= \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \, \mathbb{E}\big(2m-2h-1+Z\big)^n. \tag{552}$$

Proof. Because $\mathscr{A}(X) = \mathbb{E}(\mathrm{e}^{XY})$,

$$[X^n]\big(\mathrm{e}^{aX}\mathscr{A}(X)\big) = \frac{1}{n!}\mathbb{E}(a+Y)^n.$$

Apply this termwise to B_m. The second form follows from $m-1-h+Y = (2m-2h-1+Z)/2$. ☐

Expanding the centered expectation in (552), using symmetry to discard the odd moments, recovers (539) immediately. In this sense the Thue–Morse/moment formula is simply the binomial theorem applied inside one probability expectation.

Corollary 52.3 (Cauchy coefficient representation). *For any $\rho > 0$,*

$$\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{2\pi \mathrm{i}} \int_{|X|=\rho} \frac{B_m(X)\mathscr{A}(X)}{X^{n+1}} \, \mathrm{d}X. \tag{553}$$

This contour formula is elementary, but useful conceptually: the q-binomial formula will turn out to be a second residue computation, now in an auxiliary geometric node variable.

52.3 The numerator and the law separate completely

Expanding (550) without centering gives another exact finite form:

$$\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \sum_{j=0}^{n} \binom{n}{j} d_j (m-1-h)^{n-j}. \tag{554}$$

The d_j are universal and may be precomputed once. The numerator enters only through the finite power sums $\sum_{h<m} \varepsilon_h (m-1-h)^r$. This is often a more transparent bridge to the Taylor reduction series than the centered c_k formula, because the reduction polynomials themselves are written in the d_j.

Remark 52.4 (Two independent cancellations). The Thue–Morse prefix in (554) may annihilate low ordinary powers when m is a complete binary block. The q-binomial formula introduces a second cancellation across the geometric scale index. These cancellations live in different variables and should not be conflated: one acts on h or r inside a binary block, the other on the scale node -2^k.

53 Moment coordinates and alternative exact representations

53.1 The triangular transforms between c_k and d_n

The affine relation $Z = 2Y - 1$ gives both directions of the moment change of coordinates.

Proposition 53.1 (Moment transforms)**.** *For every $n, k \in \mathbb{N}$,*

$$d_n = 2^{-n} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k, \tag{555}$$

$$c_k = \sum_{j=0}^{2k} (-1)^j 2^j \binom{2k}{j} d_j. \tag{556}$$

Proof. Since $Y = (1+Z)/2$, expand Y^n and use the vanishing odd moments of Z. Conversely expand $(2Y-1)^{2k}$; because $2k$ is even, the sign of the term of degree j is $(-1)^j$. □

The first formula explains the odd-looking shift $2m - 2h - 1$ in (539): it is precisely the centered coordinate obtained after converting the half moments to the even moments of up.

53.2 An infinite-cube composition formula

Each factor in (543) has the expansion

$$E_0(X/2^j) = \sum_{r \geq 0} \frac{X^r}{2^{jr}(r+1)!}.$$

Taking the coefficient of X^n gives an exact positive series over finite-support weak compositions.

Proposition 53.2 (Composition representation of the half moments)**.** *For every $n \in \mathbb{N}$,*

$$d_n = n! \sum_{\substack{r_1, r_2, \ldots \geq 0 \\ r_1 + r_2 + \cdots = n}} \prod_{j \geq 1} \frac{2^{-jr_j}}{(r_j+1)!}. \tag{557}$$

Only finitely many r_j are nonzero in each summand, and the series converges absolutely. Equivalently,

$$d_n = \lim_{J \to \infty} \int_{[0,1]^J} \left(\sum_{j=1}^{J} \frac{u_j}{2^j} \right)^n \mathrm{d}u_1 \cdots \mathrm{d}u_J. \tag{558}$$

This gives a q-free, manifestly positive representation of every inverse-dyadic value:

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} d_n = 2^{-\binom{n}{2}} \sum_{r_1 + r_2 + \cdots = n} \prod_{j \geq 1} \frac{2^{-jr_j}}{(r_j+1)!}. \tag{559}$$

53.3 Cumulants, Bernoulli numbers, and Bell polynomials

The infinite product also gives a compact partition formula. With Bernoulli numbers in the convention $B_2 = 1/6$, $B_4 = -1/30$, one has

$$\log \frac{\mathrm{e}^X - 1}{X} = \frac{X}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} X^{2r}. \tag{560}$$

Summing this identity over the dyadic scales in (543) yields

$$\log \mathscr{A}(X) = \frac{X}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!(2^{2r}-1)} X^{2r}. \tag{561}$$

Thus the cumulants κ_j of Y are

$$\kappa_1 = \frac{1}{2}, \qquad \kappa_{2r} = \frac{B_{2r}}{2r(2^{2r}-1)}, \qquad \kappa_{2r+1} = 0 \quad (r \geq 1). \tag{562}$$

Theorem 53.3 (Bell-polynomial representation)**.** *Let $\mathcal{B}_n$ be the complete exponential Bell polynomial. Then*

$$d_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \tag{563}$$

and explicitly

$$d_n = n! \sum_{\substack{m_1,\dots,m_n \ge 0 \\ 1m_1+2m_2+\cdots+nm_n=n}} \prod_{r=1}^{n} \frac{1}{m_r!} \left(\frac{\kappa_r}{r!}\right)^{m_r}. \tag{564}$$

Consequently

$$\boxed{F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} \mathcal{B}_n(\kappa_1, \dots, \kappa_n).} \tag{565}$$

Proof. Equation (561) is the standard cumulant expansion $\log \mathscr{A}(X) = \sum_{j\ge1} \kappa_j X^j/j!$. Exponentiating and comparing coefficients is exactly the defining generating identity for the complete Bell polynomials. □

The representation is useful when Bernoulli arithmetic is already available. It also gives the moment recurrence

$$d_{n+1} = \sum_{j=0}^{n} \binom{n}{j} \kappa_{j+1} d_{n-j}, \tag{566}$$

while (544) gives the rational recurrence

$$(2^n - 1)d_n = \sum_{r=1}^{n} \binom{n}{r} \frac{d_{n-r}}{r+1} \qquad (n \ge 1). \tag{567}$$

n	d_n	c_n	$F(2^{-n})$
0	1	1	1
1	1/2	1/9	1/2
2	5/18	19/675	5/72
3	1/6	583/59535	1/288
4	143/1350	132809/32531625	143/2073600
5	19/270	46840699/24405225075	19/33177600

Table 9: The columns use different index conventions: c_n is the $2n$th centered moment, whereas d_n is the nth half moment.

53.4 Exact moment correction of the matched spline

The centered finite spline of degree $p \ge 1$ is

$$S_p(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left(r - 2^p x + \frac{1}{2}\right)^p, \qquad N_p(x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right). \tag{568}$$

At degree zero set

$$S_0(x) = \sum_{r=0}^{N_0(x)-1} \varepsilon_r. \tag{569}$$

The companion article proves the global estimate

$$\sup_{x\in\mathbb{R}} |S_p(x) - \mathcal{F}(x)| \le 2^{-p}. \tag{570}$$

At the matched grid point $m/2^p$ the cutoff is m and

$$S_p(m/2^p) = \frac{1}{2^{\binom{p+1}{2}} p!} \sum_{h=0}^{m-1} \varepsilon_h (2m - 2h - 1)^p. \tag{571}$$

The $k = 0$ part of (539) is exactly this matched spline. The remaining moments supply a finite correction hierarchy.

Theorem 53.4 (Moment-corrected matched-spline identity)**.** *For all $m, n \in \mathbb{N}$,*

$$\boxed{\mathcal{F}(m/2^n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{c_k}{(2k)!\, 2^{k(2n-2k+1)}} S_{n-2k}\left(\frac{m}{2^{n-2k}}\right).} \tag{572}$$

For $0 \le m \le 2^n$ the left side is $F(m/2^n)$. The spline arguments on the right are understood globally and may lie outside $[0, 1]$.

Proof. In (539) put $p = n - 2k$. By (571),

$$\sum_{h=0}^{m-1} \varepsilon_h (2m - 2h - 1)^p = 2^{\binom{p+1}{2}} p!\, S_p(m/2^p).$$

Moreover

$$\frac{1}{n!} \binom{n}{2k} p! = \frac{1}{(2k)!},$$

and

$$\binom{n+1}{2} - \binom{p+1}{2} = k(2n - 2k + 1).$$

Substitution gives (572). □

The formula has three noteworthy features.

1. Its first term is the obvious finite-spline approximation $S_n(m/2^n)$.
2. The correction drops the spline degree by two at each step, reflecting the evenness of up and the disappearance of odd centered moments.
3. The numerator m remains fixed, while the denominator is coarsened from 2^n to 2^{n-2k}. Thus the formula couples degree reduction to dyadic rescaling.

Corollary 53.5 (Exact defect of the matched spline)**.** *For every $m, n \in \mathbb{N}$,*

$$\mathcal{F}(m/2^n) - S_n(m/2^n) = \sum_{k=1}^{\lfloor n/2 \rfloor} \frac{c_k}{(2k)!\, 2^{k(2n-2k+1)}} S_{n-2k}\left(\frac{m}{2^{n-2k}}\right). \tag{573}$$

This is stronger than an error estimate: it identifies the error exactly as a finite sequence of lower-degree global splines. The small factor in the first correction is $c_1/(2 \cdot 2^{2n-1}) = 1/(9 \cdot 2^{2n})$.

53.5 The reciprocal correction transform

The previous relation is triangular and can be inverted. Define reciprocal centered moments γ_k by

$$\frac{1}{\mathscr{C}(T)} = \sum_{k\geq 0} \gamma_k \frac{T^{2k}}{(2k)!}. \tag{574}$$

Thus

$$\gamma_0 = 1, \qquad \gamma_n = -\sum_{k=0}^{n-1} \binom{2n}{2k} \gamma_k c_{n-k} \quad (n \geq 1), \tag{575}$$

and the first values are

$$1, \quad -\frac{1}{9}, \quad \frac{31}{675}, \quad -\frac{2347}{59535}, \quad \frac{1904369}{32531625}, \ldots. \tag{576}$$

Theorem 53.6 (Reciprocal moment deconvolution)**.** *For every $m, n \in \mathbb{N}$,*

$$\boxed{S_n(m/2^n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\gamma_k}{(2k)!\, 2^{k(2n-2k+1)}} \mathcal{F}\left(\frac{m}{2^{n-2k}}\right).} \tag{577}$$

Proof. For a fixed numerator m, introduce

$$P_p(m) = \sum_{h=0}^{m-1} \varepsilon_h (2m - 2h - 1)^p, \qquad G_p(m) = 2^{\binom{p+1}{2}} p!\, \mathcal{F}(m/2^p).$$

The exact moment formula says, in exponential generating functions,

$$\sum_{p\geq 0} G_p(m) \frac{T^p}{p!} = \mathscr{C}(T) \sum_{p\geq 0} P_p(m) \frac{T^p}{p!}.$$

Multiply by $1/\mathscr{C}(T)$, compare coefficients, and then use $P_p(m) = 2^{\binom{p+1}{2}} p! S_p(m/2^p)$. The same triangular exponent simplification as in theorem 53.4 gives (577). ☐

Equations (572) and (577) are a finite convolution/deconvolution pair. They show that the exact dyadic values and the matched centered splines carry precisely the same information once the universal moment series $\mathscr{C}$ is known.

First bridge to the discrete theory. The centered spline is not merely an approximation from which exact dyadic arithmetic is separate. At a matched grid point, the exact value is obtained by a finite universal moment correction of that spline, and the correction is invertible. Later, the half-base q-Pochhammer row will give a second exact correction, using consecutive spline degrees at the *same* argument rather than parity-separated degrees at rescaled arguments.

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54 The half-base q-formula as geometric coefficient extraction

54.1 The Gaussian row is a divided difference

For a general base q, write

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \tag{578}$$

and let $\left[{n \atop k}\right]_q$ denote the Gaussian binomial coefficient. At $q = 1/2$ the finite q-binomial theorem is

$$(z; 1/2)_n = \sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \tag{579}$$

Put

$$x_k = -2^k, \qquad w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}, \qquad \mathfrak{m}_n = \prod_{\ell=1}^{n} (2^\ell - 1). \tag{580}$$

Then

$$\sum_{k=0}^{n} w_{n,k} x_k^d = \begin{cases} 0, & 0 \le d < n, \\ \mathfrak{m}_n, & d = n, \end{cases} \tag{581}$$

and

$$\frac{w_{n,k}}{\mathfrak{m}_n} = \frac{1}{\prod_{\ell \ne k} (x_k - x_\ell)}, \qquad (1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n. \tag{582}$$

Thus the functional

$$\mathfrak{D}_n[P] = \frac{1}{\mathfrak{m}_n} \sum_{k=0}^{n} w_{n,k} P(-2^k) \tag{583}$$

is the divided difference $[-1, -2, \ldots, -2^n]P$. For every polynomial of degree at most n, it returns the leading coefficient.

Proposition 54.1 (Residue form of the geometric extractor)**.** *Let*

$$\Pi_n(z) = \prod_{k=0}^{n} (z + 2^k),$$

and let P be a polynomial of degree at most n. If Γ encloses all nodes $-1, -2, \ldots, -2^n$, then

$$[z^n]P(z) = \mathfrak{D}_n[P] = \frac{1}{2\pi \mathrm{i}} \int_\Gamma \frac{P(z)}{\Pi_n(z)} \, \mathrm{d}z. \tag{584}$$

Proof. The residues at the simple poles $x_k = -2^k$ are $P(x_k)/\Pi_n'(x_k) = P(x_k)/\prod_{\ell \ne k}(x_k - x_\ell)$, giving the barycentric sum in (583). Lagrange interpolation identifies that sum with the leading coefficient. □

The q-symbols in the dyadic formula therefore have a concrete and limited task: they are the closed form of a divided-difference stencil on a geometric grid. No infinite basic-hypergeometric summation is involved.

54.2 The parametric coefficient polynomial

Introduce an auxiliary node variable z and define

$$\mathscr{R}_m(z; X) = \frac{\mathrm{e}^{-X} B_m(zX) \mathscr{A}(zX)}{\mathscr{A}(-X)}, \qquad r_{m,n}(z) = [X^n]\mathscr{R}_m(z; X). \tag{585}$$

Because $\mathscr{A}(-X)$ is a unit with constant term 1, this is a well-defined formal series and an entire analytic germ. More importantly, $r_{m,n}$ is a polynomial in z.

Lemma 54.2 (Node polynomial and its leading coefficient)**.** *For fixed m, n, the polynomial $r_{m,n}(z)$ has degree at most n and*

$$[z^n] r_{m,n}(z) = [X^n](B_m(X)\mathscr{A}(X)) = 2^{\binom{n}{2}} \mathcal{F}(m/2^n). \tag{586}$$

Proof. Write $\mathrm{e}^{-X}/\mathscr{A}(-X)=\sum_{j\ge 0}u_jX^j$ with $u_0=1$. Since $B_m(zX)\mathscr{A}(zX)=(B_m\mathscr{A})(zX)$, the coefficient of X^n is a polynomial in z of degree at most n. Its z^n term can only use the constant u_0, so its leading coefficient is $[X^n](B_m\mathscr{A})$. Apply theorem 52.1. □

The exact dyadic value is now the leading coefficient of a degree-n polynomial sampled at $-1,-2,\dots,-2^n$. Applying proposition 54.1 gives a compact alternative to the displayed q-sum.

Corollary 54.3 (Geometric-node contour representation). *For every $m,n\in\mathbb{N}$,*

$$\boxed{\mathcal{F}(m/2^n)=\frac{2^{-\binom{n}{2}}}{2\pi\mathrm{i}}\int_\Gamma \frac{r_{m,n}(z)}{\prod_{k=0}^{n}(z+2^k)}\,\mathrm{d}z,} \tag{587}$$

where Γ encloses the geometric nodes.

Together, (553) and (587) exhibit two independent Cauchy extractions:

coefficient in the analytic variable X	leading coefficient in the geometric node z
$[X^n](B_m\mathscr{A})$	$[z^n]r_{m,n}(z)$
ordinary Cauchy kernel X^{-n-1}	barycentric kernel $\Pi_n(z)^{-1}$.

The finite q-binomial formula is what results when the second contour is evaluated by residues and each node value is converted to a Thue–Morse block sum.

54.3 Why the node values are Thue–Morse power sums

For a complete binary block set

$$C_k(X)=\sum_{r=0}^{2^k-1}\varepsilon_r\mathrm{e}^{(r-2^k)X}. \tag{588}$$

The digit factorization $\sum_{r<2^k}\varepsilon_r z^r=\prod_{j=0}^{k-1}(1-z^{2^j})$ gives

$$C_k(X)=(-1)^k2^{\binom{k}{2}}X^k\mathrm{e}^{-X}\prod_{j=0}^{k-1}E_0(-2^jX). \tag{589}$$

Hence C_k has a zero of exact order k. The refinement (544), iterated k times, gives

$$\frac{\mathrm{e}^{-X}\mathscr{A}(-2^kX)}{\mathscr{A}(-X)}=\mathrm{e}^{-X}\prod_{j=0}^{k-1}E_0(-2^jX). \tag{590}$$

Combining (589) and (590) identifies the node specialization of (585) with a normalized complete block.

For a general numerator, the long block factors as

$$\sum_{r=0}^{m2^k-1}\varepsilon_r\mathrm{e}^{(r-m2^k)X}=B_m(-2^kX)C_k(X). \tag{591}$$

After division by the forced factor $(-1)^k2^{\binom{k}{2}}X^k$, the coefficient of X^n is

$$\frac{(-1)^k2^{-\binom{k}{2}}}{(n+k)!}\sum_{r=0}^{m2^k-1}\varepsilon_r(r-m2^k)^{n+k}. \tag{592}$$

Multiplication by the interpolation weight $w_{n,k}$ cancels the sign and contributes a second factor $2^{-\binom{k}{2}}$. Their product is the characteristic $4^{-\binom{k}{2}}$ of (540). Finally (582) converts $\mathfrak{m}_n$ into $2^{n^2}(1/2;1/2)_n$ after the factor $2^{\binom{n}{2}}$ from lemma 54.2 is included. This recovers the exact q-formula with $Q=0$.

54.4 Translation invariance is lower-degree annihilation

The shift Q enters a block generating function by multiplication with e^{QX}. If $P_{m,k}(X)$ denotes the normalized long-block series after its forced X^k has been removed, then the q-numerator is a weighted sum of $[X^n](e^{QX}P_{m,k}(X))$. Expanding the exponential,

$$[X^n](e^{QX}P_{m,k}(X)) = \sum_{s=0}^{n} \frac{Q^{n-s}}{(n-s)!}[X^s]P_{m,k}(X). \tag{593}$$

For each $s < n$, the Gaussian row sees a polynomial of geometric-node degree at most s and annihilates it. Only the $s = n$ term survives; it has no Q factor.

Proposition 54.4 (Finite translation invariance)**.** *For fixed m, n, the polynomial*

$$Q \longmapsto \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + Q)^{n+k} \tag{594}$$

is constant.

This is stronger than saying that a limit is independent of Q: every coefficient of $Q, Q^2, \ldots$ vanishes in the finite expression itself. The distinction becomes important in section 56, where the finite discrete rows generally do depend on Q.

54.5 A useful abstract summary

The exact q-formula is a composition of two annihilators:

1. the complete Thue–Morse block has a zero of order k, removing all inner polynomial degrees below k;
2. the Gaussian row on -2^k removes all remaining node degrees below n and returns the leading degree n.

The exponent $n + k$ in the inner power is therefore forced: k degrees are consumed by the binary block, leaving exactly n degrees for the geometric extractor. The formula is a hybrid of automatic-sequence cancellation, ordinary exponential generating functions, and finite q-calculus; it is not a terminating basic-hypergeometric summation in the usual ${}_r\phi_s$ sense [7, 9].

55 Binary Taylor reduction and the global series

55.1 The finite reduction polynomial

The inverse-dyadic moments assemble into the polynomial

$$\mathcal{T}_m(y) = \sum_{j=0}^{m} 2^{\binom{j+1}{2} - \binom{m-j}{2}} \frac{d_{m-j}}{(m-j)!} \frac{y^j}{j!}. \tag{595}$$

The signed global extension satisfies the exact Taylor-block identity

$$\mathcal{F}(2^{-m} + y) = -\mathcal{F}(y) + \mathcal{T}_m(y), \qquad 0 \le y < 2^{-m}. \tag{596}$$

This is the elementary step behind both the terminating dyadic recursion and the infinite binary telescope.

For $x \geq 0$ set

$$p_m(x) = \lfloor 2^m x \rfloor, \qquad y_m(x) = x - \frac{p_m(x)}{2^m}, \qquad 0 \leq y_m(x) < 2^{-m}, \tag{597}$$

and put

$$p^*_{m-1}(x) = \begin{cases} \lfloor x/2 \rfloor, & m = 0, \\ p_{m-1}(x), & m \geq 1. \end{cases} \tag{598}$$

Define

$$\alpha_m(x) = (-1)^{\mathrm{wt}_2(p_m(x))}\big(2p^*_{m-1}(x) - p_m(x)\big). \tag{599}$$

This coefficient is 0 unless the newly exposed binary digit is 1, in which case it is a sign.

Theorem 55.1 (Binary reduction series)**.** *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x)\mathcal{T}_m(y_m(x)).} \tag{600}$$

The series converges absolutely, and its partial sum through $m = N$ has the uniform remainder bound

$$\left|\mathcal{F}(x) - \sum_{m=0}^{N} \alpha_m(x)\mathcal{T}_m(y_m(x))\right| \leq 2^{1-N} \qquad (N \geq 1). \tag{601}$$

The proof is an exact residual telescope. If $R_m(x) = (-1)^{\mathrm{wt}_2(p_m(x))}\mathcal{F}(y_m(x))$, then

$$\alpha_m(x)\mathcal{T}_m(y_m(x)) = R_{m-1}(x) - R_m(x) \quad (m \geq 1),$$

with a separately correct scale-zero term. Since $0 \leq y_m < 2^{-m}$ and $|\mathcal{F}(y)| \leq 2y$ near the origin, $|R_m| \leq 2^{1-m}$.

55.2 The sparse digit form on the unit interval

For $0 \leq x < 1$, write the floor-selected binary expansion

$$x = 0.\delta_1\delta_2\delta_3\cdots = \sum_{m\geq 1} \delta_m 2^{-m}, \qquad \delta_m \in \{0, 1\}, \tag{602}$$

where p_m is the integer represented by the first m digits. Let $s_m = \delta_1 + \cdots + \delta_m$ and let

$$t_m = \sum_{\ell > m} \delta_\ell 2^{-\ell} = y_m(x) \tag{603}$$

be the unscaled tail. If $\delta_m = 1$, then $p_m = 2p_{m-1} + 1$ and

$$\alpha_m = (-1)^{s_m}(-1) = (-1)^{s_{m-1}};$$

if $\delta_m = 0$, then $\alpha_m = 0$.

Corollary 55.2 (Digit-sparse series)**.** *For $0 \leq x < 1$,*

$$\boxed{F(x) = \sum_{\substack{m\geq 1 \\ \delta_m = 1}} (-1)^{s_{m-1}}\mathcal{T}_m(t_m).} \tag{604}$$

At a dyadic rational the floor convention chooses the terminating binary expansion, so the sum is finite and contains exactly one term for each set bit of the numerator.

This is the most direct connection between the fast bit-recursive evaluator and the arbitrary argument series: the finite recursion is not a separate algorithm pasted onto the infinite formula; it is literally the same telescope when the binary tail eventually becomes zero.

55.3 Replacing each Taylor coefficient by a finite q-identity

For $Q \in \mathbb{C}$ define the finite inverse-block numerator

$$\mathcal{N}_n^{(Q)} = \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k + Q)^{n+k}. \tag{605}$$

The exact one-block q-formula is equivalent to

$$\frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}}(1/2;1/2)_n} = \frac{2^n d_n}{n!}, \tag{606}$$

which is independent of Q. Insert these finite identities into the polynomial (595):

$$\mathcal{Q}_m^{(Q)}(y) = 2^{-\binom{m+1}{2}} \sum_{n=0}^{m} \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}}(1/2;1/2)_n}. \tag{607}$$

Theorem 55.3 (q-Taylor identity and global q-series)**.** *For every m, y, and $Q \in \mathbb{C}$,*

$$\mathcal{Q}_m^{(Q)}(y) = \mathcal{T}_m(y). \tag{608}$$

Consequently, for fixed Q and every $x \geq 0$,

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{Q}_m^{(Q)}(y_m(x)),} \tag{609}$$

with absolute convergence.

The key word is *termwise*: every finite polynomial $\mathcal{Q}_m^{(Q)}$ already equals $\mathcal{T}_m$ and is already independent of Q before the outer infinite sum is considered. The q-symbols do not create the convergence of the outer series. They merely provide an alternative finite presentation of each Taylor block.

55.4 The dyadic triangle of finite representations

Suppose $x = m/2^s \in [0,1]$ is written with a terminating binary expansion. Then three different finite mechanisms evaluate the same rational number:

1. (539) sums the Thue–Morse prefix $0 \leq h < m$ and applies the even-moment correction in one shot;
2. (540) uses one geometric row $0 \leq k \leq s$ and two nested cancellations;
3. (604) walks only through the set bits of m, using inverse-dyadic Taylor data.

The finite sums have no natural term-by-term identification. Their equality is mediated by the master coefficient (550) and by the exact Taylor reduction (596). A fourth finite representation, obtained from the discrete limit, appears in the next section.

56 Discrete q-limits, exact stabilization, and extrapolation

56.1 Shifted splines and the finite row

For $p \geq 1$, $Q \in \mathbb{C}$, and $x \in \mathbb{R}$, define

$$S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r (r - 2^p x + Q)^p. \tag{610}$$

At $Q = 1/2$ this is the centered spline S_p. The formal development proves the exact fixed-cutoff Taylor relation and the uniform bound

$$\left|S_p^{(Q)}(x) - S_p(x)\right| \le 2^{-(p-1)}\left(\mathrm{e}^{|Q-1/2|} - 1\right). \tag{611}$$

Together with (570),

$$\left|S_p^{(Q)}(x) - \mathcal{F}(x)\right| \le (2\mathrm{e}^{|Q-1/2|} - 1)2^{-p}. \tag{612}$$

The discrete q-approximant is

$$\mathcal{D}_n^{(Q)}(x) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \times \sum_{r=0}^{N_{n+k}(x)-1} \varepsilon_r (r - 2^{n+k}x + Q)^{n+k}. \tag{613}$$

For every fixed Q and every real x, the Lean development proves

$$\mathcal{D}_n^{(Q)}(x) \longrightarrow \mathcal{F}(x). \tag{614}$$

The notation strongly resembles the exact dyadic formula (540). The following result makes the resemblance literal.

56.2 At dyadic inputs the limit stabilizes exactly

Theorem 56.1 (Finite stabilization on the dyadic grid)**.** *Let $x = m/2^s$ with $m, s \in \mathbb{N}$. Then, for every $n \ge s$ and every $Q \in \mathbb{C}$,*

$$\boxed{\mathcal{D}_n^{(Q)}(x) = \mathcal{F}(x).} \tag{615}$$

If $0 \le x \le 1$, the right side is $F(x)$.

Proof. For $n \ge s$ put $M = m2^{n-s}$, so that $x = M/2^n$. For every $0 \le k \le n$,

$$2^{n+k}x = M2^k \in \mathbb{N}, \qquad N_{n+k}(x) = \left\lfloor M2^k + \frac{1}{2} \right\rfloor = M2^k.$$

Therefore (613) becomes exactly the dyadic formula (540) with numerator M and denominator exponent n. That formula is valid for unreduced dyadic representations and for every complex translation Q. □

This is the sharpest connection between the exact and limiting formulae: the discrete formula extends the exact dyadic row to arbitrary x, and on a dyadic grid it needs no limiting process once the row depth reaches the denominator depth.

Corollary 56.2 (Exact finite shifted-spline extrapolation)**.** *For $x = m/2^s$, $n \ge s$, and every $Q \in \mathbb{C}$,*

$$\mathcal{F}(x) = \sum_{j=0}^{n} \omega_{n,j} S_{2n-j}^{(Q)}(x), \tag{616}$$

where the weights $\omega_{n,j}$ are defined in (618) below.

Thus every dyadic value admits an exact representation as a finite signed combination of consecutive spline degrees, all evaluated at the same point. Each shifted spline may depend strongly on Q, but the complete combination does not.

56.3 The row polynomial

Reindexing (613) by $j = n - k$ gives the exact identity

$$\mathcal{D}_n^{(Q)}(x) = \sum_{j=0}^{n} \omega_{n,j} S_{2n-j}^{(Q)}(x), \tag{617}$$

with

$$\omega_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}}{(1/2;1/2)_n}. \tag{618}$$

The full row is encoded by one normalized q-Pochhammer polynomial.

Proposition 56.3 (Generating polynomial of the Toeplitz row)**.** *Define*

$$W_n(z) = \sum_{j=0}^{n} \omega_{n,j} z^j. \tag{619}$$

Then

$$\boxed{W_n(z) = \frac{(z/2;1/2)_n}{(1/2;1/2)_n} = \prod_{\ell=1}^{n} \frac{1 - z/2^\ell}{1 - 1/2^\ell}.} \tag{620}$$

Consequently

$$W_n(1) = 1, \tag{621}$$

$$W_n(2^r) = 0 \qquad (1 \le r \le n), \tag{622}$$

$$\sum_{j=0}^{n} |\omega_{n,j}| = W_n(-1) = \frac{(-1/2;1/2)_n}{(1/2;1/2)_n}, \tag{623}$$

$$\sum_{j=0}^{n} |\omega_{n,j}| 2^j = W_n(-2) = \frac{(-1;1/2)_n}{(1/2;1/2)_n}. \tag{624}$$

Proof. Apply the finite q-binomial theorem to $(z/2;1/2)_n$. Its coefficient of z^j is

$$(-1)^j 2^{-\binom{j}{2}} 2^{-j} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} = (-1)^j 2^{-\binom{j+1}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2},$$

which gives (620) after division by $(1/2;1/2)_n$. The evaluations at $1, 2^r, -1, -2$ follow from the product and from the alternating signs of the coefficients. □

The familiar row-sum identity and bounded variation are therefore only two evaluations of a single polynomial. The roots $2, 4, \dots, 2^n$ contain additional information that the basic Toeplitz proof does not need.

56.4 A q-Pochhammer filter in the degree variable

Fix x and Q and regard $p \mapsto S_p^{(Q)}(x)$ as a sequence. Let the backward shift L act by $\mathsf{L} S_p^{(Q)} = S_{p-1}^{(Q)}$. Then (617) and (620) can be written as

$$\boxed{\mathcal{D}_n^{(Q)}(x) = W_n(\mathsf{L}) S_{2n}^{(Q)}(x) = \prod_{\ell=1}^{n} \frac{1 - 2^{-\ell}\mathsf{L}}{1 - 2^{-\ell}} S_{2n}^{(Q)}(x).} \tag{625}$$

Each factor preserves a constant sequence because its value at $\mathsf{L} = 1$ is 1. Meanwhile the zero $W_n(2^r) = 0$ means that it removes a geometric error mode with ratio 2^{-r} in the degree.

Proposition 56.4 (q-Richardson exactness)**.** *Suppose a model sequence has the form*

$$u_p = L + \sum_{r=1}^{n} a_r 2^{-rp}. \tag{626}$$

Then

$$\sum_{j=0}^{n} \omega_{n,j} u_{2n-j} = L. \tag{627}$$

Proof. The constant contributes $LW_n(1) = L$. The rth error contributes

$$a_r 2^{-2nr} \sum_{j=0}^{n} \omega_{n,j} 2^{rj} = a_r 2^{-2nr} W_n(2^r) = 0.$$

□

No claim is needed that the actual spline error has a finite expansion of the form (626). The proposition explains the algebraic design of the row: it is a geometric-grid analogue of Richardson extrapolation, with the normalized q-Pochhammer polynomial as its error-annihilating filter.

56.5 A quantitative 4^{-n} convergence bound

The bounded-variation proof of (614) uses only that all sampled degrees $2n - j$ tend to infinity. Retaining their actual sizes gives a stronger quantitative consequence.

Theorem 56.5 (Explicit discrete-row error)**.** *For $n \geq 1$, $Q \in \mathbb{C}$, and every $x \in \mathbb{R}$, let*

$$H_n = \frac{(-1;1/2)_n}{(1/2;1/2)_n}. \tag{628}$$

Then

$$\boxed{|\mathcal{D}_n^{(Q)}(x) - \mathcal{F}(x)| \leq H_n(2\mathrm{e}^{|Q-1/2|} - 1)4^{-n}.} \tag{629}$$

Moreover $H_n \leq 64$, so the convergence is uniformly $\mathcal{O}_Q(4^{-n})$ on the whole real line. At the centered shift,

$$|\mathcal{D}_n^{(1/2)}(x) - \mathcal{F}(x)| \leq H_n 4^{-n}. \tag{630}$$

Proof. Apply (612) with $p = 2n - j$ in (617):

$$\begin{aligned}
|\mathcal{D}_n^{(Q)}(x) - \mathcal{F}(x)| &\leq (2\mathrm{e}^{|Q-1/2|} - 1) \sum_{j=0}^{n} |\omega_{n,j}| 2^{-(2n-j)} \\
&= (2\mathrm{e}^{|Q-1/2|} - 1)4^{-n} \sum_{j=0}^{n} |\omega_{n,j}| 2^{j}.
\end{aligned}$$

Use (624). Finally

$$H_n = \frac{2}{1 - 2^{-n}} \frac{(-1/2;1/2)_{n-1}}{(1/2;1/2)_{n-1}} \leq 4 \cdot 16 = 64,$$

where the last factor is the uniform variation bound proved in the formal development. □

The same argument isolates the finite translation dependence.

Corollary 56.6 (Decay of the shift dependence)**.** *For every $x \in \mathbb{R}$ and $n \geq 1$,*

$$|\mathcal{D}_n^{(Q)}(x) - \mathcal{D}_n^{(1/2)}(x)| \leq 2H_n(\mathrm{e}^{|Q-1/2|} - 1)4^{-n}. \tag{631}$$

Thus the finite rows can depend dramatically on Q at small n, but the dependence is uniformly suppressed at a geometric rate. This is analytically different from proposition 54.4, where the finite dyadic expression is exactly constant in Q.

56.6 A synchronized nearest-dyadic q-limit

There is a simpler way to turn the exact dyadic formula into a limit at arbitrary argument. For $x \geq 0$ define the nearest nonnegative dyadic numerator

$$M_n(x) = \max\left(0, \left\lfloor 2^n x + \frac{1}{2} \right\rfloor\right) = N_n(x), \qquad \widetilde{x}_n = \frac{M_n(x)}{2^n}. \tag{632}$$

Use this one numerator at every inner scale:

$$\widetilde{\mathcal{D}}_n^{(Q)}(x) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \times \sum_{r=0}^{M_n(x)2^k-1} \varepsilon_r (r - M_n(x)2^k + Q)^{n+k}. \tag{633}$$

Unlike (613), every finite row is now an exact dyadic formula.

Theorem 56.7 (Synchronized exact-grid representation)**.** *For every n, $Q \in \mathbb{C}$, and $x \geq 0$,*

$$\boxed{\widetilde{\mathcal{D}}_n^{(Q)}(x) = \mathcal{F}\left(\frac{M_n(x)}{2^n}\right).} \tag{634}$$

In particular, the left side is exactly independent of Q. On $0 \leq x \leq 1$,

$$\left|\widetilde{\mathcal{D}}_n^{(Q)}(x) - F(x)\right| \leq 2^{-n}, \tag{635}$$

and hence

$$F(x) = \lim_{n\to\infty} \widetilde{\mathcal{D}}_n^{(Q)}(x). \tag{636}$$

Proof. Equation (634) is (540) with numerator $M_n(x)$. For $x \in [0,1]$, $|\widetilde{x}_n - x| \leq 2^{-n-1}$ and the bounded Fabius function is 2-Lipschitz, giving (635). □

The same construction applied to (539) gives a purely moment/Thue–Morse limit:

$$F(x) = \lim_{n\to\infty} \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{M_n(x)-1} \varepsilon_h \times \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2M_n(x) - 2h - 1)^{n-2k}, \tag{637}$$

again with error at most 2^{-n} on $[0,1]$. This representation contains no q-symbols at all: it is exact dyadic arithmetic followed by nearest-grid convergence.

The synchronized limit is conceptually simple but converges at the basic grid rate $\mathcal{O}(2^{-n})$. The asynchronous row (613) is adapted to the natural spline degree $n+k$ at each scale and, by theorem 56.5, achieves the stronger uniform bound $\mathcal{O}_Q(4^{-n})$. The normalized q-Pochhammer row acts as an acceleration filter.

56.7 What fails away from a dyadic point

Let

$$\theta_n(x) = 2^n x - M_n(x), \qquad -\frac{1}{2} \leq \theta_n(x) < \frac{1}{2}. \tag{638}$$

Then for every $k \geq 0$,

$$N_{n+k}(x) = M_n(x)2^k + \left\lfloor 2^k \theta_n(x) + \frac{1}{2} \right\rfloor, \tag{639}$$

and the shifted coordinate in the inner power is

$$r - 2^{n+k}x + Q = r - M_n(x)2^k + (Q - 2^k\theta_n(x)). \tag{640}$$

Thus the source discrete row differs from the synchronized exact dyadic row in exactly two ways:

1. the complete block of length $M_n 2^k$ acquires a boundary defect $\lfloor 2^k\theta_n + 1/2\rfloor$;
2. the common translation Q is replaced by the scale-dependent translation $Q - 2^k\theta_n$.

At a resolved dyadic point, $\theta_n = 0$, so both defects disappear simultaneously; this is another proof of theorem 56.1. Away from the dyadic grid the shifts vary with k, so the finite common-translation invariance of proposition 54.4 cannot be applied across the row.

56.8 Two infinite constructions that share a limit but not terms

The global binary q-series (609) and the discrete limit (614) both evaluate $\mathcal{F}$, but their relation is not that of an infinite series and its partial sums.

Feature	Binary q-series	Discrete q-limit
Outer index	binary scale m	approximation row n
Finite inner object	$\mathcal{Q}_m^{(Q)} = \mathcal{T}_m$	shifted splines $S_{2n-j}^{(Q)}$
Shift dependence	absent in every summand	generally present in every finite row
Convergence source	exact residual telescope	Toeplitz summability of spline convergence
At dyadic x	outer series terminates at the last set bit	rows stabilize exactly once n reaches the denominator depth
Natural error bound	2^{1-N} after scale N	$H_n(2\mathrm{e}^{\lvert Q-1/2\rvert} - 1)4^{-n}$

Table 10: The two arbitrary-argument q constructions.

Warning 56.8. It is therefore incorrect to call $\mathcal{D}_n^{(Q)}$ the nth partial sum of the global q-binomial series. The two families use the same finite half-base identities, but they insert them into different analytic frameworks.

56.9 The first rows

For reference, the first normalized Toeplitz rows are

$$\begin{aligned}
(\omega_{1,j})_{j=0}^{1} &= (2, -1),\\
(\omega_{2,j})_{j=0}^{2} &= (8/3, -2, 1/3),\\
(\omega_{3,j})_{j=0}^{3} &= (64/21, -8/3, 2/3, -1/21),\\
(\omega_{4,j})_{j=0}^{4} &= (1024/315, -64/21, 8/9, -2/21, 1/315).
\end{aligned}$$

Their ordinary and 2^j-weighted variations are shown in table 11.

n	$\sum_j \lvert\omega_{n,j}\rvert$	$H_n = \sum_j \lvert\omega_{n,j}\rvert 2^j$
1	3	4
2	5	8
3	45/7	80/7
4	51/7	96/7
5	1683/217	3264/217

Table 11: Two evaluations of the row polynomial: $W_n(-1)$ and $W_n(-2)$.

57 A unified dyadic identity and a worked example

57.1 The complete equality chain

Let $x = m/2^s \in [0,1]$, use the terminating binary digits $x = 0.\delta_1 \cdots \delta_s$, and let $N \geq s$. Combining the preceding results gives the following chain of exact finite expressions:

$$F(x) = 2^{-\binom{s}{2}}[X^s](B_m(X)\mathscr{A}(X)) \tag{641}$$

$$= \frac{2^{-\binom{s+1}{2}}}{s!} \sum_{h=0}^{m-1} \varepsilon_h \sum_{k=0}^{\lfloor s/2 \rfloor} \binom{s}{2k} c_k (2m - 2h - 1)^{s-2k} \tag{642}$$

$$= \frac{1}{2^{s^2}(1/2;1/2)_s} \sum_{k=0}^{s} \frac{\left[{s \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(s+k)!} \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + Q)^{s+k} \tag{643}$$

$$= \sum_{k=0}^{\lfloor s/2 \rfloor} \frac{c_k}{(2k)!2^{k(2s-2k+1)}} S_{s-2k}\left(\frac{m}{2^{s-2k}}\right) \tag{644}$$

$$= \sum_{\substack{1 \leq j \leq s \\ \delta_j = 1}} (-1)^{\delta_1 + \cdots + \delta_{j-1}} \mathcal{T}_j\left(\sum_{\ell > j} \delta_\ell 2^{-\ell}\right) \tag{645}$$

$$= \sum_{j=0}^{N} \omega_{N,j} S^{(Q)}_{2N-j}(x). \tag{646}$$

The letter k has unrelated local meanings in different lines; this is unavoidable because one line indexes even moments and another indexes geometric scales. The equalities are not termwise transformations. They are different factorizations of the same exact value:

- (641) is the common scalar invariant;
- (642) expands the probability law in centered moments;
- (643) extracts the same coefficient by geometric interpolation;
- (644) corrects one matched spline through even moments;
- (645) telescopes through the set bits;
- (646) extrapolates consecutive spline degrees at the same point.

57.2 The value at 5/16

Take

$$x = \frac{5}{16} = 0.0101_2, \qquad F(x) = \frac{305857}{2073600} \approx 0.1475004822530864.$$

This one value illustrates every bridge above.

57.2.1 Master coefficient and moment formula

The prefix polynomial is

$$B_5(X) = \mathrm{e}^{4X} - \mathrm{e}^{3X} - \mathrm{e}^{2X} + \mathrm{e}^{X} - 1. \tag{647}$$

Using $d_0 = 1, d_1 = 1/2, d_2 = 5/18, d_3 = 1/6, d_4 = 143/1350$ in $\mathscr{A}$ gives

$$[X^4](B_5\mathscr{A}) = \frac{305857}{32400}. \tag{648}$$

Since $2^{-\binom{4}{2}} = 1/64$, (550) gives (57.2). Equivalently, with $c_1 = 1/9$ and $c_2 = 19/675$,

$$F(5/16) = \frac{2^{-10}}{4!} \sum_{h=0}^{4} \varepsilon_h \left[t_h^4 + 6c_1 t_h^2 + c_2\right], \qquad t_h = 9 - 2h. \tag{649}$$

57.2.2 Moment correction of the matched spline

The three spline values required by (572) are

$$S_4(5/16) = \frac{1205}{8192}, \qquad S_2(5/4) = \frac{15}{16}, \qquad S_0(5) = -1. \tag{650}$$

Hence

$$\begin{aligned} F(5/16) &= S_4(5/16) + \frac{1}{2304} S_2(5/4) + \frac{19}{16588800} S_0(5) \\ &= \frac{1205}{8192} + \frac{5}{12288} - \frac{19}{16588800} = \frac{305857}{2073600}. \end{aligned} \tag{651}$$

The matched degree-four spline is already close, but the exact discrepancy is resolved by two universal lower-degree corrections.

57.2.3 The two set bits

The binary digits equal 1 only at positions 2 and 4. The reduction polynomial at the first position is

$$\mathcal{T}_2(y) = \frac{5}{72} + y + 4y^2, \tag{652}$$

and $\mathcal{T}_4(0) = F(1/16) = 143/2073600$. The signs are $+$ at position 2 and $-$ at position 4, so

$$\begin{aligned} F(5/16) &= \mathcal{T}_2(1/16) - \mathcal{T}_4(0) \\ &= \frac{85}{576} - \frac{143}{2073600} = \frac{305857}{2073600}. \end{aligned} \tag{653}$$

The prefix formula sums five Thue–Morse terms; the binary formula needs only the two set bits.

57.2.4 Exact q-extrapolation of centered splines

At row $N = 4$, (646) reads

$$\begin{aligned} F(5/16) = \frac{1024}{315} S_8(5/16) - \frac{64}{21} S_7(5/16) + \frac{8}{9} S_6(5/16) \\ - \frac{2}{21} S_5(5/16) + \frac{1}{315} S_4(5/16). \end{aligned} \tag{654}$$

The individual spline values are

$$\begin{aligned} S_4 &= \frac{1205}{8192}, \qquad S_5 = \frac{289799}{1966080}, \qquad S_6 = \frac{13917509}{94371840}, \\ S_7 &= \frac{74236247}{503316480}, \qquad S_8 = \frac{395939357}{2684354560}, \end{aligned} \tag{655}$$

all evaluated at 5/16. None equals $F(5/16)$, but the q-Pochhammer filter combines them exactly.

57.2.5 Finite shift dependence before stabilization

The first discrete rows illustrate the difference between exact finite translation invariance and asymptotic translation independence.

n	$\mathcal{D}_n^{(0)}(5/16)$	$\mathcal{D}_n^{(1/2)}(5/16)$	$\mathcal{D}_n^{(3)}(5/16)$
1	0.156250000	0.156250000	3.906250000
2	0.151041667	0.147460938	−0.531250000
3	0.147505374	0.147501983	0.125788484
4	0.147500482	0.147500482	0.147500482

Table 12: The row becomes exactly equal to $F(5/16)$ at $n = 4$, independently of Q.

The exact dyadic q-formula itself is independent of Q already at row four, although its outer summands vary substantially. Their decimal contributions are:

scale k	$Q = 0$	$Q = 1/2$	$Q = 3$
0	0.000626240	0.000466967	0.000000000
1	−0.016345021	−0.014038037	−0.005184307
2	0.141657926	0.131089095	0.084488329
3	−0.467452954	−0.449506045	−0.365062120
4	0.489014292	0.479488502	0.433258580
sum	0.147500482	0.147500482	0.147500482

Table 13: Different translated decompositions of the same exact dyadic value.

58 Computational and conceptual consequences

58.1 Which representation is best for which task?

The formulae are mathematically equivalent at dyadic inputs, but not computationally interchangeable.

The exact q-formula contains inner blocks of length $m2^k$, so a direct implementation can be expensive. Its value is primarily structural: it exposes geometric interpolation, translation invariance, and the extrapolation filter. For exact numerical evaluation, the bit recursion or the master coefficient with precomputed moments is generally preferable.

58.2 A compact exact-arithmetic scheme

Equation (567) computes all half moments with rational arithmetic, after which (554) evaluates a dyadic value.

Algorithm 58.1 (Moment-prefix evaluator)**.** Given $m, n \in \mathbb{N}$:

Task	Recommended representation	Reason
Prove rationality or denominator facts	master coefficient or moment formula	all arithmetic is finite and the universal moments are explicit rationals
Evaluate one dyadic with a sparse numerator	binary Taylor reduction	the number of reductions is the number of set bits
Evaluate many numerators at one depth	precompute d_j or c_k, then use prefix recurrences	universal moment data are reused
Explain the q-coefficients	geometric divided difference and residue form	Gaussian coefficients are recognized as interpolation weights, not mysterious special-function decorations
Approximate arbitrary x with a simple certified error	centered spline S_p	direct global error 2^{-p} and one finite Thue–Morse sum
Accelerate arbitrary-argument spline convergence	discrete row $\mathcal{D}_n^{(Q)}$	exact q-Pochhammer filter and uniform $\mathcal{O}_Q(4^{-n})$ bound
Use only exact dyadic arithmetic in a limiting scheme	synchronized row $\widetilde{\mathcal{D}}_n^{(Q)}$ or synchronized moment limit	every finite stage is an exact dyadic value and is translation-independent
Trace dependence on binary digits	digit-sparse series	one nonzero summand per exposed 1-bit

Table 14: Representation selection. The q-formula is structurally illuminating but is usually not the cheapest naive exact evaluator.

1. Set $d_0 = 1$ and, for $1 \le j \le n$, compute

$$d_j = \frac{1}{2^j - 1} \sum_{r=1}^{j} \binom{j}{r} \frac{d_{j-r}}{r+1}.$$

2. Form

$$V = \sum_{h=0}^{m-1} \varepsilon_h \sum_{j=0}^{n} \binom{n}{j} d_j (m-1-h)^{n-j}.$$

3. Return $2^{-\binom{n}{2}} V/n!$.

The result is $\mathcal{F}(m/2^n)$ and is $F(m/2^n)$ when $0 \le m \le 2^n$.

A literal pseudocode version is included for clarity.

Listing 3: Exact dyadic evaluation from half moments.

```
d[0] := 1
for j := 1,...,n do
    s := 0
    for r := 1,...,j do
        s := s + binomial(j,r) * d[j-r] / (r+1)
    end
    d[j] := s / (2^j - 1)
end

V := 0
for h := 0,...,m-1 do
    inner := 0
    for j := 0,...,n do
```

```
        inner := inner + binomial(n,j) * d[j] * (m-1-h)^(n-j)
    end
    V := V + (-1)^(binary_weight(h)) * inner
end
return V / (2^(n*(n-1)/2) * n!)
```

The centered-moment formula can be evaluated similarly. Its advantage is parity: only $\lfloor n/2 \rfloor + 1$ moments occur. The half-moment formula aligns more directly with the binary Taylor polynomials.

58.3 Exact arithmetic checks implied by the connections

The network of representations supplies a strong verification suite. For chosen m, n one can independently check:

1. the master coefficient against the centered-moment formula;
2. the half-base q expression at two or more translations Q;
3. the terminating binary reduction when $m/2^n \in [0,1]$;
4. the moment-corrected matched-spline identity;
5. the exact q-extrapolation row at any order $N \geq n$;
6. representation independence under $(m,n) \mapsto (2m, n+1)$.

These checks stress different cancellation patterns. Agreement is therefore more informative than recomputing the same formula in a rearranged order.

58.4 Endpoint and convention warnings

Several small conventions are mathematically active.

Warning 58.2 (The scale-zero term). The global binary series begins at $m = 0$. It happens to vanish for $0 \leq x < 1$, but it equals 1 at $x = 1$ and controls the signed behavior on integer blocks. A series beginning at $m = 1$ is valid only on the half-open unit interval.

Warning 58.3 (Integer exponents). The exponent $\binom{j+1}{2} - \binom{m-j}{2}$ in (595) is an integer and may be negative. Replacing integer subtraction by truncated natural subtraction changes the polynomial.

Warning 58.4 (Centered powers are signed). Expressions such as $(r - m2^k + Q)^{n+k}$ live in a characteristic-zero scalar field. The quantity $r - m2^k$ is deliberately negative on much of the block. Truncated subtraction in $\mathbb{N}$ destroys the formula.

Warning 58.5 (Binary expansions at dyadics). The floor prefixes $p_m = \lfloor 2^m x \rfloor$ select the terminating expansion at a dyadic rational. Replacing it by the alternative expansion ending in infinitely many 1-digits changes the termwise digit series even though the represented real number is unchanged.

59 Relation to the Lean development

The principal source theorems are formalized in the following modules. Filenames are relative to `Analysis/FabiusFunction`.

Module	Role in the present article
`DyadicClosedForm.lean`	Exact Thue–Morse/even-moment formula for arbitrary dyadic numerators.
`HalfQBinomial.lean`	Finite q-Pochhammer and Gaussian-binomial calculus at $q = 1/2$, including the interpolation annihilation row.
`ThueMorseExponential.lean`	Complete-block exponential factorization, Prouhet cancellation, and translation of inner powers.
`FabiusQBinomialFormula.lean`	Centered and translated finite q-binomial formulas, arbitrary numerators, translation invariance, block/refinement normal forms, and representation independence.
`FabiusQBinomialTaylor.lean`	Identification of the finite q expression with the Taylor reduction polynomial.
`FabiusQBinomialScalarFormula.lean`	Scalar normalizations and real/complex translated forms.
`FabiusRawQBinomialFormula.lean` and `FabiusRawQBinomialScalar.lean`	Raw inverse-dyadic forms and their scalar versions.
`FabiusBinaryReductionSeries.lean`	Exact binary residual telescope, absolute convergence, and uniform tail bound.
`FabiusParityPowerSeries.lean`	Corrected scale-zero parity-power presentation.
`FabiusGlobalQBinomialSeries.lean`	Termwise substitution of finite q-identities into the binary telescope.
`FabiusComplexShiftSpline.lean`	Shifted finite splines, fixed-cutoff Taylor expansion, and uniform shift bound.
`FabiusDiscreteLimitToeplitz.lean`	Row mass, exact total variation, uniform variation bound, and finite-row Toeplitz theorem.
`FabiusDiscreteLimitIntegration.lean`	Reindexing of the discrete formula and convergence to the signed global function.

The following statements are direct paper-level corollaries or reorganizations of those formalized results:

- the expectation and two contour representations (551), (553), and (587);
- the composition and Bell-polynomial formulae for d_n;
- the moment-corrected spline transform and its reciprocal (572)–(577);
- exact finite stabilization of the discrete row at dyadic arguments;
- the row polynomial W_n, the shift-operator form, and the q-Richardson interpretation;
- the explicit $\mathcal{O}_Q(4^{-n})$ bound (629);
- the synchronized nearest-dyadic q- and moment-limits.

They require no new unproved property of the Fabius function; each follows by finite algebra, coefficient extraction, or the already formalized uniform spline estimates. They are marked here to distinguish conceptual reorganization from theorem names that already occur verbatim in the repository.

60 Conclusion

The exact dyadic, global-series, and discrete-limit formulae are best understood not as a collection of special tricks but as a small number of reusable operators acting in different variables.

The product $B_m\mathscr{A}$ separates numerator arithmetic from the universal probability law. Expanding its coefficient in centered moments gives the classical Thue–Morse formula; reading it as an expectation gives a probabilistic representation; taking a Cauchy integral gives an analytic one. Introducing the geometric node z turns the same coefficient into the leading term of a degree-n polynomial. The Gaussian row at base $1/2$ is then simply the barycentric divided difference on $-1, -2, \dots, -2^n$, and the q-Pochhammer symbol is the node polynomial of that grid.

The centered finite spline sits directly inside the exact arithmetic. Its matched-grid value is the zeroth centered-moment term, and the remaining moments form an exact, invertible defect correction. A second correction uses the normalized q-Pochhammer polynomial in the spline degree: it preserves constants, annihilates geometric error modes, and yields exact recovery at dyadic points. The resulting discrete row is therefore naturally viewed as a q-Richardson extrapolator, not merely as an opaque Toeplitz average.

Finally, the two arbitrary-argument q constructions have distinct logical roles. The global q-series is the binary residual telescope with an alternative finite formula inserted into each summand. The discrete limit is a finite extrapolation row whose shift dependence disappears only asymptotically. Their meeting point is the dyadic grid: there the binary series terminates, the discrete row stabilizes, the synchronized row is exact, and all roads return to the same master coefficient.

Supplementary material from this dossier

61 A compact identity sheet

For convenience, the principal transformations are collected here.

Probability and moments

$$
\begin{aligned}
Y &= \sum_{j\geq 1} U_j/2^j, \qquad Z = 2Y-1,\\
\mathscr{A}(X) &= \mathbb{E}(\mathrm{e}^{XY}) = \prod_{j\geq 1} \frac{\mathrm{e}^{X/2^j}-1}{X/2^j},\\
\mathscr{A}(X) &= \mathrm{e}^{X/2}\mathscr{C}(X/2), \qquad \mathscr{C}(T) = \sum_{k\geq 0} c_k T^{2k}/(2k)!,\\
d_n &= 2^{-n} \sum_{k\leq n/2} \binom{n}{2k} c_k,\\
c_k &= \sum_{j=0}^{2k} (-1)^j 2^j \binom{2k}{j} d_j.
\end{aligned}
$$

Master coefficient

$$
\begin{aligned}
B_m(X) &= \sum_{h<m} \varepsilon_h \mathrm{e}^{(m-1-h)X},\\
\mathcal{F}(m/2^n) &= 2^{-\binom{n}{2}} [X^n](B_m\mathscr{A}),\\
\mathcal{F}(m/2^n) &= \frac{2^{-\binom{n}{2}}}{n!} \sum_{h<m} \varepsilon_h \mathbb{E}(m-1-h+Y)^n.
\end{aligned}
$$

Geometric extraction

$$
\begin{aligned}
w_{n,k} &= (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n\\ k\end{bmatrix}_{1/2}, \qquad x_k = -2^k,\\
\frac{1}{\mathfrak{m}_n} \sum_{k=0}^{n} w_{n,k} P(x_k) &= [z^n]P(z) \quad (\deg P \leq n),\\
\mathfrak{m}_n &= 2^{\binom{n+1}{2}} (1/2;1/2)_n.
\end{aligned}
$$

Spline corrections

$$
\begin{aligned}
\mathcal{F}(m/2^n) &= \sum_{k\leq n/2} \frac{c_k}{(2k)!2^{k(2n-2k+1)}} S_{n-2k}(m/2^{n-2k}),\\
S_n(m/2^n) &= \sum_{k\leq n/2} \frac{\gamma_k}{(2k)!2^{k(2n-2k+1)}} \mathcal{F}(m/2^{n-2k}),\\
\sum_{k\geq 0} \gamma_k T^{2k}/(2k)! &= 1/\mathscr{C}(T).
\end{aligned}
$$

Discrete filter

$$\mathcal{D}_n^{(Q)}(x) = \sum_{j=0}^{n} \omega_{n,j} S_{2n-j}^{(Q)}(x),$$
$$W_n(z) = \sum_j \omega_{n,j} z^j = \frac{(z/2;1/2)_n}{(1/2;1/2)_n},$$
$$|\mathcal{D}_n^{(Q)}(x) - \mathcal{F}(x)| \le \frac{(-1;1/2)_n}{(1/2;1/2)_n}(2\mathrm{e}^{|Q-1/2|} - 1)4^{-n},$$
$$x = m/2^s,\ n \ge s \Longrightarrow \mathcal{D}_n^{(Q)}(x) = \mathcal{F}(x).$$

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Part VI
From Exact Dyadic Arithmetic to Endpoint Asymptotics

Source provenance and status. Former path: `Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex`. Coefficient-extraction and fixed-numerator bridge from exact arithmetic to the lower-Lambert expansion. It overlaps both the Dyadic Web and the later dyadic-asymptotic dossier below; retain it for its vertical-saddle, phase-locking, and Bromwich-to-coefficient derivations. Claims lacking an exact named declaration remain formal obligations.

Original source SHA-256: `c5ad7c5298d958ab63f459a3e246c2293d3020525223bc05ef2d0e1edab58f10`.

Dossier abstract

The exact arithmetic theory of the Fabius function and its sharp endpoint asymptotic are usually presented as nearly disjoint subjects. The former gives rational formulae for values at dyadic arguments, including a half-moment recurrence, a finite sum over ordered compositions, a closed Thue–Morse formula, and a two-level formula involving q-Pochhammer symbols and Gaussian binomial coefficients. The latter starts from the negative Laplace product, applies Mellin analysis, introduces the lower Lambert coordinate, and performs a saddle-point inversion. This report closes the structural gap between them.

The central observation is that the exact reciprocal-power identity

$$F(2^{-n}) = 2^{-\binom{n}{2}}[z^n]G(z)$$

turns the endpoint problem into a coefficient problem for the entire moment function

$$G(z) = \prod_{j\geq 1} \frac{e^{z/2^j} - 1}{z/2^j}.$$

If λ_n is the large solution of $\lambda_n 2^{-\lambda_n} = 2^{-n}$, then $n = \lambda_n - \log_2 \lambda_n$. Vertical coefficient extraction at $z = \lambda_n$ produces a local phase that is, up to an exponentially small tail, *identical* to the local phase in the continuous Bromwich proof. The apparently different periodic arguments also agree because

$$\log_2 \lambda_n = \lambda_n - n \equiv \lambda_n \pmod 1.$$

Consequently the exact dyadic formula re-derives not only the quadratic logarithmic decay but the corrected Gamma–zeta periodic fluctuation, the constant term, the first inverse-Lambert correction, and the complete all-orders expansion.

The q-binomial formula leads to the same object after an exact finite cancellation: its Mersenne/q-Pochhammer normalizer disappears, and the remaining numerator is a coefficient of $B_m(z)G(z)$, where B_m is a finite Thue–Morse prefix exponential. This gives the same all-orders asymptotic on every fixed-numerator dyadic ray $m2^{-n}$. A rescaled Bromwich identity supplies a second, exact explanation: n applications of the refinement equation map the continuous saddle 2^{λ_n} to the coefficient saddle λ_n.

The report also separates what follows for arbitrary real x from what is special to exact dyadic rays. Monotonicity and reciprocal-power values recover the global leading law

$$\log F(x) \sim -\frac{(\log x)^2}{2\log 2},$$

but the full $O(1/|\log x|)$ expansion for every real argument requires a uniform two-parameter analysis of the prefix factor B_m when its numerator grows. Thus the exact formulae do contain the endpoint asymptotic, but they reveal it only after the correct algebraic resummation and saddle scaling.

62 The missing bridge

The formal development in the repository [1] contains two powerful bodies of results. On the arithmetic side, it proves exact rational formulae for $F(m/2^n)$, develops half moments and their recurrences, gives a finite composition formula for $F(2^{-n})$, and derives a global Gaussian-binomial formula at base $q = 1/2$. On the analytic side, it derives a corrected sharp expansion of $F(x)$ as $x \downarrow 0$, including a nonconstant periodic function with explicit Gamma–zeta Fourier coefficients and an all-orders saddle series. The accompanying exposition [2] develops both sides in detail, but it does not turn the exact dyadic identities themselves into a derivation of the endpoint asymptotic.

That omission is conspicuous for three reasons.

1. The reciprocal powers 2^{-n} are simultaneously the cleanest exact arguments and a canonical sequence tending to the endpoint.
2. The exact formulae already contain geometric scales $1, 2, 4, \ldots$ through Mersenne factors, Thue–Morse blocks, and q-binomial coefficients at $q = 1/2$; the periodic term in the asymptotic also comes from a geometric dyadic harmonic sum.
3. The exact values provide rational data of arbitrarily high precision, so one expects more than a numerical comparison with an independently derived asymptotic.

The purpose of this report is to exhibit the missing mechanism. Its shortest form is the chain

$$\boxed{\begin{aligned}\text{exact dyadic value} &\longrightarrow 2^{-\binom{n}{2}}[z^n]G(z)\\ &\longrightarrow \text{vertical coefficient saddle at } z = \lambda_n\\ &\longrightarrow \text{negative-product decomposition at } \lambda_n\\ &\longrightarrow \text{the same periodic Lambert asymptotic.}\end{aligned}} \tag{656}$$

The algebraic formulae are not merely compatible with the endpoint result. After resummation they generate the same analytic phase.

62.1 Main conclusions

Let

$$L = \log 2, \qquad \lambda(x) = -\frac{1}{L}W_{-1}(-Lx), \qquad \lambda(x)2^{-\lambda(x)} = x, \tag{657}$$

where W_{-1} is the lower real branch of Lambert's W-function [7]. Define

$$\mathcal{M}(x) = -\frac{L}{2}\lambda(x)^2 + \left(1 + \frac{L}{2}\right)\lambda(x) - \frac{1}{2}\log\lambda(x) + C_F + \Psi(\lambda(x)), \tag{658}$$

where C_F and the smooth mean-zero 1-periodic function Ψ are specified in section 65.

The report proves the following statements from the exact dyadic coefficient formula.

Theorem 62.1 (Reciprocal-power asymptotic from exact coefficients). *Let $\lambda_n = \lambda(2^{-n})$, equivalently*

$$\lambda_n - \log_2 \lambda_n = n. \tag{659}$$

Then, as $n \to \infty$,

$$\log F(2^{-n}) = \mathcal{M}(2^{-n}) + \mathcal{O}(\lambda_n^{-1}). \tag{660}$$

More precisely, for every fixed $N \geq 1$ there are smooth 1-periodic functions A_j, independent of the dyadic sequence, such that

$$\log F(2^{-n}) = \mathcal{M}(2^{-n}) + \sum_{j=1}^{N-1} \frac{A_j(\lambda_n)}{\lambda_n^j} + \mathcal{O}_N(\lambda_n^{-N}). \tag{661}$$

The first coefficient is

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \tag{662}$$

These are exactly the correction functions in the continuous endpoint expansion.

Theorem 62.2 (Fixed-numerator dyadic rays). *Fix an integer $m \geq 1$. Put $x_{m,n} = m2^{-n}$ and $\lambda_{m,n} = \lambda(x_{m,n})$. Then for every fixed $N \geq 1$,*

$$\log F(m2^{-n}) = \mathcal{M}(m2^{-n}) + \sum_{j=1}^{N-1} \frac{A_j(\lambda_{m,n})}{\lambda_{m,n}^j} + \mathcal{O}_{m,N}(\lambda_{m,n}^{-N}) \tag{663}$$

as $n \to \infty$, with the same functions A_j as in theorem 62.1.

Corollary 62.3 (What exact dyadic values imply globally). *The reciprocal-power asymptotic, together with monotonicity of F, implies*

$$\frac{\log F(x)}{(\log x)^2} \longrightarrow -\frac{1}{2\log 2} \qquad (x \downarrow 0). \tag{664}$$

The first two statements retain the constant, the periodic fluctuation, and every inverse-power correction. The third statement is weaker because the mesh between consecutive reciprocal powers is too coarse for a sharp additive asymptotic in $\log F$; this distinction is analyzed in section 73.

63 Exact arithmetic in a coefficient-ready normalization

63.1 The probability law and its entire moment function

Let $F : [0, 1] \to [0, 1]$ be the Fabius function in its cumulative-distribution normalization, and let X be a random variable with distribution function F. Equivalently, one may take the convergent random series

$$X = \sum_{j \geq 0} \frac{U_j}{2^{j+1}}, \tag{665}$$

where the U_j are independent uniform random variables on $[0, 1]$. The corresponding density is the translated and rescaled Rvachev up-function. The support is $[0, 1]$, and the law is symmetric about $1/2$.

Define the half moments

$$d_0 = 1, \qquad d_n = n \int_0^1 t^{n-1}\, \mathrm{up}(t)\, \mathrm{d}t = \mathbb{E}X^n \quad (n \geq 1), \tag{666}$$

and the entire moment-generating function

$$G(z) = \mathbb{E}e^{zX} = \sum_{n\geq 0} d_n \frac{z^n}{n!}. \tag{667}$$

The refinement equation gives

$$G(2z) = \frac{e^z - 1}{z} G(z). \tag{668}$$

Symmetry of X about $1/2$ gives the second exact relation

$$G(z) = e^z G(-z). \tag{669}$$

It is convenient to remove the factorial from the moments:

$$a_n = \frac{d_n}{n!} = [z^n]G(z). \tag{670}$$

The half-moment recurrence becomes

$$(2^n - 1)a_n = \sum_{j=0}^{n-1} \frac{a_j}{(n-j+1)!}, \qquad a_0 = 1. \tag{671}$$

63.2 Reciprocal powers and the triangular factor

The exact bridge between moments and endpoint values is

$$\boxed{F(2^{-n}) = \frac{d_n}{n!\,2^{\binom{n}{2}}} = 2^{-\binom{n}{2}} a_n.} \tag{672}$$

The factor $2^{-\binom{n}{2}}$ already accounts for the leading quadratic term $-(\log 2)n^2/2$ in the logarithm. It does *not*, by itself, account for the $n \log n$ term, the Lambert phase, the constant, or the periodic fluctuation. Those are hidden in the much subtler coefficient a_n.

Writing $V_n = F(2^{-n})$ and normalizing $A_n = 2^{\binom{n}{2}} V_n = a_n$, one obtains the triangular recurrence

$$A_n = \sum_{k=0}^{n-1} \frac{A_k}{(2^n - 1)(n-k+1)!}. \tag{673}$$

Iterating it over all directed paths from 0 to n gives the finite composition formula

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \models n} \prod_{j=1}^{\mathrm{len}(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}, \tag{674}$$

where $\rho = (r_1, \ldots, r_\ell)$ is an ordered composition of n and $s_j = r_1 + \cdots + r_j$. Formula (674) is exact and positive, but it contains 2^{n-1} terms. Its asymptotic structure becomes visible only after the path sum is resummed into a generating function; see section 64.1.

63.3 The global q-binomial formula

For

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j) \tag{675}$$

and the Gaussian coefficient $\left[{n \atop k}\right]_q$, the formal development proves, for the signed global Fabius extension $\mathcal{F}$,

$$\mathcal{F}(m/2^n) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\left[{n \atop k}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \times \sum_{r=0}^{m2^k-1} (-1)^{\mathrm{wt}_2(r)}(r - m2^k + Q)^{n+k}, \tag{676}$$

independently of the complex translation parameter Q. For $0 \le m \le 2^n$, the left side is the bounded Fabius value $F(m/2^n)$.

At first sight (676) seems to offer a direct route to an asymptotic: the prefactor contains 2^{-n^2} and the finite product $(1/2;1/2)_n$. That reading is misleading. The double sum undergoes a cancellation of exactly the same scale. The correct first step is an exact algebraic collapse, not termwise estimation.

Define the Thue–Morse signs and prefix exponential

$$\varepsilon_h = (-1)^{\mathrm{wt}_2(h)}, \qquad B_m(z) = \sum_{h=0}^{m-1} \varepsilon_h \mathrm{e}^{(m-1-h)z}, \qquad B_0 = 0, \tag{677}$$

and the Mersenne product

$$\mathfrak{m}_n = \prod_{\ell=1}^{n} (2^\ell - 1). \tag{678}$$

The exact finite q-binomial extraction theorem gives

$$\boxed{F(m/2^n) = 2^{-\binom{n}{2}}[z^n]\big(B_m(z)G(z)\big).} \tag{679}$$

For $m = 1$, $B_1 = 1$ and this reduces to (672). The reason the normalizer disappears is

$$(1/2;1/2)_n = 2^{-\binom{n+1}{2}}\mathfrak{m}_n, \tag{680}$$

and the centered numerator of (676) is exactly

$$\mathfrak{m}_n[z^n]\big(B_m(z)G(z)\big). \tag{681}$$

Thus

$$\frac{\mathfrak{m}_n}{2^{n^2}(1/2;1/2)_n} = 2^{-\binom{n}{2}}. \tag{682}$$

The q-Pochhammer factor supplies the same triangular power already present in (672); the remaining endpoint asymptotic resides in a coefficient of the entire product G.

Remark 63.1 (Why termwise estimates fail). The individual terms in (676) are vastly larger than their sum. The inner Thue–Morse block annihilates low polynomial degrees, and the outer Gaussian row annihilates the remaining lower powers on the geometric node set $-1, -2, \ldots, -2^n$. Estimating before these exact cancellations discards the quantity one is trying to compute. Formula (679) is therefore not a cosmetic rewrite; it is the asymptotically meaningful normal form.

64 Resumming the exact recurrences

64.1 From the triangular recurrence to the entire product

Let

$$\mathscr{A}(z) = \sum_{n\geq 0} a_n z^n = G(z). \tag{683}$$

Multiplying (671) by z^n and summing over $n \geq 1$ gives

$$\begin{aligned}\mathscr{A}(2z) - \mathscr{A}(z) &= \mathscr{A}(z) \sum_{r\geq 1} \frac{z^r}{(r+1)!} \\ &= \mathscr{A}(z)\left(\frac{\mathrm{e}^z - 1}{z} - 1\right),\end{aligned}$$

so

$$\mathscr{A}(2z) = \frac{\mathrm{e}^z - 1}{z} \mathscr{A}(z). \tag{684}$$

This is exactly (668). Iteration toward the origin yields the locally uniformly convergent product

$$\boxed{G(z) = \prod_{j=1}^{\infty} \frac{\mathrm{e}^{z/2^j} - 1}{z/2^j}.} \tag{685}$$

The path sum (674), the recurrence (673), the functional equation (684), and the product (685) are four presentations of one object. The first two are finite and exact at each n; the last two are the forms suited to asymptotics.

64.2 A probabilistic interpretation of the product

Each factor is a uniform moment-generating function:

$$\frac{\mathrm{e}^w - 1}{w} = \int_0^1 \mathrm{e}^{wu}\, \mathrm{d}u. \tag{686}$$

Thus (685) is precisely the moment-generating function of the random series (665). This gives several useful estimates for free. For real $r > 0$ and real t,

$$|G(r + \mathrm{i}t)| \leq G(r), \tag{687}$$

because $|\mathrm{e}^{(r+\mathrm{i}t)X}| = \mathrm{e}^{rX}$. The bound will control the noncentral part of the coefficient integral without a separate zero analysis.

64.3 Positive and negative arguments

Define, for $s > 0$,

$$\Lambda(s) = \log G(-s). \tag{688}$$

Using (669),

$$\log G(s) = s + \Lambda(s). \tag{689}$$

At the product level,

$$G(-s) = \prod_{j=1}^{\infty} \frac{1 - \mathrm{e}^{-s/2^j}}{s/2^j}, \tag{690}$$

$$\Lambda(s) = \sum_{j=1}^{\infty} \kappa(s/2^j), \qquad \kappa(u) = \log \frac{1 - \mathrm{e}^{-u}}{u}. \tag{691}$$

This identity is the point where the exact moment product meets the endpoint asymptotic. No new transform has appeared: the negative Laplace product used in the continuous proof is the negative-real restriction of the entire product forced by the exact recurrence.

65 The periodic decomposition of the negative product

This section records the analytic information about (691) needed by the coefficient saddle. The derivation is included because it identifies exactly where the $q = 1/2$ geometry reappears in the asymptotic.

65.1 Mellin transform of the dyadic kernel

For $-1 < \Re z < 0$,

$$\widetilde{\kappa}(z) = \int_0^\infty \kappa(u)u^{z-1}\,\mathrm{d}u = -\Gamma(z)\zeta(1+z). \tag{692}$$

Indeed, $\kappa(u) = -u/2 + \mathcal{O}(u^2)$ at the origin and $\kappa(u) = -\log u + \mathcal{O}(\mathrm{e}^{-u})$ at infinity. Integration by parts, followed by the subtracted Bose integral, gives (692).

The geometric harmonic sum contributes

$$\widetilde{\Lambda}(z) = \frac{2^z}{1-2^z}\bigl(-\Gamma(z)\zeta(1+z)\bigr), \qquad -1 < \Re z < 0. \tag{693}$$

The factor

$$\frac{2^z}{1-2^z} \tag{694}$$

is the analytic image of the same dyadic scale that produced the Mersenne factors and the half-base Gaussian coefficients in the exact arithmetic. Its poles are

$$\chi_k = \frac{2\pi\mathrm{i}k}{L}, \qquad k \in \mathbb{Z}. \tag{695}$$

The triple pole at 0 produces a quadratic polynomial in $\log s$ and a constant; the nonzero poles produce a periodic Fourier series in $\log_2 s$.

65.2 Exact decomposition, mean, and oscillation

There is a smooth 1-periodic function P and a positive forward-tail term $E(s)$ such that

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s), \tag{696}$$

where

$$0 \le E(s) \le 4\mathrm{e}^{-s} \tag{697}$$

for all sufficiently large s. Put

$$\overline{P} = \int_0^1 P(t)\,\mathrm{d}t, \qquad \Psi(t) = P(t) - \overline{P}. \tag{698}$$

Then

$$\Psi(t) = \sum_{k\neq 0} \widehat{\Psi}(k)\mathrm{e}^{2\pi\mathrm{i}kt}, \qquad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}. \tag{699}$$

The Fourier coefficients decay exponentially in $|k|$, so Ψ is real analytic. It is nonconstant, although its amplitude is only a few parts in 10^6.

The finite part at the Mellin origin is

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12}, \tag{700}$$

where γ is Euler's constant and γ_1 is the first Stieltjes constant. One has

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}, \tag{701}$$

and the endpoint constant after the Gaussian normalization is

$$C_F = \overline{P} - \frac{1}{2}\log(2\pi) = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log\pi \approx -2.02798389863885942397. \tag{702}$$

Remark 65.1 (The same dyadic geometry twice)*.* In the exact formula, the geometric nodes $-1, -2, \ldots, -2^n$ are isolated by the finite half-base q-binomial theorem. In the asymptotic, the infinite scale family $s/2, s/4, \ldots$ is summed by the Mellin factor $2^z/(1-2^z)$. Finite q-interpolation and Mellin periodicity are therefore two transforms of the same refinement grid.

66 Vertical coefficient extraction

The usual Cauchy coefficient integral on a circle is enough for a saddle analysis. A vertical representation is better here because it makes the coefficient phase identical to the continuous Bromwich phase.

Lemma 66.1 (Vertical extraction of moments)**.** *For every $n \geq 1$ and every $c > 0$,*

$$a_n = [z^n]G(z) = \frac{1}{2\pi \mathrm{i}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{G(z)}{z^{n+1}}\,\mathrm{d}z. \tag{703}$$

The integral is absolutely convergent.

Proof. On the line $\Re z = c$, (687) gives $|G(z)| \leq G(c)$, while $|z|^{-n-1}$ is integrable for $n \geq 1$. Using $G(z) = \mathbb{E}\mathrm{e}^{zX}$, Fubini's theorem and the classical inverse-Laplace formula give

$$\begin{aligned} \frac{1}{2\pi \mathrm{i}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{G(z)}{z^{n+1}}\,\mathrm{d}z &= \mathbb{E}\left[\frac{1}{2\pi \mathrm{i}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{\mathrm{e}^{zX}}{z^{n+1}}\,\mathrm{d}z\right] \\ &= \mathbb{E}\frac{X^n}{n!} = \frac{d_n}{n!} = a_n. \end{aligned}$$

□

Combining (672) and (703) gives the exact contour formula

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{2\pi \mathrm{i}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{G(z)}{z^{n+1}}\,\mathrm{d}z. \tag{704}$$

This is obtained entirely from the exact moment identity; no inversion formula for F has been used.

67 The reciprocal-power saddle

67.1 The Lambert coordinate is forced by coefficient balance

Let $\lambda = \lambda_n$ be the large solution of (659). It is not introduced by analogy with the continuous proof. It is the leading solution of the coefficient saddle equation.

Indeed, by (689) and (696),

$$\log G(r) = r - \frac{(\log r)^2}{2L} + \frac{\log r}{2} + P(\log_2 r) + E(r). \tag{705}$$

The logarithmic derivative satisfies

$$r\frac{G'(r)}{G(r)} = r - \log_2 r + \frac{1}{2} + \frac{\Psi'(\log_2 r)}{L} + \mathcal{O}(re^{-r}). \tag{706}$$

The exact Hayman radius differs by only $\mathcal{O}(1)$ from the solution of $r - \log_2 r = n$, and using the latter changes the logarithm of the coefficient only by $\mathcal{O}(1/r)$. The Lambert coordinate is therefore intrinsic to the coefficient problem.

67.2 Exact coefficient saddle mass

Set $c = \lambda$ in (704) and write $z = \lambda(1 + \mathrm{i}\theta)$. Define

$$\mathcal{C}_n = \sqrt{\frac{\lambda}{2\pi}} \int_{-\infty}^{\infty} \frac{G(\lambda(1+\mathrm{i}\theta))}{G(\lambda)} (1+\mathrm{i}\theta)^{-n-1}\,\mathrm{d}\theta. \tag{707}$$

Then

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}} G(\lambda)\lambda^{-n}}{\sqrt{2\pi\lambda}} \mathcal{C}_n. \tag{708}$$

Using $G(\lambda) = \exp(\lambda + \Lambda(\lambda))$,

$$\log F(2^{-n}) = -L\binom{n}{2} + \lambda + \Lambda(\lambda) - n\log\lambda - \frac{1}{2}\log(2\pi\lambda) + \log \mathcal{C}_n. \tag{709}$$

67.3 Phase locking

The Lambert equation gives the exact integer relation

$$\log_2 \lambda = \lambda - n. \tag{710}$$

Since Ψ is 1-periodic,

$$\boxed{\Psi(\log_2 \lambda) = \Psi(\lambda).} \tag{711}$$

This identity is the missing connection between the phase produced by the moment product and the phase used in the endpoint expansion. A naive coefficient estimate sees the periodic environment at $\log_2 \lambda$; the exact dyadic saddle equation shifts that argument by the integer n.

Substitute (696) into (709). Put $q = \log\lambda = L(\lambda - n)$. The nonconstant part becomes $\Psi(\lambda)$ by (711), and the polynomial terms satisfy the exact cancellation

$$\begin{aligned} &-\frac{L}{2}(n^2 - n) + \lambda - \frac{q^2}{2L} + \frac{q}{2} - nq \\ &\qquad = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda. \end{aligned} \tag{712}$$

Together with (702), this gives an exact separation analogous to the one in the continuous saddle proof.

Proposition 67.1 (Exact dyadic separation)**.** *For $\lambda = \lambda_n$,*

$$\boxed{\log F(2^{-n}) - \mathcal{M}(2^{-n}) = E(\lambda) + \log \mathcal{C}_n.} \tag{713}$$

All algebraic normalization, the constant, and the periodic term have now been extracted from an exact rational value. The only remaining task is to show that the dimensionless coefficient saddle mass is $1 + \mathcal{O}(1/\lambda)$ and to continue its expansion.

67.4 The local phase is the continuous phase

Let

$$u = \operatorname{Log}(1 + \mathrm{i}\theta). \tag{714}$$

On a fixed neighborhood of $\theta = 0$, the complex continuation of (696) is valid. The exponent in (707) is

$$\Phi_\lambda^{\mathrm{coef}}(\theta) = \log \frac{G(\lambda(1+\mathrm{i}\theta))}{G(\lambda)} - (n+1)u. \tag{715}$$

Using $G(z) = \mathrm{e}^z G(-z)$, the periodic decomposition, and $n + \log_2 \lambda = \lambda$, one obtains

$$\Phi_\lambda^{\mathrm{coef}}(\theta) = \lambda(\mathrm{i}\theta - u) - \frac{u}{2} - \frac{u^2}{2L} + \Psi\left(\lambda + \frac{u}{L}\right) - \Psi(\lambda) + \mathcal{E}_\lambda^{\mathrm{coef}}(\theta), \tag{716}$$

where $\mathcal{E}_\lambda^{\mathrm{coef}}(0) = 0$ and every derivative of the error is $\mathcal{O}(\mathrm{e}^{-\lambda})$ on a fixed central neighborhood.

Formula (716) is the decisive identity. Apart from an exponentially small tail, it is exactly the phase obtained by scaling the continuous Bromwich integral at its saddle. Therefore every local Gaussian polynomial and every all-orders correction coefficient must coincide.

67.5 Gaussian domination and the leading mass

The central scale is $\theta = v/\sqrt{\lambda}$. Since

$$\operatorname{Log}(1 + \mathrm{i}\theta) = \mathrm{i}\theta + \frac{\theta^2}{2} - \frac{\mathrm{i}\theta^3}{3} - \frac{\theta^4}{4} + \mathcal{O}(\theta^5), \tag{717}$$

the leading term in (716) is $-\lambda\theta^2/2$.

The tails are especially simple in the vertical representation. By (687),

$$\left| \frac{G(\lambda(1+\mathrm{i}\theta))}{G(\lambda)} (1+\mathrm{i}\theta)^{-n-1} \right| \le (1+\theta^2)^{-(n+1)/2}. \tag{718}$$

Since $n \sim \lambda$, this gives Gaussian decay near the origin and exponential decay outside any fixed neighborhood. Taylor expansion on a window growing more slowly than $\lambda^{1/6}$, followed by (718), gives

$$\mathcal{C}_n = 1 + \mathcal{O}(\lambda^{-1}). \tag{719}$$

The order $\lambda^{-1/2}$ term vanishes after integration because it is odd and purely imaginary.

Combining (713), (697), and (719) proves (660).

67.6 First correction

Substitute $\theta = v/\sqrt{\lambda}$ into (716). Through order λ^{-1},

$$\Phi_\lambda^{\mathrm{coef}}(v/\sqrt{\lambda}) = -\frac{v^2}{2} + \lambda^{-1/2} P_1(v;\lambda) + \lambda^{-1} P_2(v;\lambda) + \mathcal{O}\left(\lambda^{-3/2}(1 + |v|^9)\right), \tag{720}$$

where

$$P_1(v;u) = \mathrm{i}\left[\left(\frac{\Psi'(u)}{L} - \frac{1}{2}\right)v + \frac{v^3}{3}\right], \tag{721}$$

$$P_2(v;u) = \frac{v^4}{4} + \left[-\frac{1}{4} + \frac{1}{2L} + \frac{\Psi'(u)}{2L} - \frac{\Psi''(u)}{2L^2}\right]v^2. \tag{722}$$

If Z is standard normal, the relative coefficient of $1/\lambda$ is

$$\mathbb{E}\left[P_2(Z;u) + \frac{1}{2}P_1(Z;u)^2\right] = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \tag{723}$$

This is precisely $A_1(u)$ in (662). Since the logarithm of $1 + A_1/\lambda + \mathcal{O}(\lambda^{-2})$ has the same first coefficient,

$$\log \mathcal{C}_n = \frac{A_1(\lambda_n)}{\lambda_n} + \mathcal{O}(\lambda_n^{-2}). \tag{724}$$

Thus the first correction in the continuous asymptotic has been recovered from the exact coefficient formula.

67.7 All orders

At order N, expand (716) after the central scaling through λ^{-N}. The coefficients are polynomials in v whose scalar coefficients are differential polynomials in

$$\Psi'(\lambda), \Psi''(\lambda), \ldots, \Psi^{(2N)}(\lambda). \tag{725}$$

After exponentiation, all odd powers of $\lambda^{-1/2}$ integrate to zero by parity. Gaussian moments convert the remaining terms into integer powers of $1/\lambda$. The tail bound (718) justifies termwise integration and gives an arbitrarily small remainder. Consequently

$$\log \mathcal{C}_n = \sum_{j=1}^{N-1} \frac{A_j(\lambda_n)}{\lambda_n^j} + \mathcal{O}_N(\lambda_n^{-N}), \tag{726}$$

with the same A_j as in the continuous saddle expansion, because the local phase is the same. This completes the proof of theorem 62.1.

68 The elementary n-scale expansion

The Lambert form is the cleanest because it preserves the exact periodic argument. Expanding the phase in the integer parameter gives a more familiar formula. The large solution of (659) satisfies

$$\lambda_n = n + \frac{\log n}{L} + \frac{\log n}{L^2 n} + \frac{\log n - \frac{1}{2}\log^2 n}{L^3 n^2} + \mathcal{O}\left(\frac{\log^3 n}{n^3}\right). \tag{727}$$

Substitution into (660), with the exact $\Psi(\lambda_n)$ retained, gives

$$\begin{aligned} \log F(2^{-n}) = &-\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ &- \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \end{aligned} \tag{728}$$

This is now a consequence of the exact rational coefficient sequence rather than an independent specialization of the continuous theorem.

Formula (728) also clarifies the roles of the factors in (672):

- $2^{-\binom{n}{2}}$ gives the quadratic term and part of the linear term;
- $a_n = d_n/n!$ supplies the cancellation that creates $-n\log n$ and the remaining drift;
- the dyadic product for G supplies C_F and Ψ;
- Gaussian coefficient extraction supplies $-\frac{1}{2}\log\lambda_n$ and the functions A_j.

69 Fixed-numerator dyadic rays

The coefficient form (679) makes it possible to repeat the derivation for $m2^{-n}$ with fixed m.

69.1 The prefix factor near the saddle

Write

$$B_m(z) = \mathrm{e}^{(m-1)z}\mathcal{S}_m(z), \qquad \mathcal{S}_m(z) = \sum_{h=0}^{m-1} \varepsilon_h \mathrm{e}^{-hz}. \tag{729}$$

For $\Re z = r > 0$,

$$|\mathcal{S}_m(z) - 1| \le \sum_{h\ge1} \mathrm{e}^{-hr} = \frac{\mathrm{e}^{-r}}{1-\mathrm{e}^{-r}}. \tag{730}$$

For fixed m and $r \to \infty$, the entire prefix is therefore dominated by its first exponential:

$$B_m(z) = \mathrm{e}^{(m-1)z}\left(1 + \mathcal{O}_m(\mathrm{e}^{-r})\right), \tag{731}$$

uniformly on vertical lines and, with derivatives, on a fixed central window.

69.2 The ray saddle and its phase lock

Let

$$x = m2^{-n}, \qquad \lambda = \lambda(x), \qquad r = \frac{\lambda}{m}. \tag{732}$$

The Lambert equation implies

$$n = \lambda - \log_2\lambda + \log_2 m = mr - \log_2 r. \tag{733}$$

Equivalently,

$$\log_2 r = \lambda - n, \qquad \Psi(\log_2 r) = \Psi(\lambda). \tag{734}$$

Thus the exact periodic phase locks in the same way as for $m = 1$.

For fixed m and all sufficiently large n, $\mathcal{S}_m(r) > 0$, so the following real logarithms are well defined. Apply vertical coefficient extraction to $H_m(z) = B_m(z)G(z)$:

$$[z^n]H_m(z) = \frac{1}{2\pi\mathrm{i}}\int_{r-\mathrm{i}\infty}^{r+\mathrm{i}\infty} \frac{H_m(z)}{z^{n+1}}\,\mathrm{d}z. \tag{735}$$

Define the normalized ray saddle mass

$$\mathcal{C}_{m,n} = \sqrt{\frac{\lambda}{2\pi}}\int_{-\infty}^{\infty} \frac{B_m(r(1+\mathrm{i}\theta))G(r(1+\mathrm{i}\theta))}{B_m(r)G(r)}(1+\mathrm{i}\theta)^{-n-1}\,\mathrm{d}\theta. \tag{736}$$

Then the exact coefficient formula gives

$$\begin{aligned}\log F(m2^{-n}) = &-L\binom{n}{2} + \log B_m(r) + \log G(r) - n\log r \\ &-\frac{1}{2}\log(2\pi\lambda) + \log\mathcal{C}_{m,n}.\end{aligned} \tag{737}$$

Using (729) and (689),

$$\log B_m(r) + \log G(r) = \lambda + \Lambda(r) + \log \mathcal{S}_m(r). \tag{738}$$

The same algebra as (712), now with $\log r = L(\lambda - n)$, yields the exact separation

$$\boxed{\log F(m2^{-n}) - \mathcal{M}(m2^{-n}) = E(r) + \log \mathcal{S}_m(r) + \log \mathcal{C}_{m,n}.} \tag{739}$$

69.3 Identity of local phases

On the central window, (731) gives

$$\begin{aligned}\log \frac{B_m(r(1+\mathrm{i}\theta))G(r(1+\mathrm{i}\theta))}{B_m(r)G(r)} &- (n+1)u \\ = \lambda(\mathrm{i}\theta - u) - \frac{u}{2} - \frac{u^2}{2L} &+ \Psi\left(\lambda + \frac{u}{L}\right) - \Psi(\lambda) + \mathcal{E}_{m,\lambda}(\theta),\end{aligned}$$

where every derivative of $\mathcal{E}_{m,\lambda}$ is $\mathcal{O}_m(\mathrm{e}^{-\lambda/m})$. This is again (716). The vertical tails are bounded by the denominator $(1+\theta^2)^{-(n+1)/2}$ times a fixed m-dependent constant. Therefore

$$\log \mathcal{C}_{m,n} = \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}_{m,N}(\lambda^{-N}). \tag{740}$$

The two additional terms in (739) are exponentially small for fixed m. This proves theorem 62.2.

69.4 A moderate-numerator extension

The estimate (730) is independent of m. Its derivatives on the central scale introduce only fixed powers of the reduced saddle $r = \lambda/m$. Consequently the same argument remains uniform whenever the exponential prefix error dominates every such polynomial loss.

Proposition 69.1 (A useful uniform regime)**.** *Let $m = m(n)$, put $\lambda = \lambda(m2^{-n})$, and suppose*

$$\frac{\lambda/m}{\log \lambda} \longrightarrow \infty. \tag{741}$$

Then, for every fixed $N \geq 1$, the expansion in (663) holds through order λ^{-N} uniformly along that sequence.

Indeed, every fixed collection of derivatives of the prefix correction is bounded by $\mathcal{O}_N(r^{K_N}\mathrm{e}^{-r})$ for some finite K_N, and (741) makes this smaller than every power of $1/\lambda$. The condition is equivalent to $m = o(\lambda/\log \lambda)$. It does not cover the fine dyadic meshes needed to approximate every real x at the sharp logarithmic scale; see section 73.

70 An exact Bromwich-to-coefficient collapse

The coefficient derivation above is already sufficient. A second identity shows directly how the continuous endpoint saddle and the exact dyadic coefficient saddle are the same object viewed before and after n refinement steps.

70.1 Iterating the negative refinement equation

From (668),

$$G(-2z) = \frac{1-\mathrm{e}^{-z}}{z} G(-z). \tag{742}$$

After n iterations,

$$G(-2^n z) = 2^{-\binom{n}{2}} z^{-n} G(-z) \prod_{j=0}^{n-1} (1-\mathrm{e}^{-2^j z}). \tag{743}$$

The finite product is the Thue–Morse block polynomial

$$P_n(z) = \prod_{j=0}^{n-1} (1-\mathrm{e}^{-2^j z}) = \sum_{h=0}^{2^n-1} \varepsilon_h \mathrm{e}^{-hz}. \tag{744}$$

For $0 < x < 1$, the Laplace–Stieltjes inversion formula is

$$F(x) = \frac{1}{2\pi\mathrm{i}} \int_{R-\mathrm{i}\infty}^{R+\mathrm{i}\infty} \frac{\mathrm{e}^{sx} G(-s)}{s} \,\mathrm{d}s, \qquad R > 0. \tag{745}$$

Set $x = 2^{-n}$ and substitute $s = 2^n z$. Then

$$\begin{aligned} F(2^{-n}) &= \frac{1}{2\pi\mathrm{i}} \int \frac{\mathrm{e}^{z} G(-2^n z)}{z} \,\mathrm{d}z \\ &= \frac{2^{-\binom{n}{2}}}{2\pi\mathrm{i}} \int \frac{G(z) P_n(z)}{z^{n+1}} \,\mathrm{d}z, \end{aligned} \tag{746}$$

where (669) was used in the second line.

70.2 Why the finite product disappears exactly

Insert (744) and $G(z) = \mathbb{E}\mathrm{e}^{zX}$ into (746). The inverse-Laplace kernel gives

$$F(2^{-n}) = 2^{-\binom{n}{2}} \mathbb{E}\left[\sum_{h=0}^{2^n-1} \varepsilon_h \frac{(X-h)_+^n}{n!} \right]. \tag{747}$$

Because $0 \le X \le 1$, all terms with $h \ge 1$ vanish almost surely. The $h = 0$ term is $\mathbb{E}X^n/n! = a_n$, and (672) follows. Thus the exact reciprocal-power bridge itself can be read as a support-induced collapse of the rescaled Bromwich integral.

Near the saddle this collapse has an even simpler form. On the line $\Re z = \lambda_n$,

$$P_n(z) = 1 + \mathcal{O}(\mathrm{e}^{-\lambda_n}) \tag{748}$$

uniformly in $\Im z$, because $\sum_{j\ge 0} \mathrm{e}^{-2^j \lambda_n} = \mathcal{O}(\mathrm{e}^{-\lambda_n})$. The continuous saddle radius is

$$R = \frac{\lambda_n}{2^{-n}} = 2^n \lambda_n = 2^{\lambda_n}, \tag{749}$$

where the last equality is (659). The substitution $s = 2^n z$ maps R exactly to $z = \lambda_n$, and (743) maps the continuous integrand to the coefficient integrand, up to the beyond-all-orders factor $P_n(z)$. This is the most literal form of the connection sought in this report.

71 What the different exact formulae contribute

The exact formulae are not interchangeable as asymptotic tools. Their roles can be summarized as follows.

Exact representation	What it exposes directly	What must be done before asymptotics
Half-moment identity $F(2^{-n}) = 2^{-\binom{n}{2}} d_n/n!$	Separates the universal triangular power of two and identifies the residual as a moment coefficient.	Replace $d_n/n!$ by $[z^n]G(z)$ and perform coefficient extraction.
Triangular recurrence (673)	Gives exact rational computation and a positive recursion.	Sum over n to recover $G(2z) = ((e^z - 1)/z)G(z)$.
Composition/path formula (674)	Gives a finite nonrecursive sum and a combinatorial interpretation of the triangular factor.	Resum the 2^{n-1} paths into the same functional equation; termwise asymptotics obscure the collective saddle.
q-binomial formula (676)	Makes the geometric interpolation and two levels of exact cancellation explicit.	Perform the finite Gaussian-binomial extraction first. The Mersenne and q-Pochhammer factors then cancel, leaving (679).
Closed Thue–Morse formula	Gives every dyadic numerator and reveals binary blocks.	Convert it to $[z^n](B_m G)$; for fixed m, the leading exponential in B_m supplies the ray saddle.
Bit/Taylor recursion	Is efficient for exact evaluation at a specified dyadic argument.	It is locally subtractive and hides the global saddle; it is best used for numerical verification, not as a termwise asymptotic expansion.

71.1 Why the q-Pochhammer factor is not the periodic term

A tempting but incorrect heuristic is to take logarithms of $(1/2;1/2)_n$ and seek the endpoint constant or oscillation there. The finite product converges to a nonzero constant after its explicit triangular normalization, but in (676) it is paired with a numerator containing the same Mersenne product. Their exact cancellation occurs before any limit. The nonconstant periodic term instead comes from the *infinite* product for G and the nonzero imaginary-axis poles of the Mellin factor (694).

71.2 Why the composition sum must be resummed

The composition weights are positive, so one might hope for a direct probabilistic limit theorem on random compositions. The difficulty is that the endpoint index s_j appears in the Mersenne factor $2^{s_j} - 1$, while the jump size r_j appears in $(r_j + 1)!$. The dominant path length and jump profile drift with n, and no fixed finite family of compositions controls the sum. The generating function performs the exact renewal resummation and turns that drifting ensemble into the single analytic saddle of G.

72 Exact-rational numerical checks

The following computations use the recurrence (671) in exact rational arithmetic. Floating-point conversion occurs only after the rational value has been formed. The periodic function and its derivatives are evaluated from the Fourier series (699).

Let

$$R_n = \log F(2^{-n}) - \mathcal{M}(2^{-n}). \tag{750}$$

The first-order prediction is $R_n = A_1(\lambda_n)/\lambda_n + \mathcal{O}(\lambda_n^{-2})$.

n	λ_n	$\log F(2^{-n})$	R_n	$\lambda_n R_n$	$\lambda_n^2(R_n - A_1/\lambda_n)$
12	16.0000000000	−70.54224031962472	0.04969809560	0.7951695295	0.5134917380
24	28.8505257022	−253.3041353903491	0.02706676636	0.7808904386	0.5161104586
48	53.7481430054	−932.8335151659519	0.01437385195	0.7725678503	0.5168295087
96	102.682040054	−3520.201176273548	0.007479125975	0.7679719130	0.5174474457

Table 17: Reciprocal-power values computed exactly. The scaled residual $\lambda_n R_n$ approaches the periodic first coefficient $A_1(\lambda_n)$, which is approximately 0.763 on these phases. After subtracting that term, multiplication by λ_n^2 produces a bounded second-order quantity.

The fixed-numerator theorem can be tested independently by extracting $[z^n](B_m G)$ exactly. The table reports

$$R_{m,n} = \log F(m2^{-n}) - \mathcal{M}(m2^{-n}). \tag{751}$$

m	n	$\lambda_{m,n}$	$\lambda_{m,n}R_{m,n}$	$A_1(\lambda_{m,n})$	$\lambda_{m,n}^2(R_{m,n} - A_1/\lambda_{m,n})$
1	20	24.62186833	0.7836542295	0.7629268732	0.5104
1	40	45.50804986	0.7742689337	0.7629490545	0.5151
1	80	86.43351899	0.7689739455	0.7629823190	0.5179
3	20	22.93448405	0.7855611616	0.7630465477	0.5164
3	40	43.87020711	0.7748252957	0.7630120722	0.5182
3	80	84.82139378	0.7691025485	0.7629858145	0.5188
5	20	22.14711904	0.7864070706	0.7631005942	0.5162
5	40	43.10795409	0.7751701258	0.7631010621	0.5203
5	80	84.07161885	0.7692992153	0.7630968112	0.5214

Table 18: Exact coefficient checks on the rays $m2^{-n}$. The same first correction A_1 works for $m = 1, 3, 5$, and the twice-scaled remainder stays bounded, as predicted by theorem 62.2.

The numerical convergence is deliberately slow at first order: if $R = A_1/\lambda + A_2/\lambda^2 + \cdots$, then $\lambda R = A_1 + A_2/\lambda + \cdots$. The tables show exactly this drift and then stabilize it by subtracting A_1/λ.

73 From exact dyadics to arbitrary real arguments

73.1 The leading law follows by squeezing

Let $x = 2^{-t}$ and choose the integer n with $n \le t < n + 1$. Monotonicity gives

$$F(2^{-(n+1)}) \le F(x) \le F(2^{-n}). \tag{752}$$

From (728),

$$\log F(2^{-n}) = -\frac{L}{2}n^2 + \mathcal{O}(n \log n). \tag{753}$$

The same estimate holds with $n + 1$, and $t = n + \mathcal{O}(1)$. Dividing the logarithm of (752) by t^2 gives

$$\log F(2^{-t}) \sim -\frac{L}{2}t^2. \tag{754}$$

Since $\log x = -Lt$, this is (664). Thus the exact reciprocal-power sequence does recover the universal leading endpoint behavior for every real argument.

73.2 Why squeezing cannot recover the sharp periodic formula

The logarithmic gap between neighboring exact samples is of order n:

$$\log F(2^{-n}) - \log F(2^{-(n+1)}) = Ln + \mathcal{O}(\log n). \tag{755}$$

It diverges. Therefore monotonicity between consecutive reciprocal powers cannot determine even the constant term in $\log F(x)$, much less an error of order $1/\lambda$.

To approximate x finely enough for an additive $o(1)$ error in $\log F(x)$, a dyadic grid needs relative spacing much smaller than $1/\lambda$. At level n, that means a reduced numerator of size much larger than λ. The fixed- or moderate-numerator analysis in section 69 instead assumes that $r = \lambda/m$ is large. The two regimes do not overlap.

73.3 The remaining two-parameter problem

For a general dyadic sequence $x_n = m_n 2^{-n}$, the exact coefficient is

$$F(x_n) = 2^{-\binom{n}{2}}[z^n]\big(B_{m_n}(z)G(z)\big). \tag{756}$$

When $m_n \gg \lambda_n$, the candidate radius $r_n = \lambda_n/m_n$ is small and the prefix polynomial

$$\mathcal{S}_{m_n}(r_n) = \sum_{h<m_n} \varepsilon_h \mathrm{e}^{-hr_n} \tag{757}$$

can no longer be replaced by 1. Its behavior depends on a long Thue–Morse prefix and, through binary refinement, on the numerator's digits. A uniform saddle theorem in this regime would amount to a two-parameter de-Poissonization of the exact dyadic formula. It is a legitimate route to the full continuous asymptotic, but it is no simpler than retaining the negative Laplace transform and applying the existing Bromwich analysis directly.

This is the precise boundary of the present re-derivation:

- exact coefficients recover the complete sharp expansion on reciprocal powers and all fixed-numerator rays;
- exact coefficients plus monotonicity recover the leading quadratic law for all real x;
- the full uniform sharp expansion for arbitrary x requires control of growing prefix factors, which is the continuous problem in another coordinate system.

74 A formalization roadmap

Most algebraic ingredients are already formalized in the repository. The new bridge can be organized into a comparatively small analytic layer.

1. **Normalized coefficient sequence.** Package $a_n = d_n/n! = 2^{\binom{n}{2}}F(2^{-n})$ and identify its ordinary generating function with G.
2. **Vertical coefficient extraction.** Prove (703) from the bounded-support moment representation. For $n \geq 1$, absolute convergence follows from $|G(c+\mathrm{i}t)| \leq G(c)$.
3. **Exact dyadic saddle mass.** Define $\mathcal{C}_n$ by (707) and prove the exact factorization (708).
4. **Phase locking.** From $\lambda_n 2^{-\lambda_n} = 2^{-n}$ prove $\log_2 \lambda_n = \lambda_n - n$, and use periodicity to rewrite $\Psi(\log_2 \lambda_n)$ as $\Psi(\lambda_n)$.
5. **Exact separation.** Formalize the algebraic identity (712) and derive (713).

6. **Reuse the generic saddle machinery.** Prove (716). Its main part is literally the phase already used by the all-orders endpoint modules; the only new tail is exponentially small in λ rather than in 2^{λ}.
7. **Prefix rays.** Use the elementary bound (730) to lift the same proof to fixed m, yielding (739) and theorem 62.2.
8. **Optional rescaled-Bromwich theorem.** Formalize (743) and the support collapse (747). This provides an exact commutative square between the moment and endpoint inversion proofs.

The conceptual gain is substantial: the asymptotic theorem at reciprocal powers would no longer be merely checked against exact values. It would be derived from the exact-value infrastructure, with the continuous and discrete saddle phases proved equal.

75 Conclusions

The exact arithmetic and endpoint asymptotic of the Fabius function are connected more tightly than their usual presentations suggest.

The reciprocal-power formula first removes a universal factor $2^{-\binom{n}{2}}$ and leaves a coefficient $[z^n]G(z)$. The recurrence and the composition formula resum into the entire dyadic product for G. Its negative-real restriction is exactly the Laplace product whose Mellin transform produces the quadratic drift, the constant, and the Gamma–zeta periodic term. Coefficient extraction at the Lambert radius then supplies the Gaussian normalization and the inverse-power corrections.

The crucial arithmetic identity

$$n = \lambda_n - \log_2 \lambda_n$$

has two simultaneous consequences. It makes λ_n the coefficient saddle, and it locks the log-periodic phase: $\Psi(\log_2 \lambda_n) = \Psi(\lambda_n)$. Once this is noticed, the local coefficient phase becomes identical to the local continuous Bromwich phase. The equality persists to every order.

The Gaussian-binomial formula contributes a complementary lesson. Its conspicuous q-Pochhammer normalizer is not itself the endpoint asymptotic. Exact geometric interpolation cancels it against the Mersenne factor in the numerator and exposes the coefficient $[z^n](B_m G)$. For fixed m, the prefix factor has one dominant exponential, and the same saddle analysis gives the full sharp expansion on $m2^{-n}$.

Finally, rescaling the continuous inversion at 2^{-n} and iterating the negative refinement equation maps the saddle 2^{λ_n} to λ_n exactly. The finite Thue–Morse product introduced by those refinement steps is exponentially close to 1 at the saddle and disappears exactly after integration because the probability law is supported on $[0, 1]$. This identity supplies a direct analytic explanation of why the discrete and continuous derivations agree.

Thus the missing connection can be stated succinctly:

$$\boxed{\begin{array}{c} \text{finite dyadic arithmetic} \\ \Big\downarrow \ \text{exact cancellation and resummation} \\ \text{the same infinite dyadic product} \\ \Big\downarrow \ \text{coefficient saddle} \\ \text{the corrected endpoint asymptotic} \end{array}}$$

Supplementary material from this dossier

76 The key algebraic cancellation

For completeness, let $p = L(\lambda - n)$. Then $n = \lambda - p/L$, and

$$\begin{aligned} -\frac{L}{2}(n^2 - n) &= -\frac{L}{2}\lambda^2 + \lambda p - \frac{p^2}{2L} + \frac{L}{2}\lambda - \frac{p}{2}, \\ -np &= -\lambda p + \frac{p^2}{L}. \end{aligned}$$

Therefore

$$\begin{aligned} &-\frac{L}{2}(n^2 - n) + \lambda - \frac{p^2}{2L} + \frac{p}{2} - np \\ &= -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda. \end{aligned}$$

For reciprocal powers, $p = \log\lambda$. For a fixed numerator m, the same calculation uses $p = \log(\lambda/m)$, because $\log_2(\lambda/m) = \lambda - n$.

77 Exact arithmetic used in the numerical tables

The reciprocal coefficients are generated by

$$a_0 = 1, \qquad a_n = \frac{1}{2^n - 1}\sum_{j=0}^{n-1}\frac{a_j}{(n-j+1)!}. \tag{758}$$

All a_n are stored as reduced rational numbers, and

$$F(2^{-n}) = 2^{-\binom{n}{2}}a_n. \tag{759}$$

For a fixed numerator,

$$[z^n](B_m(z)G(z)) = \sum_{h=0}^{m-1}\varepsilon_h\sum_{j=0}^{n}a_j\frac{(m-1-h)^{n-j}}{(n-j)!}, \tag{760}$$

$$F(m2^{-n}) = 2^{-\binom{n}{2}}[z^n](B_mG). \tag{761}$$

Equations (760)–(761) involve rational arithmetic only. The Lambert coordinate is then evaluated at high precision from (657); Ψ and its derivatives are evaluated from (699). Because the Fourier coefficients decay exponentially, a small number of modes suffices for the digits printed in tables 17 and 18.

78 Bibliographic note

The exact arithmetic identities used here are developed in the ProveIt repository and its accompanying exposition [1, 2], building in particular on the compact-support and arithmetic papers of Arias de Reyna [3, 4]. Haugland [5] gives another exact-evaluation perspective. The Mellin treatment of harmonic sums follows the general framework of Flajolet, Gourdon, and Dumas [6], while the Lambert coordinate uses the branch conventions of Corless et al. [7]. The contribution of this report is the explicit coefficient-saddle bridge, its phase-locking identity, its extension to fixed dyadic numerators, and the exact rescaled-Bromwich collapse.

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Part VII
From Exact Dyadic Formulae to the Small-Argument Asymptotic

Source provenance and status. Former path: `Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex`. Newest and preferred dyadic-to-asymptotic synthesis. It decompresses the half-base q-binomial formula, gives a rigorous moment-to-Laplace route, and separates formalized inputs from its independent coefficient-saddle derivation. Prefer its normalizations and status ledger where it overlaps the preceding bridge.

Original source SHA-256: `6e98efd3afc402159a7f080e8604c951d5bc51be4a73383da67c1241d46d54ca`.

Dossier abstract

The exact arithmetic of the Fabius function and its endpoint asymptotics are usually presented as two nearly disjoint subjects. At reciprocal powers of two one has rational recurrences, composition sums, Thue–Morse power sums, and a finite half-base q-binomial formula; near the origin one has a negative-Laplace product, a lower-Lambert saddle, and a logarithmically periodic correction. This report supplies the missing bridge.

The central exact observation is that the apparently formidable q-binomial numerator at $F(2^{-n})$ is not a new asymptotic object. After choosing a convenient scalar translation, the inner Thue–Morse block becomes a coefficient of the ratio $A(-2^kX)/A(-X)$, where

$$A(z) = \sum_{n\geq 0} a_n z^n = \prod_{j\geq 1} \frac{\mathrm{e}^{z/2^j}-1}{z/2^j}, \qquad F(2^{-n}) = 2^{-\binom{n}{2}} a_n.$$

The Gaussian row at $q = 1/2$ is exactly the divided-difference functional on the geometric nodes $-1, -2, \ldots, -2^n$; it annihilates all lower powers and extracts the leading coefficient. Consequently the complete finite numerator collapses identically to $\mathfrak{m}_n a_n$, where $\mathfrak{m}_n = \prod_{j=1}^{n}(2^j-1)$. This "decompression" proves that the q-Pochhammer, q-binomial, Thue–Morse, recurrence, composition, and generating-product formulae all encode the same coefficient sequence.

Two asymptotic derivations are then developed. The fully rigorous real-variable route writes the half moment as $d_n = \mathbb{E}(1-X)^n$, compares it through second order with the negative Laplace transform $A(-n) = \mathbb{E}\mathrm{e}^{-nX}$, and uses the exact quadratic-plus-periodic decomposition of $\log A(-s)$. A complementary coefficient-saddle route applies Cauchy's formula directly to $a_n = [z^n]A(z)$; it reaches the same lower-Lambert coordinate and provides an independent explanation of every cancellation. If $L = \log 2$ and λ_n is the large solution of

$$\lambda_n - \frac{\log \lambda_n}{L} = n, \qquad \lambda_n = -\frac{1}{L} W_{-1}(-L2^{-n}),$$

then the resulting sharp dyadic formula is

$$\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1+\frac{L}{2}\right)\lambda_n - \frac{1}{2}\log \lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}),$$

where Ψ is a smooth, nonconstant, mean-zero, one-periodic function with explicit Gamma–zeta Fourier coefficients. Expanding the Lambert phase yields

$$\log F(2^{-n}) = -\frac{L}{2}n^2 - n\log n + \left(1+\frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).$$

The exact phase is essential: replacing it by $\log_2 n$ leaves a generally larger $(\log n)/n$ term. Corollaries give equally sharp asymptotics for the recurrence sequence, the half moments, the composition sum, and the literal q-binomial numerator. Numerical checks against exact rational values confirm the constant, phase, and first saddle correction.

79 The missing bridge

The Fabius function admits two descriptions that look, at first sight, almost antagonistic. Its values at dyadic rationals are algebraic and finite: every value is rational; there are terminating recurrences; the Thue–Morse sequence provides finite signed power sums; and a half-base Gaussian-binomial row packages those sums into a compact formula. Its endpoint asymptotic is analytic and infinite: one studies a Laplace product, separates a periodic renormalization of a dyadic harmonic sum, and evaluates a saddle determined by the lower real branch of Lambert's W-function.

The two subjects are not merely compatible. They are two coordinate systems on the same object. The exact formulae contain the asymptotic information in a highly compressed form, and the purpose of this report is to make the decompression explicit.

79.1 What is proved

There are four levels of connection.

1. The reciprocal-power value $F(2^{-n})$ is exactly a half moment divided by a factorial and a triangular power of two. This already gives the quadratic logarithmic decay.
2. After factorial normalization, all reciprocal-power values are coefficients of one entire function A. The recurrence, composition sum, and infinite product are equivalent presentations of A.
3. The finite q-binomial/Thue–Morse formula can be reduced identically to the same coefficient $a_n = [z^n]A(z)$. No limiting argument is involved. The q-binomial row is a geometric divided difference; its role is to undo the blockwise Thue–Morse cancellation.
4. The product for $A(-s)$ has an exact quadratic-plus-periodic logarithm. Transferring from $A(-n)$ to the endpoint moment, or applying a coefficient saddle directly to A, yields the corrected lower-Lambert asymptotic.

The first, second, third, and the full real-variable fourth level are represented in the Lean formalization under `Analysis/FabiusFunction`. The coefficient-saddle calculation in Section 86 is included as an independent conventional-mathematics derivation; it is useful because it starts from the coefficient formula itself and exposes the same cancellations with very little probability language.

79.2 Headline notation and result

Throughout,

$$L = \log 2. \tag{762}$$

Let P be the periodic correction in the exact negative-Laplace decomposition, let

$$\overline{P} = \int_0^1 P(u)\,\mathrm{d}u, \qquad \Psi(u) = P(u) - \overline{P}, \tag{763}$$

and put

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12}, \qquad C_F = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log \pi. \tag{764}$$

Here γ is Euler's constant and γ_1 is the first Stieltjes constant. For every sufficiently large integer n, let λ_n be the large positive solution of

$$\lambda_n 2^{-\lambda_n} = 2^{-n}, \qquad \text{equivalently} \qquad \lambda_n - \frac{\log \lambda_n}{L} = n. \tag{765}$$

Theorem 79.1 (Sharp reciprocal-power asymptotic)**.** *As $n \to \infty$,*

$$\boxed{\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log \lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).} \tag{766}$$

Equivalently,

$$\begin{aligned}\log F(2^{-n}) = & -\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ & - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{767}$$

The function Ψ is smooth, mean-zero, one-periodic, and nonconstant. Its Fourier coefficients are explicit.

The rest of the report derives Theorem 79.1 from the exact arithmetic formulae rather than taking it as an externally supplied endpoint theorem.

The most important conceptual point is that the periodic term is already present in the exact dyadic values. It does not arise because one interpolates between dyadic points. The fractional parts of the exact saddle coordinates λ_n drift through the logarithmic period, so the single rational sequence $F(2^{-n})$ samples the entire periodic correction.

80 One coefficient sequence behind the exact formulae

80.1 The half moments and the endpoint identity

Let F denote the bounded Fabius distribution function and let up be Rvachev's even compactly supported density in the normalization

$$\mathrm{up}(t) = F(1 - |t|), \qquad |t| \le 1. \tag{768}$$

Define the half moments

$$d_0 = 1, \qquad d_n = n\int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t \quad (n \ge 1). \tag{769}$$

They are rational and positive. Their first few values are

$$d_0 = 1, \qquad d_1 = \frac{1}{2}, \qquad d_2 = \frac{5}{18}, \qquad d_3 = \frac{1}{6}, \qquad d_4 = \frac{143}{1350}. \tag{770}$$

Theorem 80.1 (Exact endpoint–moment bridge)**.** *For every $n \ge 0$,*

$$d_n = n!\,2^{\binom{n}{2}}F(2^{-n}), \qquad F(2^{-n}) = \frac{d_n}{n!\,2^{\binom{n}{2}}}. \tag{771}$$

Proof. The primitive-chain proof is especially revealing. For $r \geq 0$ define

$$f_r(t) = 2^{\binom{r}{2}} \operatorname{up}((t+1)2^{-r} - 1), \qquad -1 \leq t \leq 1. \tag{772}$$

The functional–differential equation of up implies

$$f'_{r+1}(t) = f_r(t), \qquad f_r(-1) = 0. \tag{773}$$

Put $J_n = \int_{-1}^{0} t^{n-1} \operatorname{up}(t)\, \mathrm{d}t$. Repeated integration by parts, using (773), gives

$$J_n = (-1)^{n-1}(n-1)! f_n(0). \tag{774}$$

Evenness of up turns the left side into $(-1)^{n-1} \int_0^1 t^{n-1} \operatorname{up}(t)\, \mathrm{d}t$, while

$$f_n(0) = 2^{\binom{n}{2}} \operatorname{up}(2^{-n} - 1) = 2^{\binom{n}{2}} F(2^{-n}).$$

Multiplication by n proves (771). The case $n = 0$ is $F(1) = d_0 = 1$. □

There is also a probability interpretation that will be decisive later. Let X be a random variable with distribution function F. Integrating by parts against the distribution measure gives

$$d_n = \mathbb{E}(1 - X)^n. \tag{775}$$

Thus d_n is simultaneously an exact rational endpoint value and an endpoint moment of the probability law.

80.2 Factorial normalization and the recurrence

Set

$$a_n = \frac{d_n}{n!}, \qquad A(z) = \sum_{n=0}^{\infty} a_n z^n. \tag{776}$$

Then Theorem 80.1 becomes the particularly economical identity

$$\boxed{F(2^{-n}) = 2^{-\binom{n}{2}} a_n.} \tag{777}$$

The half-moment recurrence is equivalent to

$$(2^n - 1)a_n = \sum_{k=0}^{n-1} \frac{a_k}{(n-k+1)!}, \qquad n \geq 1, \qquad a_0 = 1. \tag{778}$$

Equivalently, directly in terms of endpoint values,

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} F(2^{-k}). \tag{779}$$

The recurrence is not merely an evaluation algorithm. It is already a coefficient equation. Let

$$E(z) = \frac{\mathrm{e}^z - 1}{z}, \qquad E(0) = 1. \tag{780}$$

Multiplying (778) by z^n and summing gives

$$\boxed{A(2z) = E(z)A(z).} \tag{781}$$

Conversely, coefficient comparison in (781) recovers (778). Thus the exact recurrence and the analytic refinement equation are the same statement.

80.3 Composition sum

Repeated substitution in (778) gives a finite closed form. A composition $\rho \vDash n$ is an ordered tuple $\rho = (r_1, \dots, r_m)$ of positive integers with sum n; write $s_j = r_1 + \cdots + r_j$ for its prefix sums.

Proposition 80.2 (Exact composition formula)**.** *For $n \geq 1$,*

$$a_n = \sum_{\rho \vDash n} \prod_{j=1}^{\ell(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \tag{782}$$

Consequently

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \vDash n} \prod_{j=1}^{\ell(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \tag{783}$$

Proof. Every use of (778) replaces a current index s_j by a smaller prefix s_{j-1} and contributes $((2^{s_j} - 1)(s_j - s_{j-1} + 1)!)^{-1}$. A complete descent from n to 0 is therefore the same as a composition of n, with increments $r_j = s_j - s_{j-1}$. Summing over all descents gives (782). □

This formula looks combinatorial, but asymptotically it is not a sum over independently comparable contributions. The number of compositions is 2^{n-1}, and the dominant family moves with n. The correct way to analyze the sum is to reassemble it into A.

80.4 The entire product

Iterating (781) toward the origin gives

$$\boxed{A(z) = \prod_{j=1}^{\infty} E(z/2^j) = \prod_{j=1}^{\infty} \frac{e^{z/2^j} - 1}{z/2^j}.} \tag{784}$$

The product converges locally uniformly and defines an entire function. It is the moment generating function of X:

$$A(z) = \mathbb{E}e^{zX}. \tag{785}$$

Since $E(z) = e^z E(-z)$ and $\sum_{j \geq 1} 2^{-j} = 1$, the product also satisfies the exact reflection identity

$$A(z) = e^z A(-z). \tag{786}$$

The equivalence graph is now

$$\boxed{\begin{array}{c} \text{endpoint values} \longleftrightarrow d_n \longleftrightarrow a_n \longleftrightarrow \text{recurrence} \\ \longleftrightarrow \text{composition sum} \longleftrightarrow A(z) \longleftrightarrow \text{entire product} \end{array}}$$

The next section inserts the q-binomial and Thue–Morse formula into this same graph.

81 Decompressing the half-base q-binomial formula

The q-binomial formula is the most conspicuous exact representation and the least transparent asymptotic one. Its finite sum contains two distinct annihilation mechanisms:

- inside each dyadic block, Thue–Morse signs annihilate low polynomial degrees;
- across the block sizes, a Gaussian-binomial divided difference annihilates the remaining low powers of the geometric scale.

Once both cancellations are written as coefficient extraction, the entire numerator reduces to $\mathfrak{m}_n a_n$.

81.1 The geometric interpolation functional

For $q = 1/2$, write

$$(a; 1/2)_n = \prod_{j=0}^{n-1} (1 - a2^{-j}), \tag{787}$$

and let $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ denote the Gaussian binomial coefficient [3, 9, 12, 6]. The finite q-binomial theorem is

$$(z; 1/2)_n = \sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \tag{788}$$

Define

$$x_k = -2^k, \qquad w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}, \qquad \mathfrak{m}_n = \prod_{j=1}^{n} (2^j - 1). \tag{789}$$

Lemma 81.1 (Leading-coefficient extraction)**.** *If R is a polynomial of degree at most n, then*

$$\sum_{k=0}^{n} w_{n,k} R(-2^k) = \mathfrak{m}_n [z^n] R(z). \tag{790}$$

Equivalently,

$$\sum_{k=0}^{n} w_{n,k} x_k^d = \begin{cases} 0, & 0 \le d < n, \\ \mathfrak{m}_n, & d = n. \end{cases} \tag{791}$$

Proof. The normalized weights are the barycentric weights for the nodes $x_0, \dots, x_n$:

$$\frac{w_{n,k}}{\mathfrak{m}_n} = \frac{1}{\prod_{j \ne k} (x_k - x_j)}. \tag{792}$$

The coefficient of z^n in the Lagrange interpolation formula for R is therefore the left side of (790) divided by $\mathfrak{m}_n$. Formula (792) can also be checked directly from (788). □

The Pochhammer normalization is exactly the same Mersenne product:

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n. \tag{793}$$

This identity will make the last power-of-two cancellation automatic.

81.2 The finite Thue–Morse blocks

Let

$$\varepsilon_r = (-1)^{\mathrm{wt}_2(r)}. \tag{794}$$

For $k, N \ge 0$ and a scalar translation Q, define

$$S_{k,N}(Q) = \sum_{r=0}^{2^k - 1} \varepsilon_r (r - 2^k + Q)^N. \tag{795}$$

The complete-block exponential generating function is

$$C_k(X) = \sum_{r < 2^k} \varepsilon_r \mathrm{e}^{(r - 2^k) X}. \tag{796}$$

The elementary binary product

$$\sum_{r < 2^k} \varepsilon_r z^r = \prod_{j=0}^{k-1} (1 - z^{2^j}) \tag{797}$$

implies

$$\frac{C_k(X)}{X^k} = (-1)^k 2^{\binom{k}{2}} \mathrm{e}^{-X} \prod_{j=0}^{k-1} E(-2^j X). \tag{798}$$

In particular, C_k has a zero of order k at the origin. Equivalently, $S_{k,N}(Q) = 0$ for $N < k$, independently of Q.

Iterating (781) on the negative dyadic orbit gives

$$\frac{A(-2^k X)}{A(-X)} = \prod_{j=0}^{k-1} E(-2^j X). \tag{799}$$

Translation invariance permits the especially convenient choice $Q = 1$. Since

$$\sum_{N\geq 0} S_{k,N}(1) \frac{X^N}{N!} = \mathrm{e}^X C_k(X), \tag{800}$$

(798) and (799) yield the exact coefficient identity

$$\boxed{\frac{S_{k,n+k}(1)}{(n+k)!} = (-1)^k 2^{\binom{k}{2}} [X^n] \frac{A(-2^k X)}{A(-X)}.} \tag{801}$$

This is the first decompression: an inner Thue–Morse power sum is a coefficient of the same entire function that generates the half moments.

81.3 The literal numerator collapses

At $m = 1$, the global q-binomial formula reads

$$F(2^{-n}) = \frac{\mathcal{N}_n}{2^{n^2} (1/2; 1/2)_n}, \tag{802}$$

where, using the shift $Q = 1$,

$$\mathcal{N}_n = \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}} (n+k)!} S_{k,n+k}(1). \tag{803}$$

Theorem 81.2 (q-binomial decompression)**.** *For every $n \geq 0$,*

$$\boxed{\mathcal{N}_n = \mathfrak{m}_n a_n.} \tag{804}$$

Consequently the finite q-binomial formula is identically the endpoint formula $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$.

Proof. For fixed n, introduce the polynomial

$$R_n(z) = [X^n] \frac{A(zX)}{A(-X)}. \tag{805}$$

It has degree at most n. Since $A(-X)^{-1} = 1 + \mathcal{O}(X)$ and $A(zX) = \sum_{j\geq 0} a_j z^j X^j$, its leading coefficient is

$$[z^n] R_n(z) = a_n. \tag{806}$$

Substitution of (801) into (803) gives

$$\mathcal{N}_n = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} 2^{-2\binom{k}{2}}(-1)^k 2^{\binom{k}{2}} R_n(-2^k)$$
$$= \sum_{k=0}^{n} w_{n,k} R_n(-2^k). \tag{807}$$

By Lemma 81.1, this is $\mathfrak{m}_n[z^n]R_n(z) = \mathfrak{m}_n a_n$, proving (804). Finally, (793) gives

$$\frac{\mathfrak{m}_n a_n}{2^{n^2}(1/2;1/2)_n} = \frac{\mathfrak{m}_n a_n}{2^{n^2-\binom{n+1}{2}}\mathfrak{m}_n} = 2^{-\binom{n}{2}} a_n.$$

□

The two finite cancellations have now been separated completely. The Thue–Morse block removes k powers of X and converts a power sum into $R_n(-2^k)$. The half-base q-binomial row then removes all powers of the scale node below degree n and returns the coefficient a_n. The q-Pochhammer factor does not introduce a new special-function asymptotic; it is the normalization of this divided difference.

81.4 General dyadic numerators

For completeness, the exact global formula is

$$\mathcal{F}(m/2^n) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{1/2}}{4^{\binom{k}{2}}(n+k)!}$$
$$\times \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + Q)^{n+k}. \tag{808}$$

It is independent of Q. Splitting $r = h2^k + s$ factors the long block into the complete block C_k and a finite prefix polynomial. The same interpolation functional then extracts one coefficient of a product of that prefix polynomial with A. Thus even the general formula contains no asymptotically unrelated object. What changes when m varies with n is the coefficient multiplier, not the underlying dyadic product.

82 The first asymptotic already visible in the exact values

Before resolving the periodic fluctuation, the endpoint identity alone gives the principal quadratic law. Taking logarithms in (771),

$$\log F(2^{-n}) = \log d_n - \log(n!) - \binom{n}{2} L. \tag{809}$$

Because $0 \le \text{up} \le 1$,

$$d_n \le n \int_0^1 t^{n-1}\,\mathrm{d}t = 1. \tag{810}$$

A fixed positive portion of the mass on $[0, 1/2]$ gives, for $n \ge 1$,

$$2^{-(n+1)} \le d_n. \tag{811}$$

Combining these bounds with elementary factorial estimates yields the explicit inequality

$$\left|\log F(2^{-n}) + \frac{L}{2}n^2\right| \leq 3n\log(n+1), \qquad n \geq 1. \tag{812}$$

Therefore

$$\lim_{n\to\infty} \frac{\log F(2^{-n})}{n^2} = -\frac{L}{2}. \tag{813}$$

This is already a derivation of the leading endpoint asymptotic from exact arithmetic. It also explains why neither the factorial nor the half moment can change the coefficient of n^2: $\log(n!) = \mathcal{O}(n\log n)$ and $\log d_n = \mathcal{O}(n)$ under the crude bounds. To recover the $-n\log n$ term, the $\log^2 n$ term, the constant, and the periodic fluctuation, however, one must understand d_n much more sharply.

83 The exact negative-Laplace decomposition

The sharp asymptotic enters through the same function A. For $s > 0$ put

$$B(s) = A(-s) = \mathbb{E}\mathrm{e}^{-sX}, \qquad \Lambda(s) = \log B(s). \tag{814}$$

From (784),

$$B(s) = \prod_{j=1}^{\infty} \frac{1-\mathrm{e}^{-s/2^j}}{s/2^j}, \tag{815}$$

and hence

$$\Lambda(s) = \sum_{j=1}^{\infty} \kappa(s/2^j), \qquad \kappa(u) = \log\frac{1-\mathrm{e}^{-u}}{u}. \tag{816}$$

This is a dyadic harmonic sum. Its dilation equation is exact:

$$\Lambda(2s) = \Lambda(s) + \log(1-\mathrm{e}^{-s}) - \log s. \tag{817}$$

83.1 Periodization of the dilation equation

Define the forward logarithmic tail

$$T(s) = \sum_{m=0}^{\infty} \log(1-\mathrm{e}^{-s2^m}) < 0, \qquad \mathcal{E}(s) = -T(s) > 0. \tag{818}$$

Since $2^m \geq m+1$,

$$0 \leq \mathcal{E}(s) \leq \frac{\mathrm{e}^{-s}}{(1-\mathrm{e}^{-s})^2}, \tag{819}$$

and in particular $\mathcal{E}(s) \leq 4\mathrm{e}^{-s}$ for $s \geq L$. For $u \in \mathbb{R}$, set

$$P(u) = \Lambda(2^u) + \frac{L}{2}(u^2-u) + T(2^u). \tag{820}$$

Theorem 83.1 (Exact quadratic-plus-periodic decomposition)**.** *The function P is smooth and one-periodic. For every $s > 0$,*

$$\boxed{\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2}\log s + P(\log_2 s) + \mathcal{E}(s).} \tag{821}$$

All derivatives of P are bounded.

Proof. The forward tail satisfies

$$T(2s) = T(s) - \log(1 - \mathrm{e}^{-s}).$$

Under $u \mapsto u + 1$, the change in the quadratic term of (820) is

$$\frac{L}{2}((u+1)^2 - (u+1) - (u^2 - u)) = Lu = \log s.$$

This cancels the changes in (817) and in T, proving $P(u+1) = P(u)$. Rearrangement gives (821). Local uniform convergence of the defining series permits differentiation; periodicity then makes every derivative bounded. □

The drift $-(\log s)^2/(2L)$ is the analytic counterpart of the triangular power $2^{-\binom{n}{2}}$ in the exact endpoint formula. The periodic remainder is the residual freedom in solving the dilation equation on logarithmic scale.

83.2 Mean, Fourier modes, and the asymptotic constant

Let $\overline{P}$ and Ψ be as in (763). The Mellin transform of the kernel gives

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}, \tag{822}$$

where A_0 is defined in (764). With

$$\chi_k = \frac{2\pi \mathrm{i} k}{L}, \tag{823}$$

the Fourier coefficients of the centered function are

$$\widehat{\Psi}(0) = 0, \qquad \boxed{\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}} \quad (k \neq 0). \tag{824}$$

The Gamma factor has no zeros, and ζ has no zeros on $\Re z = 1$ away from its pole at 1, so every nonzero Fourier coefficient is nonzero. Thus Ψ is genuinely nonconstant. Stirling's vertical estimate for Γ makes the coefficients exponentially small; in fact the oscillation has amplitude only a few parts in 10^6, but it cannot be deleted at an error scale tending to zero.

The Gaussian normalization later contributes $-\frac{1}{2}\log(2\pi)$. Therefore

$$\overline{P} - \frac{1}{2}\log(2\pi) = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log \pi = C_F. \tag{825}$$

Numerically,

$$\overline{P} = -1.10904536543418668\ldots, \qquad C_F = -2.02798389863885942\ldots. \tag{826}$$

84 Sharp route I: endpoint moments versus Laplace moments

This route begins with the exact rational object $d_n = \mathbb{E}(1 - X)^n$ and converts it into the negative-Laplace product at the natural scale $s = n$. It is the route closest to the current Lean bridge modules.

84.1 Second-order kernel comparison

For $0 \le x \le 1$ and $n \ge 1$, the logarithmic expansion

$$n \log(1-x) = -nx - \frac{n}{2}x^2 + \mathcal{O}(nx^3)$$

is useful only after the large-x region is controlled uniformly. A convenient global form is

$$\left|(1-x)^n - \mathrm{e}^{-nx}\left(1 - \frac{n}{2}x^2\right)\right| \le 16\mathrm{e}^{-nx}(nx^3 + n^2x^4). \tag{827}$$

Integrating against the law of X gives

$$d_n = B(n) - \frac{n}{2}\mathbb{E}(X^2\mathrm{e}^{-nX}) + R_n, \tag{828}$$

$$|R_n| \le 16\left(n\mathbb{E}(X^3\mathrm{e}^{-nX}) + n^2\mathbb{E}(X^4\mathrm{e}^{-nX})\right). \tag{829}$$

Let $q(s) = \Lambda(s) = \log B(s)$. The normalized tilted second moment is

$$\frac{\mathbb{E}(X^2\mathrm{e}^{-sX})}{B(s)} = q''(s) + q'(s)^2. \tag{830}$$

The corresponding third- and fourth-moment identities are polynomials in $q', q'', q^{(3)}, q^{(4)}$. Differentiation of (821), together with the exponentially small tail, shows that these normalized moments have precisely the inverse powers needed to make (829) a relative $\mathcal{O}(n^{-1})$ error after the second term is retained.

Proposition 84.1 (Endpoint–Laplace logarithmic transfer)**.** *As $n \to \infty$,*

$$\boxed{\log d_n = q(n) - \frac{n}{2}(q''(n) + q'(n)^2) + \mathcal{O}(n^{-1}).} \tag{831}$$

Proof architecture. Divide (828) by $B(n)$. The second-order displacement is

$$\alpha_n = \frac{n}{2}(q''(n) + q'(n)^2) = \mathcal{O}(\log^2 n/n),$$

and the normalized remainder is $\mathcal{O}(n^{-1})$. Thus $d_n/B(n) = 1 - \alpha_n + \mathcal{O}(n^{-1})$. The local estimate $\log(1-y) = -y + \mathcal{O}(y^2)$ applies; since $\alpha_n^2 = \mathcal{O}(\log^4 n/n^2) = \mathcal{O}(n^{-1})$, it gives (831). □

84.2 Differentiating the exact periodic decomposition

Write

$$\ell = \log s, \qquad u = \log_2 s = \frac{\ell}{L}. \tag{832}$$

Ignoring an exponentially small differentiated tail, (821) gives

$$q'(s) = \frac{-\ell/L + 1/2 + \Psi'(u)/L}{s} + \mathcal{O}(\mathrm{e}^{-s}), \tag{833}$$

and

$$q''(s) = \frac{\ell/L - 1/2 - 1/L - \Psi'(u)/L + \Psi''(u)/L^2}{s^2} + \mathcal{O}(\mathrm{e}^{-s}). \tag{834}$$

Squaring (833) and adding (834), the terms linear in ℓ that do not involve Ψ' cancel. Since Ψ' and Ψ'' are bounded,

$$\frac{s}{2}(q''(s) + q'(s)^2) = \frac{\ell^2}{2L^2 s} - \frac{\ell\,\Psi'(u)}{L^2 s} + \mathcal{O}(s^{-1}). \tag{835}$$

This cancellation is the analytic reason that the first nonperiodic endpoint correction is $-\log^2 n/(2L^2 n)$ rather than a mixture of $\log^2 n/n$ and $\log n/n$ terms.

Substituting (821) and (835) into (831) yields the first sharp form.

Theorem 84.2 (Half moments at the logarithmic phase). *Let $u_n = \log_2 n$. Then*

$$\begin{aligned}\log d_n = &-\frac{\log^2 n}{2L} + \frac{1}{2}\log n + \overline{P} + \Psi(u_n)\\ &-\frac{\log^2 n}{2L^2 n} + \frac{\log n}{L^2 n}\Psi'(u_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{836}$$

Combining (836) with Stirling's formula and (809) gives

$$\begin{aligned}\log F(2^{-n}) = &-\frac{L}{2}n^2 - n\log n + \left(1+\frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F\\ &-\frac{\log^2 n}{2L^2 n} + \Psi(u_n) + \frac{\log n}{L^2 n}\Psi'(u_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{837}$$

Formula (837) has already recovered the constant and the periodic function from the exact endpoint values. The derivative term is not an extra oscillation; it is the first-order displacement from the naive phase u_n to the exact Lambert phase.

85 The lower-Lambert phase and the derivative cancellation

85.1 The exact phase

The natural saddle coordinate at $x = 2^{-n}$ is

$$\lambda_n = -\frac{1}{L}W_{-1}(-L2^{-n}), \tag{838}$$

where W_{-1} is the lower real branch of Lambert's function [5]. It is the large solution of (765). Its fixed-point form is

$$\lambda_n = n + \log_2 \lambda_n. \tag{839}$$

The standard lower-Lambert expansion gives

$$\begin{aligned}\lambda_n = n &+ \frac{\log n}{L} + \frac{\log n}{L^2 n} + \frac{\log n - \frac{1}{2}\log^2 n}{L^3 n^2}\\ &+ \frac{\log n - \frac{3}{2}\log^2 n + \frac{1}{3}\log^3 n}{L^4 n^3} + \mathcal{O}\left(\frac{\log^4 n}{n^4}\right).\end{aligned} \tag{840}$$

For the periodic phase transfer only the first displacement is needed. Define

$$\delta_n = \log_2 \lambda_n - \log_2 n. \tag{841}$$

Then

$$\delta_n = \frac{\log n}{L^2 n} + \mathcal{O}\left(\frac{\log^2 n}{n^2}\right). \tag{842}$$

85.2 Why Ψ is evaluated at λ_n

Since n is an integer, periodicity and (839) imply the exact identity

$$\Psi(\lambda_n) = \Psi(\lambda_n - n) = \Psi(\log_2 \lambda_n). \tag{843}$$

Taylor expansion at $u_n = \log_2 n$, with uniformly bounded second derivative, gives

$$\begin{aligned}\Psi(\lambda_n) &= \Psi(u_n + \delta_n)\\ &= \Psi(u_n) + \frac{\log n}{L^2 n}\Psi'(u_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{844}$$

Indeed, the error in (842) contributes $\mathcal{O}(\log^2 n/n^2) = \mathcal{O}(n^{-1})$, and the quadratic Taylor remainder contributes the same or less. Substitution of (844) into (837) proves (767).

This calculation explains exactly why the phase may not be replaced by $\log_2 n$ at the claimed precision. One would omit the term

$$\frac{\log n}{L^2 n}\Psi'(\log_2 n), \tag{845}$$

which is generally of order $(\log n)/n$, not $1/n$. The tiny amplitude of Ψ affects numerical visibility, but not the asymptotic order.

Warning 85.1 (Phase precision is amplified). The leading exponent contains $-L\lambda^2/2$. An error $\Delta\lambda$ in the phase changes the exponent by approximately $-L\lambda\Delta\lambda$. Therefore deriving an elementary expansion with remainder $\mathcal{O}(n^{-1})$ requires the Lambert phase to one order beyond a naive term count. Keeping λ_n exact inside Ψ avoids this bookkeeping hazard.

85.3 Compact versus expanded form

Define

$$\mathcal{M}_n = -\frac{L}{2}\lambda_n^2 + \left(1+\frac{L}{2}\right)\lambda_n - \frac{1}{2}\log\lambda_n + C_F + \Psi(\lambda_n). \tag{846}$$

Substitution of (840) and collection of terms gives

$$\begin{aligned}\mathcal{M}_n = &-\frac{L}{2}n^2 - n\log n + \left(1+\frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ &- \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{847}$$

Thus (766) and (767) are equivalent at the stated precision. The compact formula is structurally cleaner and is the one that survives unchanged for arbitrary small real arguments.

86 Sharp route II: a coefficient saddle from the exact generating product

The endpoint–Laplace route starts from $d_n = \mathbb{E}(1-X)^n$. There is a second route beginning directly with the recurrence coefficient $a_n = [z^n]A(z)$. It is particularly well suited to the question posed here because a_n is exactly what remains after the q-binomial formula is decompressed.

86.1 Positive-axis growth of A

The reflection identity (786) and Theorem 83.1 give, for $r > 0$,

$$\begin{aligned}K(r) := \log A(r) &= r + \Lambda(r) \\ &= r - \frac{\log^2 r}{2L} + \frac{1}{2}\log r + P(\log_2 r) + \mathcal{E}(r).\end{aligned} \tag{848}$$

The reference saddle is chosen by ignoring the bounded derivative of the periodic term:

$$r - \frac{\log r}{L} = n. \tag{849}$$

Its large solution is exactly $r = \lambda_n$. The exact coefficient saddle for $rK'(r) = n$ differs from λ_n by only $\mathcal{O}(1)$, because

$$rK'(r) = r - \frac{\log r}{L} + \frac{1}{2} + \frac{1}{L}P'(\log_2 r) + \mathcal{O}(re^{-r}). \tag{850}$$

An $\mathcal{O}(1)$ displacement of a saddle of curvature $\asymp 1/r$ changes the logarithmic coefficient estimate by only $\mathcal{O}(1/r)$.

86.2 Cauchy extraction

Cauchy's formula gives

$$a_n = \frac{A(r)}{2\pi r^n} \int_{-\pi}^{\pi} \frac{A(re^{i\theta})}{A(r)} e^{-in\theta}\, d\theta. \tag{851}$$

At $r = \lambda_n$, split the product (784) near the dyadic index $j \approx \log_2 r$. In a central window, Taylor expansion of the logarithm gives a Gaussian with variance parameter $r + \mathcal{O}(1)$; outside that window, a positive proportion of factors has modulus strictly below its positive-axis value, and the remaining factors supply arbitrary polynomial decay. The standard coefficient-saddle estimate [10, 8] is therefore

$$a_n = \frac{A(\lambda_n)}{\lambda_n^n \sqrt{2\pi\lambda_n}} \left(1 + \mathcal{O}(\lambda_n^{-1})\right). \tag{852}$$

The use of the reference rather than exact saddle is absorbed by the same relative error.

Taking logarithms and applying (848),

$$\log a_n = \lambda_n - n\log\lambda_n - \frac{\log^2\lambda_n}{2L} + P(\log_2\lambda_n) - \frac{1}{2}\log(2\pi) + \mathcal{O}(n^{-1}). \tag{853}$$

The $+\frac{1}{2}\log\lambda_n$ in K has canceled the Gaussian denominator. Now use $n = \lambda_n - (\log\lambda_n)/L$:

$$\log a_n = \lambda_n - \lambda_n\log\lambda_n + \frac{\log^2\lambda_n}{2L} + P(\lambda_n) - \frac{1}{2}\log(2\pi) + \mathcal{O}(n^{-1}), \tag{854}$$

where periodicity was used in the form $P(\log_2\lambda_n) = P(\lambda_n - n) = P(\lambda_n)$.

Finally, subtract $\binom{n}{2}L$ according to (777). Writing $\ell_n = \log\lambda_n$ and $n = \lambda_n - \ell_n/L$, one has

$$-\binom{n}{2}L = -\frac{L}{2}\lambda_n^2 + \lambda_n\ell_n - \frac{\ell_n^2}{2L} + \frac{L}{2}\lambda_n - \frac{1}{2}\ell_n.$$

Adding (854) makes the terms $\lambda_n\ell_n$ and $\ell_n^2/(2L)$ cancel exactly. Using (825) gives

$$\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log\lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \tag{855}$$

This is (766).

The coefficient saddle provides a particularly direct answer to the original question. Starting from the exact q-binomial value, Theorem 81.2 produces a_n; Cauchy extraction of that coefficient from the exact product produces the lower-Lambert asymptotic. The endpoint–Laplace route and the coefficient route are not the same proof in disguise: one compares $(1-X)^n$ with e^{-nX} on the real line, while the other analyzes a complex coefficient integral. Their agreement fixes the normalization and phase very rigidly.

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87 Consequences for every exact arithmetic representation

87.1 The normalized recurrence and composition sum

Since $a_n = 2^{\binom{n}{2}} F(2^{-n})$, (767) immediately gives

$$\begin{aligned}\log a_n = {}& -n\log n + n - \frac{\log^2 n}{2L} + C_F \\ & - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{856}$$

Thus the exact composition sum (782) has the asymptotic

$$\begin{aligned}&\sum_{\rho\vDash n}\prod_{j=1}^{\ell(\rho)} \frac{1}{(2^{s_j}-1)(r_j+1)!} \\ &= \exp\left(-n\log n + n - \frac{\log^2 n}{2L} + C_F - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1})\right).\end{aligned} \tag{857}$$

The recurrence therefore has a moving periodic multiplier; no ansatz with a single fixed constant can be sharp.

87.2 The half moments

Adding Stirling's expansion to (856),

$$\begin{aligned}\log d_n = {}& \frac{1}{2}\log(2\pi n) + C_F + \Psi(\lambda_n) - \frac{\log^2 n}{2L} \\ & - \frac{\log^2 n}{2L^2 n} + \mathcal{O}(n^{-1}).\end{aligned} \tag{858}$$

In particular,

$$d_n \sim \sqrt{2\pi n}\,\exp\left(C_F + \Psi(\lambda_n) - \frac{\log^2 n}{2L}\right). \tag{859}$$

The notation "$\sim$" is legitimate because the omitted logarithmic error tends to zero. The periodic factor $\exp\Psi(\lambda_n)$ remains part of the equivalent.

87.3 The literal q-binomial numerator

By Theorem 81.2,

$$\mathcal{N}_n = \mathfrak{m}_n a_n. \tag{860}$$

Furthermore,

$$\mathfrak{m}_n = 2^{\binom{n+1}{2}}(1/2;1/2)_n \tag{861}$$

and

$$\log(1/2;1/2)_n = \log(1/2;1/2)_\infty + \mathcal{O}(2^{-n}). \tag{862}$$

Consequently

$$\begin{aligned}\log\mathcal{N}_n = {}& \frac{L}{2}n^2 - n\log n + \left(1+\frac{L}{2}\right)n - \frac{\log^2 n}{2L} \\ & + C_F + \log(1/2;1/2)_\infty - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{863}$$

This formula is the sharp asymptotic of the finite signed q-binomial/Thue–Morse sum itself. It makes the scale cancellation visible: the numerator grows like $\exp(Ln^2/2)$, while the explicit denominator in (802) contains 2^{n^2}. The residual triangular factor is exactly the decay $2^{-\binom{n}{2}}$ in (777).

87.4 A first all-orders corollary

The general small-argument saddle expansion supplies a first correction

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \tag{864}$$

At reciprocal powers this gives

$$\log F(2^{-n}) = \mathcal{M}_n + \frac{A_1(\lambda_n)}{\lambda_n} + \mathcal{O}(\lambda_n^{-2}). \tag{865}$$

The same correction can be transported to a_n, d_n, the composition sum, and $\mathcal{N}_n$ by the exact multiplicative normalizations above. Higher orders have the same structure: a Poincaré series in inverse powers of λ_n with smooth periodic coefficient functions evaluated at the exact phase.

88 What each exact formula contributes asymptotically

It is useful to separate exact equivalence from asymptotic convenience. The formulae compute the same numbers, but they expose different structures.

Exact representation	What it exposes	Best asymptotic use
Endpoint identity $F(2^{-n}) = d_n/(n!2^{\binom{n}{2}})$	The triangular dyadic factor, factorial drift, and one unknown half moment	Immediately proves the coefficient $-L/2$ of n^2; becomes sharp after d_n is compared with $A(-n)$
Recurrence for a_n	A positive triangular coefficient recursion	Reassemble into $A(2z) = E(z)A(z)$; direct iteration term by term obscures the moving saddle
Composition sum	A positive sum over all ordered descents of the recurrence	Identifies a_n exactly; asymptotics are best obtained after resummation into A, not by selecting a few compositions
Entire product $A(z) = \prod E(z/2^j)$	The dyadic scale at every analytic level	Primary source for both the negative-Laplace harmonic sum and the coefficient saddle
Thue–Morse block	A zero of exact order k and a finite product over dyadic scales	Converts the inner signed power sum into a coefficient of $A(-2^k X)/A(-X)$
Half-base q-binomial row	A divided difference on $-1, -2, \ldots, -2^n$	Extracts the leading coefficient a_n after the block cancellation; termwise asymptotics are badly conditioned
q-Pochhammer normalization	The Mersenne product $\mathfrak{m}_n$ in normalized form	Contributes the elementary limit $(1/2;1/2)_n \to (1/2;1/2)_\infty$ and cancels $\mathfrak{m}_n$ exactly

Exact representation	What it exposes	Best asymptotic use
Denominator and valuation formulae	Prime-adic arithmetic of the exact rationals	Strong arithmetic information, but by themselves insufficient to determine Archimedean magnitude or the periodic saddle phase

88.1 Why direct termwise analysis of the q-sum is misleading

The formula

$$F(2^{-n}) = \frac{\mathcal{N}_n}{2^{n^2}(1/2;1/2)_n}$$

contains an explicit logarithmic contribution $-Ln^2$, whereas the final leading term is only $-Ln^2/2$. The missing $+Ln^2/2$ is generated inside the alternating finite numerator. There is no contradiction: the outer Gaussian-binomial row is designed to annihilate all lower scale powers exactly, and the surviving leading coefficient is exponentially larger than a naive bounded-summand estimate would suggest.

The cancellation can be quantified without inspecting any individual summand. From (863),

$$\log \mathcal{N}_n = \frac{L}{2}n^2 + \mathcal{O}(n \log n), \tag{866}$$

while

$$-\log(2^{n^2}(1/2;1/2)_n) = -Ln^2 - \log(1/2;1/2)_\infty + o(1). \tag{867}$$

Their sum is $-Ln^2/2 + \mathcal{O}(n \log n)$, as required.

Remark 88.1 (Numerical conditioning). A floating-point implementation of the literal signed formula is normally inferior to the positive recurrence. It must resolve the two exact annihilations before dividing by a factor of size 2^{n^2}. The decompressed identity $\mathcal{N}_n = \mathfrak{m}_n a_n$ is therefore not only a proof device; it is the stable evaluation strategy.

89 How much of the full small-argument asymptotic is encoded by inverse dyadics?

89.1 The arbitrary-argument formula

For $x \downarrow 0$, let

$$\lambda(x) = -\frac{1}{L}W_{-1}(-Lx), \qquad \lambda(x)2^{-\lambda(x)} = x. \tag{868}$$

The full endpoint saddle has the same compact main term:

$$\log F(x) = -\frac{L}{2}\lambda(x)^2 + \left(1 + \frac{L}{2}\right)\lambda(x) - \frac{1}{2}\log \lambda(x) + C_F + \Psi(\lambda(x)) + \mathcal{O}(\lambda(x)^{-1}). \tag{869}$$

At $x = 2^{-n}$ this becomes (766). Thus the exact dyadic arithmetic recovers the restriction of the entire arbitrary-argument asymptotic, including the same constant and the same periodic function at the same exact phase.

89.2 Dense sampling of the periodic correction

The sequence of inverse dyadics does not freeze the periodic phase. Put

$$y_n = \lambda_n - n = \log_2 \lambda_n. \tag{870}$$

Then y_n is increasing and unbounded, while

$$y_{n+1} - y_n \longrightarrow 0. \tag{871}$$

Indeed, $\lambda_n \sim n$ and $y_{n+1} - y_n = \log_2(\lambda_{n+1}/\lambda_n) = \mathcal{O}(n^{-1})$.

Lemma 89.1 (Density of the dyadic Lambert phases)**.** *The fractional parts* $\{\lambda_n\} = \{y_n\}$ *are dense in* $[0, 1]$.

Proof. Fix $\theta \in [0, 1]$. For every sufficiently large integer m, let $n(m)$ be the first index for which $y_{n(m)} \geq m + \theta$. Then

$$0 \leq y_{n(m)} - (m + \theta) < y_{n(m)} - y_{n(m)-1}.$$

The right side tends to zero by (871). Hence the fractional parts of $y_{n(m)}$ tend to θ. □

After removing the nonperiodic terms in (767), the exact rational sequence therefore determines Ψ uniquely. More precisely, if

$$\begin{aligned} R_n = {} & \log F(2^{-n}) + \frac{L}{2} n^2 + n \log n - \left(1 + \frac{L}{2}\right) n + \frac{\log^2 n}{2L} \\ & - C_F + \frac{\log^2 n}{2L^2 n}, \end{aligned} \tag{872}$$

then

$$R_n = \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \tag{873}$$

For any $u \in [0, 1]$, choose a subsequence with $\{\lambda_{n_j}\} \to u$; continuity gives $R_{n_j} \to \Psi(u)$. In this precise sense the inverse-power values encode the whole periodic correction, not just a single phase sample.

89.3 What inverse powers alone do not prove

The density result does not by itself extend (766) uniformly to all $x \downarrow 0$. Monotonicity only brackets a point $2^{-(n+1)} \leq x \leq 2^{-n}$ between two endpoint values whose logarithms differ by order n:

$$\log F(2^{-n}) - \log F(2^{-(n+1)}) = Ln + \mathcal{O}(\log n). \tag{874}$$

That uncertainty is enormous compared with the desired $\mathcal{O}(n^{-1})$ remainder.

The global formula (808) contains every dyadic rational and could, in principle, be used on meshes much finer than one reciprocal-power interval. To derive (869) by that route one would need a two-parameter asymptotic uniform in both numerator and denominator. The arbitrary-argument Bromwich saddle avoids that difficult finite-sum uniformity. Accordingly, the exact inverse-power formulae recover the full phase function and all constants, but a genuinely uniform analytic argument remains necessary to pass from the dense arithmetic set to a sharp theorem on a continuum.

90 Numerical verification from exact rational values

The recurrence (778) computes a_n, hence $F(2^{-n})$, in exact rational arithmetic. For example,

$$F(2^{-10}) = \frac{134926369}{1246394851358539387238350848000}, \tag{875}$$

so

$$\log F(2^{-10}) = -50.5775682823421608125\ldots.$$

Let

$$\rho_n = \log F(2^{-n}) - \mathcal{M}_n. \tag{876}$$

The leading theorem predicts $\rho_n = \mathcal{O}(1/\lambda_n)$, and the first correction predicts $\lambda_n \rho_n = A_1(\lambda_n) + \mathcal{O}(1/\lambda_n)$. The table was computed by exact rational recurrence followed by one high-precision logarithm; P and its derivatives were evaluated from the absolutely convergent product and forward-tail series.

n	λ_n	ρ_n	$\lambda_n \rho_n$	$A_1(\lambda_n)$	$\lambda_n^2(\rho_n - A_1/\lambda_n)$
10	13.78503056	0.05801333439	0.7997155875	0.7629678468	0.5065687288
20	24.62186833	0.03182756966	0.7836542295	0.7629268731	0.5103462408
40	45.50804986	0.01701388954	0.7742689337	0.7629490545	0.5151456258
80	86.43351899	0.008896709914	0.7689739453	0.7629823191	0.5178773369
160	167.3870441	0.004576872612	0.7661091776	0.7630070725	0.5192522062

Table 20: Exact reciprocal-power values against the compact Lambert main and first correction.

The fourth and fifth columns approach one another, while the last column remains bounded and appears to settle, exactly as (865) predicts.

A second check displays the cancellation in the literal q-formula. Since

$$\log F(2^{-n}) = \log \mathcal{N}_n - n^2 L - \log(1/2; 1/2)_n,$$

one obtains:

n	$\log \mathcal{N}_n$	$n^2 L$	$\log(1/2;1/2)_n$	$\log F(2^{-n})$
10	17.4960644003	69.3147180560	-1.2410853733	-50.5775682823
20	95.4684669707	277.2588722240	-1.2420611411	-180.5483441121
40	447.4055728988	1109.0354888959	-1.2420620948	-660.3878539023
80	1957.8744511397	4436.1419555836	-1.2420620948	-2477.0254423491

Table 21: Logarithmic cancellation in the finite q-binomial formula.

The Pochhammer logarithm has already converged to

$$\log(1/2; 1/2)_\infty = -1.2420620948124149458\ldots. \tag{877}$$

The nontrivial compensation is between $\log \mathcal{N}_n \sim Ln^2/2$ and the explicit $-Ln^2$ denominator.

91 Correspondence with the Lean development

The relevant modules in the 25 August 2026 repository snapshot [14] form a fairly literal proof graph. The table records the role of each group rather than every import edge.

Module	Role in the bridge
`FabiusRecurrenceSequence.lean`	Defines $a_n = d_n/n!$, proves its recurrence, identifies $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$, and connects its ordinary series to the analytic generating function and dyadic product.
`FabiusInverseDyadicClosedForm.lean`	Unfolds the inverse-dyadic recurrence into the exact composition formula.

Module	Role in the bridge
`FabiusRawQBinomialFormula.lean`, `FabiusQBinomialFormula.lean`, `FabiusQBinomialScalarFormula.lean`, `FabiusDyadicQBinomialScalar.lean`	Develop the Thue–Morse block factorization, half-base Gaussian interpolation, scalar translation invariance, and the global exact dyadic formula. The coefficient-extraction argument in Section 81 is the ordinary-mathematics unpacking of this layer.
`FabiusDyadicLogBounds.lean`	Combines the exact endpoint identity with elementary half-moment bounds to prove the $-Ln^2/2$ law and an explicit $3n\log(n+1)$ error.
`NegativeLaplace.lean`, `PeriodicCorrection.lean`, `PeriodicFourier.lean`	Construct the negative-Laplace logarithm, its exact quadratic-plus-periodic decomposition, the forward-tail bound, regularity, mean normalization, and Fourier data.
`EndpointLaplaceComparison.lean`	Proves the second-order pointwise and integrated comparison between $(1-x)^n$ and $\mathrm{e}^{-nx}(1-nx^2/2)$ and transfers the estimate through the logarithm.
`DyadicSharpDecomposition.lean`, `FabiusDyadicSharpCumulant.lean`	Assemble the exact logarithmic decomposition of $F(2^{-n})$, including Stirling, the periodic term, the endpoint/Laplace error, and the exact second tilted cumulant.
`LaplacePeriodicSecondOrder.lean`	Differentiates the periodic Laplace decomposition and proves the explicit $\Psi'(\log_2 n)$ correction with an $\mathcal{O}(1/n)$ remainder.
`FabiusLambertPhase.lean`, `FabiusDyadicSharpAsymptotic.lean`	Construct the exact lower-Lambert phase, transfer the periodic derivative term to $\Psi(\lambda_n)$, and prove the corrected sharp dyadic theorem.
`FabiusSharpLambertMain.lean`, `FabiusSharpAsymptotic.lean`, `FabiusFullAsymptoticExpansion.lean`	Prove the arbitrary-small-argument Lambert main and continue the saddle expansion to all orders.

The coefficient-saddle derivation in Section 86 is deliberately marked as an alternative exposition. The formal dyadic proof uses the endpoint–Laplace comparison; the formal arbitrary-argument proof uses the Bromwich/saddle kernel. No claim is made that the Cauchy coefficient route appears as a separate theorem under that exact name.

92 Conclusions

The conspicuous gap between exact dyadic evaluation and endpoint asymptotics can be closed without adding a new special function or an independent heuristic ansatz.

First, the exact reciprocal-power values are controlled by a single coefficient sequence:

$$F(2^{-n}) = 2^{-\binom{n}{2}} a_n, \qquad A(z) = \sum_{n\geq 0} a_n z^n = \prod_{j\geq 1} E(z/2^j).$$

The recurrence and composition formula are simply two expansions of this identity.

Second, the q-binomial formula reduces exactly to the same sequence. The inner Thue–Morse sum is a coefficient of $A(-2^k X)/A(-X)$, and the outer half-base Gaussian row is the leading-coefficient functional on the nodes -2^k. Therefore

$$\mathcal{N}_n = \mathfrak{m}_n a_n$$

with no approximation. This is the direct algebraic connection that makes asymptotic analysis of the finite q-formula feasible.

Third, the negative-axis product of A satisfies an exact dyadic dilation equation. Removing its quadratic drift leaves a smooth periodic function. A second-order comparison between the endpoint kernel $(1-X)^n$ and the Laplace kernel e^{-nX} supplies the first cumulant correction; differentiating the exact periodic decomposition identifies that correction; and the lower-Lambert fixed point moves it to the exact phase. The result is

$$\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1+\frac{L}{2}\right)\lambda_n - \frac{1}{2}\log\lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).$$

Finally, the inverse dyadic sequence is richer than a sparse one-phase sample: the fractional parts of λ_n are dense, so the exact rational values determine the entire continuous periodic correction after renormalization. What they do not provide by monotonicity alone is a uniform continuum theorem at the sharp error scale; that step still requires a uniform saddle argument or a two-parameter analysis of the general dyadic formula.

The answer to the motivating question is therefore affirmative in a strong sense. The small-argument asymptotic can be re-derived from the exact calculation formulae, including the constant, the logarithmic correction, the exact lower-Lambert phase, and the nonconstant periodic fluctuation. The only essential move is to respect the exact cancellations built into those formulae rather than attempting to estimate their finite summands independently.

Supplementary material from this dossier

93 Algebra behind the expanded dyadic formula

This appendix records the expansion needed to pass from the compact Lambert form to the explicit expression in n. Put $\ell = \log n$ and seek

$$\lambda_n = n + \frac{\ell}{L} + \frac{c_1}{n} + \frac{c_2}{n^2} + \mathcal{O}(\ell^3/n^3). \tag{878}$$

The fixed-point equation is

$$\lambda_n = n + \frac{1}{L}\log\lambda_n. \tag{879}$$

Writing

$$\lambda_n = n\left(1 + \frac{\ell}{Ln} + \frac{c_1}{n^2} + \frac{c_2}{n^3} + \cdots\right)$$

and expanding the logarithm gives

$$\log\lambda_n = \ell + \frac{\ell}{Ln} + \frac{c_1 - \ell^2/(2L^2)}{n^2} + \mathcal{O}(\ell^3/n^3).$$

Comparison in (879) yields

$$c_1 = \frac{\ell}{L^2}, \qquad c_2 = \frac{\ell - \ell^2/2}{L^3}, \tag{880}$$

which gives the first four terms of (840).

Now set

$$H(\lambda) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda.$$

Substitution of (878) with (880) and collection through order $1/n$ gives

$$\begin{aligned} H(\lambda_n) = &-\frac{L}{2}n^2 - n\ell + \left(1 + \frac{L}{2}\right)n - \frac{\ell^2}{2L} \\ &- \frac{\ell^2}{2L^2 n} + \mathcal{O}(n^{-1}). \end{aligned} \tag{881}$$

The notation $\mathcal{O}(n^{-1})$ in (881) includes bounded multiples of $1/n$; the displayed ℓ^2/n term is separated because it is asymptotically larger. Adding $C_F + \Psi(\lambda_n)$ proves (847).

94 A stable numerical recipe

The following procedure reproduces Table 20 without evaluating the unstable signed q-sum.

Algorithm 94.1 (Exact arithmetic plus convergent products)**.**

1. Compute $a_0 = 1$ and, for $1 \le j \le n$,

   $$a_j = \frac{1}{2^j - 1}\sum_{k<j}\frac{a_k}{(j-k+1)!}$$

   in rational arithmetic.

2. Form the exact rational

   $$F(2^{-n}) = 2^{-\binom{n}{2}}a_n$$

   and take its logarithm only after conversion to the desired floating-point precision.

3. Solve $\lambda - \log\lambda/L = n$ on the large branch, either by $\lambda = -W_{-1}(-L2^{-n})/L$ or by Newton iteration initialized with $n + \log n/L$.
4. Reduce $u = \lambda - \lfloor\lambda\rfloor$ and put $s = 2^u \in [1,2)$. Evaluate
$$P(u) = \sum_{j\ge1} \log\frac{1-\mathrm{e}^{-s/2^j}}{s/2^j} + \frac{L}{2}(u^2-u) + \sum_{m\ge0}\log(1-\mathrm{e}^{-s2^m}).$$
The first series has a geometric small-argument tail and the second a doubly exponential tail. Use stable primitives such as `expm1` for the first factors.
5. Subtract $\overline{P}$ from (822) to obtain $\Psi(u)$. Differentiate the locally uniformly convergent series termwise, or use the rapidly convergent Fourier series (824), to obtain Ψ' and Ψ''.
6. Evaluate $\mathcal{M}_n$ from (846) and, when desired, A_1 from (864).

For a direct verification of the q-formula one may compute $\mathcal{N}_n = \mathfrak{m}_n a_n$ exactly and compare it with the finite signed expression. The equality is a better test than comparing floating-point approximations of the two sides because it checks both cancellation layers over the rationals.

95 Notation cross-reference

Symbol	Meaning
L	$\log 2$
F	bounded Fabius function / distribution function on $[0,1]$
up	Rvachev up-function, with $\mathrm{up}(t) = F(1-\lvert t\rvert)$ on $[-1,1]$
X	random variable with distribution function F
d_n	half moment $\mathbb{E}(1-X)^n$
a_n	factorially normalized half moment $d_n/n!$
$A(z)$	$\sum a_n z^n = \mathbb{E}\mathrm{e}^{zX}$
$E(z)$	$(\mathrm{e}^z-1)/z$
$\mathfrak{m}_n$	$\prod_{j=1}^{n}(2^j-1)$
$\mathcal{N}_n$	finite q-binomial/Thue–Morse numerator for $F(2^{-n})$
$B(s)$	$A(-s) = \mathbb{E}\mathrm{e}^{-sX}$
$\Lambda(s)$	$\log B(s)$
P	exact one-periodic correction in Λ
Ψ	$P - \overline{P}$, mean-zero periodic correction
C_F	constant $\overline{P} - \frac{1}{2}\log(2\pi)$
λ_n	large solution of $\lambda - \log\lambda/L = n$
$\mathcal{M}_n$	compact Lambert main term for $\log F(2^{-n})$

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Part VIII
Iterated Thue–Morse Prefix Sums and the Fabius Function

Source provenance and status. Former path: `Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex`. Elementary quantitative route through discrete curvature and cardinal moments. It overlaps the next dossier but retains distinct finite bounds, binary-phase expansions, exact special values, plots, and a formalization roadmap. The exact discrete bridge has Lean-backed ingredients; every quantitative rate, constant, and interpolation estimate in this dossier remains a natural-language proof and formal obligation.

Original source SHA-256: `577c5f3426def68a774fedf3fce61552d32200c8da526ace44178f2a8995a6d3`.

Dossier abstract

Let $\varepsilon_m = (-1)^{\mathrm{wt}_2(m)}$ be the signed Thue–Morse sequence and let $\sigma_{k+1}(m) = \sum_{j=0}^{m} \sigma_k(j)$ be its iterated *inclusive* prefix sums. The formal development in the `ProveIt` repository proves that

$$\mathcal{A}_n(x) = \frac{\sigma_{n+1}(\lfloor 2^n x \rfloor)}{2^{\binom{n}{2}}}, \qquad 0 \le x \le 1,$$

converges pointwise to the bounded Fabius function $F(x)$. The existing exposition deliberately asserts no rate. This report supplies one.

The coefficient carrying $\mathcal{A}_n(x)$ is the value of a polynomial histogram at a cell center, but that center is not $x-1$. Its exact displacement is

$$\delta_n(x) = \frac{n + 2 - 2\{2^n x\}}{2^{n+1}} = \frac{n}{2^{n+1}} + \frac{1 - \{2^n x\}}{2^n}.$$

At the histogram centers themselves, a Fourier-tail factorization and a cardinal moment calculation give the uniform estimate

$$|\mathcal{W}_n(\xi_{n,m}) - \mathrm{up}(\xi_{n,m})| \le \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n}.$$

Consequently, uniformly for $x \in [0,1]$ and $n \ge 3$,

$$\mathcal{A}_n(x) = F(x) + F'(x)\delta_n(x) + \frac{1}{2}F''(x)\delta_n(x)^2 + \rho_n(x),$$

where

$$|\rho_n(x)| \le \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n} + \frac{4}{3}(n+2)^3 8^{-n}.$$

In fact the entire normalized error profile converges uniformly:

$$\frac{2^n}{n}(\mathcal{A}_n - F) \longrightarrow \frac{1}{2}F' \qquad \text{in } L^\infty[0,1].$$

Consequently the uniform error has the sharp constant

$$\boxed{\lim_{n\to\infty} \frac{2^n}{n} \|\mathcal{A}_n - F\|_{L^\infty[0,1]} = 1.}$$

The factor n is therefore not an intrinsic weakness of the Thue–Morse coefficients. It is a deterministic centering drift. The histogram itself has uniform error asymptotic to 2^{-n}, while linear interpolation through the correct coefficient centers has the explicit faster bound $(4n/3+25/9)4^{-n}$. The report also records exact endpoint and midpoint formulas, a three-scale asymptotic expansion containing the binary phase $\{2^n x\}$, and a roadmap for formalizing the new quantitative theorem in Lean.

96 Scope, conventions, and the quantitative question

The purpose of this report is narrow but structurally revealing. The existing Fabius–Rvachev synthesis develops the discrete Thue–Morse construction in full: the prefix hierarchy, its binomial convolution kernel, the dyadic zero runs, the formal generating series, the corrected normalization, the exact identification with the polynomial histogram, and the resulting pointwise convergence [5, 6]. The corresponding Lean module `ThueMorseApproximation.lean` explicitly says that no quantitative uniform rate is asserted. We keep every convention of that development and add the missing estimate.

Throughout, F is the bounded Fabius distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$, and up is Rvachev's even compactly supported bump,

$$\operatorname{up}(y) = F(1-|y|), \qquad F(x) = \operatorname{up}(x-1) \quad (0 \le x \le 1). \tag{882}$$

We use the Fourier transform

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi \mathrm{i} x\xi}\,\mathrm{d}x, \qquad \operatorname{sinc} z = \frac{\sin z}{z}. \tag{883}$$

The derivative bounds proved sharply in the formal development are

$$\left\|F^{(r)}\right\|_\infty \le 2^{\binom{r+1}{2}}, \qquad \left\|\operatorname{up}^{(r)}\right\|_\infty \le 2^{\binom{r+1}{2}}. \tag{884}$$

We shall use, in particular,

$$0 \le F' \le 2, \qquad \|F''\|_\infty \le 8, \qquad \|F'''\|_\infty \le 64, \tag{885}$$

and $F'(1/2) = 2$.

The central question is not merely whether the prefix sequence converges, but which of three mechanisms controls its error:

1. the difference between the finite coefficient histogram and up at the exact coefficient centers;
2. the discretization incurred by replacing a smooth function with a step histogram;
3. the displacement between the center selected by the published floor index and the intended evaluation point.

The answer is unexpectedly clean. These effects have orders $n4^{-n}$, 2^{-n}, and $n2^{-n}$, respectively. The third dominates the particular floor-sampled approximation to F.

Status of formal verification. The exact prefix–coefficient identity and the qualitative convergence used here are already formalized in Lean. The quantitative center estimate, the asymptotic expansion, and the sharp constant 1 are new deductions in this report. They are presented with complete conventional proofs, but they have not been added to the repository's Lean files as of the snapshot date.

97 The discrete hierarchy

97.1 Signed Thue–Morse and inclusive summation

Let

$$\varepsilon_m = (-1)^{\mathrm{wt}_2(m)}, \tag{886}$$

where $\mathrm{wt}_2(m)$ is the number of 1's in the binary expansion of m. Define

$$\sigma_0(m) = \varepsilon_m, \qquad \sigma_{k+1}(m) = \sum_{j=0}^{m} \sigma_k(j). \tag{887}$$

The summation is inclusive. Thus

$$\sigma_{k+1}(m+1) - \sigma_{k+1}(m) = \sigma_k(m+1), \tag{888}$$

with the lower-order value evaluated at the new index. This one-place shift is responsible for the order correction that appears later.

Repeated summation gives the discrete B-spline convolution law

$$\sigma_k(m) = \sum_{r=0}^{m} \binom{m-r+k-1}{k-1} \varepsilon_r, \qquad k \geq 1. \tag{889}$$

Equivalently,

$$\sigma_{k+1}(m) = \sum_{r=0}^{m} \binom{m-r+k}{k} \varepsilon_r. \tag{890}$$

The kernel is the coefficient sequence of $(1-z)^{-k}$.

The formal generating functions make the same statement in one line. On the first dyadic block,

$$K_r(z) = \sum_{m=0}^{2^r-1} \varepsilon_m z^m = \prod_{j=0}^{r-1} (1 - z^{2^j}), \tag{891}$$

and coefficientwise the infinite signed series is

$$K(z) = \prod_{j \geq 0} (1 - z^{2^j}). \tag{892}$$

Hence the k-fold prefix series is formally

$$S_k(z) = \sum_{m \geq 0} \sigma_k(m) z^m = \frac{K(z)}{(1-z)^k}. \tag{893}$$

No analytic convergence is involved in this identity.

97.2 Dyadic annihilation

The product in (891) has a zero of order r at $z = 1$. Equivalently, a complete block of length 2^r annihilates every polynomial in m of degree below r. Applied to (889), this gives the exact zero run

$$\sigma_k(m) = 0 \quad \text{for} \quad 2^r - k \leq m \leq 2^r - 1, \qquad 1 \leq k \leq r, \tag{894}$$

with sharp boundary values

$$\sigma_k(2^r - k - 1) = (-1)^{r-k}, \qquad \sigma_k(2^r) = -1. \tag{895}$$

These identities explain both the long flat regions of the iterated prefixes and the exceptional right endpoint of the unit-interval grid.

98 The exact polynomial bridge

98.1 Approximation polynomials

Put

$$P_n(z) = \prod_{j=1}^{n}(1 + z + \cdots + z^{2^j-1}) = \sum_{m=0}^{g_n} a_{n,m} z^m, \tag{896}$$

with $P_0 = 1$. Its degree and total mass are

$$g_n = \sum_{j=1}^{n}(2^j - 1) = 2^{n+1} - n - 2, \qquad P_n(1) = 2^{\binom{n+1}{2}}. \tag{897}$$

The coefficients count restricted sums

$$a_{n,m} = \#\left\{(s_1, \dots, s_n) : 0 \le s_j < 2^j,\ \sum_{j=1}^{n} s_j = m\right\}. \tag{898}$$

They are nonnegative and symmetric: $a_{n,m} = a_{n,g_n-m}$.

The denominator-cleared product identity is

$$(1 - z)^{n+1} P_n(z) = \prod_{j=0}^{n}(1 - z^{2^j}). \tag{899}$$

Comparing (899) with (893) gives the exact coefficient bridge

$$\boxed{\sigma_{n+1}(m) = a_{n,m} \quad \text{whenever} \quad 0 \le m < 2^{n+1}.} \tag{900}$$

The restriction is real: at $m = 2^{n+1}$ the left side is -1 by (895), while the polynomial coefficient is zero.

98.2 The centered probability measure and histogram

Let

$$h_n = 2^{-n}, \qquad \xi_{n,m} = \frac{2m - g_n}{2^{n+1}} = h_n\left(m - \frac{g_n}{2}\right). \tag{901}$$

The centers form an arithmetic progression of mesh h_n. Define the atomic probability measure

$$\mu_n = \sum_{m\in\mathbb{Z}} p_{n,m}\,\delta_{\xi_{n,m}}, \qquad p_{n,m} = \frac{a_{n,m}}{2^{\binom{n+1}{2}}}, \tag{902}$$

where $a_{n,m} = 0$ outside $0 \le m \le g_n$. Let

$$\beta_h(x) = \frac{1}{h}\mathbf{1}_{[-h/2,h/2]}(x) \tag{903}$$

with the harmless half-weight convention at the two endpoints. The polynomial histogram is

$$\mathcal{W}_n = \mu_n * \beta_{h_n}. \tag{904}$$

At a center only its own atom contributes, and therefore

$$\mathcal{W}_n(\xi_{n,m}) = \frac{p_{n,m}}{h_n} = \frac{a_{n,m}}{2^{\binom{n}{2}}}. \tag{905}$$

Combining (900) and (905) yields the formalized exact identification

$$\frac{\sigma_{n+1}(m)}{2^{\binom{n}{2}}} = \mathcal{W}_n(\xi_{n,m}), \qquad 0 \le m < 2^{n+1}. \tag{906}$$

99 What the floor index actually samples

The corrected prefix sample for the bump is

$$\mathcal{J}_n(t) = \frac{\sigma_{n+1}(\lfloor 2^{n+1}t \rfloor)}{2^{\binom{n}{2}}}, \qquad 0 \le t \le 1. \tag{907}$$

For $t < 1$, (906) gives

$$\mathcal{J}_n(t) = \mathcal{W}_n\left(\xi_{n,\lfloor 2^{n+1}t \rfloor}\right). \tag{908}$$

The Fabius version is the first-half rescaling

$$\mathcal{A}_n(x) = \mathcal{J}_n(x/2) = \frac{\sigma_{n+1}(\lfloor 2^n x \rfloor)}{2^{\binom{n}{2}}}, \qquad 0 \le x \le 1. \tag{909}$$

Lemma 99.1 (Exact center drift). *For $x \in [0,1]$, put*

$$\theta_n(x) = \{2^n x\}, \qquad \delta_n(x) = \frac{n+2-2\theta_n(x)}{2^{n+1}}. \tag{910}$$

Then

$$\xi_{n,\lfloor 2^n x \rfloor} = x - 1 + \delta_n(x), \tag{911}$$

and

$$\delta_n(x) = \frac{n}{2^{n+1}} + \frac{1-\theta_n(x)}{2^n}, \qquad \frac{n}{2^{n+1}} < \delta_n(x) \le \frac{n+2}{2^{n+1}}. \tag{912}$$

Proof. Use $g_n = 2^{n+1} - n - 2$ and $\lfloor 2^n x \rfloor = 2^n x - \theta_n(x)$ in (901):

$$\begin{aligned}\xi_{n,\lfloor 2^n x \rfloor} &= \frac{2(2^n x - \theta_n(x)) - (2^{n+1} - n - 2)}{2^{n+1}} \\ &= x - 1 + \frac{n+2-2\theta_n(x)}{2^{n+1}}.\end{aligned}$$

The decomposition and bounds follow from $0 \le \theta_n(x) < 1$. □

The dominant term in (912) is $n2^{-n-1}$. It comes from the deficit $n+2$ in the degree $g_n = 2^{n+1} - n - 2$, not from random rounding. The binary fractional part contributes the lower-order oscillation $(1-\theta_n(x))2^{-n}$.

The source of the rate. The approximation does not read the histogram near $x-1$ and then incur a single-cell error. It reads a center roughly $n/2$ cells to the right. Since F' is generally nonzero, a smooth function changes by order $(n/2)h_n = n2^{-n-1}$ across that displacement. Any proof that ignores the exact center formula can at best obscure this leading term.

100 Fourier factorization of the omitted tail

100.1 The cosine products

The centered uniform variable on the 2^j lattice points of spacing h_n has Fourier transform

$$\frac{\sin(\pi 2^j h_n \xi)}{2^j \sin(\pi h_n \xi)} = \prod_{r=0}^{j-1} \cos(\pi 2^r h_n \xi). \tag{913}$$

Multiplying over $j = 1, \dots, n$ and collecting equal cosine factors gives

$$\widehat{\mu}_n(\xi) = \prod_{\ell=1}^{n} \cos\left(\frac{\pi\xi}{2^\ell}\right)^\ell. \tag{914}$$

The Rvachev bump has the infinite product

$$\widehat{\mathrm{up}}(\xi) = \prod_{\ell=1}^{\infty} \cos\left(\frac{\pi\xi}{2^\ell}\right)^\ell. \tag{915}$$

This is equivalent to the more familiar infinite sinc product.

Set $h = h_n$. Reindexing the tail of (915) gives

$$\begin{aligned}\prod_{\ell=n+1}^{\infty} \cos\left(\frac{\pi\xi}{2^\ell}\right)^\ell &= \prod_{r=1}^{\infty} \cos\left(\frac{\pi h\xi}{2^r}\right)^{n+r} \\ &= \widehat{\mathrm{up}}(h\xi)\left(\prod_{r=1}^{\infty} \cos\left(\frac{\pi h\xi}{2^r}\right)\right)^n \\ &= \widehat{\mathrm{up}}(h\xi)\,\mathrm{sinc}(\pi h\xi)^n. \end{aligned} \tag{916}$$

In the last line we used $\prod_{r\geq 1} \cos(z/2^r) = \sin z/z$.

Let $D_h \,\mathrm{up}(x) = h^{-1}\,\mathrm{up}(x/h)$ and define the tail density

$$q_n = \beta_h^{*n} * D_h \,\mathrm{up}\,. \tag{917}$$

The Fourier identity (916) is precisely

$$\boxed{\mathrm{up} = \mu_n * q_n.} \tag{918}$$

Thus the limit is obtained from the discrete polynomial law by adding n independent uniform jitters of width h and an independent scaled copy of the whole Rvachev law.

100.2 A cardinal probability distribution

At the mesh points define

$$\lambda_{n,k} = hq_n(kh), \qquad k \in \mathbb{Z}. \tag{919}$$

The $\lambda_{n,k}$ form a symmetric probability distribution on $\mathbb{Z}$. Indeed, write $q_n = \beta_h * \rho_n$. Since translates of the box tile the line,

$$h\sum_{k\in\mathbb{Z}} \beta_h(kh - s) = 1 \tag{920}$$

for almost every s. Integration against ρ_n yields

$$\sum_{k\in\mathbb{Z}} \lambda_{n,k} = 1, \qquad \lambda_{n,-k} = \lambda_{n,k}. \tag{921}$$

Equation (918), together with $p_{n,m} = h\mathcal{W}_n(\xi_{n,m})$, now becomes the exact cardinal averaging identity

$$\mathrm{up}(\xi_{n,m}) = \sum_{k\in\mathbb{Z}} \lambda_{n,k}\mathcal{W}_n(\xi_{n,m-k}). \tag{922}$$

The value of the smooth limit at a center is therefore a symmetric local average of the neighboring coefficient heights.

The second moment of this discrete averaging law is also exact.

Lemma 100.1 (Cardinal second moment)**.** *For $n \geq 3$,*

$$\sum_{k\in\mathbb{Z}} k^2\lambda_{n,k} = \frac{n}{12} + \frac{1}{9}. \tag{923}$$

Proof. After scaling by h, let

$$\kappa_n = \beta_1^{*n} * \mathrm{up}, \qquad \lambda_{n,k} = \kappa_n(k).$$

Its Fourier transform is $\widehat{\kappa}_n(\xi) = \mathrm{sinc}(\pi\xi)^n\widehat{\mathrm{up}}(\xi)$. Poisson summation, differentiated twice, gives

$$-4\pi^2\sum_{k\in\mathbb{Z}} k^2\kappa_n(k) = \sum_{j\in\mathbb{Z}} \widehat{\kappa}_n''(j).$$

At every nonzero integer j, the factor $\mathrm{sinc}(\pi\xi)^n$ has a zero of order $n \geq 3$, so the second derivative vanishes there. Only $j = 0$ remains, and hence the discrete second moment equals the ordinary second moment of κ_n.

The variance of a uniform variable on $[-1/2, 1/2]$ is $1/12$. The cosine product (915) represents the Rvachev law as the sum of ℓ independent signs of amplitude $2^{-\ell-1}$ at every level ℓ; therefore

$$\mathrm{Var}(\mathrm{up}) = \sum_{\ell=1}^{\infty}\frac{\ell}{2^{2\ell+2}} = \frac{1}{4}\sum_{\ell=1}^{\infty}\ell 4^{-\ell} = \frac{1}{9}.$$

Variances add under convolution, giving (923). □

101 Discrete curvature of the coefficient rows

Write

$$w_{n,m} = \frac{a_{n,m}}{2^{\binom{n}{2}}}, \tag{924}$$

with $w_{n,m} = 0$ outside the polynomial support. Thus $w_{n,m} = \mathcal{W}_n(\xi_{n,m})$.

We first need a uniform coefficient bound. The recurrence

$$P_r(z) = P_{r-1}(z)(1 + z + \cdots + z^{2^r-1}) \tag{925}$$

shows that every coefficient of P_r is a partial sum of coefficients of P_{r-1}. Consequently

$$0 \leq a_{r,m} \leq P_{r-1}(1) = 2^{\binom{r}{2}}. \tag{926}$$

Equality holds on the central plateau $g_{r-1} \leq m \leq 2^r - 1$.

The relevant smoothness is measured by the second finite difference.

Lemma 101.1 (Uniform discrete curvature)**.** *For $n \geq 3$ and every $m \in \mathbb{Z}$,*

$$|w_{n,m} - 2w_{n,m-1} + w_{n,m-2}| \leq 32\,4^{-n}. \tag{927}$$

Proof. Factor off the last two geometric polynomials:

$$(1-z)^2P_n(z) = (1 - z^{2^{n-1}})(1 - z^{2^n})P_{n-2}(z). \tag{928}$$

The coefficient on the left is $a_{n,m} - 2a_{n,m-1} + a_{n,m-2}$. The coefficient on the right is a signed sum of four coefficients of P_{n-2}. By (926), each has absolute value at most $2^{\binom{n-2}{2}}$. Division by $2^{\binom{n}{2}}$ gives

$$4\,2^{\binom{n-2}{2}-\binom{n}{2}} = 4\,2^{-(2n-3)} = 32\,4^{-n}.$$

□

Lemma 101.2 (Symmetric discrete Taylor bound). *If a doubly infinite sequence* (v_m) *satisfies* $|v_m - 2v_{m-1} + v_{m-2}| \le M$ *for all* m*, then*

$$|v_{m+k} + v_{m-k} - 2v_m| \le Mk^2 \qquad (k \ge 0). \tag{929}$$

Proof. Let $d_m = v_m - v_{m-1}$. Consecutive first differences satisfy $|d_m - d_{m-1}| \le M$. Expanding the two increments from m to $m+k$ and from $m-k$ to m, and pairing their first differences, yields a sum of $1 + 3 + \cdots + (2k-1) = k^2$ second differences. □

102 The intrinsic error at coefficient centers

We can now quantify the part of the approximation that the qualitative moving argument proof leaves hidden.

Theorem 102.1 (Uniform center estimate). *For every* $n \ge 3$ *and every* $m \in \mathbb{Z}$*,*

$$\boxed{|\mathcal{W}_n(\xi_{n,m}) - \mathrm{up}(\xi_{n,m})| \le \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n}.} \tag{930}$$

Proof. Use (922), symmetry of $\lambda_{n,k}$, and $\sum_k \lambda_{n,k} = 1$:

$$\mathrm{up}(\xi_{n,m}) - w_{n,m} = \sum_{k\ge 1} \lambda_{n,k}\big(w_{n,m+k} + w_{n,m-k} - 2w_{n,m}\big).$$

By lemmas 101.1 and 101.2, the absolute value of the bracket is at most $32\,4^{-n}k^2$. Therefore

$$\begin{aligned}
|\mathrm{up}(\xi_{n,m}) - w_{n,m}| &\le 32\,4^{-n} \sum_{k\ge 1} k^2 \lambda_{n,k} \\
&= 16\,4^{-n} \sum_{k\in\mathbb{Z}} k^2 \lambda_{n,k} \\
&= 16\,4^{-n} \left(\frac{n}{12} + \frac{1}{9}\right),
\end{aligned}$$

which is (930). □

The proof separates two independent facts. The coefficient row has curvature $\mathcal{O}(4^{-n})$ per lattice step, while the omitted Fourier tail averages over a discrete variance of order n. Their product is $\mathcal{O}(n4^{-n})$.

Remark 102.2 (An elementary but weaker constant). If one avoids the Poisson calculation in lemma 100.1, the support of q_n gives $\lambda_{n,k} = 0$ for $|k| > (n+2)/2$. The same argument then yields

$$|\mathcal{W}_n(\xi_{n,m}) - \mathrm{up}(\xi_{n,m})| \le 4(n+2)^2 4^{-n}.$$

This already suffices for the sharp $n2^{-n}$ law. The cardinal moment is what improves the intrinsic estimate from $\mathcal{O}(n^2 4^{-n})$ to $\mathcal{O}(n4^{-n})$.

103 The sharp expansion for the Fabius sample

We now combine the exact drift with the center estimate. Recall $\mathcal{A}_n$ from (909) and δ_n from (910).

Theorem 103.1 (Uniform second-order expansion)**.** *For every $n \geq 3$ and every $x \in [0,1]$,*

$$\boxed{\mathcal{A}_n(x) = F(x) + F'(x)\delta_n(x) + \frac{1}{2}F''(x)\delta_n(x)^2 + \rho_n(x),} \tag{931}$$

where

$$|\rho_n(x)| \leq \left(\frac{4n}{3} + \frac{16}{9}\right)4^{-n} + \frac{4}{3}(n+2)^3 8^{-n}. \tag{932}$$

Proof. Put $m = \lfloor 2^n x \rfloor$ and $y = x - 1$. Since $m < 2^{n+1}$, (906) and lemma 99.1 give

$$\mathcal{A}_n(x) = \mathcal{W}_n(\xi_{n,m}), \qquad \xi_{n,m} = y + \delta_n(x).$$

By theorem 102.1, replacing $\mathcal{W}_n$ by up at this center costs at most $(4n/3 + 16/9)4^{-n}$. Taylor's theorem and $\|\mathrm{up}'''\|_\infty \leq 64$ give

$$\mathrm{up}(y+\delta) = \mathrm{up}(y) + \mathrm{up}'(y)\delta + \frac{1}{2}\,\mathrm{up}''(y)\delta^2 + R, \qquad |R| \leq \frac{64}{6}\delta^3.$$

By (912), $\delta^3 \leq (n+2)^3/2^{3n+3}$, so $|R| \leq \frac{4}{3}(n+2)^3 8^{-n}$. Finally $\mathrm{up}^{(r)}(x-1) = F^{(r)}(x)$ on $[0,1]$ by (882). □

For the leading rate it is convenient to absorb the quadratic Taylor term.

Corollary 103.2 (Uniform first-order expansion)**.** *For $n \geq 3$ and $x \in [0,1]$,*

$$\mathcal{A}_n(x) = F(x) + F'(x)\delta_n(x) + r_n(x), \tag{933}$$

with

$$|r_n(x)| \leq \left(n^2 + \frac{16}{3}n + \frac{52}{9}\right)4^{-n} < (n+3)^2 4^{-n}. \tag{934}$$

Proof. Taylor's theorem to first order, $\|F''\|_\infty \leq 8$, and $\delta_n(x) \leq (n+2)2^{-n-1}$ give a quadratic remainder at most $(n+2)^2 4^{-n}$. Add (930). □

Substituting (912) into (931) exposes three distinct scales.

Corollary 103.3 (Binary-phase expansion)**.** *Uniformly for $x \in [0,1]$,*

$$\begin{aligned}\mathcal{A}_n(x) - F(x) = {} & \frac{1}{2}F'(x)\,n2^{-n} + F'(x)(1-\theta_n(x))2^{-n} \\ & + \frac{1}{8}F''(x)\,n^2 4^{-n} + \mathcal{O}(n4^{-n}).\end{aligned} \tag{935}$$

The implied constant is absolute.

Proof. Expand $\delta_n = (n/2 + 1 - \theta_n)2^{-n}$ and its square in (931). The cross term in δ_n^2 is $\mathcal{O}(n4^{-n})$, the constant term is $\mathcal{O}(4^{-n})$, and the remainder in (932) is uniformly $\mathcal{O}(n4^{-n})$ because $n^3 8^{-n} = o(n4^{-n})$. □

The second term in (935) is not periodic in n for a general real x, but it is a literal binary shift: $\theta_n(x)$ is the fractional tail left after moving the binary point n places. For a dyadic x, it vanishes eventually and the coefficient of 2^{-n} becomes exactly $F'(x)$.

104 Pointwise and uniform sharpness

104.1 Uniform error profile and pointwise law

Theorem 104.1 (Uniform normalized error profile)**.** *For every $n \geq 3$,*

$$\boxed{\sup_{x\in[0,1]} \left| \frac{2^n}{n}(\mathcal{A}_n(x) - F(x)) - \frac{1}{2}F'(x) \right| \leq \frac{2}{n} + \left(n + \frac{16}{3} + \frac{52}{9n}\right) 2^{-n}.} \tag{936}$$

Consequently

$$\frac{2^n}{n}(\mathcal{A}_n - F) \longrightarrow \frac{1}{2}F' \qquad \textit{uniformly on } [0,1]. \tag{937}$$

In particular, for every fixed $x \in (0,1)$,

$$\lim_{n\to\infty} \frac{2^n}{n}(\mathcal{A}_n(x) - F(x)) = \frac{1}{2}F'(x). \tag{938}$$

Since $F'(x) > 0$ on $(0,1)$, the order $n2^{-n}$ is sharp at every interior point.

Proof. From (912),

$$\frac{2^n}{n}\delta_n(x) = \frac{1}{2} + \frac{1 - \theta_n(x)}{n}.$$

Insert this in (933). Since $0 \leq F' \leq 2$ and $0 < 1 - \theta_n(x) \leq 1$, the phase term is bounded by $2/n$. Multiplying (934) by $2^n/n$ gives the second term on the right of (936). □

At the endpoints the limiting profile vanishes. Their unscaled errors obey a different, much faster regime; see section 105.

104.2 The exact uniform constant

The upper bound follows immediately from $0 \leq F' \leq 2$ and corollary 103.2:

$$\|\mathcal{A}_n - F\|_{L^\infty[0,1]} \leq (n+2)2^{-n} + (n+3)^2 4^{-n}. \tag{939}$$

To prove that the coefficient is not an artifact of this estimate, we evaluate the midpoint exactly in proposition 105.1. That formula gives

$$\|\mathcal{A}_n - F\|_\infty \geq (n+2)2^{-n} - 2^{-\binom{n}{2}}. \tag{940}$$

Theorem 104.2 (Sharp uniform convergence rate)**.** *For every $n \geq 3$, the corrected Thue–Morse prefix approximation satisfies the finite-level bracket*

$$\boxed{-2^{-\binom{n}{2}} \leq \|\mathcal{A}_n - F\|_{L^\infty[0,1]} - (n+2)2^{-n} \leq (n+3)^2 4^{-n}.} \tag{941}$$

Consequently

$$\boxed{\|\mathcal{A}_n - F\|_{L^\infty[0,1]} = (n+2)2^{-n} + \mathcal{O}(n^2 4^{-n}) \sim n2^{-n}, \qquad \lim_{n\to\infty} \frac{2^n}{n} \|\mathcal{A}_n - F\|_{L^\infty[0,1]} = 1.} \tag{942}$$

Proof. Subtract $(n+2)2^{-n}$ from (939) and (940). This gives (941); the asymptotic statements follow immediately. Alternatively, the last limit also follows by taking sup norms in the uniform profile limit (937), since $\|F'/2\|_\infty = 1$. □

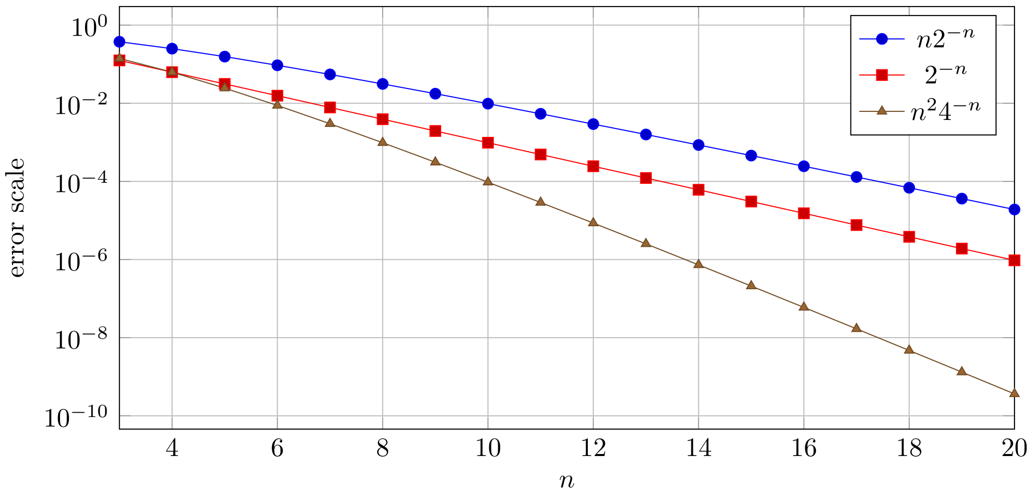

Figure 4: The three scales in (935). The deterministic center drift $n2^{-n}$ dominates the one-cell scale 2^{-n} and the quadratic Taylor scale $n^2 4^{-n}$.

105 Exact special values

The general estimates become especially transparent at 0, 1/2, and 1.

105.1 The endpoints

Since $a_{n,0} = 1$,

$$\mathcal{A}_n(0) = 2^{-\binom{n}{2}}. \tag{943}$$

At $x = 1$, the index is 2^n. Using $P_n = P_{n-1}(1 + z + \cdots + z^{2^n-1})$ and $g_{n-1} < 2^n$, the coefficient at z^{2^n} is the sum of all coefficients of P_{n-1} except the constant term. Thus

$$\mathcal{A}_n(1) = 1 - 2^{-\binom{n}{2}}. \tag{944}$$

Hence

$$\mathcal{A}_n(0) - F(0) = 2^{-\binom{n}{2}}, \qquad \mathcal{A}_n(1) - F(1) = -2^{-\binom{n}{2}}. \tag{945}$$

The endpoint convergence is superexponential in n, vastly faster than the interior $n2^{-n}$ law.

105.2 An exact midpoint coefficient

Proposition 105.1 (Exact midpoint error)**.** *For every $n \geq 2$,*

$$\boxed{\mathcal{A}_n\left(\frac{1}{2}\right) = \frac{1}{2} + \frac{n+2}{2^n} - 2^{-\binom{n}{2}}.} \tag{946}$$

Consequently

$$\frac{2^n}{n}\left(\mathcal{A}_n\left(\frac{1}{2}\right) - \frac{1}{2}\right) = 1 + \frac{2}{n} - \frac{2^{n-\binom{n}{2}}}{n}. \tag{947}$$

Proof. Let $b_r = [z^r]P_{n-1}(z)$ and $M = 2^{n-1}$. Since $M < 2^n$,

$$[z^M]P_n(z) = \sum_{r=0}^{M} b_r.$$

Put $D = g_{n-2} = 2^{n-1} - n$ and $C = P_{n-2}(1) = 2^{\binom{n-1}{2}}$. From $P_{n-1} = P_{n-2}(1 + \cdots + z^{2^{n-1}-1})$, the central coefficients are

$$b_{D-1} = C - 1, \qquad b_D = b_{D+1} = \cdots = b_{2^{n-1}-1} = C, \qquad b_{2^{n-1}} = C - 1.$$

The polynomial P_{n-1} is symmetric, has degree $g_{n-1} = 2^n - n - 1$, and total mass $2^{\binom{n}{2}}$. The interval $[g_{n-1} - M, M] = [D-1, M]$ is precisely the displayed central block. Pairing symmetric coefficients gives

$$\begin{aligned} 2\sum_{r=0}^{M} b_r &= 2^{\binom{n}{2}} + \sum_{r=D-1}^{M} b_r \\ &= 2^{\binom{n}{2}} + (n+2)C - 2. \end{aligned}$$

Divide by $2^{\binom{n}{2}}$ and use $C/2^{\binom{n}{2}} = 2^{-(n-1)}$. □

n	$\mathcal{A}_n(1/2) - 1/2$	$(2^n/n)(\mathcal{A}_n(1/2) - 1/2)$	leading value $1 + 2/n$
4	0.3593750000	1.4375000000	1.5000000000
6	0.1249694824	1.3330078125	1.3333333333
8	0.0390624963	1.2499998808	1.2500000000
10	0.0117187500	1.2000000000	1.2000000000
12	0.0034179688	1.1666666667	1.1666666667
14	0.0009765625	1.1428571429	1.1428571429
16	0.0002746582	1.1250000000	1.1250000000
20	0.0000209808	1.1000000000	1.1000000000

The tiny discrepancy from $1 + 2/n$ is exactly the superexponential last term in (947).

106 The corresponding result for Rvachev's bump

For $0 \le t < 1$, let

$$\vartheta_n(t) = \left\{2^{n+1}t\right\}, \qquad \Delta_n(t) = \frac{n + 2 - 2\vartheta_n(t)}{2^{n+1}}. \tag{948}$$

Then

$$\xi_{n,\lfloor 2^{n+1}t\rfloor} = 2t - 1 + \Delta_n(t). \tag{949}$$

The proof of theorem 103.1 gives, uniformly for $0 \le t < 1$,

$$\begin{aligned} \mathcal{J}_n(t) &= \mathrm{up}(2t-1) + \mathrm{up}'(2t-1)\Delta_n(t) \\ &\quad + \frac{1}{2}\,\mathrm{up}''(2t-1)\Delta_n(t)^2 + \rho_n^{\mathrm{up}}(t), \end{aligned} \tag{950}$$

with the same bound as (932). The right endpoint is exceptional because the coefficient identity stops one index earlier:

$$\mathcal{J}_n(1) = -2^{-\binom{n}{2}}, \qquad \mathrm{up}(1) = 0. \tag{951}$$

This exceptional error is negligible on every exponential scale discussed here.

Corollary 106.1 (Pointwise and uniform rate for up)**.** *For fixed $t \in (0, 1)$,*

$$\frac{2^n}{n}\left(\mathcal{J}_n(t) - \mathrm{up}(2t-1)\right) \longrightarrow \frac{1}{2}\,\mathrm{up}'(2t-1). \tag{952}$$

At $t = 1/2$, both the derivative term and the actual error collapse:

$$\mathcal{J}_n(1/2) = 1 - 2^{-\binom{n}{2}}. \tag{953}$$

Moreover,

$$\lim_{n\to\infty} \frac{2^n}{n} \left\|\mathcal{J}_n - \mathrm{up}(2\,\cdot -1)\right\|_{L^\infty[0,1]} = 1. \tag{954}$$

The uniform lower bound may be read at $t = 1/4$, since $\mathcal{J}_n(1/4) = \mathcal{A}_n(1/2)$ and $\mathrm{up}(-1/2) = F(1/2) = 1/2$.

107 Rates for the histogram and for recentered interpolation

The sharp $n2^{-n}$ rate belongs to the particular moving sample (909). It should not be attributed to the coefficient histogram itself.

107.1 The step histogram

Every interior point of a histogram cell lies within $h_n/2$ of its center. Since up is 2-Lipschitz and the center error is given by theorem 102.1,

$$\|\mathcal{W}_n - \mathrm{up}\|_\infty \le 2^{-n} + \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n}. \tag{955}$$

The constant in front of 2^{-n} is sharp. Near $y = -1/2$, where $\mathrm{up}'(y) = 2$, move from a cell center to a point arbitrarily close to the right edge of the same cell. The histogram remains constant while up rises by $h_n + o(h_n)$; the center error is $o(h_n)$. Therefore

$$\boxed{\lim_{n\to\infty} 2^n \|\mathcal{W}_n - \mathrm{up}\|_\infty = 1.} \tag{956}$$

Thus the step representation has the expected one-cell rate 2^{-n}.

107.2 Linear interpolation through the true centers

Extend $w_{n,m}$ by zero for all $m \in \mathbb{Z}$, and let $\mathcal{L}_n$ be the piecewise-linear function satisfying

$$\mathcal{L}_n(\xi_{n,m}) = w_{n,m} \qquad (m \in \mathbb{Z}). \tag{957}$$

Let $I_n\,\mathrm{up}$ be the ordinary linear interpolant of the exact values $\mathrm{up}(\xi_{n,m})$ on the same mesh. Convexity of interpolation weights and theorem 102.1 give

$$\|\mathcal{L}_n - I_n\,\mathrm{up}\|_\infty \le \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n}.$$

For a C^2 function on a mesh of width h, the linear interpolation error is at most $h^2 \|f''\|_\infty /8$. Since $\|\mathrm{up}''\|_\infty \le 8$,

$$\boxed{\|\mathcal{L}_n - \mathrm{up}\|_\infty \le \left(\frac{4n}{3} + \frac{25}{9}\right) 4^{-n}.} \tag{958}$$

Translating by one gives the same bound for an interpolant of F on $[0, 1]$.

This estimate removes both the step error and the deterministic $n/2$-cell drift. It is the most direct quantitative evidence that the iterated Thue–Morse coefficients are intrinsically accurate; the slower floor-sample rate is a coordinate choice.

Approximation	Construction	Uniform scale
Center values	$w_{n,m}$ compared with $\mathrm{up}(\xi_{n,m})$	$\mathcal{O}(n4^{-n})$
Linear center interpolant	interpolate $(\xi_{n,m}, w_{n,m})$	$\mathcal{O}(n4^{-n})$
Step histogram	keep $w_{n,m}$ constant on its cell	$\sim 2^{-n}$
Floor sample for F	read $m = \lfloor 2^n x \rfloor$ without recentering	$\sim n2^{-n}$
Centered finite spline	different convolution/CDF scheme in the synthesis	at most 2^{-n}
Literal unshifted prefix grid	use order n rather than $n+1$	fails at the endpoint

108 Why the constants take these values

The proof contains four constants, each with a transparent origin.

1. The coefficient curvature constant 32 is
$$4 \cdot 2^{\binom{n-2}{2}-\binom{n}{2}} = 32\,4^{-n}.$$
The factor 4 counts the four shifted copies of P_{n-2} in (928).
2. The discrete variance $n/12 + 1/9$ consists of n unit-width box jitters, each of variance $1/12$, plus the variance $1/9$ of the scaled Rvachev tail.
3. The leading displacement is $n2^{-n-1}$. Multiplication by the maximal slope $F'(1/2) = 2$ gives the uniform leading error $n2^{-n}$ and hence the sharp constant 1.
4. The quadratic Taylor coefficient uses the sharp bound $\|F''\|_\infty \leq 8$, while the cubic remainder uses $\|F'''\|_\infty \leq 64$. These are the $r = 2$ and $r = 3$ instances of (884), not fitted numerical estimates.

The hierarchy can be summarized schematically as
$$\underbrace{n2^{-n}}_{\text{degree/center drift}} \gg \underbrace{2^{-n}}_{\text{binary floor phase and one-cell error}} \gg \underbrace{n^2 4^{-n}}_{\text{quadratic Taylor term}} \gg \underbrace{n4^{-n}}_{\text{intrinsic center error}}. \tag{959}$$
For fixed x, the first two terms are explicitly visible in (935). The third is the first smooth nonlinear correction. The fourth is where the finite coefficient law itself differs from the infinite Rvachev law.

109 A Lean formalization roadmap

The new theorem is well aligned with the current module decomposition. The existing formalized chain is:

1. `ThueMorsePrefix.lean`: `iteratedPrefix`, the binomial convolution law, Prouhet cancellation, and exact dyadic zero runs;
2. `ThueMorseGenerating.lean`: formal power-series identities and the corrected normalized grid;
3. `ThueMorseApproximation.lean`: `iteratedPrefix_eq_approximationPolynomial_coeff`, `correctedPrefixCoefficient_eq_stepApproximant`, and the qualitative convergence theorems;
4. `StepApproximationLimit.lean`: coefficient monotonicity and the moving-argument squeeze used for pointwise convergence.

A formal proof of theorem 104.2 can be organized into the following small additions.

1. Define the zero-extended normalized coefficient row $w_{n,m}$ on $\mathbb{Z}$ and prove (927) by coefficient extraction from (928). This is algebraic and should require no measure theory.
2. Reuse the existing cosine-product theorem to define q_n and prove $\mathrm{up} = \mu_n * q_n$. The identity (916) is a direct reindexing plus the standard infinite cosine product.
3. Prove the cardinal partition $\sum_k hq_n(kh) = 1$ by factoring one box density. The center averaging identity (922) then follows from convolution with an atomic measure.
4. For a first formal rate, use the compact support of q_n and obtain the elementary $4(n+2)^2 4^{-n}$ center bound. This avoids differentiated Poisson summation and already proves the sharp $n2^{-n}$ limit.
5. In a second refinement, invoke the repository's Poisson-summation layer to prove the exact second moment (923); this upgrades the center estimate to (930).
6. Formalize the exact drift identity (911), apply the repository's derivative bounds, and finish with a uniform Taylor estimate. The exact midpoint formula (946) supplies the matching lower bound without any compactness argument.

The logical core of the sharp constant requires only the weaker center estimate. Thus the project can be split cleanly: first formalize the asymptotic constant 1 with elementary support bounds, then strengthen the remainder by adding the cardinal second-moment calculation.

110 Conclusion

The iterated signed Thue–Morse prefixes approach the Fabius function through a sequence of exact algebraic identifications:

$$\text{prefix sums} \longleftrightarrow \text{coefficients of } P_n \longleftrightarrow \text{center heights of } \mathcal{W}_n \longrightarrow \mathrm{up} \longrightarrow F.$$

The first two arrows are exact below the first omitted dyadic coefficient. The last two were already known qualitatively. The quantitative analysis shows that the finite coefficient law is exceptionally accurate at its natural centers: its error is only $\mathcal{O}(n4^{-n})$. The floor sample used to recover F reads those centers at a deterministic displacement

$$\delta_n(x) = \frac{n}{2^{n+1}} + \frac{1-\{2^n x\}}{2^n}.$$

That displacement produces the complete leading behavior.

The final result is therefore both sharp and explanatory:

$$\mathcal{A}_n(x) - F(x) = F'(x)\delta_n(x) + \mathcal{O}(n^2 4^{-n}), \qquad \frac{2^n}{n}(\mathcal{A}_n - F) \longrightarrow \frac{1}{2}F' \quad \text{uniformly.}$$

Moreover

$$\|\mathcal{A}_n - F\|_\infty = (n+2)2^{-n} + \mathcal{O}(n^2 4^{-n}).$$

The leading pointwise profile is $F'/2$, the leading uniform constant is 1, and the next term remembers the binary tail $\{2^n x\}$. Recenter and interpolate the same coefficients, and the factor $n2^{-n}$ disappears, leaving the much faster explicit bound $\mathcal{O}(n4^{-n})$.

Supplementary material from this dossier

111 Notation crosswalk

Symbol	Meaning
ε_m	signed Thue–Morse value $(-1)^{\mathrm{wt}_2(m)}$
$\sigma_k(m)$	k-fold inclusive prefix sum of ε_m
$P_n(z)$	$\prod_{j=1}^{n}(1+z+\cdots+z^{2^j-1})$
$a_{n,m}$	coefficient $[z^m]P_n(z)$
g_n	degree $2^{n+1}-n-2$ of P_n
h_n	mesh width 2^{-n}
$\xi_{n,m}$	centered coefficient location $(2m-g_n)/2^{n+1}$
$w_{n,m}$	normalized height $a_{n,m}/2^{\binom{n}{2}}$
μ_n	centered atomic coefficient distribution
$\mathcal{W}_n$	box-smoothed coefficient histogram $\mu_n * \beta_{h_n}$
$\mathcal{J}_n(t)$	corrected bump sample $\sigma_{n+1}(\lfloor 2^{n+1}t\rfloor)/2^{\binom{n}{2}}$
$\mathcal{A}_n(x)$	Fabius sample $\mathcal{J}_n(x/2)$
$\delta_n(x)$	exact center displacement in (910)
$\theta_n(x)$	binary phase $\{2^n x\}$
$\lambda_{n,k}$	cardinal tail weights $h_n q_n(kh_n)$

112 A compact statement of the main estimates

For rapid reference, for every $n \geq 3$:

$$\begin{aligned}
\sup_{m\in\mathbb{Z}} |\mathcal{W}_n(\xi_{n,m}) - \mathrm{up}(\xi_{n,m})| &\leq \left(\frac{4n}{3}+\frac{16}{9}\right)4^{-n},\\
\|\mathcal{W}_n - \mathrm{up}\|_\infty &\leq 2^{-n} + \left(\frac{4n}{3}+\frac{16}{9}\right)4^{-n},\\
\|\mathcal{L}_n - \mathrm{up}\|_\infty &\leq \left(\frac{4n}{3}+\frac{25}{9}\right)4^{-n},\\
\sup_{x\in[0,1]} \left|\mathcal{A}_n(x) - F(x) - F'(x)\delta_n(x)\right| &\leq \left(n^2+\frac{16}{3}n+\frac{52}{9}\right)4^{-n},\\
\sup_{x\in[0,1]} \left|\mathcal{A}_n(x) - F(x) - F'(x)\delta_n(x) - \tfrac{1}{2}F''(x)\delta_n(x)^2\right| &\leq \left(\frac{4n}{3}+\frac{16}{9}\right)4^{-n} + \frac{4}{3}(n+2)^3 8^{-n}.
\end{aligned}$$

The normalized profile and norm satisfy

$$\begin{aligned}
&\sup_{x\in[0,1]} \left|\frac{2^n}{n}(\mathcal{A}_n(x)-F(x)) - \frac{1}{2}F'(x)\right| \leq \frac{2}{n} + \left(n+\frac{16}{3}+\frac{52}{9n}\right)2^{-n},\\
&-2^{-\binom{n}{2}} \leq \|\mathcal{A}_n - F\|_\infty - (n+2)2^{-n} \leq (n+3)^2 4^{-n}.
\end{aligned}$$

The exact asymptotic conclusions are

$$\begin{aligned}
\frac{2^n}{n}(\mathcal{A}_n - F) &\longrightarrow \frac{1}{2}F' && \text{uniformly on } [0,1],\\
\frac{2^n}{n}\|\mathcal{A}_n - F\|_\infty &\longrightarrow 1,\\
2^n\|\mathcal{W}_n - \mathrm{up}\|_\infty &\longrightarrow 1.
\end{aligned}$$

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Part IX
Signed Thue–Morse Prefix Sums and the Fabius Function

Source provenance and status. Former path: `Fabius_Thue_Morse_Convergence_Rate-2/Fabius_Thue_Morse_Convergence_Rates.tex`. Sharper Fourier/deconvolution treatment of the same approximation family, including center-grid expansions, sharp constants, all-orders patterns, and accelerated reconstructions. Its qualitative bridge reuses formal inputs, but its deconvolution estimates, quantitative rates, sharp constants, all-orders pattern, and acceleration results remain natural-language proofs and formal obligations.

Original source SHA-256: `384d69b461cb94af33f1c080703e269ea77104405d1940413f194417 2b3312c0`.

Dossier abstract

Let $\varepsilon_m = (-1)^{\mathrm{wt}_2(m)}$ be the signed Thue–Morse sequence and let $\sigma_{k+1}(m) = \sum_{j=0}^{m} \sigma_k(j)$ be its inclusive iterated prefix sums. The formal development in `Analysis/FabiusFunction` proves that a shifted and normalized order-$(n+1)$ prefix row is exactly the coefficient row of a finite product of discrete uniform laws, and that its associated histogram converges pointwise to Rvachev's compactly supported function up. Sampling the same row at $\lfloor 2^n x \rfloor$ consequently converges to the bounded Fabius function $F(x)$. The existing argument is deliberately qualitative and supplies no rate.

This report makes the convergence quantitative. At the natural cell centers, the normalized prefix coefficients admit a uniform deconvolution expansion

$$H_n(\lambda) = \mathrm{up}(\lambda) - \frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda) + \left(\frac{15n+16}{43200\,16^n} + \frac{(3n+4)^2}{10368\,16^n}\right)\mathrm{up}^{(4)}(\lambda) + \mathcal{O}\left(\frac{n^3}{64^n}\right).$$

It follows that the sharp uniform center-grid error is $\frac{1}{3}n4^{-n}(1+o(1))$; the same rate holds after piecewise-linear interpolation. The piecewise-constant histogram is less accurate because of cell quantization: $\|\mathcal{W}_n - \mathrm{up}\,\|_\infty = 2^{-n} + \mathcal{O}(n4^{-n})$. Most strikingly, the literal dyadic-floor approximation

$$A_n(x) = \frac{\sigma_{n+1}(\lfloor 2^n x \rfloor)}{2^{\binom{n}{2}}}$$

has a still larger, entirely geometric first-order error. Uniformly on $[0,1]$,

$$A_n(x) = F(x) + \frac{n+2-2\{2^n x\}}{2^{n+1}}F'(x) + \mathcal{O}\left(\frac{n^2}{4^n}\right),$$

and therefore

$$\left\|\frac{2^{n+1}}{n}(A_n - F) - F'\right\|_\infty \longrightarrow 0, \qquad \frac{2^n}{n}\|A_n - F\|_\infty \longrightarrow 1.$$

The factor n comes from the exact inward displacement of the finite support, not from slow weak convergence. At $x = 1/2$ the phenomenon has the closed form $A_n(1/2) - 1/2 = (n+2)2^{-n} - 2^{-\binom{n}{2}}$.

The bridge identities used as input are formalized in Lean. The Fourier asymptotic expansion, its sharp constants, and the accelerated finite-difference constructions developed here are new analytic deductions and are not claimed to have been formalized in the repository.

113 Scope, conventions, and the answer in one table

The source article [4] isolates the connection between iterated prefix sums and up, corrects two index/normalization defects in a naive formulation, and proves pointwise convergence. It also says explicitly that its monotonicity-and-continuity argument gives no uniform rate. The purpose of this supplement is to determine that rate, including its leading constant, and to explain why several superficially similar readings of the same integer row have different orders of accuracy.

We use the source article's normalization throughout:

- F is the bounded distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported density on $[-1, 1]$, with $\mathrm{up}(x-1) = F(x)$ for $0 \le x \le 1$;
- the Fourier transform is $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi i x\xi}\,\mathrm{d}x$.

The sharp derivative theorem from the source gives

$$\|F^{(r)}\|_\infty = \|\mathrm{up}^{(r)}\|_\infty = 2^{\binom{r+1}{2}}. \tag{960}$$

In particular $\|F'\|_\infty = \|\mathrm{up}'\|_\infty = 2$ and $\|\mathrm{up}''\|_\infty = 8$.

The three natural rates. Put $h_n = 2^{-n}$. The same normalized prefix row produces three different errors.

Reading of the row	Sharp uniform error	Dominant mechanism
Natural center values, or their linear interpolant	$\frac{1}{3}nh_n^2(1+o(1))$	Deconvolution of the omitted mean-zero Bernoulli tail; its variance is of order nh_n^2.
Piecewise-constant histogram $\mathcal{W}_n$	$h_n + \mathcal{O}(nh_n^2)$	Horizontal quantization by half a cell, amplified by the optimal slope 2.
Dyadic-floor Fabius sample A_n	$nh_n(1+o(1))$	The finite support is shorter than $[-1,1]$ by $(n+2)h_n/2$ at each end, producing an inward drift.

Thus the pointwise floor formula is not the most accurate reconstruction of the function from the prefix row. It is worse than the histogram by a factor asymptotic to n, while a centered linear reconstruction is better than the histogram by a factor asymptotic to $3 \cdot 2^n/n$.

113.1 What is formalized and what is new here

The exact prefix convolution law, Prouhet cancellation, zero runs, generating-series identity, coefficient bridge, corrected cell centers, and pointwise limit are developed in `ThueMorsePrefix.lean` and `ThueMorseApproximation.lean`; the finite atomic and step approximants are developed in `EarlyApproximants.lean`, `EarlyMeasureBridge.lean`, `StepMeasureBridge.lean`, and `StepApproximationLimit.lean` [5]. These are the algebraic and measure-theoretic inputs used below.

The new ingredient is a quantitative inversion of the finite cosine product on its natural Nyquist interval. It yields a complete asymptotic expansion in even derivatives of up. The first two nonzero terms are enough to determine all sharp rates stated in the abstract. No part of that new expansion is represented here as already machine-checked.

114 The exact discrete bridge

114.1 Signed Thue–Morse prefixes

Let

$$\varepsilon_m = (-1)^{\mathrm{wt}_2(m)}, \qquad m \geq 0, \tag{961}$$

where $\mathrm{wt}_2(m)$ is the number of ones in the binary expansion of m. Define inclusive iterated prefix sums by

$$\sigma_0(m) = \varepsilon_m, \qquad \sigma_{k+1}(m) = \sum_{j=0}^{m} \sigma_k(j). \tag{962}$$

The word *inclusive* matters: one has

$$\sigma_{k+1}(m+1) - \sigma_{k+1}(m) = \sigma_k(m+1), \tag{963}$$

with the lower-order value at the new index.

Repeated summation is convolution with the discrete truncated-power kernel. For $k \geq 1$,

$$\sigma_k(m) = \sum_{r=0}^{m} \binom{m-r+k-1}{k-1} \varepsilon_r. \tag{964}$$

The signed generating series is

$$\sum_{m\geq 0} \varepsilon_m z^m = \prod_{j\geq 0}(1 - z^{2^j}), \qquad \sum_{m\geq 0} \sigma_k(m) z^m = \frac{\prod_{j\geq 0}(1 - z^{2^j})}{(1-z)^k}. \tag{965}$$

All identities in this paragraph are formal power-series identities over $\mathbb{Z}$.

114.2 The finite polynomial and the one-order shift

For $n \geq 0$ put

$$P_n(z) = \prod_{i=1}^{n} (1 + z + \cdots + z^{2^i - 1}), \qquad g_n = \deg P_n = 2^{n+1} - n - 2. \tag{966}$$

The cancellation of the first $n+1$ factors in (965) gives the exact coefficient identity

$$[z^m]P_n(z) = \sigma_{n+1}(m), \qquad 0 \leq m < 2^{n+1}. \tag{967}$$

This is the source of the order shift: the polynomial indexed by n corresponds to the $(n+1)$st, not the nth, prefix row.

Every factor of P_n is palindromic, hence

$$[z^m]P_n = [z^{g_n - m}]P_n, \qquad P_n(1) = 2^{\binom{n+1}{2}}. \tag{968}$$

The coefficients are nonnegative integers. Thus, after normalization, they form a symmetric probability mass function.

114.3 Natural centers, atomic law, and histogram

Set

$$h_n = 2^{-n}, \qquad \lambda_{n,m} = \frac{2m - g_n}{2^{n+1}} = h_n \left(m - \frac{g_n}{2} \right). \tag{969}$$

The points $\lambda_{n,m}$ lie on a translate of the lattice $h_n\mathbb{Z}$. Define

$$p_{n,m} = 2^{-\binom{n+1}{2}}[z^m]P_n, \qquad \mu_n = \sum_{m\in\mathbb{Z}} p_{n,m}\,\delta_{\lambda_{n,m}}, \tag{970}$$

where coefficients outside $0 \le m \le g_n$ are understood to be zero. Then μ_n is a probability measure. Its mass density relative to the lattice spacing is

$$H_n(\lambda_{n,m}) = \frac{p_{n,m}}{h_n} = \frac{[z^m]P_n}{2^{\binom{n}{2}}}. \tag{971}$$

By (967), this is exactly

$$H_n(\lambda_{n,m}) = \frac{\sigma_{n+1}(m)}{2^{\binom{n}{2}}}, \qquad 0 \le m < 2^{n+1}. \tag{972}$$

Let $I_{n,m}$ be the cell of width h_n centered at $\lambda_{n,m}$. Spreading each atom of μ_n uniformly over its cell gives the step density

$$\mathcal{W}_n(x) = \sum_m H_n(\lambda_{n,m})\mathbf{1}_{I_{n,m}}(x), \tag{973}$$

with half weight from each adjacent cell at a shared endpoint. In the interior of a cell, $\mathcal{W}_n$ is simply the corresponding normalized prefix coefficient. The formal repository proves

$$\mathcal{W}_n(x) \longrightarrow \mathrm{up}(x) \tag{974}$$

pointwise. The rest of this report quantifies (974) and the moving-center sample derived from it.

115 The hidden Bernoulli tail

115.1 Rademacher factorization

Let $R_{\ell,r}$ be independent Rademacher variables, each taking ±1 with probability $1/2$. The binary factorization of every discrete uniform variable in (966) gives an equivalent representation of (970):

$$X_n = \sum_{\ell=1}^{n}\sum_{r=1}^{\ell} 2^{-\ell-1}R_{\ell,r}, \qquad \mathcal{L}(X_n) = \mu_n. \tag{975}$$

Consequently

$$\widehat{\mu_n}(\xi) = \prod_{\ell=1}^{n}\left(\cos\frac{\pi\xi}{2^\ell}\right)^{\ell}. \tag{976}$$

The complete series

$$Y = \sum_{\ell=1}^{\infty}\sum_{r=1}^{\ell} 2^{-\ell-1}R_{\ell,r} \tag{977}$$

converges absolutely, and its density is up. Equivalently,

$$\widehat{\mathrm{up}}(\xi) = \prod_{\ell=1}^{\infty}\left(\cos\frac{\pi\xi}{2^\ell}\right)^{\ell}. \tag{978}$$

This cosine representation is equivalent to the more familiar infinite sinc product.

Write

$$T_n = Y - X_n = \sum_{\ell>n} \sum_{r=1}^{\ell} 2^{-\ell-1} R_{\ell,r}. \tag{979}$$

Then X_n and T_n are independent and

$$\mathrm{up}(x)\,\mathrm{d}x = \mu_n * \mathcal{L}(T_n). \tag{980}$$

Thus μ_n is obtained by deleting a small, symmetric smoothing tail from the target law. The center-grid asymptotics are an exact lattice version of deconvolving that tail.

115.2 Exact support radius and variance

Proposition 115.1 (Geometry and second moment of the omitted tail)**.** *The tail T_n has mean zero, support contained in $[-r_n, r_n]$, and variance v_n, where*

$$r_n = \sum_{\ell>n} \ell 2^{-\ell-1} = \frac{n+2}{2^{n+1}}, \tag{981}$$

$$v_n = \sum_{\ell>n} \ell 2^{-2\ell-2} = \frac{3n+4}{36\,4^n}. \tag{982}$$

Both support endpoints occur.

Proof. The mean vanishes by symmetry. The maximal possible absolute value is obtained by taking all signs equal, so it is the first displayed sum. The geometric-tail identity

$$\sum_{\ell=n+1}^{\infty} \ell q^{\ell} = q^{n+1} \frac{(n+1) - nq}{(1-q)^2}$$

with $q = 1/2$ gives (981). Independence and $\mathrm{Var}(R_{\ell,r}) = 1$ give the variance sum, and the same identity with $q = 1/4$, followed by the extra factor $1/4$, gives (982). □

The support of μ_n is therefore

$$\operatorname{supp} \mu_n = [-1 + r_n, 1 - r_n] \cap \big(h_n(\mathbb{Z} - g_n/2)\big), \tag{983}$$

because $g_n/2^{n+1} = 1 - r_n$. Two scales should be distinguished:

$$r_n \sim \frac{n}{2} h_n, \qquad \sqrt{v_n} \sim \sqrt{\frac{n}{12}}\, h_n. \tag{984}$$

The support radius is a worst-case all-signs quantity, while the standard deviation is the typical scale. The floor-indexed approximation will inherit the former; the correctly centered deconvolution error will inherit the variance itself.

115.3 Why the leading center correction is second order

If a smooth density f is convolved with a small mean-zero law of variance v, then formally

$$f * \nu = f + \frac{v}{2} f'' + \cdots .$$

Solving backward gives $f - \frac{v}{2} f'' + \cdots$. Since (980) smooths the atomic row by the tail law, one should expect

$$H_n(\lambda) - \mathrm{up}(\lambda) \approx -\frac{v_n}{2} \mathrm{up}''(\lambda) = -\frac{3n+4}{72\,4^n} \mathrm{up}''(\lambda). \tag{985}$$

The sign is that of deconvolution: in a convex region the unsmoothed profile lies below the smoothed one. The next sections make (985) uniform and rigorous and compute the fourth-derivative term.

116 Fourier inversion on the moving lattice

The main technical point is that one must invert the finite atomic law on exactly one period of its lattice Fourier series. Ordinary Fourier inversion of a density cannot be applied directly to μ_n.

116.1 A shifted-lattice inversion formula

Lemma 116.1 (Nyquist inversion for a shifted lattice)**.** *Let $\mu = \sum_{k\in\mathbb{Z}} p_k\delta_{a+hk}$ be a finite measure on a translate of $h\mathbb{Z}$, and let $\widehat{\mu}(\xi) = \sum_k p_k \mathrm{e}^{-2\pi\mathrm{i}(a+hk)\xi}$. Then, for every $m \in \mathbb{Z}$,*

$$\frac{p_m}{h} = \int_{-1/(2h)}^{1/(2h)} \widehat{\mu}(\xi)\mathrm{e}^{2\pi\mathrm{i}(a+hm)\xi}\,\mathrm{d}\xi. \tag{986}$$

Proof. Insert the finite sum defining $\widehat{\mu}$ and integrate term by term. The kth term is

$$p_k \int_{-1/(2h)}^{1/(2h)} \mathrm{e}^{2\pi\mathrm{i}h(m-k)\xi}\,\mathrm{d}\xi,$$

which is p_m/h when $k = m$ and zero otherwise. The shift a cancels from the phase. □

For μ_n, the interval in (986) is

$$B_n = [-2^{n-1}, 2^{n-1}]. \tag{987}$$

Combining (971) and (986) gives

$$H_n(\lambda_{n,m}) = \int_{B_n} \widehat{\mu_n}(\xi)\mathrm{e}^{2\pi\mathrm{i}\lambda_{n,m}\xi}\,\mathrm{d}\xi. \tag{988}$$

116.2 The exact deconvolution multiplier

On B_n, all factors in the omitted cosine tail are positive: for $\ell > n$, $|\pi\xi/2^\ell| \le \pi/4$. Hence one may divide (978) by (976) and write

$$\widehat{\mu_n}(\xi) = \widehat{\mathrm{up}}(\xi)R_n(\xi), \qquad R_n(\xi) = \prod_{\ell>n}\left(\sec\frac{\pi\xi}{2^\ell}\right)^\ell, \qquad \xi \in B_n. \tag{989}$$

The multiplier R_n is the reciprocal characteristic function of the omitted tail on the frequency range where lattice inversion needs it.

There is a useful global bound. Put $t = \pi\xi/2^n$, so $|t| \le \pi/2$. Reindexing $\ell = n + r$ and using Viète's product $\prod_{r\ge1}\cos(t/2^r) = \sin t/t$ gives

$$R_n(\xi) = \left(\frac{t}{\sin t}\right)^n \prod_{r=1}^{\infty} \sec(t/2^r)^r. \tag{990}$$

The second product is bounded uniformly for $|t| \le \pi/2$, while $t/\sin t \le \pi/2$. Thus there is an absolute C_* such that

$$1 \le R_n(\xi) \le C_*\left(\frac{\pi}{2}\right)^n, \qquad \xi \in B_n. \tag{991}$$

This exponential growth is harmless because $\widehat{\mathrm{up}}$ decays faster than every power. Indeed, compact support and smoothness give, for each $M \ge 1$,

$$|\widehat{\mathrm{up}}(\xi)| \le \frac{\|\,\mathrm{up}^{(M)}\,\|_1}{(2\pi|\xi|)^M} \le \frac{2^{1+\binom{M+1}{2}}}{(2\pi|\xi|)^M} \qquad (\xi \neq 0), \tag{992}$$

where (960) was used in the second inequality.

117 Sharp center-grid asymptotics

117.1 Expansion of the reciprocal tail product

For $|u| < \pi/2$,

$$\log \sec u = \frac{u^2}{2} + \frac{u^4}{12} + \mathcal{O}(u^6). \tag{993}$$

The two geometric tails needed below are

$$\frac{1}{2}\sum_{\ell>n} \ell 4^{-\ell} = \frac{3n+4}{18\,4^n} =: \alpha_n, \tag{994}$$

$$\frac{1}{12}\sum_{\ell>n} \ell 16^{-\ell} = \frac{15n+16}{2700\,16^n} =: \beta_n. \tag{995}$$

Therefore, on a low-frequency range growing sufficiently slowly with n,

$$\log R_n(\xi) = \alpha_n(\pi\xi)^2 + \beta_n(\pi\xi)^4 + \mathcal{O}\left(\frac{n}{64^n}|\xi|^6\right), \tag{996}$$

and exponentiation gives

$$R_n(\xi) = 1 + \alpha_n(\pi\xi)^2 + \left(\beta_n + \frac{\alpha_n^2}{2}\right)(\pi\xi)^4 + \mathcal{O}\left(\frac{n^3}{64^n}|\xi|^6\right). \tag{997}$$

117.2 The uniform expansion

Theorem 117.1 (Two-term lattice deconvolution expansion)**.** *Uniformly for all lattice centers $\lambda = \lambda_{n,m}$,*

$$\begin{aligned} H_n(\lambda) = {} & \mathrm{up}(\lambda) - \frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda) \\ & + \left(\frac{15n+16}{43200\,16^n} + \frac{(3n+4)^2}{10368\,16^n}\right)\mathrm{up}^{(4)}(\lambda) + \mathcal{O}\left(\frac{n^3}{64^n}\right). \end{aligned} \tag{998}$$

The implicit constant is absolute and effective. In particular,

$$H_n(\lambda) - \mathrm{up}(\lambda) = -\frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda) + \mathcal{O}\left(\frac{n^2}{16^n}\right) \tag{999}$$

uniformly on the moving lattice.

Proof. Use (988) and (989) and split the integral at $|\xi| = 2^{n/4}$. On the inner interval, (997) is uniform. Since $\int_{\mathbb{R}} |\xi|^6 |\widehat{\mathrm{up}}(\xi)|\,\mathrm{d}\xi < \infty$, its integrated remainder is $\mathcal{O}(n^3 64^{-n})$ uniformly in the phase λ.

On the complementary part of B_n, combine (991) with (992). Taking, for example, $M = 32$ makes the resulting integral $o(64^{-n})$. The tails of the three polynomial Fourier integrals outside B_n are even smaller. We may therefore replace the truncated integrals of the displayed polynomial terms by integrals over all of $\mathbb{R}$.

With the chosen Fourier convention,

$$\int_{\mathbb{R}} (\pi\xi)^2 \widehat{\mathrm{up}}(\xi) \mathrm{e}^{2\pi\mathrm{i}\lambda\xi}\,\mathrm{d}\xi = -\frac{1}{4}\,\mathrm{up}''(\lambda), \qquad \int_{\mathbb{R}} (\pi\xi)^4 \widehat{\mathrm{up}}(\xi) \mathrm{e}^{2\pi\mathrm{i}\lambda\xi}\,\mathrm{d}\xi = \frac{1}{16}\,\mathrm{up}^{(4)}(\lambda).$$

Substituting (994) and (995) gives (998). Omitting the fourth-derivative term and using (960) gives (999). □

Remark 117.2 (Variance form). Because $\alpha_n = 2v_n$, the leading term is exactly

$$H_n = \mathrm{up} - \frac{v_n}{2}\,\mathrm{up}'' + \cdots\,, \tag{1000}$$

which confirms the deconvolution heuristic (985).

117.3 Sharp norm constant

Corollary 117.3 (Sharp center-grid rate). *One has the uniform profile limit*

$$\sup_{m\in\mathbb{Z}} \left| \frac{4^n}{n}(H_n(\lambda_{n,m}) - \mathrm{up}(\lambda_{n,m})) + \frac{1}{24}\,\mathrm{up}''(\lambda_{n,m}) \right| \longrightarrow 0. \tag{1001}$$

Consequently,

$$\lim_{n\to\infty} \frac{4^n}{n} \sup_{m\in\mathbb{Z}} |H_n(\lambda_{n,m}) - \mathrm{up}(\lambda_{n,m})| = \frac{1}{3}. \tag{1002}$$

Proof. The coefficient $(3n+4)/(72n)$ tends to $1/24$, while the remainder in (999), after multiplication by $4^n/n$, tends uniformly to zero. The lattices become dense, and up'' is continuous with exact supremum 8. This supremum is attained; for example, the attained-point form of the sharp derivative theorem gives $\mathrm{up}''(-3/4) = F''(1/4) = 8$. Hence the norm limit is $8/24 = 1/3$. □

117.4 A continuous reconstruction with the same rate

Let L_n be the piecewise-linear interpolant through the points $(\lambda_{n,m}, H_n(\lambda_{n,m}))$, with zero values continued outside the finite support. The ordinary interpolation error for a C^2 function is at most $h_n^2\|\,\mathrm{up}''\,\|_\infty/8 = h_n^2$. This is smaller by a factor $1/n$ than the leading center bias.

Corollary 117.4 (Sharp continuous linear reconstruction). *Uniformly on* $\mathbb{R}$*,*

$$\frac{4^n}{n}(L_n - \mathrm{up}) \longrightarrow -\frac{1}{24}\,\mathrm{up}'', \tag{1003}$$

in the supremum norm. In particular,

$$\|L_n - \mathrm{up}\,\|_\infty = \frac{n}{3\,4^n}(1+o(1)). \tag{1004}$$

Thus iterated prefix sums do not merely converge to up after a qualitative passage through histograms. Read at their actual centers and interpolated linearly, they give a continuous approximation with a fully determined, curvature-shaped leading error.

117.5 The all-orders pattern

The same proof is not tied to fourth order. Write

$$\log\sec z = \sum_{r\ge 1} c_r z^{2r} \quad (c_1 = 1/2,\ c_2 = 1/12,\ c_3 = 1/45, \ldots) \tag{1005}$$

and put

$$s_{r,n} = c_r \sum_{\ell>n} \ell 2^{-2r\ell}. \tag{1006}$$

For a fixed M, expand

$$\exp\left(\sum_{r=1}^{M-1} s_{r,n} z^{2r}\right) = \sum_{j=0}^{M-1} b_{j,n} z^{2j} + \mathcal{O}(z^{2M}). \tag{1007}$$

Then the lattice inversion argument gives

$$H_n(\lambda) = \sum_{j=0}^{M-1} \frac{(-1)^j b_{j,n}}{4^j} \operatorname{up}^{(2j)}(\lambda) + \mathcal{O}_M\left(n^M 4^{-Mn}\right), \tag{1008}$$

uniformly on the center lattice. Each $b_{j,n}$ is an explicit polynomial in the geometric tails $s_{r,n}$. Thus the connection admits a full asymptotic differential expansion, not just a big-O convergence estimate.

118 The histogram: a sharp first-order quantization error

The center values are exceptionally accurate, but $\mathcal{W}_n$ holds each value constant across a cell. That reconstruction injects a larger first-order horizontal error.

For a point x in the interior of a cell, let $c_n(x)$ denote that cell's center and define the sawtooth phase

$$\rho_n(x) = \frac{c_n(x) - x}{h_n}, \qquad -\frac{1}{2} < \rho_n(x) < \frac{1}{2}. \tag{1009}$$

Then (999) and Taylor's theorem give, uniformly away from the shared cell endpoints,

$$\mathcal{W}_n(x) - \operatorname{up}(x) = h_n \rho_n(x) \operatorname{up}'(x) + \mathcal{O}(nh_n^2). \tag{1010}$$

At a shared endpoint, the half-weight convention averages the two adjacent center values and cancels the first-order odd displacement, leaving only $\mathcal{O}(nh_n^2)$ there.

Theorem 118.1 (Sharp histogram rate)**.** *The step approximants satisfy*

$$\|\mathcal{W}_n - \operatorname{up}\|_\infty = 2^{-n} + \mathcal{O}\left(\frac{n}{4^n}\right). \tag{1011}$$

Equivalently,

$$\lim_{n\to\infty} 2^n \|\mathcal{W}_n - \operatorname{up}\|_\infty = 1. \tag{1012}$$

Proof. For an interior point, combine (999) with the optimal Lipschitz bound $\|\operatorname{up}'\|_\infty = 2$:

$$|\mathcal{W}_n(x) - \operatorname{up}(x)| \le \mathcal{O}(nh_n^2) + 2|c_n(x) - x| \le h_n + \mathcal{O}(nh_n^2).$$

At a shared endpoint the averaged value obeys the same bound. There remains only the thin uncovered strip between the support of $\mathcal{W}_n$ and an endpoint of $[-1,1]$. Its width is at most $r_n + h_n/2 = \mathcal{O}(nh_n)$. Since up and all of its derivatives vanish at ± 1, Taylor's theorem with the sharp bounded third derivative gives

$$|\operatorname{up}(\pm 1 \mp s)| \le \frac{\|\operatorname{up}^{(3)}\|_\infty}{6} s^3 = \mathcal{O}(n^3 h_n^3) = o(nh_n^2) \qquad (0 \le s \le r_n + h_n/2).$$

Outside $[-1,1]$ both functions vanish. This completes the upper estimate globally.

For the lower estimate, choose cells whose centers tend to $-1/2$, where $\operatorname{up}'(-1/2) = 2$, and approach one edge of each chosen cell from its interior. The center error is $\mathcal{O}(nh_n^2)$, while the change of up across half a cell is $h_n + \mathcal{O}(h_n^2)$. Taking the supremum gives $\|\mathcal{W}_n - \operatorname{up}\|_\infty \ge h_n - \mathcal{O}(nh_n^2)$. □

Remark 118.2. The pointwise proof in the source article could not reveal (1011): weak convergence plus unimodality controls values through fixed comparison points but does not track a moving point at distance $h_n/2$ from a cell center. The rate is a local geometric phenomenon, and the sharp constant comes from the sharp derivative bound.

119 The floor-prefix approximation to the bounded Fabius function

119.1 Definition and exact moving-center displacement

The source theorem obtains F from the prefix row by evaluating its up approximation at half the argument. Define

$$A_n(x) = \mathcal{J}_n(x/2) = \frac{\sigma_{n+1}(\lfloor 2^n x \rfloor)}{2^{\binom{n}{2}}}, \qquad 0 \le x \le 1. \tag{1013}$$

Let

$$\theta_n(x) = \{2^n x\} = 2^n x - \lfloor 2^n x \rfloor. \tag{1014}$$

The normalized prefix value in (1013) is not located at $x-1$. Its actual center is

$$\begin{aligned} \lambda_n(x) &= \lambda_{n,\lfloor 2^n x \rfloor} \\ &= x - 1 + \delta_n(x), \end{aligned} \tag{1015}$$

$$\delta_n(x) = \frac{n+2-2\theta_n(x)}{2^{n+1}}. \tag{1016}$$

This identity, not merely the fact that $\lambda_n(x) \to x-1$, determines the dominant error. It also gives

$$\frac{n}{2^{n+1}} < \delta_n(x) \le \frac{n+2}{2^{n+1}}. \tag{1017}$$

In terms of the omitted-tail support radius (981),

$$\delta_n(x) = r_n - \theta_n(x) h_n. \tag{1018}$$

Thus the index $\lfloor 2^n x \rfloor$ compensates for the ordinary fractional cell position but not for the shortening of the finite support. The residual is roughly $r_n \sim n h_n/2$, which is much larger than either the cell width error of the histogram or the variance-scale center bias.

119.2 First- and second-order expansions

Theorem 119.1 (Uniform floor-prefix expansion)**.** *Uniformly for $x \in [0,1]$,*

$$A_n(x) = F(x) + \delta_n(x) F'(x) + \mathcal{O}\left(\frac{n^2}{4^n}\right). \tag{1019}$$

More precisely,

$$\begin{aligned} A_n(x) &= F(x) + \delta_n(x) F'(x) \\ &\quad + \left(\frac{\delta_n(x)^2}{2} - \frac{3n+4}{72\, 4^n}\right) F''(x) + \mathcal{O}\left(\frac{n^3}{8^n}\right). \end{aligned} \tag{1020}$$

All remainder constants are independent of x and n.

Proof. By the exact coefficient identification, $A_n(x) = H_n(\lambda_n(x))$. Apply (999) at $\lambda_n(x) = x - 1 + \delta_n(x)$, use $\mathrm{up}(x-1) = F(x)$ and the same identity for derivatives, and Taylor-expand around $x-1$. Since $\delta_n = \mathcal{O}(n 2^{-n})$, the first omitted Taylor term in (1019) is $\mathcal{O}(n^2 4^{-n})$.

For (1020), retain the quadratic Taylor term and the leading center-bias term. The remainders are $\mathcal{O}(\delta_n^3)$, $\mathcal{O}(n 4^{-n} \delta_n)$, and $\mathcal{O}(n^2 16^{-n})$, all absorbed by $\mathcal{O}(n^3 8^{-n})$. □

Corollary 119.2 (Uniform profile and sharp norm rate)**.** *The entire rescaled error profile converges uniformly:*

$$\left\| \frac{2^{n+1}}{n}(A_n - F) - F' \right\|_\infty \longrightarrow 0. \tag{1021}$$

Consequently,

$$\lim_{n\to\infty} \frac{2^n}{n} \|A_n - F\|_\infty = 1. \tag{1022}$$

For each fixed x,

$$\lim_{n\to\infty} \frac{2^{n+1}}{n}(A_n(x) - F(x)) = F'(x). \tag{1023}$$

Proof. By (1016),

$$\frac{2^{n+1}}{n}\delta_n(x) = 1 + \frac{2 - 2\theta_n(x)}{n} \longrightarrow 1$$

uniformly in x. Multiplying the remainder in (1019) by $2^{n+1}/n$ gives $\mathcal{O}(n2^{-n})$. This proves (1021). Taking norms and using $\|F'\|_\infty = 2$ gives (1022). $\square$

119.3 The equivalent statement for the up sample

The source notation uses

$$\mathcal{J}_n(u) = \frac{\sigma_{n+1}(\lfloor 2^{n+1}u \rfloor)}{2^{\binom{n}{2}}}. \tag{1024}$$

For $0 \le u < 1$ put $\vartheta_n(u) = \{2^{n+1}u\}$ and

$$d_n(u) = \frac{n + 2 - 2\vartheta_n(u)}{2^{n+1}}. \tag{1025}$$

Then

$$\mathcal{J}_n(u) = \mathrm{up}(2u - 1) + d_n(u)\,\mathrm{up}'(2u - 1) + \mathcal{O}\left(\frac{n^2}{4^n}\right). \tag{1026}$$

At $u = 1$, the prefix coefficient identity is outside its range and the exact zero-run value is $-2^{-\binom{n}{2}}$; since $\mathrm{up}(1) = \mathrm{up}'(1) = 0$, this exceptional endpoint is much smaller than the uniform principal scale and does not change the norm asymptotic.

120 An exact midpoint identity

The sharp constant in (1022) can be seen without asymptotic analysis at the single point where F' attains its maximum.

Theorem 120.1 (Exact floor-prefix error at $x = 1/2$)**.** *For every $n \ge 2$,*

$$A_n\left(\frac{1}{2}\right) = \frac{1}{2} + \frac{n+2}{2^n} - 2^{-\binom{n}{2}}. \tag{1027}$$

Equivalently,

$$\frac{2^n}{n}\left(A_n\left(\frac{1}{2}\right) - \frac{1}{2}\right) = 1 + \frac{2}{n} - \frac{2^{n-\binom{n}{2}}}{n}. \tag{1028}$$

Proof. Put $M = 2^{n-1}$ and write $Q = P_{n-1} = \sum_{r=0}^{d} q_r z^r$, where $d = g_{n-1} = 2^n - n - 1$, extending q_r by zero outside $0 \le r \le d$. Since $M < 2^n$, multiplication by the last factor of P_n gives

$$[z^M]P_n = \sum_{r=0}^{M} q_r =: C. \tag{1029}$$

The polynomial Q is palindromic and has total mass

$$T = Q(1) = 2^{\binom{n}{2}}. \tag{1030}$$

By symmetry,

$$2C - T = \sum_{r=d-M}^{M} q_r. \tag{1031}$$

The band in (1031) has $n+2$ coefficients. To evaluate them, write $Q = P_{n-2}(1 + \cdots + z^{2^{n-1}-1})$. Put $T_0 = P_{n-2}(1) = 2^{\binom{n-1}{2}}$. The n interior coefficients of the band are all T_0, because the convolution window contains all coefficients of P_{n-2}. Each of the two boundary coefficients is $T_0 - 1$: its window omits only the monic leading coefficient of P_{n-2}, and palindromicity gives the same value at the opposite boundary. Hence

$$2C - T = (n+2)T_0 - 2. \tag{1032}$$

Since $T/T_0 = 2^{n-1}$,

$$\frac{C}{T} = \frac{1}{2} + \frac{n+2}{2^n} - \frac{1}{T}.$$

Finally, (967) and (1013) identify C/T with $A_n(1/2)$, proving (1027). □

Table 24: Exact midpoint error and its sharp-rate normalization.

n	$A_n(1/2) - 1/2$	$2^n(A_n(1/2) - 1/2)/n$
2	0.5	1.000000000000
3	0.5	1.333333333333
4	0.359375	1.437500000000
5	0.2177734375	1.393750000000
6	0.124969482421875	1.333007812500
8	0.0390624962747097	1.249999880791
10	0.0117187499999716	1.199999999997
12	0.0034179687500000	1.166666666667
16	0.0002746582031250	1.125000000000
20	0.00002098083496094	1.100000000000

The exact identity also explains an otherwise puzzling numerical observation: after a few levels, the error is almost indistinguishable from the tangent displacement $2\delta_n(1/2) = (n+2)2^{-n}$. The remaining defect is the super-exponentially small term $2^{-\binom{n}{2}}$.

120.1 Correct centering can be exact at the same point

The large error in (1027) is not an unavoidable defect of the prefix row. It is caused by reading the wrong index as the point $-1/2$ on the up axis. When n is even, $-1/2$ itself is a lattice center, with index

$$m_n^* = 2^{n-1} - \frac{n+2}{2}. \tag{1033}$$

For this index, $m_n^* = (g_{n-1} - 1)/2$. Since P_{n-1} is palindromic of odd degree, exactly half of its total coefficient mass lies at indices at most m_n^*. The last-factor prefix identity therefore gives

$$H_n(-1/2) = \frac{1}{2} = \mathrm{up}(-1/2) \qquad (n \text{ even}). \tag{1034}$$

The floor rule instead uses $M = 2^{n-1}$, which is $(n+2)/2$ whole cells to the right of m_n^*. This is the discrete form of the support drift.

121 Bias removal and accelerated prefix reconstructions

The expansion (998) can be used constructively. It tells us how to remove the leading deconvolution bias using only neighboring prefix coefficients.

For a function on the center lattice, define the centered second difference

$$\Delta_h^2 H(\lambda) = \frac{H(\lambda + h_n) - 2H(\lambda) + H(\lambda - h_n)}{h_n^2}. \tag{1035}$$

Put

$$c_n = \frac{3n+4}{72} h_n^2 \tag{1036}$$

and define the corrected grid

$$H_n^{[1]}(\lambda) = H_n(\lambda) + c_n \Delta_h^2 H_n(\lambda). \tag{1037}$$

Every value in (1037) is a rational linear combination of three normalized iterated-prefix values.

Theorem 121.1 (One-step deconvolution acceleration)**.** *Uniformly on the center lattice,*

$$H_n^{[1]}(\lambda) - \mathrm{up}(\lambda) = -\frac{n^2}{1152\,16^n}\,\mathrm{up}^{(4)}(\lambda) + o\left(\frac{n^2}{16^n}\right). \tag{1038}$$

Consequently,

$$\lim_{n\to\infty} \frac{16^n}{n^2} \sup_{\lambda \in h_n(\mathbb{Z} - g_n/2)} |H_n^{[1]}(\lambda) - \mathrm{up}(\lambda)| = \frac{8}{9}. \tag{1039}$$

Proof. For a smooth function,

$$\Delta_h^2 f = f'' + \frac{h_n^2}{12} f^{(4)} + \mathcal{O}(h_n^4). \tag{1040}$$

Write (998) as $H_n = \mathrm{up} - c_n\,\mathrm{up}'' + d_n\,\mathrm{up}^{(4)} + \mathcal{O}(n^3 h_n^6)$, where

$$d_n = \left(\frac{15n+16}{43200} + \frac{(3n+4)^2}{10368}\right) h_n^4.$$

Substitute this into (1037) and use (1040). If the uniform remainder is denoted by R_n, then $\|\Delta_h^2 R_n\|_\infty \le 4h_n^{-2}\|R_n\|_\infty$; hence $c_n \Delta_h^2 R_n = \mathcal{O}(n^4 h_n^6) = o(n^2 h_n^4)$. Thus the remainder remains negligible at the claimed scale. The second-derivative terms cancel, and the coefficient of $\mathrm{up}^{(4)}$ is

$$d_n - c_n^2 + \frac{c_n h_n^2}{12} = -\frac{n^2}{1152} h_n^4 + \mathcal{O}(n h_n^4).$$

This proves (1038). By (960), $\|\mathrm{up}^{(4)}\|_\infty = 2^{10} = 1024$, so the sharp norm constant is $1024/1152 = 8/9$. □

Higher corrections follow by replacing the reciprocal-tail multiplier with successively longer polynomials in the lattice Laplacian. The all-orders expansion (1008) supplies the coefficients recursively. To preserve the accelerated rate in a continuous reconstruction, one should use interpolation of sufficiently high order: piecewise-linear interpolation has an $\mathcal{O}(h_n^2)$ geometric error and would mask the $\mathcal{O}(n^2 h_n^4)$ corrected lattice error, whereas local cubic interpolation has $\mathcal{O}(h_n^4)$ error and retains the latter as the dominant term.

122 Conceptual synthesis

122.1 One row, three coordinate choices

The normalized integer row

$$m \longmapsto \frac{\sigma_{n+1}(m)}{2^{\binom{n}{2}}} \tag{1041}$$

contains highly accurate samples, but the meaning of m must be specified.

1. The intrinsic coordinate is the center $\lambda_{n,m} = h_n(m - g_n/2)$. At that coordinate the row differs from up by a curvature term of order nh_n^2.
2. Holding a center value constant across its width-h_n cell changes the coordinate by at most $h_n/2$. The target has maximal slope 2, so the histogram error is h_n.
3. Replacing the intrinsic coordinate by the naive dyadic coordinate implicit in $m = \lfloor 2^n x \rfloor$ overlooks the support deficit r_n. It changes the coordinate by roughly $nh_n/2$, so the Fabius error is roughly $nh_n F'(x)/2$.

The three rates are therefore compatible, not contradictory. They measure distinct reconstructions of the same discrete data.

122.2 Variance versus support radius

The center expansion sees the omitted tail only through its even cumulants. Its mean is zero, so the first nonzero effect is its variance $v_n \asymp nh_n^2$. The floor-coordinate drift, by contrast, sees the extreme support loss $r_n \asymp nh_n$. This is why a centered analytic argument is exponentially more accurate than a support-aligned but miscentered index rule.

This distinction is common in lattice approximation. Weak convergence controls averages, and moment matching controls smooth centered observables, but an endpoint-derived coordinate can retain a worst-case support defect long after the bulk distribution has become extremely close.

122.3 Why the n is structural

The degree

$$g_n = 2^{n+1} - n - 2 \tag{1042}$$

is shorter than the naive value 2^{n+1} by exactly $n+2$. After scaling by 2^{n+1} and centering, half of this deficit appears at each endpoint. Hence the factor n in the floor rate is already encoded in the elementary degree formula for P_n. No delicate property of F is needed to predict its presence; the analytic work is needed to turn the horizontal deficit into the sharp vertical profile F' and to control the smaller terms.

122.4 A practical recommendation

For numerical reconstruction from a computed prefix row:

- do not use $m/2^n$ as the physical coordinate;
- attach each value to $\lambda_{n,m}$ from (969);
- use at least linear interpolation rather than a histogram;
- if another factor of roughly $4^n/n$ is valuable, apply the three-point correction (1037) and use cubic interpolation.

The coefficients remain exact rationals until the final interpolation or evaluation stage.

123 A Lean formalization roadmap

The new proof separates naturally into reusable lemmas. A formalization need not begin with the full asymptotic theorem.

1. **Shifted-lattice Fourier inversion.** Formalize lemma 116.1 for a finitely supported measure on $a + h\mathbb{Z}$. This is a finite-sum orthogonality calculation.
2. **Rademacher realization.** Refine the existing finite-measure bridge to the triangular Rademacher array (975); obtain (976) and the independent tail decomposition (979).
3. **Exact geometric tails.** Add the closed forms (981), (982), (994), and (995). These reduce to finite manipulations of geometric series.
4. **Uniform** $\log\sec$ **remainder.** Prove a sixth-order Taylor bound on $[-1/4, 1/4]$ and use it scale by scale. The low-frequency cutoff can be chosen as $2^{n/4}$ or replaced by any convenient dyadic integer bound.
5. **Rapid Fourier decay.** Package repeated integration by parts for compactly supported C^∞ functions and instantiate it with the already formalized sharp derivative bounds.
6. **Center expansion.** Combine the preceding lemmas with Fourier derivative identities. The first-order version (999) is substantially shorter than the fourth-order theorem and already proves all three basic rates.
7. **Exact drift.** Formalize (1015) by integer-floor arithmetic. Then (1019) is a direct Taylor estimate.
8. **Midpoint identity.** Prove theorem 120.1 independently of Fourier analysis from palindromicity and the central coefficient plateau. It is an especially useful regression test for index conventions.

The formal theorem statements should distinguish three domains: values at the center lattice, the step function on all reals, and the floor sample on $[0, 1]$. Combining them into one statement would obscure exactly the coordinate issue responsible for the different rates.

124 Conclusion

Iterated summation of the signed Thue–Morse sequence does more than produce a sequence that happens to resemble the Fabius function. After the correct one-order shift, its finite row is an exact lattice deconvolution of up. The omitted part is an explicit triangular Bernoulli tail with support radius $(n + 2)2^{-n-1}$ and variance $(3n + 4)/(36\,4^n)$. Those two exact numbers govern two different approximation regimes.

At the natural centers, the mean-zero tail leaves a variance-scale curvature bias, yielding the sharp rate $n/(3\,4^n)$. Holding the values constant across cells introduces the larger sharp rate 2^{-n}. Reading the same row at the naive dyadic floor leaves the full support deficit as a coordinate drift and yields the still larger sharp rate $n2^{-n}$. The latter has the uniform error profile F' and an exact midpoint formula.

The main conceptual correction is therefore simple but consequential: the integer index is not itself the continuum coordinate. Once the genuine center map is retained, the Thue–Morse prefix construction is far more accurate than its original pointwise formulation suggests, and its error admits a complete, explicitly computable differential expansion.

Supplementary material from this dossier

125 Elementary geometric-tail formulas

For $|q| < 1$ and $n \geq 0$,

$$\sum_{\ell=n+1}^{\infty} \ell q^{\ell} = q \frac{\mathrm{d}}{\mathrm{d}q} \left(\frac{q^{n+1}}{1-q} \right) = q^{n+1} \frac{(n+1) - nq}{(1-q)^2}. \tag{1043}$$

The specializations used in the text are

$$\sum_{\ell > n} \ell 2^{-\ell} = (n+2) 2^{-n}, \tag{1044}$$

$$\sum_{\ell > n} \ell 4^{-\ell} = \frac{3n+4}{9\, 4^n}, \tag{1045}$$

$$\sum_{\ell > n} \ell 16^{-\ell} = \frac{15n+16}{225\, 16^n}, \tag{1046}$$

$$\sum_{\ell > n} \ell 64^{-\ell} = \frac{63n+64}{3969\, 64^n}. \tag{1047}$$

These identities make every coefficient in the all-orders expansion elementary: no unevaluated infinite tail sum is necessary.

126 A short exact-arithmetic algorithm

The coefficient row $[z^m]P_n$ can be built without polynomial multiplication. Multiplication by $1 + \cdots + z^{2^i - 1}$ is a sliding-window sum. The following pseudocode maintains exact integers and costs $\mathcal{O}(2^n)$ arithmetic operations per level.

```
coeff := [1]
for i := 1,...,n:
    width := 2^i
    nextLength := length(coeff) + width - 1
    next := array(nextLength, 0)
    window := 0
    for m := 0,...,nextLength-1:
        if m < length(coeff):
            window := window + coeff[m]
        if m >= width:
            window := window - coeff[m-width]
        next[m] := window
    coeff := next

# Normalized prefix sample:
H_n(m) := coeff[m] / 2^(n(n-1)/2)
# Physical coordinate:
lambda_n(m) := (2m - (2^(n+1)-n-2)) / 2^(n+1)
```

The midpoint identity (1027), symmetry $\mathtt{coeff}[\mathtt{m}] = \mathtt{coeff}[\mathtt{g_n} - \mathtt{m}]$, the total mass $2^{\binom{n+1}{2}}$, and the central maximum $2^{\binom{n}{2}}$ provide independent exact checks of an implementation.

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Part X
Fabius Inverse and Saddle Research Frontiers

Source provenance and status. Former path: `Fabius_Inverse_and_Saddle_Research_Frontiers/Fabius_Inverse_and_Saddle_Research_Frontiers.tex`. Specialist inverse and saddle dossier. Formalized inputs are distinguished from derived consequences; inverse endpoint asymptotics and the logarithmic highest-derivative transfer remain explicit obligations.

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Dossier abstract

The Fabius function is unusually rich: a single compactly supported C^∞ density produces a self-similar distribution function, exact dyadic arithmetic, Fourier products, a probability model, and a small-argument expansion whose coefficients are smooth periodic functions. The companion article *The Fabius Function and Rvachev's Up-Function* develops most of that landscape. The present paper is deliberately different. It does not repeat the introductory theory. Instead it fills two identifiable gaps in the exposition by turning recently formalized material into ordinary mathematics.

The first gap is the inverse. We construct the totalized inverse $G = F^{-1}$ on the whole real line, prove its global monotonicity, continuity, reflection law, comparison calculus, and exact values at all outputs of the dyadic evaluator, and show how every flatness estimate for F becomes an effective steepness estimate for G. We then derive the classical interior calculus of the inverse and invert the sharp Lambert-phase asymptotic. The result is the explicit endpoint law

$$G(y) \sim \frac{\sqrt{\log(1/y)}}{\mathrm{e}\sqrt{\log 2}} \exp\Bigl(-\sqrt{2\log 2\,\log(1/y)}\Bigr) \qquad (y \downarrow 0),$$

together with its reflected analogue at 1. This explains in one formula why the inverse is steeper than every positive power of y, although it still tends to zero faster than every negative power of $\log(1/y)$.

The second gap concerns the internal machinery of the all-orders saddle expansion. We give complete derivations of the closed periodic jet formula, its shifted signed-Stirling conversion, the collapse of the centered exponent coefficients, the formal mass-to-logarithm recursions, the leading-derivative law, and the full second logarithmic correction. Finally, we give a proof-level account of the arbitrary-order analytic remainder: the order-dependent central window, vertical Taylor control, finite exponential substitution, Gaussian contraction, polynomial-tail replacement, minor-arc estimates, logarithmic transfer, and return from the dyadic logarithmic scale to $x \downarrow 0$.

Statements explicitly represented by theorem declarations in the Lean development are marked Lean. Consequences proved here from those statements by standard real or analytic arguments are marked Derived.

Formalization boundary. This document is a research-frontier notebook, not part of the project's formalization-backed primary exposition. It deliberately preserves a mixture of proved Lean inputs, ordinary mathematical deductions, symbolic coefficient calculations, and numerical evidence so that the unformalized steps are not lost. A statement here is a project theorem only when an exact current Lean declaration is named for the full statement; the labels LEAN and DERIVED record the draft's provenance but do not themselves certify that every displayed consequence has been formalized. In particular, Hölder consequences beyond the now-formalized inverse-curvature identities, inverse endpoint asymptotics, the ordered-composition coefficient formulas and the logarithmic highest-derivative law remain research obligations.

127 Purpose, scope, and status conventions

127.1 A supplement rather than a second survey

Let F denote the bounded, totalized Fabius function: $F(x) = 0$ for $x \le 0$, $F(x) = 1$ for $x \ge 1$, and on $[0, 1]$ it is the unique strictly increasing C^∞ function associated with Rvachev's compactly supported up-function. We use only a small portion of the foundational theory:

$$F(0) = 0, \qquad F(1) = 1, \qquad F(1 - x) = 1 - F(x), \tag{1048}$$

$$F'(x) = 2\,\mathrm{up}(2x - 1), \qquad 0 < x < 1, \tag{1049}$$

where up is positive on $(-1, 1)$ and $\mathrm{up}(0) = 1$. Consequently F is a continuous strictly increasing bijection of $[0, 1]$ onto itself.

The main companion article already explains the random-series model, infinite convolutions, the Fourier transform, moment recurrences, Thue–Morse identities, dyadic rationality, and the broad shape of the Lambert saddle analysis. Repeating those chapters would obscure the missing material. The division of labor adopted here is summarized in table 25.

Table 25: Expository audit and the role of this companion.

Topic	Situation in the main article	Treatment here
Inverse function	The totalized inverse module and its endpoint estimates are absent.	Full construction, exact identities, dyadic inversion, effective steepness, endpoint limits, regularity, and inverse asymptotics.
Closed saddle jets	Closed formulas are displayed, but the operator calculus and the shifted Stirling conversion are compressed.	Complete induction, forcing-term propagation, coefficient table, and the reason for the index shift $s(n + 1, m + 1)$.
Mass and logarithmic coefficients	The all-orders result is stated and some coefficients are listed.	Formal exponential recursion, Gaussian contraction, composition formulas, structural bounds, and the leading derivative law.
Second correction	The final polynomial appears, while much of the order-four algebra is easy to miss.	A line-by-line calculation from $E_1, \ldots, E_4$ through B_4, Gaussian moments, a_2, and A_2.

Topic	Situation in the main article	Treatment here
Arbitrary-order remainder	The main article gives a high-level proof architecture.	The order-dependent contour window and every analytic error mechanism are separated and justified.
Probability, Fourier theory, dyadic arithmetic	Already developed in substantial depth.	Used as input only; not duplicated.

127.2 What the labels mean

A paragraph headed Lean restates a mathematical theorem whose content is represented directly in the Lean files under

`Analysis/FabiusFunction/Lean/FabiusFunction`.

A paragraph headed Derived gives a paper-level consequence. Its proof uses formalized results plus a standard theorem such as the inverse function theorem, ordinary asymptotic inversion, or elementary formal power-series algebra. The distinction matters especially in sections 131 and 135: those sections contain new synthesis, but they do not pretend that every displayed corollary has a declaration of its own.

The exact Lambert phase is never silently replaced inside a periodic coefficient. If $\lambda = \lambda(x)$ is the exact lower solution of $x = \lambda 2^{-\lambda}$, then a term such as $A_j(\lambda)$ remains at that exact phase. Replacing λ by an asymptotic approximation inside A_j is a separate composition problem, because a bounded phase error does not become small modulo the period.

128 The inverse as a global mathematical object

128.1 The order isomorphism on the unit interval

Set

$$I = [0, 1], \qquad \mathrm{clamp}_{[0,1]}(y) = \min\{1, \max\{0, y\}\}.$$

Because $F|_I$ is a continuous strictly increasing bijection $I \to I$, it is an order isomorphism. Its ordinary inverse will first be denoted by $F_I^{-1} : I \to I$.

Definition 128.1 (Totalized inverse)**.** The totalized inverse of the Fabius function is

$$G(y) = F_I^{-1}(\mathrm{clamp}_{[0,1]}(y)), \qquad y \in \mathbb{R}.$$

Thus the inverse is extended by constants outside its natural range, exactly as F itself is extended by constants outside its natural domain.

Theorem 128.2 (Global inverse package; Lean)**.** *The function G has the following properties.*

1. $0 \le G(y) \le 1$ *for every* $y \in \mathbb{R}$.
2. *If* $0 \le y \le 1$, *then*
$$F(G(y)) = y.$$
If $0 \le x \le 1$, *then*
$$G(F(x)) = x.$$

3. *G is strictly increasing on $[0,1]$, monotone on $\mathbb{R}$, and continuous on $\mathbb{R}$.*

4. *The restriction $G|_I$ is a bijection $I \to I$.*

Proof. The clamp maps $\mathbb{R}$ continuously and monotonically into I. The inverse of an order isomorphism between compact real intervals is itself an order isomorphism, hence is continuous and strictly increasing. Their composition is therefore continuous and monotone on the whole line and takes values in I. If $y \in I$, then $\mathrm{clamp}_{[0,1]}(y) = y$, so the usual right-inverse identity for F_I^{-1} gives $F(G(y)) = y$. Likewise, $F(x) \in I$ for $x \in I$, and the left-inverse identity gives $G(F(x)) = x$. Strict increase and bijectivity on I are inherited from F_I^{-1}. □

The totalization is not cosmetic. It produces a globally continuous monotone function and makes the reflection law valid without restricting the variable to $[0,1]$.

128.2 A four-way comparison calculus

The order-isomorphism viewpoint turns inverse inequalities into direct inequalities without any numerical inversion.

Theorem 128.3 (Comparison equivalences; Lean)**.** *For $x, y \in [0,1]$,*

$$G(y) \le x \iff y \le F(x), \tag{1050}$$
$$F(x) \le y \iff x \le G(y), \tag{1051}$$
$$G(y) < x \iff y < F(x), \tag{1052}$$
$$F(x) < y \iff x < G(y). \tag{1053}$$

Proof. Apply the strictly increasing function F to the two sides of an inequality involving $G(y)$, then use $F(G(y)) = y$. For example,

$$G(y) \le x \iff F(G(y)) \le F(x) \iff y \le F(x).$$

The other three equivalences are identical, with the order reversed in placement but not in direction because F and G are increasing. □

Remark 128.4. The equivalences are often more useful than an explicit inverse formula. They let one convert a known dyadic or asymptotic bound for F into a location bound for G with no loss in constants.

128.3 Tails, endpoints, and reflection

Theorem 128.5 (Constant tails and symmetry; Lean)**.** *For every $y \in \mathbb{R}$,*

$$y \le 0 \implies G(y) = 0, \tag{1054}$$
$$y \ge 1 \implies G(y) = 1, \tag{1055}$$
$$G(1-y) = 1 - G(y). \tag{1056}$$

In particular,

$$G(0) = 0, \qquad G(1) = 1, \qquad G(1/2) = 1/2.$$

Proof. The two tail identities follow immediately from the clamp. Suppose first that $0 \le y \le 1$. Then $G(y) \in [0,1]$, and by the symmetry of F,

$$F(1 - G(y)) = 1 - F(G(y)) = 1 - y.$$

The point $1 - G(y)$ lies in $[0,1]$, so uniqueness of the inverse gives $G(1-y) = 1 - G(y)$. If $y \le 0$ or $y \ge 1$, the same equation follows from the two constant tails. Setting $y = 0, 1,$ and $1/2$ gives the stated special values. □

129 Exact inversion of the dyadic evaluator

129.1 The exact statement

Let $\mathsf{F}_n(a)$ denote the exact bounded dyadic evaluator from the formal development: for $0 \le a \le 2^n$,

$$\mathsf{F}_n(a) = F\left(\frac{a}{2^n}\right),$$

and for larger a it is clamped at the endpoint value 1.

Theorem 129.1 (Dyadic inversion; Lean)**.** *For all $n, a \in \mathbb{N}$,*

$$G(\mathsf{F}_n(a)) = \frac{\min\{a, 2^n\}}{2^n}.$$

In particular, if $0 \le a \le 2^n$, then

$$G\left(F\left(\frac{a}{2^n}\right)\right) = \frac{a}{2^n}.$$

Proof. When $a \le 2^n$, the argument $a/2^n$ lies in I, so the left-inverse identity in theorem 128.2 applies. When $a \ge 2^n$, the bounded evaluator equals 1, the minimum equals 2^n, and both sides are 1. □

The theorem is stronger than a table of examples: every exact rational produced by the dyadic algorithm immediately gives an exact algebraic input for the inverse.

129.2 A small exact table

x	$F(x)$	exact inverse statement
0	0	$G(0) = 0$
$\frac{1}{16}$	$\frac{143}{2073600}$	$G\left(\frac{143}{2073600}\right) = \frac{1}{16}$
$\frac{1}{8}$	$\frac{1}{288}$	$G\left(\frac{1}{288}\right) = \frac{1}{8}$
$\frac{1}{4}$	$\frac{5}{72}$	$G\left(\frac{5}{72}\right) = \frac{1}{4}$
$\frac{3}{8}$	$\frac{73}{288}$	$G\left(\frac{73}{288}\right) = \frac{3}{8}$
$\frac{1}{2}$	$\frac{1}{2}$	$G\left(\frac{1}{2}\right) = \frac{1}{2}$
$\frac{5}{8}$	$\frac{215}{288}$	$G\left(\frac{215}{288}\right) = \frac{5}{8}$
$\frac{3}{4}$	$\frac{67}{72}$	$G\left(\frac{67}{72}\right) = \frac{3}{4}$
$\frac{7}{8}$	$\frac{287}{288}$	$G\left(\frac{287}{288}\right) = \frac{7}{8}$
$\frac{15}{16}$	$\frac{2073457}{2073600}$	$G\left(\frac{2073457}{2073600}\right) = \frac{15}{16}$

Table 26: Exact inverse values obtained from the dyadic evaluator and reflection.

Remark 129.2. The threshold $F(1/4) = 5/72$ will reappear in the effective endpoint steepness theorem. Its occurrence is structural rather than arbitrary: $1/4 = 2^{-2}$ is precisely the scale on which the quadratic flatness bound is valid.

130 Flatness of F becomes steepness of G

130.1 The transported power inequalities

Write

$$C_n = 2^{\binom{n+1}{2}}.$$

The effective flatness theorem for F says, in one of its standard forms, that

$$0 \le x \le 2^{-n} \quad \Longrightarrow \quad F(x) \le C_n x^n.$$

The sharp version improves the right-hand side by the factor $n!$.

Theorem 130.1 (Inverted effective and sharp flatness; Lean)**.** *Let $n \in \mathbb{N}$ and $0 \le y \le 1$. If*

$$2^n G(y) \le 1,$$

then

$$y \le C_n G(y)^n, \tag{1057}$$

$$y \le \frac{C_n}{n!} G(y)^n. \tag{1058}$$

Moreover, the scale condition follows from the argument condition

$$0 \le y \le F(2^{-n}).$$

Proof. Put $x = G(y)$. The scale hypothesis says $0 \le x \le 2^{-n}$, while $F(x) = y$. Substituting x into the effective and sharp flatness bounds for F gives (1057) and (1058).

Now suppose $y \le F(2^{-n})$. By (1050), $G(y) \le 2^{-n}$, which is exactly the required scale condition. □

For $n \ge 1$, taking positive nth roots yields the more geometric form.

Corollary 130.2 (Root lower bounds; Lean input, Derived form)**.** *If $n \ge 1$, $0 < y \le F(2^{-n})$, then*

$$G(y) \ge C_n^{-1/n} y^{1/n}, \tag{1059}$$

$$G(y) \ge \left(\frac{n!}{C_n}\right)^{1/n} y^{1/n}. \tag{1060}$$

Proof. All quantities are nonnegative. Rearrange equations (1057) and (1058) and apply the monotonicity of the positive nth-root function. □

130.2 An effective bound against every line

The quadratic case already detects infinite endpoint slope.

Theorem 130.3 (Effective linear steepness; Lean)**.** *Let $M, y \in \mathbb{R}$. If*

$$0 < y, \qquad y \le F(1/4) = \frac{5}{72}, \qquad 8M^2 y < 1,$$

then

$$My < G(y).$$

Proof. The comparison theorem gives $G(y) \le 1/4$, so the quadratic inverse flatness estimate applies:

$$y \le 8G(y)^2.$$

If $M \le 0$, then $My \le 0 < G(y)$ because $y > 0$ and G is strictly increasing on $[0, 1]$. Suppose $M > 0$ and, toward a contradiction, $G(y) \le My$. Then

$$y \le 8G(y)^2 \le 8M^2y^2.$$

Dividing by $y > 0$ gives $1 \le 8M^2y$, contrary to the last hypothesis. □

130.3 Infinite one-sided slope

Theorem 130.4 (Endpoint quotient divergence; Lean)**.** *As $y \downarrow 0$,*

$$\frac{G(y)}{y} \longrightarrow +\infty.$$

By reflection, as $y \uparrow 1$,

$$\frac{1 - G(y)}{1 - y} \longrightarrow +\infty.$$

Proof. Fix a real target A. Put $B = |A| + 1$. For all sufficiently small positive y we have simultaneously

$$y < F(1/4), \qquad 8B^2y < 1.$$

By theorem 130.3, $By < G(y)$, and hence

$$A \le B < \frac{G(y)}{y}.$$

This is exactly the filter definition of divergence to $+\infty$.

For the right endpoint use (1056):

$$\frac{1 - G(y)}{1 - y} = \frac{G(1 - y)}{1 - y},$$

and let $1 - y \downarrow 0$. □

130.4 Failure of every positive Hölder estimate

Theorem 130.5 (Stronger than non-Lipschitz; Derived)**.** *For every $\alpha > 0$,*

$$\frac{G(y)}{y^\alpha} \longrightarrow +\infty \qquad (y \downarrow 0).$$

Consequently there are no $C, \delta > 0$ for which

$$G(y) \le Cy^\alpha \qquad (0 \le y \le \delta).$$

The reflected assertion holds at 1:

$$\frac{1 - G(y)}{(1 - y)^\alpha} \longrightarrow +\infty \qquad (y \uparrow 1).$$

Proof. Choose an integer $n > 1/\alpha$. On the window $0 < y \le F(2^{-n})$, equation (1059) gives

$$\frac{G(y)}{y^\alpha} \ge C_n^{-1/n} y^{1/n-\alpha}.$$

The exponent $1/n - \alpha$ is negative, so the right-hand side tends to $+\infty$. A local Hölder upper bound would keep the quotient bounded and is therefore impossible. The statement at 1 follows from reflection. □

Each fixed n supplies a root lower bound only on the shrinking window $y \le F(2^{-n})$. There is no bound uniform in n. This is not a defect: it is exactly the scale dependence needed to prove that G outruns *every fixed* positive power.

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131 Classical calculus and the exact regularity of the inverse

This section uses the formalized inverse package together with standard one-variable analysis. Smoothness, the reciprocal first-derivative formula, the exact second-derivative formula, and the strict closed-half shape are now marked LEAN; the general higher inverse-derivative formulas remain DERIVED.

131.1 Smoothness and differential identities in the interior

Theorem 131.1 (Interior inverse calculus through first order; LEAN)**.** *The inverse G is C^∞ on $(0,1)$. For $0 < y < 1$,*

$$G'(y) = \frac{1}{F'(G(y))} = \frac{1}{2\,\mathrm{up}(2G(y)-1)} > 0. \tag{1061}$$

Proof. For $x \in (0,1)$, (1049) and positivity of up on $(-1,1)$ imply $F'(x) > 0$. The C^∞ inverse function theorem therefore applies at every interior point and gives a smooth local inverse. Uniqueness identifies all these local inverses with G. Differentiating $F(G(y)) = y$ and substituting (1049) proves (1061). □

The corresponding Lean declarations are `fabiusInv_contDiffOn_Ioo`, `fabiusInv_hasDerivAt`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, and `deriv_fabiusInv_pos`.

Remark 131.2 (Current global smoothness locus; LEAN). The totalized inverse has locally constant tails. The newer declarations `fabiusInv_differentiableAt_iff` and `fabiusInv_contDiffAt_infty_iff` sharpen the interior statement: the inverse is differentiable, and indeed C^∞, exactly at the points $y \ne 0, 1$. At each clamping endpoint it is not differentiable. More generally, `fabiusInv_contDiffAt_iff` says that order-zero smoothness is global, whereas every positive finite order and C^∞ have precisely those two exceptions. Its order side condition deliberately excludes Mathlib's stronger analytic order; the analytic locus is formalized separately by `fabiusInv_analyticAt_iff` in `InverseNotElementary.lean`.

Proposition 131.3 (Second inverse derivative; LEAN)**.** *For $0 < y < 1$,*

$$G''(y) = -\frac{F''(G(y))}{F'(G(y))^3}. \tag{1062}$$

Proof. Differentiating the identity behind (1061) once more gives

$$F''(G(y))G'(y)^2 + F'(G(y))G''(y) = 0.$$

Substitution of (1061) yields (1062). This is `deriv_fabiusInv_hasDerivAt` and its pointwise corollary `deriv_deriv_fabiusInv`. □

Remark 131.4 (Higher inverse derivatives; Derived)*.* Repeated differentiation expresses every higher derivative of G as a rational polynomial in the derivatives of F evaluated at $G(y)$, with a power of $F'(G(y))$ in the denominator. These general Faà di Bruno inverse-derivative polynomials are not separately packaged in the Lean API.

Corollary 131.5 (Midpoint differential data; Derived)**.** *At the fixed point* $1/2$*,*

$$G(1/2) = 1/2, \qquad G'(1/2) = \frac{1}{2}, \qquad G''(1/2) = 0.$$

Proof. The value follows from reflection. Since $\mathrm{up}(0) = 1$, (1061) gives $G'(1/2) = 1/2$. The up-function is even, so $\mathrm{up}'(0) = 0$, hence $F''(1/2) = 4\,\mathrm{up}'(0) = 0$; now use (1062). □

131.2 Shape reversal under inversion

The formal development proves that F is strictly convex on $(0, 1/2)$ and strictly concave on $(1/2, 1)$. Inversion reverses these two curvature signs.

Proposition 131.6 (Concavity and convexity of the inverse; Lean)**.** *The function G is strictly concave on* $[0, 1/2]$ *and strictly convex on* $[1/2, 1]$*.*

Proof. The inverse is strictly increasing and preserves the midpoint. It therefore maps the first half into itself, where F' is strictly increasing and positive. Its reciprocal in (1061) is strictly decreasing, so the derivative criterion and continuity give strict concavity on the closed first half. On the second half F' is strictly decreasing and positive, hence the reciprocal derivative of the inverse is strictly increasing and the same criterion gives strict convexity.

The closed-interval conclusions are `strictConcaveOn_fabiusInv_firstHalf` and `strictConvexOn_fabiusInv_secondHalf`. Their derivative criteria inspect only the two open interval interiors, while global continuity supplies the endpoints; no finite endpoint derivative is asserted. □

Remark 131.7 (Endpoint slope; Derived)*.* The secant slopes from the origin diverge by theorem 130.4, and reflection gives the right-endpoint counterpart. Interpreting this together with concavity as an extended one-sided derivative statement remains ordinary analysis rather than a separately named Lean theorem.

131.3 The C^∞ and analytic loci

Theorem 131.8 (Exact regularity locus; Lean)**.** *The totalized inverse has the following regularity.*

1. *G is C^∞ precisely on*
$$\mathbb{R} \setminus \{0, 1\}.$$
2. *It has no finite derivative at* 0 *or at* 1*.*
3. *It is real analytic precisely on*
$$\mathbb{R} \setminus [0, 1].$$

Proof. On $(-\infty, 0)$ and $(1, \infty)$ the function is constant, hence smooth and analytic. On $(0, 1)$ smoothness follows from theorem 131.1. At 0, the left derivative is 0 because the function is constant to the left, whereas the right difference quotient $G(y)/y$ tends to $+\infty$. Thus no finite derivative exists. Reflection gives the same conclusion at 1.

It remains to exclude analyticity in the open unit interval. Suppose G were analytic near some $y_0 \in (0, 1)$. Its derivative there is nonzero by (1061); the analytic inverse function theorem would then make its local inverse F analytic near $x_0 = G(y_0)$. But the Fabius function is nowhere analytic on the interior of its transition interval. This contradiction proves the claim.

The exact smoothness and differentiability classifications are `fabiusInv_contDiffAt_infty_iff` and `fabiusInv_differentiableAt_iff`; the analytic classification is `fabiusInv_analyticAt_iff`. □

132 Inverting the sharp small-argument asymptotic

The power inequalities show qualitative infinite steepness. The sharp Lambert expansion gives the actual scale. This section extracts that scale without needing the detailed value of the periodic fluctuation.

132.1 The exact lower Lambert phase

Put

$$L = \log 2.$$

For sufficiently small $x > 0$, let $\lambda = \lambda(x)$ be the large positive solution of

$$x = \lambda 2^{-\lambda} = \lambda e^{-L\lambda}. \tag{1063}$$

Equivalently,

$$\lambda(x) = -\frac{1}{L} W_{-1}(-Lx),$$

but (1063) is the identity used in the proofs.

The sharp expansion established for F has the form

$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) + \mathcal{O}(\lambda^{-1}), \tag{1064}$$

where Ψ is smooth, bounded, and one-periodic. The exact value of C_F and the Fourier description of Ψ are important for F itself, but surprisingly they do not affect the leading equivalent of the inverse.

132.2 Asymptotic solution for the phase

Let $y = F(x)$ and

$$T = \log\frac{1}{y}.$$

Negating (1064) gives

$$T = \frac{L}{2}\lambda^2 - B\lambda + \frac{1}{2}\log\lambda - C_F - \Psi(\lambda) + \mathcal{O}(\lambda^{-1}), \qquad B = 1 + \frac{L}{2}. \tag{1065}$$

Lemma 132.1 (Asymptotic inversion of the quadratic phase; Derived)**.** *As $T \to +\infty$,*

$$\lambda = \sqrt{\frac{2T}{L}} + \frac{B}{L} + \mathcal{O}\left(\frac{\log T}{\sqrt{T}}\right).$$

Proof. Since Ψ is bounded, (1065) first gives

$$T = \frac{L}{2}\lambda^2 + \mathcal{O}(\lambda + \log\lambda).$$

Hence $\lambda \to \infty$ and, with

$$q = \sqrt{\frac{2T}{L}},$$

we have $\lambda/q \to 1$. Dividing

$$T - \frac{L}{2}\lambda^2 = -\frac{L}{2}(\lambda - q)(\lambda + q)$$

by $\lambda + q \asymp q$ and using the preceding error estimate improves this to $\lambda = q + \mathcal{O}(1)$.

Write

$$\lambda = q + \frac{B}{L} + \rho.$$

Insert this into (1065). The terms proportional to q cancel because $L(B/L)q = Bq$. What remains is

$$Lq\rho = -\frac{1}{2}\log\left(q + \frac{B}{L} + \rho\right) + \mathcal{O}(1) + \mathcal{O}(\rho^2).$$

The already known bound $\rho = \mathcal{O}(1)$ makes the final term bounded. Since $q \asymp \sqrt{T}$ and $\log q \asymp \log T$, division by Lq gives

$$\rho = \mathcal{O}\left(\frac{\log T}{\sqrt{T}}\right).$$

□

132.3 The endpoint equivalent for the inverse

Theorem 132.2 (Sharp inverse asymptotic; Derived)**.** *As $y \downarrow 0$,*

$$\boxed{G(y) \sim \frac{\sqrt{\log(1/y)}}{\mathrm{e}\sqrt{\log 2}} \exp\left[-\sqrt{2\log 2\,\log(1/y)}\right].} \tag{1066}$$

Equivalently,

$$\log G(y) = -\sqrt{2L\log(1/y)} + \frac{1}{2}\log\log(1/y) - 1 - \frac{1}{2}\log L + o(1). \tag{1067}$$

At the right endpoint,

$$1 - G(y) \sim \frac{\sqrt{\log(1/(1-y))}}{\mathrm{e}\sqrt{\log 2}} \exp\left[-\sqrt{2\log 2\,\log(1/(1-y))}\right] \quad (y \uparrow 1). \tag{1068}$$

Proof. Let $x = G(y)$. Then $y = F(x)$ and x is related to its exact phase by $x = \lambda\mathrm{e}^{-L\lambda}$. Therefore

$$\log x = \log\lambda - L\lambda.$$

With $T = \log(1/y)$, lemma 132.1 gives

$$\lambda = q + \frac{B}{L} + o(1), \qquad q = \sqrt{\frac{2T}{L}},$$

because $(\log T)/\sqrt{T} = o(1)$. Also $\log\lambda = \log q + o(1)$. Consequently

$$\begin{aligned}\log x &= \log q - Lq - B + o(1)\\ &= -\sqrt{2LT} + \frac{1}{2}\log T + \frac{1}{2}\log\frac{2}{L} - 1 - \frac{L}{2} + o(1)\\ &= -\sqrt{2LT} + \frac{1}{2}\log T - 1 - \frac{1}{2}\log L + o(1),\end{aligned}$$

because $L = \log 2$. This is (1067). Exponentiating proves (1066). Finally, (1056) gives $1 - G(y) = G(1-y)$ and hence (1068). □

Remark 132.3 (Why the periodic fluctuation disappears at leading order). The bounded term $\Psi(\lambda)$ changes the solution λ of (1065) only by $\mathcal{O}(1/\lambda)$. Since $\log x = \log\lambda - L\lambda$, this changes $\log x$ by $o(1)$ and therefore changes x by a factor tending to 1. Periodicity returns in the next relative correction, but not in the leading equivalent (1066).

132.4 The complete hierarchy of elementary scales

Corollary 132.4 (Between powers and logarithms; Derived)**.** *For every $\alpha > 0$ and every $m > 0$,*

$$y^\alpha = o(G(y)), \tag{1069}$$
$$G(y) = o\big((\log(1/y))^{-m}\big) \tag{1070}$$

as $y \downarrow 0$.

Proof. Set $T = \log(1/y)$. Formula (1067) gives

$$\log\frac{G(y)}{y^\alpha} = \alpha T - \sqrt{2LT} + \mathcal{O}(\log T) \longrightarrow +\infty,$$

which proves (1069). Likewise,

$$\log(G(y)T^m) = -\sqrt{2LT} + \mathcal{O}(\log T) \longrightarrow -\infty,$$

which proves (1070). ☐

The inverse endpoint is therefore neither power-like nor logarithmic. It lies in the stretched-logarithmic regime $\exp(-c\sqrt{\log(1/y)})$ with an essential $\sqrt{\log(1/y)}$ prefactor.

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133 Closed periodic saddle jets

We now turn from the inverse to the coefficient machinery hidden inside the all-orders endpoint expansion. Throughout this part,

$$L = \log 2, \qquad D = \frac{\mathrm{d}}{\mathrm{d}u},$$

and Ψ is the smooth one-periodic function occurring in the sharp Lambert main term.

133.1 The recursive jets

Define the periodic jets p_n by

$$p_0(u) = \frac{1}{2} + \frac{\Psi'(u)}{L}, \tag{1071}$$
$$p_{n+1}(u) = \left(\frac{D}{L} - (n+1)\right)p_n(u) + \frac{(-1)^{n+1}n!}{L}. \tag{1072}$$

The bounded exponent jets are

$$d_n(u) = p_n(u) + (-1)^{n+1}n!. \tag{1073}$$

The subtraction in (1073) removes the factorial part that would otherwise obscure the centered saddle expansion.

For $n \geq 0$, write

$$H_n = \sum_{k=1}^{n} \frac{1}{k}, \qquad H_0 = 0,$$

and define integers $c(n, m)$ by

$$\prod_{k=1}^{n} (z - k) = \sum_{m=0}^{n} c(n, m) z^m. \tag{1074}$$

Theorem 133.1 (Closed periodic jets; Lean)**.** *For every $n \geq 0$,*

$$p_n(u) = (-1)^n n! \left(\frac{1}{2} + \frac{H_n}{L} \right) + \sum_{m=0}^{n} c(n, m) \frac{\Psi^{(m+1)}(u)}{L^{m+1}}, \tag{1075}$$

$$d_n(u) = (-1)^n n! \left(\frac{H_n}{L} - \frac{1}{2} \right) + \sum_{m=0}^{n} c(n, m) \frac{\Psi^{(m+1)}(u)}{L^{m+1}}. \tag{1076}$$

Proof. Let

$$T_k = \frac{D}{L} - k.$$

The operators T_k commute because each is a polynomial in D. Iterating (1072) gives

$$p_n = T_n T_{n-1} \cdots T_1 p_0 + \sum_{j=0}^{n-1} T_n T_{n-1} \cdots T_{j+2} \left(\frac{(-1)^{j+1} j!}{L} \right), \tag{1077}$$

where the product is empty when $j = n - 1$.

First consider the homogeneous term. Since

$$T_n \cdots T_1 = \prod_{k=1}^{n} \left(\frac{D}{L} - k \right),$$

its action on the constant $1/2$ is

$$\frac{1}{2} \prod_{k=1}^{n} (-k) = \frac{(-1)^n n!}{2}.$$

Its action on Ψ'/L is found by expanding the operator polynomial with (1074):

$$\prod_{k=1}^{n} \left(\frac{D}{L} - k \right) \frac{\Psi'}{L} = \sum_{m=0}^{n} c(n, m) \frac{\Psi^{(m+1)}}{L^{m+1}}.$$

Now propagate the forcing term inserted at stage j. It is constant, so each subsequent T_k acts on it as multiplication by $-k$. Hence its contribution at stage n is

$$\frac{(-1)^{j+1} j!}{L} \prod_{k=j+2}^{n} (-k) = \frac{(-1)^{j+1} j!}{L} \frac{(-1)^{n-j-1} n!}{(j+1)!} = \frac{(-1)^n n!}{(j+1)L}.$$

Summing over $j = 0, \ldots, n - 1$ gives

$$\frac{(-1)^n n!}{L} \sum_{j=0}^{n-1} \frac{1}{j+1} = \frac{(-1)^n n! H_n}{L}.$$

Combining the constant, derivative, and forcing contributions proves (1075). Finally,

$$p_n + (-1)^{n+1} n! = (-1)^n n! \left(\frac{1}{2} + \frac{H_n}{L} - 1 \right) + \text{derivative part},$$

which is exactly (1076). □

133.2 Why the Stirling indices are shifted

Let $s(n,k)$ be the signed Stirling number of the first kind, defined by

$$z(z-1)\cdots(z-n+1)=\sum_{k=0}^{n} s(n,k)z^k.$$

Proposition 133.2 (Shifted Stirling conversion; Lean)**.** *For* $0 \le m \le n$*,*

$$c(n,m)=s(n+1,m+1).$$

Proof. Apply the defining identity with $n+1$:

$$z(z-1)\cdots(z-n)=\sum_{k=0}^{n+1} s(n+1,k)z^k.$$

Both sides have zero constant term, so divide by z:

$$(z-1)\cdots(z-n)=\sum_{m=0}^{n} s(n+1,m+1)z^m.$$

Comparison with (1074) proves the formula. □

Warning 133.3. The coefficients are not $s(n,m)$. The product starts with $(z-1)$ rather than z, and this forces the simultaneous shift $(n,m)\mapsto(n+1,m+1)$. In particular, the nonzero constant column $c(n,0)=(-1)^n n!$ is essential.

133.3 Coefficient rows and the first jets

The first rows of $c(n,m)$, listed from $m=0$ to $m=n$, are

n	$c(n,0)$	$c(n,1)$	$c(n,2)$	$c(n,3)$	$c(n,4)$	$c(n,5)$
0	1					
1	−1	1				
2	2	−3	1			
3	−6	11	−6	1		
4	24	−50	35	−10	1	
5	−120	274	−225	85	−15	1

and (1076) gives

$$d_0 = -\frac{1}{2}+\frac{\Psi'}{L}, \tag{1078}$$

$$d_1 = \frac{1}{2}-\frac{1}{L}-\frac{\Psi'}{L}+\frac{\Psi''}{L^2}, \tag{1079}$$

$$d_2 = -1+\frac{3}{L}+\frac{2\Psi'}{L}-\frac{3\Psi''}{L^2}+\frac{\Psi'''}{L^3}, \tag{1080}$$

$$d_3 = 3-\frac{11}{L}-\frac{6\Psi'}{L}+\frac{11\Psi''}{L^2}-\frac{6\Psi'''}{L^3}+\frac{\Psi^{(4)}}{L^4}. \tag{1081}$$

Corollary 133.4 (Structural properties; Lean)**.** *Every p_n and d_n is smooth, one-periodic, and bounded on $\mathbb{R}$. Moreover, d_n involves derivatives of Ψ only through order $n+1$, and the coefficient of the highest derivative is $L^{-(n+1)}$.*

Proof. The closed formulas are finite linear combinations of derivatives of the smooth periodic function Ψ and constants. Smoothness, periodicity, and boundedness follow. Since $c(n,n)=1$, the coefficient of $\Psi^{(n+1)}$ is $L^{-(n+1)}$. □

134 The centered exponent and its algebraic collapse

134.1 From a long Taylor expansion to one short formula

At the saddle, introduce the half-power parameter

$$\varepsilon = \lambda^{-1/2}$$

and the Gaussian variable v. After removing the exact quadratic term, the centered exponent has a formal expansion

$$\mathcal{E}(u, v; \varepsilon) = \sum_{m \geq 1} E_m(u, v)\varepsilon^m.$$

The original construction of E_m contains separate contributions from the vertical negative-Laplace jet and from the Taylor expansion of $\log(1 + \mathrm{i}\varepsilon v)$. The bounded exponent jets make these two pieces collapse.

Theorem 134.1 (Closed exponent coefficient; Lean)**.** *For every $m \geq 1$,*

$$E_m(u, v) = \mathrm{i}^m \left(\frac{d_{m-1}(u)}{m!} v^m + \frac{(-1)^{m+1}}{m+2} v^{m+2} \right). \tag{1082}$$

In particular,

$$E_1 = \mathrm{i}v \, \frac{3d_0 + v^2}{3}, \tag{1083}$$

$$E_2 = v^2 \, \frac{-2d_1 + v^2}{4}, \tag{1084}$$

$$E_3 = \mathrm{i}v^3 \left(-\frac{d_2}{6} - \frac{v^2}{5} \right), \tag{1085}$$

$$E_4 = \frac{v^4}{24}(d_3 - 4v^2). \tag{1086}$$

Proof. Before centering, the coefficient at order m has two contributions. The vertical jet contributes $\mathrm{i}^m p_{m-1}(u)v^m/m!$, while the logarithmic saddle coordinate contributes a factorial term to the same monomial and a universal v^{m+2} term. Using

$$d_{m-1} = p_{m-1} + (-1)^m (m-1)!$$

combines the two v^m coefficients into $\mathrm{i}^m d_{m-1}/m!$. The remaining logarithmic term is

$$\frac{\mathrm{i}^{m+2}(-1)^m}{m+2} v^{m+2} = \mathrm{i}^m \frac{(-1)^{m+1}}{m+2} v^{m+2},$$

because $\mathrm{i}^2 = -1$. This is (1082). Substitution of $m = 1, 2, 3, 4$ gives (1083)–(1086). □

Corollary 134.2 (Parity; Lean)**.** *For every $m \geq 1$,*

$$E_m(u, -v) = (-1)^m E_m(u, v).$$

Thus the odd-indexed E_m are odd polynomials in v and the even-indexed E_m are even.

Proof. Both monomials in (1082) have parity m, because $m + 2$ has the same parity as m. □

135 Formal exponential, Gaussian contraction, and logarithm

135.1 The exponential recursion

Define polynomials $B_n(u, v)$ by the formal identity

$$\exp\left(\sum_{m\geq 1} E_m(u, v)\varepsilon^m\right) = \sum_{n\geq 0} B_n(u, v)\varepsilon^n. \tag{1087}$$

Thus $B_0 = 1$.

Proposition 135.1 (Finite exponential recursion; Lean)**.** *For $n \geq 1$,*

$$nB_n = \sum_{m=1}^{n} mE_m B_{n-m}. \tag{1088}$$

Equivalently,

$$B_n = \sum_{\substack{r\geq 1\\ m_1+\cdots+m_r=n}} \frac{1}{r!} E_{m_1} \cdots E_{m_r},$$

where the inner sum is over ordered compositions of n into r positive parts.

Proof. Differentiate (1087) with respect to ε. The derivative of the left side is

$$\left(\sum_{m\geq 1} mE_m\varepsilon^{m-1}\right)\left(\sum_{k\geq 0} B_k\varepsilon^k\right),$$

whereas the derivative of the right side is $\sum_{n\geq 1} nB_n\varepsilon^{n-1}$. Comparing the coefficient of ε^{n-1} gives (1088). Expanding the exponential as

$$\sum_{r\geq 0} \frac{1}{r!}\left(\sum_{m\geq 1} E_m\varepsilon^m\right)^r$$

and collecting total degree n gives the composition formula. □

Corollary 135.2 (Parity of the exponential coefficients; Lean)**.** *For all $n \geq 0$,*

$$B_n(u, -v) = (-1)^n B_n(u, v).$$

Proof. Every product contributing to B_n has indices $m_1 + \cdots + m_r = n$. By corollary 134.2, its parity is $(-1)^{m_1+\cdots+m_r} = (-1)^n$. □

135.2 Gaussian contraction and mass coefficients

For a polynomial P define normalized Gaussian contraction by

$$\mathfrak{G}[P] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \mathrm{e}^{-v^2/2} P(v)\, \mathrm{d}v.$$

It is determined by

$$\mathfrak{G}[v^{2k}] = (2k-1)!!, \qquad \mathfrak{G}[v^{2k+1}] = 0. \tag{1089}$$

The mass coefficients are

$$a_j(u) = \mathfrak{G}[B_{2j}(u, \cdot)], \qquad j \geq 0. \tag{1090}$$

In particular $a_0 = 1$. The odd half-powers vanish automatically:

$$\mathfrak{G}[B_{2j+1}] = 0$$

by corollary 135.2.

Theorem 135.3 (Closed composition formula for a_j; Derived)**.** *For $j \geq 1$,*

$$a_j(u) = (-1)^j \sum_{r=1}^{2j} \frac{1}{r!} \sum_{\substack{m_1+\cdots+m_r=2j \\ m_i\geq 1}} \sum_{S\subseteq\{1,\dots,r\}} (2j+2|S|-1)!! \\ \times \prod_{i\in S} \frac{(-1)^{m_i+1}}{m_i+2} \prod_{i\notin S} \frac{d_{m_i-1}(u)}{m_i!}. \tag{1091}$$

Consequently a_j is a polynomial with rational coefficients in $d_0, \dots, d_{2j-1}$.

Proof. Use the ordered-composition formula in proposition 135.1 for B_{2j}. Each factor

$$E_{m_i} = \mathrm{i}^{m_i} \left(\frac{d_{m_i-1}}{m_i!} v^{m_i} + \frac{(-1)^{m_i+1}}{m_i+2} v^{m_i+2} \right)$$

offers two choices. Let S be the set of factors from which the second, universal monomial is chosen. Since $\sum_i m_i = 2j$, the product of the powers of i is

$$\mathrm{i}^{2j} = (-1)^j,$$

and the total power of v is $2j + 2|S|$. Its Gaussian contraction is $(2j + 2|S| - 1)!!$. Multiplying the remaining scalar factors gives (1091). The largest possible jet index is $m_i - 1 \leq 2j - 1$. $\square$

Formula (1091) is a human-readable algebraic consequence of the formal exponent and exponential-recursion theorems. The repository machine-checks the underlying recursion, the structural top-jet theorem, and explicit low orders. It does not package this exact threefold summation as a single general Lean declaration.

135.3 Logarithmic coefficients

Define $A_j(u)$ by

$$1 + \sum_{j\geq 1} a_j(u) z^j = \exp\left(\sum_{j\geq 1} A_j(u) z^j \right). \tag{1092}$$

These are the coefficients that occur in $\log F$.

Proposition 135.4 (Mass-to-logarithm conversion; Lean)**.** *For $n \geq 1$,*

$$A_n = a_n - \frac{1}{n} \sum_{k=1}^{n-1} k A_k\, a_{n-k}. \tag{1093}$$

Equivalently,

$$A_n = \sum_{r=1}^{n} \frac{(-1)^{r-1}}{r} \sum_{\substack{q_1+\cdots+q_r=n \\ q_i\geq 1}} a_{q_1} \cdots a_{q_r}. \tag{1094}$$

Proof. Differentiate (1092) logarithmically:

$$\sum_{n\geq 1} n a_n z^{n-1} = \left(1 + \sum_{j\geq 1} a_j z^j \right) \left(\sum_{k\geq 1} k A_k z^{k-1} \right).$$

The coefficient of z^{n-1} gives

$$na_n = \sum_{k=1}^{n} kA_k a_{n-k}, \qquad a_0 = 1,$$

and solving for A_n proves (1093). Formula (1094) is the coefficient of z^n in the ordinary series

$$\log(1+w) = \sum_{r\geq 1} \frac{(-1)^{r-1}}{r} w^r, \qquad w = \sum_{j\geq 1} a_j z^j.$$

□

Corollary 135.5 (Regularity and derivative order; Lean input, Derived conclusion)**.** *For every j, the functions a_j and A_j are smooth, one-periodic, and bounded. Neither involves a derivative of Ψ of order greater than $2j$.*

Proof. The function a_j is a polynomial in $d_0, \ldots, d_{2j-1}$, and d_n is smooth, periodic, and depends on Ψ only through order $n+1$. Thus a_j has the stated properties and uses derivatives only through order $2j$. The recursion (1093) transfers the same assertions to A_j. □

135.4 The top jet and the highest derivative

The most complicated-looking coefficient has a very simple dependence on its newest jet.

Theorem 135.6 (Leading mass-jet law; Leaninput, Derivedlogarithmic consequence)**.** *For $j \geq 1$, the mass coefficient a_j is affine in d_{2j-1}, and the coefficient of that jet is*

$$\frac{(-1)^j}{2^j j!}. \tag{1095}$$

The logarithmic coefficient A_j is derived here to have the same top-jet coefficient; this transfer is not asserted by the cited mass-coefficient theorem. Consequently the coefficient of $\Psi^{(2j)}(u)$ in $A_j(u)$ is

$$\boxed{\frac{(-1)^j}{2^j j! L^{2j}}.} \tag{1096}$$

Proof. The jet d_{2j-1} occurs only in E_{2j}. Therefore the only term of B_{2j} that contains it is the linear term E_{2j} itself. From (1082), its jet part is

$$\mathrm{i}^{2j} \frac{d_{2j-1}}{(2j)!} v^{2j} = (-1)^j \frac{d_{2j-1}}{(2j)!} v^{2j}.$$

Gaussian contraction multiplies this by $(2j-1)!!$. Since

$$(2j)! = 2^j j!(2j-1)!!,$$

the coefficient is (1095).

In (1093), every correction term $A_k a_{j-k}$ has $k < j$ and $j - k < j$; it involves jets only through d_{2j-3} and therefore cannot alter the coefficient of d_{2j-1}. Thus A_j has the same top-jet coefficient. Finally, corollary 133.4 says that the coefficient of $\Psi^{(2j)}$ in d_{2j-1} is L^{-2j}, giving (1096). □

Remark 135.7. The theorem provides a quick consistency check for every explicit coefficient. For example, A_1 must contain $-\Psi''/(2L^2)$, while A_2 must contain $+\Psi^{(4)}/(8L^4)$.

136 The first and second logarithmic corrections

136.1 Order two: the first correction

From equations (1083) and (1084),

$$B_2 = E_2 + \frac{1}{2}E_1^2.$$

A direct expansion gives

$$B_2 = -\frac{d_0^2 + d_1}{2}v^2 + \left(\frac{1}{4} - \frac{d_0}{3}\right)v^4 - \frac{1}{18}v^6.$$

Using $\mathfrak{G}[v^2] = 1$, $\mathfrak{G}[v^4] = 3$, and $\mathfrak{G}[v^6] = 15$ yields

$$a_1 = A_1 = -\frac{d_0^2}{2} - d_0 - \frac{d_1}{2} - \frac{1}{12}. \tag{1097}$$

Equivalently,

$$A_1 = -\frac{6d_0^2 + 12d_0 + 6d_1 + 1}{12}.$$

Substituting equations (1078) and (1079) recovers the first periodic correction in terms of Ψ' and Ψ''.

136.2 Order four of the formal exponential

The fourth exponential coefficient is

$$B_4 = E_4 + E_3E_1 + \frac{1}{2}E_2^2 + \frac{1}{2}E_2E_1^2 + \frac{1}{24}E_1^4. \tag{1098}$$

This follows either from the composition formula or from the recurrence (1088). Substitution of equations (1083) to (1086) and collection of even powers of v gives the following polynomial.

Proposition 136.1 (The order-four reference polynomial; Lean)**.** *Writing* $d_k = d_k(u)$*,*

$$\begin{aligned} B_4(u,v) = {} & \left(\frac{d_0^4}{24} + \frac{d_0^2 d_1}{4} + \frac{d_0 d_2}{6} + \frac{d_1^2}{8} + \frac{d_3}{24}\right)v^4 \\ & + \left(\frac{d_0^3}{18} - \frac{d_0^2}{8} + \frac{d_0 d_1}{6} + \frac{d_0}{5} - \frac{d_1}{8} + \frac{d_2}{18} - \frac{1}{6}\right)v^6 \\ & + \left(\frac{d_0^2}{36} - \frac{d_0}{12} + \frac{d_1}{36} + \frac{47}{480}\right)v^8 \\ & + \left(\frac{d_0}{162} - \frac{1}{72}\right)v^{10} + \frac{1}{1944}v^{12}. \end{aligned} \tag{1099}$$

Proof. Let

$$P_1 = d_0 v + \frac{v^3}{3}, \qquad P_3 = -\frac{d_2 v^3}{6} - \frac{v^5}{5},$$

so that $E_1 = \mathrm{i}P_1$ and $E_3 = \mathrm{i}P_3$. Then $E_3E_1 = -P_3P_1$ and $E_1^4 = P_1^4$. The even terms E_2, E_4 are already real. Insert these expressions in (1098). Each summand has degree at most 12 and is even. Multiplication gives

	v^4	v^6	v^8	v^{10}	v^{12}
E_4	$d_3/24$	$-1/6$	0	0	0
E_3E_1	$d_0d_2/6$	$d_2/18 + d_0/5$	$1/15$	0	0
$E_2^2/2$	$d_1^2/8$	$-d_1/8$	$1/32$	0	0
$E_2E_1^2/2$	$d_0^2 d_1/4$	$d_0d_1/6 - d_0^2/8$	$d_1/36 - d_0/12$	$-1/72$	0
$E_1^4/24$	$d_0^4/24$	$d_0^3/18$	$d_0^2/36$	$d_0/162$	$1/1944$

and the two universal v^8 contributions satisfy $1/15 + 1/32 = 47/480$. Summing the columns proves (1099). $\square$

136.3 Gaussian contraction and the second mass coefficient

The relevant Gaussian moments are

$$\mathfrak{G}[v^4] = 3, \quad \mathfrak{G}[v^6] = 15, \quad \mathfrak{G}[v^8] = 105, \quad \mathfrak{G}[v^{10}] = 945, \quad \mathfrak{G}[v^{12}] = 10395.$$

Theorem 136.2 (Second mass coefficient; Lean)**.** *Gaussian contraction of* (1099) *gives*

$$\begin{aligned} a_2 = \frac{d_0^4}{8} + \frac{5d_0^3}{6} + \frac{3d_0^2 d_1}{4} + \frac{25d_0^2}{24} + \frac{5d_0 d_1}{2} + \frac{d_0 d_2}{2} + \frac{d_0}{12} \\ + \frac{3d_1^2}{8} + \frac{25d_1}{24} + \frac{5d_2}{6} + \frac{d_3}{8} + \frac{1}{288}. \end{aligned} \tag{1100}$$

Proof. Multiply the coefficients of $v^4, v^6, v^8, v^{10}, v^{12}$ in (1099) by $3, 15, 105, 945, 10395$, respectively, and collect like monomials. For illustration, the coefficient of d_0 is

$$15 \cdot \frac{1}{5} + 105\left(-\frac{1}{12}\right) + 945 \cdot \frac{1}{162} = 3 - \frac{35}{4} + \frac{35}{6} = \frac{1}{12},$$

and the universal constant is

$$15\left(-\frac{1}{6}\right) + 105\frac{47}{480} + 945\left(-\frac{1}{72}\right) + 10395\frac{1}{1944} = \frac{1}{288}.$$

The remaining coefficients follow by the same arithmetic. $\square$

136.4 The second logarithmic coefficient

Since

$$\log(1 + a_1 z + a_2 z^2 + \cdots) = a_1 z + \left(a_2 - \frac{1}{2}a_1^2\right) z^2 + \cdots,$$

we have $A_2 = a_2 - a_1^2/2$.

Theorem 136.3 (Second periodic saddle correction; Lean)**.** *The second logarithmic coefficient is*

$$\boxed{A_2 = \frac{8d_0^3 + 12d_0^2 d_1 + 12d_0^2 + 48d_0 d_1 + 12d_0 d_2 + 6d_1^2 + 24d_1 + 20d_2 + 3d_3}{24}}. \tag{1101}$$

It is smooth, one-periodic, and bounded.

Proof. Insert (1097) and (1100) into $A_2 = a_2 - a_1^2/2$. The quartic term $d_0^4/8$ cancels, as do the terms $d_0/12$ and $1/288$ after the square is expanded. The remaining terms reduce to (1101). Smoothness, periodicity, and boundedness follow because A_2 is a polynomial in d_0, d_1, d_2, d_3. $\square$

Corollary 136.4 (Two explicit correction orders; Lean)**.** *Let $\lambda = \lambda(x)$ be the exact lower Lambert phase. Then, as $x \downarrow 0$,*

$$\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda)}{\lambda} + \frac{A_2(\lambda)}{\lambda^2} + \mathcal{O}(\lambda^{-3}), \tag{1102}$$

where $\mathcal{M}(x)$ is the exact sharp Lambert main term. Since $\lambda \asymp -\log x$, the remainder is also $\mathcal{O}((-\log x)^{-3})$.

Warning 136.5. The coefficients in (1102) are evaluated at the exact phase $\lambda(x)$. Substituting a truncated asymptotic formula for λ into A_1 or A_2 is not licensed merely by $\lambda - \lambda_{\text{approx}} = \mathcal{O}(1)$: periodic functions are sensitive to bounded phase shifts.

137 Why the all-orders remainder is actually rigorous

The formal coefficient algebra by itself does not prove an asymptotic expansion. One must control the exact contour integral uniformly on a window that grows with the requested order, prove that exponentiation does not magnify the Taylor error, replace the finite Gaussian window by the whole line, and show that the discarded contour is smaller than every retained inverse power. This section spells out that analytic chain.

137.1 The exact normalized saddle mass

For $x > 0$ small, let $\lambda = \lambda(x)$ be the exact phase (1063). The exact saddle reduction can be organized as

$$\log F(x) = \mathcal{M}(x) + \log \mathcal{G}(x) + \mathcal{T}(x), \tag{1103}$$

where:

- $\mathcal{M}(x)$ is the exact sharp Lambert main term, containing the quadratic phase, the Gaussian normalization, the periodic finite part, and the fixed constant;
- $\mathcal{G}(x) > 0$ is the normalized saddle-kernel mass and $\mathcal{G}(x) \to 1$;
- $\mathcal{T}(x)$ is the forward negative-Laplace tail, which is flat against every inverse power of λ.

Thus all algebraic corrections beyond $\mathcal{M}$ come from $\log \mathcal{G}$.

It is technically convenient to begin on the real dyadic logarithmic scale

$$x = 2^{-t}, \qquad t \to +\infty,$$

and to write

$$b = \lambda(2^{-t}), \qquad \varepsilon = b^{-1/2}.$$

This is not a restriction to integer t. Every sufficiently small positive x has the unique representation $x = 2^{-t}$ with $t = -\log_2 x$.

After the saddle contour is centered and rescaled by $v = \sqrt{b}\,\theta$, its central kernel has the normalized form

$$\frac{1}{\sqrt{2\pi}} \mathrm{e}^{-v^2/2} \exp R_b(v), \tag{1104}$$

where R_b is the exact centered exponent. The formal expansion of R_b is

$$R_b(v) \sim \sum_{m \geq 1} E_m(b, v) b^{-m/2}. \tag{1105}$$

The first variable of E_m is evaluated at b because all coefficient functions are periodic in the exact phase.

137.2 The order-dependent central window

For a requested order $N \geq 0$, define

$$A_N(b) = \sqrt{32(N+1)\log b}. \tag{1106}$$

Lemma 137.1 (Scale separation; Lean)**.** *For every fixed N,*

$$A_N(b) \longrightarrow +\infty, \qquad \varepsilon A_N(b) \longrightarrow 0 \qquad (b \to +\infty).$$

Moreover, for every fixed $r > 0$ and $\delta > 0$,

$$A_N(b)^r = o(b^\delta).$$

Proof. The first limit is immediate from $\log b \to \infty$. The second follows from

$$\varepsilon A_N(b) = \sqrt{\frac{32(N+1)\log b}{b}} \longrightarrow 0.$$

Finally, every fixed power of $\log b$ grows more slowly than every positive power of b. □

The window has two simultaneous purposes. It grows, so replacing a Gaussian integral over it by a whole-line Gaussian integral creates a tiny tail. Yet its angular width $A_N(b)/\sqrt{b}$ shrinks to zero, so all local Taylor estimates remain uniform.

137.3 All-order control of the exact centered exponent

The exact negative-Laplace factor contains a finite periodic part plus endpoint and forward-product corrections. On the vertical line through the saddle, the proof decomposes it into:

1. a Taylor polynomial in $\theta = v/\sqrt{b}$ whose coefficients are the periodic jets p_n;
2. the Taylor polynomial of $\log(1 + \mathrm{i}\theta)$;
3. two boundary-jet remainders from the finite part;
4. forward-product jets whose size is exponentially small in b;
5. explicit integral remainders for the vertical derivatives.

The factorial pieces in the first two items combine into the bounded jets d_n, producing (1082).

Proposition 137.2 (Integrated exponent truncation; Lean)**.** *Fix N. The Taylor depth can be chosen, depending only on N, so that on $|v| \le A_N(b)$ the exact exponent may be replaced by*

$$R_{b,N}(v) = \sum_{m=1}^{2N+1} E_m(b, v) b^{-m/2}$$

with an error which, after multiplication by the Gaussian weight and integration over the central window, is

$$\mathcal{O}(b^{-N-1}).$$

The implied constant may depend on N but not on b.

Proof. All vertical derivatives of the periodic finite part are uniformly bounded because the periodic jets are smooth and bounded. Taylor's theorem therefore bounds a vertical remainder of order M by

$$C_M|\theta|^{M+1} = C_M b^{-(M+1)/2}|v|^{M+1}.$$

The logarithmic coordinate has an analogous remainder on the shrinking angular window, because $\varepsilon A_N(b) \to 0$ keeps $1 + \mathrm{i}\theta$ uniformly away from zero. Boundary jets satisfy the same kind of estimate, while the forward-product terms are exponentially small and hence smaller than any prescribed inverse power.

Choose M with enough surplus beyond $2N + 1$. On the central window, powers of $|v|$ are bounded by powers of $A_N(b)$, hence by powers of $\log b$. By lemma 137.1, the spare powers of $b^{-1/2}$ absorb every such logarithmic loss. The pointwise exponent error is therefore integrably $\mathcal{O}(b^{-N-1})$ against $\mathrm{e}^{-v^2/2}\,\mathrm{d}v$. Removing terms above order $2N + 1$ from the finite Taylor polynomial costs the same order because each carries at least the corresponding spare half-power. This leaves $R_{b,N}$. □

137.4 Finite exponentiation without an infinite-series argument

A common informal step is to exponentiate (1105) term by term. The formal development instead uses a finite Taylor polynomial for the exponential and an explicit remainder.

Let

$$\mathcal{P}_N(b,v) = \sum_{\ell=0}^{2N+1} B_\ell(b,v) b^{-\ell/2}.$$

It is the reference polynomial obtained from the formal recursion in proposition 135.1.

Proposition 137.3 (Finite exponential substitution; Lean)**.** *For every fixed N,*

$$\frac{1}{\sqrt{2\pi}} \int_{|v|\le A_N(b)} \mathrm{e}^{-v^2/2} \left|\exp R_b(v) - \mathcal{P}_N(b,v)\right| \,\mathrm{d}v = \mathcal{O}(b^{-N-1}). \tag{1107}$$

Proof. First replace R_b by $R_{b,N}$ using proposition 137.2. On the central window the truncated exponent tends uniformly to zero after the exact quadratic term has been removed, and it is bounded by a fixed polynomial in v times $b^{-1/2}$. Apply Taylor's formula

$$\mathrm{e}^z = \sum_{r=0}^{L-1} \frac{z^r}{r!} + \frac{z^L}{(L-1)!} \int_0^1 (1-s)^{L-1} \mathrm{e}^{sz} \,\mathrm{d}s$$

with $L = 2(N+1)$. Uniform control of the real part of $R_{b,N}$ bounds the integral remainder by a Gaussian-integrable polynomial times b^{-N-1}.

Now expand the finite Taylor polynomial in the finitely many powers $b^{-m/2}$. The coefficient of $b^{-\ell/2}$ for $\ell < L$ is exactly B_ℓ, by the same recursion that defines the formal exponential coefficients. Algebraically, the difference between the finite exponential Taylor polynomial and $\mathcal{P}_N$ factors as

$$b^{-L/2} \times \{\text{a finite quotient polynomial in } b^{-1/2}, v, E_1, \ldots, E_{2N+1}\}.$$

Because $L/2 = N+1$ and the coefficient functions are bounded, Gaussian integration gives (1107). □

137.5 Gaussian contraction on the whole line

By corollary 135.2, odd ℓ contribute odd polynomials. Therefore

$$\mathfrak{G}[\mathcal{P}_N(b,\cdot)] = \sum_{j=0}^{N} a_j(b) b^{-j}. \tag{1108}$$

The exact central integral, however, only runs over $|v| \le A_N(b)$. We must show that the polynomial reference can be extended to the whole real line at the required order.

Lemma 137.4 (Gaussian polynomial tail; Lean)**.** *For every fixed polynomial P and every fixed N,*

$$\int_{|v|>A_N(b)} \mathrm{e}^{-v^2/2} |P(v)| \,\mathrm{d}v = \mathcal{O}(b^{-N-1}).$$

The same assertion holds for a finite family of polynomials whose coefficients are uniformly bounded in b.

Proof. For every integer $q \ge 0$ there is a constant C_q such that

$$\int_A^\infty v^q \mathrm{e}^{-v^2/2} \,\mathrm{d}v \le C_q (1 + A^q) \mathrm{e}^{-A^2/4} \qquad (A \ge 1).$$

At $A = A_N(b)$,

$$\mathrm{e}^{-A_N(b)^2/4} = \mathrm{e}^{-8(N+1)\log b} = b^{-8(N+1)}.$$

The polynomial factor in $A_N(b)$ is a power of $\log b$ and is absorbed by the enormous surplus of inverse powers. Sum over the finitely many monomials of P. □

Combining proposition 137.3 and lemma 137.4 with (1108), the central contour contributes

$$\sum_{j=0}^{N} a_j(b)b^{-j} + \mathcal{O}(b^{-N-1}). \tag{1109}$$

137.6 The complementary contour

The exact contour outside the central window is split into an intermediate region and a far region. The two estimates use different mechanisms.

Proposition 137.5 (All-order minor-arc bounds; Lean)**.** *For every fixed N, the normalized contour outside $|v| \le A_N(b)$ contributes $\mathcal{O}(b^{-N-1})$. More precisely:*

1. *the intermediate region has a bound with fourfold power slack, of order at most a constant times $b^{-4(N+1)}$ once the relevant dyadic index is at least $b/4$;*
2. *the far region is bounded by a constant times*

$$b \exp\left(-\frac{\log 2}{4} b\right),$$

which is flat against every inverse power of b.

Proof. On the intermediate region the modulus estimate for the saddle kernel contains a decay factor whose exponent is quadratic in the enlarged radius. Substitution of (1106) yields

$$\exp\{-4(N+1)\log b\} = b^{-4(N+1)}$$

after the standard comparison between the dyadic index and b. The remaining geometric and polynomial factors consume far less than the three spare blocks of inverse powers.

In the far region the dyadic product supplies geometric decay in the index. Summing the geometric tail gives a bound of the displayed exponential form. For every K,

$$b^K\, b\mathrm{e}^{-(\log 2)b/4} \longrightarrow 0,$$

so this contribution is $\mathcal{O}(b^{-N-1})$ for the requested fixed N. □

137.7 The all-orders mass expansion

Theorem 137.6 (Normalized mass expansion; Lean)**.** *For every fixed $N \ge 0$, as $t \to +\infty$ with $b = \lambda(2^{-t})$,*

$$\mathcal{G}(2^{-t}) = \sum_{j=0}^{N} \frac{a_j(b)}{b^j} + \mathcal{O}(b^{-N-1}). \tag{1110}$$

The coefficient functions are evaluated at the exact phase b.

Proof. The central part is (1109). Add the intermediate and far parts controlled by proposition 137.5. All three errors are $\mathcal{O}(b^{-N-1})$. □

137.8 Transfer through the logarithm

Theorem 137.7 (Logarithmic expansion on the dyadic real scale; Lean). *For every fixed $N \geq 0$,*

$$\log \mathcal{G}(2^{-t}) = \sum_{j=1}^{N} \frac{A_j(b)}{b^j} + \mathcal{O}(b^{-N-1}), \qquad b = \lambda(2^{-t}). \tag{1111}$$

Proof. The boundedness of the a_j and (1110) with $N = 0$ give $\mathcal{G}(2^{-t}) = 1 + \mathcal{O}(b^{-1})$, hence $\mathcal{G} > 0$ and $|\mathcal{G} - 1| < 1/2$ eventually. Taylor's theorem for $\log(1 + z)$ is therefore uniform. Substitute (1110) into the finite Taylor polynomial for the logarithm. The coefficient of b^{-j} is exactly A_j, either by (1094) or by the recursion (1093). The logarithmic Taylor remainder and the original mass remainder are both $\mathcal{O}(b^{-N-1})$. □

137.9 Adding the flat tail and returning to $x \downarrow 0$

The tail $\mathcal{T}$ in (1103) satisfies, for every K,

$$\mathcal{T}(x) = \mathcal{O}(\lambda(x)^{-K}).$$

It therefore does not change any coefficient in a Poincaré expansion.

Theorem 137.8 (Full exact-phase expansion; Lean). *For every $N \geq 0$,*

$$\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N-1}) \qquad (x \downarrow 0). \tag{1112}$$

Equivalently, using a truncation with indices $j < N$,

$$\log F(x) - \mathcal{M}(x) - \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} = \mathcal{O}(\lambda(x)^{-N}).$$

Proof. First combine (1103), (1111), and the flatness of $\mathcal{T}$ on the scale $x = 2^{-t}$. Next use the exact change of variables $t = -\log_2 x$. As $x \downarrow 0$, $t \to +\infty$ and $\lambda(2^{-t}) = \lambda(x)$, so the dyadic-real estimate transports without approximation. This yields (1112). □

Warning 137.9 (Poincaré does not mean convergent). The theorem is fixed-order. For each N there is an error constant depending on N, and the central radius itself depends on N. No uniform control as $N \to \infty$ is asserted, and no convergence of the infinite coefficient series is implied.

138 A proof-oriented concordance with the Lean modules

The following table records where the two missing clusters are proved. Paths are relative to

`Analysis/FabiusFunction/Lean/FabiusFunction/`.

Table 27: Module concordance for this companion.

Module	Principal role	Status
`FabiusInverse.lean`	Order isomorphism of $[0, 1]$, totalized inverse, global continuity and monotonicity, comparison equivalences, tails, reflection, exact dyadic inversion, reciprocal first and second derivatives, interior smoothness, strict closed-half curvature, transported flatness, effective steepness, exact global differentiability and C^∞ loci, and endpoint quotient divergence.	Lean
`InverseNotElementary.lean`	Exact analytic locus of the totalized inverse and the analytic-density obstructions used in its non-elementarity theorem.	Lean
`FabiusSaddleJetClosedForm.lean`	Operator solution of the recursive periodic jets and the harmonic-number constant term.	Lean
`FabiusSaddleJetStirling.lean`	Identification of the operator coefficients with $s(n+1, m+1)$ and the corresponding closed derivative formula.	Lean
`FabiusSaddleExponentClosedForm.lean`	Collapse of each centered exponent coefficient to one bounded jet monomial plus one universal monomial; parity.	Lean
`FabiusSaddleExpansionCoefficients.lean`	Formal exponential coefficients, Gaussian contraction, mass coefficients, logarithmic transfer, periodicity, smoothness, and boundedness.	Lean
`FabiusSaddleLeadingCoefficient.lean`	Affine dependence of a_j on the newest jet and coefficient $(-1)^j/(2^j j!)$.	Lean
`FabiusSecondSaddleCorrection.lean`	Order-four exponential identity, explicit B_4, Gaussian contraction, a_2, A_2, and regularity of the second correction.	Lean
`FabiusSecondSaddleExpansion.lean`	Insertion of A_2 into the exact-phase asymptotic with a third-order remainder.	Lean
`FabiusSaddleCentralAllOrders.lean`	Exact central decomposition, finite reference polynomial, odd-term cancellation, and integrated central remainders.	Lean
`FabiusSaddleTailAllOrders.lean`	Order-dependent radius and complementary intermediate/far contour bounds of arbitrary fixed inverse-power order.	Lean
`FabiusSaddleMassAllOrders.lean`	Assembly of central Taylor control, finite exponential substitution, whole-line Gaussian contraction, and minor arcs into the complete mass expansion.	Lean
`FabiusFullAsymptoticExpansion.lean`	Transfer from mass to logarithm, addition of the flat forward tail, and transport from $x = 2^{-t}$ to the full limit $x \downarrow 0$.	Lean

138.1 Named declarations worth knowing

The inverse interface is centered around

```
fabiusOrderIso
fabiusReal_fabiusInv
fabiusInv_fabiusReal
strictMonoOn_fabiusInv
monotone_fabiusInv
continuous_fabiusInv
fabiusInv_le_iff_le_fabiusReal
fabiusReal_le_iff_le_fabiusInv
fabiusInv_lt_iff_lt_fabiusReal
fabiusReal_lt_iff_lt_fabiusInv
fabiusInv_one_sub
fabiusInv_fabiusDyadicUnit
le_two_pow_mul_fabiusInv_pow
le_two_pow_div_factorial_mul_fabiusInv_pow
mul_lt_fabiusInv
tendsto_fabiusInv_div_atTop
tendsto_one_sub_fabiusInv_div_one_sub_atTop
```

The all-orders endpoint interface culminates in

```
fabiusSaddleKernelMass_dyadicLambert_hasAsymptoticExpansion
fabiusSaddleRatio_dyadicLambert_hasAsymptoticExpansion
log_fabius_dyadicReal_sub_sharpLambertMain_hasAsymptoticExpansion
log_fabius_sub_sharpLambertMain_hasAsymptoticExpansion
log_fabius_sub_sharpLambertExpansion_isBigO
```

These names are not needed to read the mathematics, but they make the article useful as a map back into the source tree.

139 What is formalized and what is newly synthesized

For clarity, the principal statements can be divided into three groups.

139.1 Directly formalized

The following content is represented directly by declarations in the repository:

- the global totalized inverse package and all four comparison equivalences;
- the constant tails, reflection, midpoint, and exact dyadic inversion;
- interior C^∞ regularity, the exact formulas for G' and G'', and strict concavity on $[0, 1/2]$ and strict convexity on $[1/2, 1]$;
- the effective and sharp inverse power inequalities, effective linear steepness, and the two endpoint quotient limits;
- the exact global differentiability, C^∞, and analytic loci of the totalized inverse;
- the closed jet formulas, shifted Stirling conversion, and closed exponent coefficients;
- the formal exponential and logarithmic coefficient machinery, top-jet coefficient, first and second corrections;
- the arbitrary-order central, tail, mass, and logarithmic asymptotic theorems.

139.2 Classical consequences derived in this article

The following statements are proved here from the formalized input:

- divergence of $G(y)/y^\alpha$ for every $\alpha > 0$;
- the sharp equivalent (1066) and the scale hierarchy (1069)–(1070);
- the explicit all-j composition sum (1091) and the highest-Ψ-derivative corollary (1096).

139.3 Claims deliberately not made

This companion does *not* claim:

- convergence or Borel summability of the all-orders series;
- an all-orders substitution of an approximate Lambert phase inside the periodic coefficients;
- a closed form for the inverse at arbitrary rational arguments;
- analyticity of either F or G inside the transition interval;
- uniformity of the remainder constants as the truncation order tends to infinity.

140 Concluding synthesis

The inverse function and the saddle coefficient machinery illuminate opposite sides of the same endpoint phenomenon.

On the geometric side, F approaches zero more rapidly than every power. Its inverse must therefore leave zero more rapidly than every power, and the formalized comparison calculus turns this slogan into effective inequalities with exact constants. The sharp Lambert expansion goes further: the inverse lives on the stretched-logarithmic scale

$$G(y) \asymp \sqrt{\log(1/y)}\exp\left(-\sqrt{2\log 2\,\log(1/y)}\right),$$

with the exact leading constant $1/(\mathrm{e}\sqrt{\log 2})$. Reflection transfers the entire picture to the endpoint 1.

On the asymptotic side, the apparent complexity of the periodic corrections is organized by three simple structures. First, the recursive jets are solved by a product of commuting first-order operators; the forcing terms sum to a harmonic number and the derivative coefficients are shifted signed Stirling numbers. Second, centering collapses every exponent coefficient to two monomials, forcing parity and eliminating all odd half-powers under Gaussian contraction. Third, the newest jet can enter only linearly, so the coefficient of the highest derivative $\Psi^{(2j)}$ is known at every order before the lower-order algebra is performed.

The analytic remainder proof then explains why these formal coefficients are not merely symbolic. The growing window $A_N(b) = \sqrt{32(N+1)\log b}$ separates the local and global tasks: locally it is still narrow enough for uniform Taylor expansion, while globally it is wide enough to make both Gaussian and contour tails smaller than the requested inverse power. Finite exponential substitution and logarithmic transfer complete a genuine Poincaré expansion at the exact Lambert phase.

Together, these missing parts close two expository loops. Flatness of F is now matched by a precise theory of the steep inverse, and the headline all-orders expansion is now matched by a transparent account of the coefficient algebra and the analytic estimates that make it true.

Supplementary material from this dossier

141 A compact inverse reference

For convenience, here is the inverse theory in one place. For $G(y) = F_I^{-1}(\mathrm{clamp}_{[0,1]}(y))$:

$$\begin{aligned}
&0 \le G(y) \le 1, \qquad G(y) = 0\ (y \le 0), \qquad G(y) = 1\ (y \ge 1),\\
&F(G(y)) = y \quad (0 \le y \le 1), \qquad G(F(x)) = x \quad (0 \le x \le 1),\\
&G(1-y) = 1 - G(y), \qquad G(1/2) = 1/2,\\
&G(y) \le x \iff y \le F(x), \qquad F(x) \le y \iff x \le G(y),\\
&G(y) < x \iff y < F(x), \qquad F(x) < y \iff x < G(y),\\
&G(\mathsf{F}_n(a)) = \min\{a, 2^n\}/2^n,\\
&y \le C_n G(y)^n, \qquad y \le C_n G(y)^n/n! \quad \text{when } 2^n G(y) \le 1,\\
&G(y)/y \to +\infty \quad (y \downarrow 0), \qquad (1 - G(y))/(1-y) \to +\infty \quad (y \uparrow 1),\\
&G(y) \sim \frac{\sqrt{\log(1/y)}}{\mathrm{e}\sqrt{\log 2}} \exp\Big[-\sqrt{2 \log 2\, \log(1/y)}\Big].
\end{aligned}$$

142 A compact coefficient reference

With $L = \log 2$,

$$\begin{aligned}
p_0 &= \frac{1}{2} + \frac{\Psi'}{L},\\
p_{n+1} &= (D/L - (n+1))p_n + \frac{(-1)^{n+1} n!}{L},\\
d_n &= p_n + (-1)^{n+1} n!,\\
c(n,m) &= s(n+1, m+1),\\
d_n &= (-1)^n n! \left(\frac{H_n}{L} - \frac{1}{2}\right) + \sum_{m=0}^{n} c(n,m) \frac{\Psi^{(m+1)}}{L^{m+1}},\\
E_m &= \mathrm{i}^m \left(\frac{d_{m-1}}{m!} v^m + \frac{(-1)^{m+1}}{m+2} v^{m+2}\right),\\
nB_n &= \sum_{m=1}^{n} m E_m B_{n-m},\\
a_j &= \mathfrak{G}[B_{2j}],\\
A_n &= a_n - \frac{1}{n} \sum_{k=1}^{n-1} k A_k a_{n-k},\\
[d_{2j-1}]\, A_j &= \frac{(-1)^j}{2^j j!},\\
[\Psi^{(2j)}]\, A_j &= \frac{(-1)^j}{2^j j! L^{2j}}.
\end{aligned}$$

The first two logarithmic coefficients are

$$\begin{aligned} A_1 &= -\frac{d_0^2}{2} - d_0 - \frac{d_1}{2} - \frac{1}{12}, \\ A_2 &= \frac{1}{24}\big(8d_0^3 + 12d_0^2 d_1 + 12d_0^2 + 48 d_0 d_1 + 12 d_0 d_2 \\ &\qquad + 6d_1^2 + 24 d_1 + 20 d_2 + 3 d_3\big). \end{aligned}$$

143 Repository paths used in the audit

```
Analysis/FabiusFunction/
  Lean/FabiusFunction/FabiusInverse.lean
  Lean/FabiusFunction/FabiusSaddleJetClosedForm.lean
  Lean/FabiusFunction/FabiusSaddleJetStirling.lean
  Lean/FabiusFunction/FabiusSaddleExponentClosedForm.lean
  Lean/FabiusFunction/FabiusSaddleLeadingCoefficient.lean
  Lean/FabiusFunction/FabiusSecondSaddleCorrection.lean
  Lean/FabiusFunction/FabiusSecondSaddleExpansion.lean
  Lean/FabiusFunction/FabiusSaddleCentralAllOrders.lean
  Lean/FabiusFunction/FabiusSaddleTailAllOrders.lean
  Lean/FabiusFunction/FabiusSaddleMassAllOrders.lean
  Lean/FabiusFunction/FabiusFullAsymptoticExpansion.lean
  docs/Fabius_Function_and_Rvachev_Up/
    Fabius_Function_and_Rvachev_Up.tex
```

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Part XI
Small-Argument Asymptotics Research Frontiers

Source provenance and status. Former path: `Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex`. Specialist explicit-coefficient dossier. Closed higher coefficients, ordered-composition formulas, Fourier-amplitude laws, numerical tables, traps, and open problems remain research targets unless an exact Lean theorem is cited.

Original source SHA-256: `85f51a20fc7b6bdf3b1d049ec4506f508aee3c4cd70554e6a68cbcc30977cb0b`.

Dossier abstract

As $x \to 0^+$ the Fabius function F is flat to infinite order. Its Taylor series at 0 is identically zero and does not represent F on any right neighborhood; the correct local description is an exponentially small asymptotic expansion in the lower-Lambert phase $\lambda(x)$, whose coefficients A_j are one-periodic functions of λ rather than constants. At the historical starting point of this dossier, those coefficients existed only as the output of a chain of recursions — a differential recursion for the saddle jets, a formal exponential recursion, a Gaussian contraction, and a formal logarithm — and only A_1 had been written down. This note closes the chain. We solve the differential recursion in closed form using shifted signed Stirling numbers of the first kind and harmonic numbers, collapse the exponent coefficients to a single power of i times a jet term plus a universal jet-free tail, and assemble a completely explicit, non-recursive formula for every A_j as a finite sum over compositions, subsets and Gaussian moments. As applications we compute A_2 and A_3 explicitly for the first time, identify the highest-derivative term of A_j as $(-1)^j \Psi^{(2j)}/(2^j j!\, L^{2j})$, deduce the Gamma–zeta Fourier form and the amplitude law of every coefficient, and verify the whole chain numerically against exact rational values of $F(2^{-n})$.

Formalization boundary. This document preserves the former small-argument draft as a research-frontier notebook. The sharp main term, recursive all-orders expansion, closed product-polynomial saddle jets, and the first two corrections have exact Lean counterparts, but several stronger claims in the pages below do not. In particular, the ordered-composition formulas, general logarithmic highest-derivative law, expanded higher coefficients, Fourier-amplitude estimates, uniqueness claims, and all numerical tables remain to be formalized. Those passages are research targets, not established results of the project.

144 Introduction

Throughout, F is the canonical bounded Fabius function on $[0, 1]$: the cumulative distribution function of

$$X = \sum_{k\geq 1} 2^{-k} U_k, \qquad U_1, U_2, \ldots \text{ independent uniform on } [0, 1].$$

It is smooth, vanishes on $(-\infty, 0]$, equals 1 on $[1, \infty)$, satisfies $F(x) + F(1 - x) = 1$, and obeys $F'(x) = 2F(2x)$ on $[0, \frac{1}{2}]$. Every derivative vanishes at the origin, so F is flat there. Its identically-zero Taylor series fails to represent F on every right neighborhood of the origin.

This note develops the behaviour of $F(x)$ as $x \to 0^+$ in detail, and in particular gives a general formula for every coefficient of its all-orders expansion. It is a companion to the formal development in `Analysis/FabiusFunction` and to the article in `docs/Fabius_Function_and_Rvachev_Up/`, and it is self-contained: everything needed to compute any coefficient by hand is stated here.

144.1 What was already known, and what is new

The development snapshot from which this dossier began already contained the corrected sharp main term $\mathcal{M}(x)$ with its genuine nonconstant periodic correction, a proof that deleting that correction makes the standard printed formula false, and an all-orders expansion

$$\log F(x) = \mathcal{M}(x) + \sum_{j\geq 1} \frac{A_j(\lambda)}{\lambda^j}$$

in which the coefficient functions A_j were *defined only by a recursion*. At that historical stage only A_1 was displayed.

What this note adds:

1. A closed form for the saddle jets (section 147). The differential recursion is solved: the n-th jet is a harmonic-number constant plus a Stirling-weighted combination of the first $n+1$ derivatives of the periodic correction. This removes the only non-algebraic recursion from the chain.
2. A closed form for the exponent coefficients (section 148), with the factorials and the spurious extra power of i cancelled.
3. A fully explicit, non-recursive formula for every coefficient (section 149): A_j is a finite triple sum over compositions, subsets and Gaussian moments.
4. The explicit coefficients A_2 and A_3 (section 150), which did not exist anywhere before, in two equivalent forms.
5. The leading-derivative law (section 151): the top term of A_j is $(-1)^j \Psi^{(2j)}(u)/(2^j j!\, L^{2j})$, which explains why the oscillation of the coefficients grows with their order.
6. The Gamma–zeta Fourier form of every coefficient and the resulting amplitude law (section 152).
7. An independent end-to-end numerical verification (section 153) against exact rational values of $F(2^{-n})$.

144.2 Notation

symbol	meaning
L	$\log 2 = 0.69314718055994530942\ldots$
γ	Euler–Mascheroni constant
γ_1	first Stieltjes constant, $\zeta(1+s) = 1/s + \gamma - \gamma_1 s + \mathcal{O}(s^2)$
H_n	n-th harmonic number, $H_0 = 0$
Ψ	the centered one-periodic correction of section 146
$\Psi^{(m)}$	m-th derivative of Ψ
$\lambda = \lambda(x)$	the lower-Lambert phase of section 145
$\mathcal{M}(x)$	the sharp main term of section 146
A_j	the j-th expansion coefficient, a one-periodic function
a_j	the j-th *mass* coefficient, before taking the logarithm
d_n	the n-th bounded exponent jet of section 147
p_n	the n-th periodic jet of section 147
s_n	$(-1)^{n+1} n!$, the jet slope
$c(n,m)$	$[z^m]\prod_{k=1}^{n}(z-k) = s(n+1, m+1)$, a shifted signed Stirling number
$(N-1)!!$	$1 \cdot 3 \cdot 5 \cdots (N-1)$, the N-th standard Gaussian moment for even N

Z denotes a standard Gaussian variable, so $\mathbb{E}[Z^N] = (N-1)!!$ for even N and 0 for odd N.

145 The lower-Lambert phase

The saddle point of the Laplace inversion sits at radius $r = 2^{\lambda}$, where λ solves

$$\lambda\, 2^{-\lambda} = x, \qquad \text{equivalently} \qquad L\lambda = -\log x + \log \lambda. \tag{1113}$$

This is the lower branch of the Lambert W function, $\lambda(x) = -W_{-1}(-Lx)/L$, defined and strictly decreasing for $0 < Lx < \mathrm{e}^{-1}$, with $\lambda(x) \to +\infty$ as $x \to 0^+$.

The expansion below is an expansion in $1/\lambda$, *not* in $1/(-\log x)$. The two agree to leading order, since $L\lambda = -\log x + \log(-\log x) - \log L + o(1)$, but they differ at every order after that; section 155 explains why substituting an expansion of λ into the coefficients is not a legitimate step.

For dyadic arguments $x = 2^{-n}$, equation equation (1113) reads $L\lambda = nL + \log \lambda$, whose solution is conveniently obtained by the iteration $\lambda \mapsto n + \log(\lambda)/L$.

146 The periodic correction and the sharp main term

Let

$$G(-s) = \prod_{j\ge 1} \frac{1 - \mathrm{e}^{-s/2^j}}{s/2^j}, \qquad \Lambda(s) = \log G(-s),$$

be the logarithm of the two-sided Laplace transform of the Fabius density. Removing the quadratic drift and the forward tail $T(s) = \sum_{m\ge 0} \log(1 - \mathrm{e}^{-s2^m})$ produces the *exactly* one-periodic function

$$P(t) = \Lambda(2^t) + \frac{L}{2}(t^2 - t) + T(2^t), \qquad P(t+1) = P(t),$$

so that

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s), \qquad E = -T, \qquad 0 \le E(s) \le \frac{\mathrm{e}^{-s}}{(1-\mathrm{e}^{-s})^2}.$$

Write $\Psi(t) = P(t) - \bar{P}$ with $\bar{P} = \int_0^1 P$. Then Ψ is one-periodic, C^∞, of mean zero, and — this is the decisive point — *not constant.* Its Fourier coefficients are

$$\widehat{\Psi}(0) = 0, \qquad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\,\zeta(1-\chi_k)}{L}, \qquad \chi_k = \frac{2\pi \mathrm{i} k}{L}, \quad k \neq 0, \tag{1114}$$

which are nonzero for every $k \neq 0$ because Γ has no zeros and ζ has none on the line $\operatorname{Re} s = 1$. Numerically

$$\begin{aligned}
\widehat{\Psi}(1) &= (\ 7.55864754098 - 7.47010355898\,\mathrm{i}\ ) \times 10^{-7}, & |\widehat{\Psi}(1)| &= 1.06271162519 \times 10^{-6},\\
\widehat{\Psi}(2) &= (\ 3.24812347723 + 5.85175898712\,\mathrm{i}\ ) \times 10^{-13}, & |\widehat{\Psi}(2)| &= 6.69278636792 \times 10^{-13},\\
\widehat{\Psi}(3) &= (-2.81236523199 - 2.58674397916\,\mathrm{i}\ ) \times 10^{-19}, & |\widehat{\Psi}(3)| &= 3.82107872359 \times 10^{-19},
\end{aligned}$$

so Ψ itself has amplitude only about 2×10^{-6} (roughly 4×10^{-6} peak to peak). Section 156 records the numerical trap that makes these values easy to get wrong.

With $A_0^{\mathrm{c}} = (6\gamma^2 + 12\gamma_1 - \pi^2)/12$ and

$$C_F = \frac{A_0^{\mathrm{c}}}{L} - \frac{7L}{12} - \frac{\log \pi}{2} = -2.0279838986388594239 71\ldots,$$

the sharp main term is

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \Big(1 + \frac{L}{2}\Big)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda), \tag{1115}$$

and $\log F(x) = \mathcal{M}(x) + \mathcal{O}(1/\lambda)$. Deleting $\Psi(\lambda)$ invalidates that error term: the ratio $F(x)/\exp(\mathcal{M}(x) - \Psi(\lambda))$ has at least two distinct subsequential limits.

147 The saddle jets and their closed form

Let $q = \log G(-\cdot\,)$ evaluated on the saddle ray $r = 2^t$. Up to the exponentially small forward tail, the $(n+1)$-st scaled derivative has the form

$$r^{\,n+1} q^{(n+1)}(r) = s_n t + p_n(t), \qquad s_n = (-1)^{n+1} n!,$$

where the periodic jets p_n are determined by the differential recursion

$$p_0(t) = \frac{1}{2} + \frac{\Psi'(t)}{L}, \qquad p_{n+1}(t) = \frac{p_n'(t)}{L} - (n+1)p_n(t) + \frac{s_n}{L}. \tag{1116}$$

The quantity that actually enters the saddle exponent is the *bounded exponent jet* $d_n = p_n + s_n$.

147.1 Solving the recursion

Write D for $\mathrm{d}/\mathrm{d}t$. The recursion equation (1116) reads $p_{n+1} = (D/L - (n+1))p_n + s_n/L$, and the operators $D/L - j$ commute, since

$$\Big(\frac{D}{L} - j\Big)\Big(\frac{D}{L} - k\Big) = \frac{D^2}{L^2} - (j+k)\frac{D}{L} + jk$$

is symmetric in j and k. Hence the homogeneous part of the solution is the falling-factorial operator $\prod_{k=1}^n (D/L - k)$ applied to p_0. Each inhomogeneous constant s_j/L is carried forward by $\prod_{k=j+2}^n (D/L - k)$, and a constant is an eigenvector of $D/L - k$ with eigenvalue $-k$, so its contribution is

$$\frac{s_j}{L}\,(-1)^{\,n-j-1}\frac{n!}{(j+1)!} = \frac{(-1)^n\,n!}{(j+1)L}.$$

Summing over $j = 0, \ldots, n-1$ produces the harmonic number. Finally $\prod_{k=1}^n (z - k) = \sum_{m=0}^n c(n,m) z^m$, so applying the operator polynomial to $p_0 = \frac{1}{2} + \Psi'/L$ gives the result.

147.2 The closed forms

Theorem 147.1 (Closed-form jets). *For every $n \geq 0$ and every t,*

$$p_n(t) = (-1)^n n!\Big(\frac{1}{2} + \frac{H_n}{L}\Big) + \sum_{m=0}^{n} c(n,m)\,\frac{\Psi^{(m+1)}(t)}{L^{m+1}},$$
$$d_n(t) = (-1)^n n!\Big(\frac{H_n}{L} - \frac{1}{2}\Big) + \sum_{m=0}^{n} c(n,m)\,\frac{\Psi^{(m+1)}(t)}{L^{m+1}},$$

where $c(n,m) = [z^m]\prod_{k=1}^{n}(z-k)$, so that $c(n,n) = 1$ and $c(n,m) = 0$ for $m > n$.

The coefficient array obeys $c(0,0) = 1$, $c(0,m+1) = 0$, $c(n+1,0) = -(n+1)c(n,0)$ and $c(n+1,m+1) = c(n,m) - (n+1)c(n,m+1)$. The first rows are:

n	$c(n,0), c(n,1), \dots$	constant part of d_n
0	1	$-1/2$
1	-1, 1	$1/2 - 1/L$
2	2, -3, 1	$-1 + 3/L$
3	-6, 11, -6, 1	$3 - 11/L$
4	24, -50, 35, -10, 1	$-12 + 50/L$
5	-120, 274, -225, 85, -15, 1	$60 - 274/L$

Warning 147.2 (The index shift). The weights $c(n,m)$ are shifted signed Stirling numbers of the first kind, not the unshifted ones. With the standard convention $x(x-1)\cdots(x-n+1) = \sum_k s(n,k)x^k$ one has

$$z\prod_{k=1}^{n}(z-k) = \prod_{k=0}^{n}(z-k) = \sum_m s(n+1,m)z^m, \qquad \text{hence} \qquad c(n,m) = s(n+1,m+1).$$

The distinction is not cosmetic: $s(n,0) = 0$ for every $n \geq 1$, whereas $c(n,0) = \prod_{k=1}^{n}(-k) = (-1)^n n!$, which is the entire constant column of the table above and carries the harmonic-number part of the jet. Reading $c(n,m)$ as $s(n,m)$ would delete it. Concretely $c(2,\cdot) = (2,-3,1)$ while $s(2,\cdot) = (0,-1,1)$, and $c(3,\cdot) = (-6,11,-6,1)$ while $s(3,\cdot) = (0,2,-3,1)$.

Explicitly, for the first four jets,

$$d_0 = \frac{\Psi'}{L} - \frac{1}{2},$$
$$d_1 = \frac{1}{2} - \frac{1}{L} - \frac{\Psi'}{L} + \frac{\Psi''}{L^2},$$
$$d_2 = -1 + \frac{3}{L} + \frac{2\Psi'}{L} - \frac{3\Psi''}{L^2} + \frac{\Psi'''}{L^3},$$
$$d_3 = 3 - \frac{11}{L} - \frac{6\Psi'}{L} + \frac{11\Psi''}{L^2} - \frac{6\Psi'''}{L^3} + \frac{\Psi''''}{L^4}.$$

Since Ψ and all of its derivatives are one-periodic, bounded and of mean zero, every jet is a bounded one-periodic function whose mean is exactly its constant part. This is the source of every "constant part" appearing below.

Remark 147.3. Theorem 147.1 is formalized as `Fabius.negativeLaplacePeriodicJet_eq_closedForm` and `Fabius.negativeLaplaceBoundedExponentJet_eq_closedForm` in `FabiusSaddleJetClosedForm.lean`.

148 The exponent coefficients in closed form

After extracting the Gaussian factor and setting $\varepsilon = \lambda^{-1/2}$, the coefficient of ε^m in the central saddle exponent is, as originally recorded,

$$E_m(t,v) = \mathrm{i}^{\,m}\, d_{m-1}(t)\frac{v^m}{m!} + \mathrm{i}^{\,m+2} s_{m+1}\frac{v^{m+2}}{(m+2)!}.$$

Both the extra power of i and both factorials in the second term cancel: $\mathrm{i}^{\,m+2} = -\mathrm{i}^{\,m}$, $s_{m+1} = (-1)^m(m+1)!$ and $(m+1)!/(m+2)! = 1/(m+2)$.

Theorem 148.1 (Closed-form exponent)**.** *$E_0 = 0$, and for every $m \geq 1$,*

$$E_m(t,v) = \mathrm{i}^{\,m}\left(d_{m-1}(t)\frac{v^m}{m!} + (-1)^{m+1}\frac{v^{m+2}}{m+2}\right).$$

So the second summand is a *universal* tail: it does not see the periodic correction at all. The first four are

$$E_1 = \frac{\mathrm{i}v(3d_0+v^2)}{3}, \quad E_2 = \frac{v^2(-2d_1+v^2)}{4}, \quad E_3 = \mathrm{i}v^3\Big(-\frac{d_2}{6} - \frac{v^2}{5}\Big), \quad E_4 = \frac{v^4(d_3-4v^2)}{24}.$$

In particular $E_m(t,-v) = (-1)^m E_m(t,v)$, which is the algebraic reason that all odd half-powers of λ disappear after Gaussian integration, so that the expansion runs over integer powers λ^{-j} only.

Remark 148.2. Theorem 148.1 is formalized as `Fabius.negativeLaplaceExponentCoefficient_succ_eq` and `Fabius.negativeLaplaceExponentPolynomial_succ_eq` in `FabiusSaddleExponentClosedForm.lean`.

149 The general coefficient formula

Write $\mathcal{C}(N,r)$ for the set of compositions of N into r positive parts, that is, ordered tuples $(m_1,\ldots,m_r)$ of positive integers with $m_1+\cdots+m_r = N$.

149.1 Step 1: the mass coefficients

The normalized saddle mass has the expansion $1+\sum_{j\geq 1} a_j(u)\lambda^{-j}$, where a_j is the Gaussian contraction of the coefficient of ε^{2j} in $\exp(\sum_{m\geq 1} E_m\varepsilon^m)$. Expanding the exponential over compositions, and then expanding the product over the two monomials of each E_{m_i}, gives the following.

Theorem 149.1 (Explicit mass coefficients)**.** *For every $j \geq 1$,*

$$a_j(u) = (-1)^j \sum_{r=1}^{2j}\frac{1}{r!}\sum_{(m_1,\ldots,m_r)\in\mathcal{C}(2j,r)}\ \sum_{S\subseteq\{1,\ldots,r\}}(2j+2|S|-1)!!\prod_{i\in S}\frac{(-1)^{m_i+1}}{m_i+2}\prod_{i\notin S}\frac{d_{m_i-1}(u)}{m_i!}.$$

Here S records, for each factor of the product, which of the two monomials of E_{m_i} was chosen: $i\in S$ picks the universal tail $v^{m_i+2}/(m_i{+}2)$, and $i\notin S$ picks the jet term $d_{m_i-1}v^{m_i}/m_i!$. The total degree is then $2j+2|S|$, always even, and its Gaussian moment is the displayed double factorial. The prefactor $(-1)^j$ is $\mathrm{i}^{\,2j}$.

149.2 Step 2: taking the logarithm

Theorem 149.2 (Explicit logarithmic coefficients)**.** *With $a_0 = 1$, for every $j \geq 1$,*

$$A_j(u) = \sum_{r=1}^{j} \frac{(-1)^{r-1}}{r} \sum_{(q_1,\ldots,q_r)\in\mathcal{C}(j,r)} a_{q_1}(u)\cdots a_{q_r}(u).$$

149.3 Step 3: expanding the jets

Substituting theorem 147.1 into theorems 149.1 and 149.2 expresses $A_j(u)$ as a finite expression with rational coefficients in $1/L$ and $\Psi'(u),\ldots,\Psi^{(2j)}(u)$, with no recursion anywhere. Both closed forms were verified symbolically against the original recursions for $j \leq 3$, with identically vanishing residuals.

For reference, the equivalent recursive presentations, which are what the Lean development uses internally, are

$$a_j = \text{Gaussian contraction of } \operatorname{expCoeff}(E)(2j),$$

$$\operatorname{expCoeff}(E)(0) = 1, \qquad (n+1)\operatorname{expCoeff}(E)(n+1) = \sum_{j\leq n}(j+1)E_{j+1}\operatorname{expCoeff}(E)(n-j),$$

$$A_0 = 0, \qquad A_n = a_n - \frac{1}{n}\sum_{k=1}^{n-1} kA_k a_{n-k}.$$

149.4 Structural consequences

- A_j involves $\Psi',\ldots,\Psi^{(2j)}$ and no higher derivative, because the highest jet reached at order $2j$ is d_{2j-1}, which reaches $\Psi^{(2j)}$.
- Every A_j is C^∞, one-periodic and bounded, being a polynomial in bounded one-periodic smooth functions.
- The mean of A_j is its "Ψ-free part" plus the mean of the terms of degree two and higher in the derivatives of Ψ. Those means do not vanish, and they are computable in closed form, because $\langle\Psi^{(a)}\Psi^{(b)}\rangle$ depends only on $a+b$ up to sign: integration by parts over a period gives $\langle\Psi'\Psi''\rangle = 0$ and $\langle\Psi'\Psi'''\rangle = -\langle(\Psi'')^2\rangle$, and generally $\langle P_aP_b\rangle = 2\sum_{k\geq1}(2\pi ik)^a(-2\pi ik)^b|\widehat{\Psi}(k)|^2$, which carries the amplification $(2\pi k)^{a+b}$ of section 152. Applying this to equation (1120) and to A_1 leaves

$$\langle A_1\rangle = \frac{1}{24} + \frac{1}{2L} - \frac{\langle(\Psi')^2\rangle}{2L^2},$$
$$\langle A_2\rangle = \frac{1}{4L^2} - \frac{\langle(\Psi')^2\rangle}{2L^3} - \frac{\langle(\Psi'')^2\rangle}{4L^4} - \frac{\langle(\Psi')^3\rangle}{6L^3},$$

every other quadratic and cubic mean cancelling. Numerically $\langle(\Psi')^2\rangle = 8.917\times10^{-11}$, $\langle(\Psi'')^2\rangle = 3.520\times10^{-9}$ and $\langle(\Psi')^3\rangle = 1.115\times10^{-21}$, so $\langle A_1\rangle = 0.7630141870184$ and $\langle A_2\rangle = 0.5203422413049$, against Ψ-free parts 0.7630141871111 and 0.5203422452514. The gaps are -9.3×10^{-11} and -3.9×10^{-9}, and at $j=3$ the same computation gives -1.14×10^{-7}: about a factor 40 per order, for the same reason the oscillation grows. What justifies calling the Ψ-free part the mean is therefore not an absolute bound but a ratio: at each order the gap sits four to five decades below the amplitude of A_j itself.

150 The explicit coefficients

It is convenient to state the coefficients first in the jets d_n and then in the derivatives of Ψ. The jet form is dramatically shorter.

150.1 Mass coefficients in the jets

$$a_1 = -\frac{6d_0^2 + 12d_0 + 6d_1 + 1}{12},$$
$$a_2 = \frac{1}{288}\big(36d_0^4 + 240d_0^3 + 216d_0^2 d_1 + 300d_0^2 + 720d_0 d_1 + 144d_0 d_2 + 24d_0 + 108d_1^2 + 300d_1 + 240d_2 + 36d_3 + 1\big).$$

150.2 Logarithmic coefficients in the jets

$$A_1 = -\frac{6d_0^2 + 12d_0 + 6d_1 + 1}{12}\ (= a_1),$$
$$A_2 = \frac{1}{24}\big(8d_0^3 + 12d_0^2 d_1 + 12d_0^2 + 48d_0 d_1 + 12d_0 d_2 + 6d_1^2 + 24d_1 + 20d_2 + 3d_3\big), \tag{1117}$$
$$A_3 = -\frac{1}{720}\big(180d_0^4 + 720d_0^3 d_1 + 120d_0^3 d_2 + 240d_0^3 + 360d_0^2 d_1^2 + 2520d_0^2 d_1 + 1440d_0^2 d_2 + 180d_0^2 d_3 + 2160d_0 d_1^2 + 720d_0 d_1 d_2 + 1440d_0 d_1 + 2520d_0 d_2 + 960d_0 d_3 + 90d_0 d_4 + 120d_1^3 + 1980d_1^2 + 1800d_1 d_2 + 180d_1 d_3 + 150d_2^2 + 720d_2 + 780d_3 + 210d_4 + 15d_5 - 2\big). \tag{1118}$$

One order further, in the same coordinates,

$$\begin{aligned} A_4 = \frac{1}{5760}\big(&1152d_0^5 + 8640d_0^4 d_1 + 2880d_0^4 d_2 + 240d_0^4 d_3 + 1440d_0^4 \\ &+ 11520d_0^3 d_1^2 + 2880d_0^3 d_1 d_2 + 28800d_0^3 d_1 + 25920d_0^3 d_2 \\ &+ 6720d_0^3 d_3 + 480d_0^3 d_4 + 2880d_0^2 d_1^3 + 63360d_0^2 d_1^2 \\ &+ 46080d_0^2 d_1 d_2 + 4320d_0^2 d_1 d_3 + 17280d_0^2 d_1 + 2880d_0^2 d_2^2 \\ &+ 46080d_0^2 d_2 + 29280d_0^2 d_3 + 6000d_0^2 d_4 + 360d_0^2 d_5 \\ &+ 23040d_0 d_1^3 + 8640d_0 d_1^2 d_2 + 74880d_0 d_1^2 + 123840d_0 d_1 d_2 \\ &+ 30720d_0 d_1 d_3 + 2160d_0 d_1 d_4 + 21600d_0 d_2^2 + 3840d_0 d_2 d_3 \\ &+ 17280d_0 d_2 + 30720d_0 d_3 + 14640d_0 d_4 + 2400d_0 d_5 + 120d_0 d_6 \\ &+ 720d_1^4 + 30720d_1^3 + 28800d_1^2 d_2 + 2160d_1^2 d_3 + 14400d_1^2 \\ &+ 3600d_1 d_2^2 + 66240d_1 d_2 + 36960d_1 d_3 + 6720d_1 d_4 + 360d_1 d_5 \\ &+ 25200d_2^2 + 12000d_2 d_3 + 840d_2 d_4 + 480d_3^2 + 5760d_3 \\ &+ 7392d_4 + 2720d_5 + 360d_6 + 15d_7\big). \end{aligned} \tag{1119}$$

Its top term is $15d_7/5760 = d_7/384$, and $(-1)^4/(2^4 \cdot 4!) = 1/384$: a fourth independent check of theorem 151.1.

150.3 The same coefficients in the derivatives of Ψ

Write $P_k = \Psi^{(k)}(u)$. Then

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{P_2 + P_1^2}{2L^2},$$
$$\begin{aligned} A_2(u) = \frac{P_1}{24L} + \frac{2P_1 + 1}{4L^2} - \frac{P_1^3 + 3P_1^2 + 3P_1 P_2 + 3P_2 + P_3}{6L^3} \\ + \frac{4P_1^2 P_2 + 4P_1 P_3 + 2P_2^2 + P_4}{8L^4}. \end{aligned} \tag{1120}$$

The corresponding form of A_3 is long. Its *linear part* is the following; the omitted terms are quadratic and higher in the derivatives of Ψ and reach 8.9×10^{-7} over a period, which is four decades below the 6.7×10^{-2} swing of A_3 but is not negligible if six digits are wanted.

$$\begin{aligned} A_3(u) = &- \frac{7}{2880} - \frac{1}{24L} - \frac{1}{4L^2} + \frac{1}{6L^3} + \frac{P_1 + P_2}{24L^2} + \frac{48P_1 + 24P_2 - P_3}{48L^3} \\ &- \frac{6P_2 + 9P_3 + P_4}{12L^4} + \frac{3P_4 + P_5}{12L^5} - \frac{P_6}{48L^6} + (\text{quadratic and higher in the } P_k). \end{aligned} \tag{1121}$$

150.4 The Ψ-free parts

j	Ψ-free part of A_j	value
1	$\frac{1}{24} + \frac{1}{2L}$	0.7630141871
2	$\frac{1}{4L^2}$	0.5203422453
3	$-\frac{7}{2880} - \frac{1}{24L} - \frac{1}{4L^2} + \frac{1}{6L^3}$	−0.0824216430
4	$\frac{-L^3 - 72L^2 - 816L + 144}{1152L^4}$	−1.7167954639

The order-two value is remarkably clean: the Ψ-free part of A_2 is exactly $1/(4\log^2 2)$, with no rational term and no $1/L$ term at all.

150.5 The resulting expansions

$$\begin{aligned} \log F(x) &= \mathcal{M}(x) + \mathcal{O}(1/\lambda), \\ \log F(x) &= \mathcal{M}(x) + \frac{A_1(\lambda)}{\lambda} + \mathcal{O}(1/\lambda^2), \\ \log F(x) &= \mathcal{M}(x) + \frac{A_1(\lambda)}{\lambda} + \frac{A_2(\lambda)}{\lambda^2} + \mathcal{O}(1/\lambda^3), \\ \log F(x) &= \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}(1/\lambda^N) \qquad \text{for every } N. \end{aligned}$$

Written out at order three, with $d_k = d_k(\lambda)$ and $\lambda = \lambda(x)$,

$$\begin{aligned} \log F(x) = &- \frac{L}{2}\lambda^2 + \Big(1 + \frac{L}{2}\Big)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) \\ &- \frac{6d_0^2 + 12d_0 + 6d_1 + 1}{12\lambda} \\ &+ \frac{8d_0^3 + 12d_0^2 d_1 + 12d_0^2 + 48d_0 d_1 + 12d_0 d_2 + 6d_1^2 + 24d_1 + 20d_2 + 3d_3}{24\lambda^2} + \mathcal{O}(\lambda^{-3}). \end{aligned}$$

The order-three statement is `log_fabius_sub_twoSaddleCorrections_isBigO` in `FabiusSecondSaddleExpansion.lean`. Before this note the formalization stopped at order two, because A_2 was not known.

150.6 The right endpoint

Nothing separate is needed at $x \to 1^-$. The symmetry $F(x) + F(1-x) = 1$, which holds on all of $\mathbb{R}$ for the bounded Fabius function, gives $1 - F(1-x) = F(x)$ identically, so the deficiency at the right endpoint is the value at the left endpoint and every expansion above transfers verbatim:

$$\log(1 - F(1-x)) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N}), \qquad x \to 0^+,$$

with the same $\mathcal{M}$, the same λ and the same coefficients. In particular $1 - F(y)$ is flat to infinite order at $y = 1$, with exactly the same periodic modulation. The reason this is worth a sentence rather than a section is that the symmetry is exact, not asymptotic; there is no error term to propagate.

151 The leading-derivative law

Theorem 151.1 (Leading term)**.** *The coefficient of d_{2j-1} in both a_j and A_j equals $(-1)^j/(2^j j!)$, so the highest-derivative term of A_j is*

$$\frac{(-1)^j\, \Psi^{(2j)}(u)}{2^j\, j!\, L^{2j}}.$$

Proof sketch. A_j reaches $\Psi^{(2j)}$ only through the jet d_{2j-1}, and d_{2j-1} occurs in the expansion only linearly, through E_{2j}. In E_{2j} the jet term is $\mathrm{i}^{2j} d_{2j-1} v^{2j}/(2j)! = (-1)^j d_{2j-1} v^{2j}/(2j)!$; contracting v^{2j} against the Gaussian gives $(2j-1)!!$; and $(2j)! = 2^j j!\,(2j-1)!!$. Nothing else at order $2j$ is linear in d_{2j-1}, and the formal logarithm does not change a linear term. Finally the top term of d_{2j-1} is $\Psi^{(2j)}/L^{2j}$ because $c(n, n) = 1$. □

Checks: $j = 1$ gives $-\Psi''/(2L^2)$, $j = 2$ gives $+\Psi''''/(8L^4)$ and $j = 3$ gives $-\Psi^{(6)}/(48L^6)$ — exactly the top terms displayed in equations (1120) and (1121).

152 Fourier form and the amplitude law

Because $\Psi^{(m)}(u) = \sum_{k\neq 0}(2\pi\mathrm{i}k)^m \widehat{\Psi}(k)\mathrm{e}^{2\pi\mathrm{i}ku}$, section 149 together with equation (1114) turns every coefficient into an explicit Gamma–zeta Fourier object. Discarding the terms of degree two and higher in the derivatives of Ψ leaves the linearized Fourier form

$$A_j(u) = \langle A_j\rangle + 2\,\mathrm{Re}\Big(\sum_{k\geq 1} c_j(k)\widehat{\Psi}(k)\mathrm{e}^{2\pi\mathrm{i}ku}\Big) + R_j(u),$$

where the discarded remainder is measured at $\max_u |R_j(u)| = 1.6 \times 10^{-8}$ for $j = 2$ and 8.9×10^{-7} for $j = 3$, growing with j like everything else here. It is negligible against the oscillation it sits inside, and it is not negligible in absolute terms; section 149 records the same caution for the means. with $c_j(k)$ the polynomial in $w = 2\pi\mathrm{i}k$ read off from equations (1120) and (1121). For the first two,

$$c_1(k) = -\frac{w^2}{2L^2}, \qquad c_2(k) = \frac{w}{24L} + \frac{w}{2L^2} - \frac{3w^2 + w^3}{6L^3} + \frac{w^4}{8L^4}.$$

Consequences for A_2, all confirmed numerically in section 153. For the first Fourier mode, the linearized amplitude is $2|c_2(1)\widehat{\Psi}(1)|$. The left column is what the linearized form gives, the right what the exact A_2 gives; they are quoted separately because the difference is real, if small, and quoting ten digits of a linearization would be an overclaim.

mean	$\frac{1}{4L^2} = 0.5203422452514$	$\langle A_2\rangle = 0.5203422413049$
$k = 1$ amplitude	1.9398782×10^{-3}	1.9398782×10^{-3}
maximum at	$\{\lambda\} = 0.1011297$	$\{\lambda\} = 0.1011281$
minimum at	$\{\lambda\} = 0.6011297$	$\{\lambda\} = 0.6011313$

The exact peak-to-peak is 3.8797564×10^{-3}, twice the linearized $k = 1$ amplitude to the displayed precision, and the extrema agree to six. The maximum of the exact A_2 is 0.522282117 and its minimum is 0.518402361.

152.1 The amplitude law

The top term of theorem 151.1 dominates the oscillation, and it multiplies the k-th Fourier mode by $(2\pi k)^{2j}$. Taking $k = 1$,

$$\text{amplitude of } A_j \;\approx\; 2|\widehat{\Psi}(1)|\, \frac{(2\pi^2/L^2)^j}{j!}, \qquad \frac{2\pi^2}{L^2} = 41.0845769104. \tag{1122}$$

j	predicted $k = 1$ amplitude
1	8.73×10^{-5}
2	1.79×10^{-3}
3	2.46×10^{-2}
4	2.52×10^{-1}
5	2.07
6	1.42×10^{1}
7	8.33×10^{1}

These are estimates from the top term alone. Comparing them with the truth requires care about a convention: an *amplitude* is half a peak-to-peak swing, and mixing the two silently doubles every second entry. Everything below is an amplitude.

j	equation (1122)	actual	ratio
1	8.73×10^{-5}	8.73×10^{-5}	1.00
2	1.79×10^{-3}	1.94×10^{-3}	1.08
3	2.46×10^{-2}	$\approx 3 \times 10^{-2}$	≈ 1.2
4	2.52×10^{-1}	≈ 0.41	≈ 1.6

At $j = 1$ the top derivative term dominates to the displayed precision, but it is not literally the only periodic term: A_1 also contains $-(\Psi')^2/(2L^2)$. The equality in the table is therefore a very accurate linearization, not an exact amplitude identity. At $j = 2$ the w^3 term of c_2 adds in phase and lifts the amplitude above the estimate; the entries at $j = 3$ and $j = 4$ are halves of the swings measured in section 153. The ratio drifts upward because the subleading terms of c_j contribute more as j grows. So equation (1122) gives the growth rate and not the constant, and should never be used to size a particular coefficient. So the periodic parts of the coefficients grow rapidly with j, even though Ψ itself has amplitude only about 2×10^{-6} (roughly 4×10^{-6} peak to peak): differentiation progressively cancels the smallness of $\widehat{\Psi}(1)$. Any numerical extraction of a coefficient beyond the first must therefore control the phase; see section 156.

152.2 Why the expansion is only asymptotic

$|\widehat{\Psi}(k)|$ decays like $\exp(-\pi|\chi_k|/2) = \exp(-\pi^2 k/L)$, that is, about $e^{-14.24k}$, while mode k is amplified by $(2\pi k)^{2j}/(2^j j!\, L^{2j})$. The k-th contribution therefore behaves like $\exp(2j \log k - 14.24k)$, maximized near $k^* = 2j/14.24 = 0.14j$. For j beyond about 8 the higher modes dominate, and $\sup |A_j|$ grows faster than geometrically.

What is worth noticing is *which* divergence this is. A saddle-point expansion normally diverges factorially, because the Gaussian moments grow like $(2j-1)!!$ against a fixed exponent.

Here that growth is exactly cancelled: the $(2j)!$ in E_{2j} absorbs it, which is the content of theorem 151.1 and the reason the coefficient of the top jet is the mild $1/(2^j j!)$. The divergence that remains is of a different origin altogether — it is driven by the high-frequency content of Ψ, through the $(2\pi k)^{2j}$ that differentiation applies to the k-th mode. A reader who expects the usual factorial mechanism will look for it in the Gaussian integration, where it is not. (This framing is due to the parallel write-up of the main article, where it also appears.)

Remark 152.1. The expansion is a Poincaré asymptotic expansion: each truncation is valid with an $\mathcal{O}(\lambda^{-N})$ remainder for fixed N, and there is no claim, here or in the formalization, that the series converges for any fixed x. The estimate in this subsection is a heuristic size estimate, not a theorem.

153 Numerical verification

$F(2^{-n})$ is computed *exactly* as a rational number from the inverse-power recursion used by the Lean evaluator (`Arithmetic.lean`, `fabiusInversePowTwoTable`):

$$v_0 = 1, \qquad v_{n+1} = \frac{1}{2^{n+1}-1} \sum_{k=0}^{n} \frac{v_k}{2^{\binom{n+1}{2}-\binom{k}{2}}\,(n-k+2)!}, \qquad F(2^{-n}) = v_n.$$

Sanity checks: $v_1 = 1/2 = F(1/2)$ and $v_2 = 5/72 = F(1/4)$. All computations below use 260-digit working precision, with Ψ and its derivatives summed over six Gamma–zeta modes and ζ evaluated by Euler–Maclaurin (section 156).

Define the successive residuals

$$r_1 = \lambda(\log F - \mathcal{M}), \quad r_2 = \lambda^2\Big(\log F - \mathcal{M} - \frac{A_1}{\lambda}\Big), \quad r_3 = \lambda^3\Big(\log F - \mathcal{M} - \frac{A_1}{\lambda} - \frac{A_2}{\lambda^2}\Big).$$

n	λ	r_1	$A_1(\lambda)$	r_2	$A_2(\lambda)$	r_3	$A_3(\lambda)$
10	13.78503056	0.7997155875	0.7629678468	0.5065687288	0.5195595404	−0.1790787344	−0.0868938679
20	24.62186833	0.7836542295	0.7629268731	0.5103462408	0.5184188039	−0.1987615843	−0.1111875606
40	45.50804986	0.7742689337	0.7629490545	0.5151456258	0.5187247759	−0.1628801414	−0.1121601297
60	66.04538587	0.7709808331	0.7630910552	0.5210834261	0.5221643489	−0.0713899656	−0.0512594569
80	86.43351899	0.7689739453	0.7629823191	0.5178773369	0.5193822939	−0.1300787245	−0.1046130809
120	126.9885547	0.7671769004	0.7630717251	0.5213102874	0.5218167515	−0.0643151389	−0.0540815552
160	167.3870441	0.7661091776	0.7630070725	0.5192522062	0.5199082072	−0.1098060709	−0.0973365576

The decisive test is that each residual, minus its predicted coefficient, is $\mathcal{O}(1/\lambda)$. Writing $e_j = r_j - A_j$:

n	$\{\lambda\}$	λe_1	λe_2	λe_3
100	0.737929	0.51802529	−0.11213991	−1.7523898
112	0.893526	0.52019699	−0.07798492	−1.3773605
124	0.033795	0.52164121	−0.06156984	−1.3168268
136	0.161500	0.52166737	−0.06826635	−1.5329675
148	0.278716	0.52062406	−0.08861040	−1.8452719
160	0.387044	0.51925221	−0.10980607	−2.0872350

λe_1 reproduces A_2 with mean 0.5203; λe_2 reproduces A_3 with mean −0.0824; and λe_3 is bounded, locating A_4 between about −1.3 and −2.1. Each column stays bounded and oscillates with the phase rather than drifting, which is precisely the assertion that the corresponding coefficient is correct and the next one is $\mathcal{O}(1)$.

Pointwise, A_2 is confirmed at the 10^{-5} level once the next order is included. For $140 \le n \le 160$:

n	$\{\lambda\}$	$A_2(\lambda)$	r_2	$r_2 - A_3(\lambda)/\lambda$
140	0.201650	0.5219078895	0.5214039042	0.5218323910
145	0.250301	0.5214906179	0.5209456288	0.5214143255
150	0.297351	0.5209853273	0.5203979284	0.5209087825
155	0.342900	0.5204425007	0.5198167900	0.5203665207
160	0.387044	0.5199082072	0.5192522062	0.5198337121

The last two columns agree to about 7.5×10^{-5}, which is A_4/λ^2 with $A_4 \approx -2$. Note that the *variation* of A_2 over this window is 2.0×10^{-3}, twenty-six times the residual, so this is a genuine pointwise verification of the periodic shape and not merely of a mean.

Remark 153.1. An independent high-precision computation performed in a separate worktree, using half-moments to $n = 400$ at 400-digit precision and a different route to λ, located the minimum of the order-λ^{-2} residual at $\{\lambda\} = 0.602$ and measured a peak-to-peak spread of about 3×10^{-3}; equation (1120) predicts a minimum at 0.6011313 and a peak-to-peak of 3.8798×10^{-3} over the full period. That is a phase prediction, not a magnitude fit, and it was made before the comparison.

153.1 The fourth coefficient

The same apparatus confirms equation (1119). Writing $r_4 = \lambda^4(\log F - \mathcal{M} - A_1/\lambda - A_2/\lambda^2 - A_3/\lambda^3)$, at 300-digit precision with eight Gamma–zeta modes:

n	$\{\lambda\}$	r_4	$A_4(\lambda)$
120	0.988555	−1.299548008	−1.305123624
136	0.161500	−1.532967481	−1.506670619
152	0.315745	−1.939809221	−1.890185255
160	0.387044	−2.087234974	−2.035031382
176	0.519792	−2.166743696	−2.124617975
192	0.641266	−1.992935176	−1.970225155
200	0.698346	−1.853347748	−1.840001848

$\lambda(r_4 - A_4)$ stays inside $[-8.8, 0.8]$ over $120 \le n \le 200$ with no drift, which places A_5 and confirms A_4. The Ψ-free part -1.7167954639 sits inside the measured range, and the swing of the A_4 column, 0.82 peak to peak, is 1.6 times the leading-term estimate 0.50 of equation (1122) — the same upward drift of that ratio seen at $j = 2$ (1.08) and $j = 3$ (1.24), because the subleading terms of $c_j(1)$ contribute more as j grows. Equation (1122) is a leading-term estimate and should be read as one.

153.2 A worked example

It is worth seeing the expansion used once, on a value that can be written down exactly. At $x = 2^{-10}$ the evaluator gives

$$F(2^{-10}) = \frac{134\,926\,369}{1\,246\,394\,851\,358\,539\,387\,238\,350\,848\,000},$$
$$\log F(2^{-10}) = -50.57756828234216081254\ldots.$$

and $\lambda = 13.7850305606$, so $\{\lambda\} = 0.785031$. The successive truncations miss $\log F$ by

truncation	error at $n = 10$	error at $n = 160$
$\mathcal{M}(x)$	5.80×10^{-2}	4.58×10^{-3}
$+\,A_1/\lambda$	2.67×10^{-3}	1.85×10^{-5}
$+\,A_2/\lambda^2$	-6.84×10^{-5}	-2.34×10^{-8}
$+\,A_3/\lambda^3$	-3.52×10^{-5}	-2.66×10^{-9}

where $n = 160$ has $\lambda = 167.387044$ and $\log F(2^{-160}) = -9489.629874649676879065\ldots$

Two things are visible. At $n = 160$ each order gains roughly a factor λ, as it should: 250, then 790. The last gain is only 8.8, and that is not a defect — the third-order error is already dominated by A_4/λ^4, which equation (1119) puts at $-1.72/167.387^4 = -2.19 \times 10^{-9}$ against a measured -2.66×10^{-9}. So the table is measuring A_4, not a failure.

At $n = 10$, by contrast, the fourth term barely helps: the error falls only from 6.8×10^{-5} to 3.5×10^{-5}. With $\lambda = 13.8$ and $A_4 \approx -1.7$, the term A_4/λ^4 is itself 4.7×10^{-5}, comparable to the error it is meant to reduce. This is the ordinary behaviour of a Poincaré expansion near the end of its useful range, and it is the practical reason section 152 declines to claim convergence: the coefficients grow, so for a fixed x there is a best truncation, and going past it makes the approximation worse. At $x = 2^{-10}$ that optimum is already at hand around the third term.

153.3 An independent confirmation to $n = 400$

The following was produced in a separate worktree, by a different agent, from a different representation of F, and is reproduced with its permission. Half-moments were computed to $n = 400$ at 400-digit working precision from

$$d_n = \frac{1}{(n+1)(2^n - 1)} \sum_{k<n} \binom{n+1}{k} d_k,$$

whose terms are all positive, so there is no cancellation; the values agree with the exact rationals to 400 digits at $n = 5, 20, 60$. Then $\log F(2^{-n}) = \log d_n - \log n! - \binom{n}{2} \log 2$, with λ from the lower Lambert branch (satisfying $\lambda 2^{-\lambda} = x$ to 10^{-21}) and Ψ from ten Gamma–zeta modes. Nothing in this table shares an implementation with section 153.

n	λ	$\{\lambda\}$	r_2 measured	$A_2(\lambda)$	r_3
100	106.73793	0.7379	0.51802529	0.51907590	−0.11214
125	132.04488	0.0449	0.52169792	0.52216224	−0.06131
150	157.29735	0.2974	0.52039793	0.52098533	−0.09240
175	182.51185	0.5119	0.51801879	0.51869969	−0.12427
200	207.69835	0.6984	0.51821775	0.51875313	−0.11120
225	232.86334	0.8633	0.52015631	0.52049092	−0.07792
250	258.01129	0.0113	0.52175813	0.52198119	−0.05755
275	283.14540	0.1454	0.52199208	0.52220755	−0.06101
300	308.26804	0.2680	0.52104845	0.52130958	−0.08050
325	333.38103	0.3810	0.51967269	0.51997997	−0.10244
350	358.48577	0.4858	0.51856496	0.51889001	−0.11653
375	383.58340	0.5834	0.51810490	0.51841439	−0.11872
400	408.67481	0.6748	0.51833670	0.51860654	−0.11028

The decisive column is the last. Over this range λ quadruples, so an error of even 10^{-4} in A_2 would appear as a drift in r_3 of order 3×10^{-2} with a definite sign. Instead r_3 is -0.112 at $n = 100$ and -0.110 at $n = 400$, never leaving $[-0.125, -0.058]$ in between, and oscillating rather than drifting — which is what a periodic A_3 requires. Its sample mean over the thirteen rows is about -0.088, against the Ψ-free part -0.0824216430 of A_3; the sample is not a uniform phase average and carries an A_4/λ term, so the gap is expected.

The $A_2(\lambda)$ column also straddles the predicted mean $1/(4L^2) = 0.5203422452514019$, and the row-by-row differences $r_2 - A_2$ match the prediction $A_3(\lambda)/\lambda$ to about 10^{-6}: at $\{\lambda\} = 0.2680$ the predicted difference is -2.62×10^{-4} against a measured -2.611×10^{-4}, and at $\{\lambda\} = 0.6748$, -2.71×10^{-4} against -2.698×10^{-4}.

Neither this table nor section 153 is machine-checked. What is machine-checked is listed in section 154.

154 Map to the Lean development

object	Lean name	module
$\lambda(x)$	`fabiusLambertPhase`	`FabiusLambertPhase.lean`
P	`negativeLaplacePeriodicCorrection`	`PeriodicCorrection.lean`
Ψ	`negativeLaplacePsi`	`PeriodicCorrection.lean`
$\widehat{\Psi}(k)$	Fourier-coefficient theorems	`PeriodicFourier.lean`
C_F	`fabiusSharpAsymptoticConstant`	`FabiusSharpConstant.lean`
$\mathcal{M}(x)$	`fabiusSharpLambertMain`	`FabiusSharpLambertMain.lean`
s_n	`negativeLaplaceJetSlope`	`FabiusSaddleCoefficientRecurrence.lean`
p_n	`negativeLaplacePeriodicJet`	`FabiusSaddleCoefficientRecurrence.lean`
d_n	`negativeLaplaceBoundedExponentJet`	`FabiusSaddleCoefficientRecurrence.lean`
$c(n,m)$	`negativeLaplaceJetStirling`	**`FabiusSaddleJetClosedForm.lean`**
$\Psi^{(m+1)}/L^{m+1}$	`negativeLaplacePsiScaledDeriv`	**`FabiusSaddleJetClosedForm.lean`**
closed-form p_n	`negativeLaplacePeriodicJet_eq_closedForm`	**`FabiusSaddleJetClosedForm.lean`**
closed-form d_n	`negativeLaplaceBoundedExponentJet_eq_closedForm`	**`FabiusSaddleJetClosedForm.lean`**
E_m	`negativeLaplaceExponentCoefficient`	`FabiusSaddleCoefficientRecurrence.lean`
closed-form E_m	`negativeLaplaceExponentCoefficient_succ_eq`	**`FabiusSaddleExponentClosedForm.lean`**
expCoeff	`SaddleExpansion.expCoeff`	`SaddleExpansionAlgebra.lean`
contraction	`SaddleExpansion.gaussianPolynomialContraction`	`GaussianPolynomialContraction.lean`
a_j	`fabiusSaddleMassCoefficient`	`FabiusSaddleExpansionCoefficients.lean`
logCoeff	`SaddleExpansion.logCoeff`	`SaddleLogExpansionAlgebra.lean`
A_j	`fabiusSaddleLogCoefficient`	`FabiusSaddleExpansionCoefficients.lean`
A_1	`fabiusFirstSaddleCorrection`	`FabiusFirstSaddleCorrection.lean`
A_2	`fabiusSecondSaddleCorrection`	**`FabiusSecondSaddleCorrection.lean`**
a_2	`fabiusSaddleMassCoefficient_two`	**`FabiusSecondSaddleCorrection.lean`**
$c(n,m)$ identified	`negativeLaplaceJetStirling_eq_coeff`	**`FabiusSaddleJetStirling.lean`**
expCoeff at $3,4$	`expCoeff_three`, `expCoeff_four`	**`FabiusSecondSaddleCorrection.lean`**
order-three expansion	`log_fabius_sub_twoSaddleCorrections_isBigO`	**`FabiusSecondSaddleExpansion.lean`**
the expansion	`log_fabius_sub_sharpLambertMain_hasAsymptoticExpansion`	`FabiusFullAsymptoticExpansion.lean`
truncation	`log_fabius_sub_sharpLambertExpansion_isBigO`	`FabiusFullAsymptoticExpansion.lean`
exact $F(2^{-n})$	`fabiusInversePowTwoTable`, `evalFabiusDyadic`	`Arithmetic.lean`

Modules in bold are the ones added with this document.

154.1 What is machine-checked, and what is not

The three kinds of evidence in this note are not interchangeable, so they are separated here.

Machine-checked. The following modules compile under Lean 4.32.0 with Mathlib `v4.32.0`, with no `sorry`, no declared axiom and no `opaque`, on top of a serialized topological build of their whole import closure:

- `FabiusSaddleJetClosedForm.lean`
- `FabiusSaddleExponentClosedForm.lean`
- `FabiusSaddleJetStirling.lean`
- `FabiusSaddleLeadingCoefficient.lean`
- `FabiusSecondSaddleCorrection.lean`
- `FabiusSecondSaddleExpansion.lean`

Verified symbolically but not formalized. Theorems 149.1 and 149.2, the composition formulas, were checked against the original recursions in SYMPY for $j \le 3$, with residuals simplifying to identically zero. The coefficients A_3 and A_4 of equations (1118) and (1119) were derived by the same pipeline, which reproduces A_1 exactly against the independently formalized `fabiusFirstSaddleCorrection`. That agreement is the pipeline's calibration, not a proof of A_3 or A_4.

Verified numerically only. Everything in section 153. High-precision agreement between an exact rational evaluation of F and a closed form is strong evidence and is not a proof; in particular it cannot distinguish a correct coefficient from one differing by a quantity below the noise floor of the sampled range.

No claim in this note rests on the numerics alone except the location of A_5, which is stated only as a bound on an observed residual.

155 What is not claimed

- **No elementary all-orders form.** The coefficients are evaluated at the exact phase $\lambda(x)$, not at a truncation of it. The phase itself has a separate all-orders expansion in $S = -\log x$ and $z = \log S - \log L$,

$$L\lambda = S + z + \sum_{j<N} \frac{p_j(z)}{S^j} + \mathcal{O}\left(\frac{(1+|z|)^N}{S^N}\right), \quad p_1 = z, \quad p_2 = z - \frac{z^2}{2}, \quad p_3 = z - \frac{3z^2}{2} + \frac{z^3}{3},$$

  with the closed Stirling form $p_j(z) = \sum_{m=1}^{j} \frac{s(j,j-m+1)}{m!} z^m$. But $\mathcal{M}(x)$ contains $-\frac{L}{2}\lambda^2$, so an error δ in λ produces an error of order $L\lambda\delta$ in $\log F$; and the periodic coefficients are evaluated at λ, where a truncation error moves the phase. Composing the two expansions therefore needs its own theorem, which is deliberately not asserted.

- **No convergence**; see section 152.

- **No effective constants.** The $\mathcal{O}(\lambda^{-N})$ remainders are proved, but their implied constants are not surfaced.

- **No claim that any A_j is nowhere zero.** A_1, A_2 and A_3 are identified; the theorems identify the coefficient functions, they do not assert that these functions have no zeros.

156 Two traps

Warning 156.1 (The phase trap). Sampling $n = 10, 20, 40, 80, 160, 320$ to estimate a coefficient is invalid. Doubling n adds about 1 to $\log_2 \lambda$, so $\{\lambda\}$ returns to nearly the same place and a geometric sample holds the phase almost fixed. Since every coefficient is one-periodic, such a sample cannot see the oscillation at all; it shows only the $\mathcal{O}(1/\lambda)$ drift in magnitude, and looks like convergence to a constant. This mistake was made independently on this project before it was caught. Always sweep n densely and report against $\{\lambda\}$, as in section 153.

Warning 156.2 (The zeta trap). Many ζ implementations route through the alternating Dirichlet eta series, dividing by $1 - 2^{1-s}$. At $s = 1 - \chi_k$ one has $2^{1-s} = e^{2\pi i k} = 1$ exactly, so that prefactor has a pole at precisely the arguments equation (1114) needs. The singularity is removable in ζ but not in that formula, and the result is either an overflow or a plausible wrong number. Concretely, `mpmath.zeta` at these arguments returns values whose seventh significant digit drifts with the working precision:

$$\begin{aligned}
\texttt{mp.dps} = 30: &\quad \widehat{\Psi}(1) = 7.55865005 \times 10^{-7} - 7.47011001 \times 10^{-7}\,\mathrm{i},\\
\texttt{mp.dps} = 50: &\quad \widehat{\Psi}(1) = 7.55864674 \times 10^{-7} - 7.47009636 \times 10^{-7}\,\mathrm{i},\\
\texttt{mp.dps} = 80: &\quad \widehat{\Psi}(1) = 7.55865060 \times 10^{-7} - 7.47008702 \times 10^{-7}\,\mathrm{i},
\end{aligned}$$

whereas a hand-rolled Euler–Maclaurin ζ ($N = 300$, 25 Bernoulli terms) is precision-stable and returns $7.55864754097769 \times 10^{-7} - 7.47010355898008 \times 10^{-7}\,\mathrm{i}$ at every working precision. Use Euler–Maclaurin or Hurwitz, and always check at two precisions.

157 Notation crosswalk

Three namings for the same objects appear across this project.

this note / the article	Wikipedia draft	Lean
$\lambda(x)$	λ	`fabiusLambertPhase x`
Ψ	Ψ	`negativeLaplacePsi`
$\mathcal{M}(x)$	main term	`fabiusSharpLambertMain x`
A_j	P_j	`fabiusSaddleLogCoefficient j`
a_j	b_j	`fabiusSaddleMassCoefficient j`
A_1	P_1	`fabiusFirstSaddleCorrection`
A_2	P_2	`fabiusSecondSaddleCorrection`
C_F	C_F	`fabiusSharpAsymptoticConstant`

The Wikipedia draft calls the coefficients merely continuous; they are in fact C^∞, one-periodic and bounded, and section 149 shows that they are polynomials in smooth periodic functions.

The jets and the exponent coefficients have no counterpart in the Wikipedia draft, and in the main article they carry different letters, because p, d and E are all taken there: p_j is the Lambert phase polynomial and p_m the binary prefix, d_n is the half moment in some sixty places, and E is both the tail error and the exponential quotient. The article's names are:

this note	the main article	Lean
p_n	Z_n	`negativeLaplacePeriodicJet`
d_n	$\widetilde{Z}_n$	`negativeLaplaceBoundedExponentJet`
E_m	Ξ_m	`negativeLaplaceExponentCoefficient`
$\mathrm{d}/\mathrm{d}t$	D	—

The tilde in $\widetilde{Z}_n$ is apt: the two jet families differ by the single constant $(-1)^{n+1}n!$.

158 Open items

1. Formalize theorems 149.1 and 149.2 for the generic expCoeff and logCoeff recursions. This is the one part of the general formula that remains unformalized, so it is worth recording what it would actually take. Mathlib has `Composition n` with a `Fintype` instance, but no first-block decomposition equivalence $\mathcal{C}(n+1) \simeq \coprod_{b=1}^{n+1} \mathcal{C}(n+1-b)$, which is what an induction against the recursion needs.

   The route that avoids `Composition` altogether is to work with the k-fold convolution powers E^{*k} directly, as plain finite sums, and prove $\mathrm{expCoeff}(E)_n = \sum_{k \le n} (k!)^{-1} (E^{*k})_n$ when $E_0 = 0$. The pleasant part is that the product rule is a triviality at the level of coefficients: writing $(\mathrm{D}A)_n = nA_n$, $\mathrm{D}(A * B) = (\mathrm{D}A) * B + A * (\mathrm{D}B)$ is just $n = j + (n-j)$ inside the convolution sum. The unpleasant part is that promoting it to $\mathrm{D}(E^{*(k+1)}) = (k+1)\, E^{*k} * \mathrm{D}E$ needs associativity of convolution, which is a double-sum reindexing that Mathlib does not state for bare sequences either. Either ingredient is a self-contained piece of `Finset` combinatorics; neither is deep, and both are larger than the theorem they support.

2. Formalize theorem 151.1 for general j.

3. Surface effective constants in the $\mathcal{O}(\lambda^{-N})$ remainders.

4. Prove or refute convergence of the series in section 152.

5. Formalize or independently certify the displayed A_3 and A_4 formulas and their numerical checks. The order-six expCoeff, jets $d_0, \ldots, d_5$, and Gaussian moments through v^{18} described above are the data used for A_3, not a recipe for an as-yet uncomputed A_4.

Consolidated promotion rule

When a frontier claim is formalized, promotion is claim-by-claim rather than dossier-by-dossier. The required evidence is an exact current Lean declaration whose hypotheses, normalization, domain, and conclusion match the mathematical sentence. The primary exposition should cite that name and module, while this volume should remove or relabel the discharged obligation without erasing any still-unformalized surrounding argument. Until that audit is complete, the claim remains here.